Board of Studies
B. Tech Data Sciences
Curriculum Structure
Applicable from Academic Year 2024-25

Intake: 120
Duration: 4 Year

Eligibility:
H.S.C (10+2) Science with Mathematics

I. PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Graduates can:

PEO1. Utilize their proficiencies in the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volumes of data.
PEO2. Advance their technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3. Think logically, pursue lifelong learning and collaborate with an ethical attitude in a
Multidisciplinary team.
PEO4. Design and model AI based solutions to critical problem domains in the real world.
PEO5. Exhibit innovative thoughts and creative ideas for effective contribution towards economy building.

II. PROGRAM OUTCOMES (POs)

PO# Graduate Attribute

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex
engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate
consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4 Conduct investigations of complex problems: Use research-based knowledge
and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6 The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 Individual and team work: Function effectively as an individual, and as a member or
leader in diverse teams, and in multidisciplinary settings.

PO10 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 Project management and finance: Demonstrate knowledge and understanding of the
engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

III. PROGRAM SPECIFIC OUTCOMES (PSOs)

Graduates should be able to:

PSO1 Efficient AI-based Processes: Evolve AI-based efficient domain-specific processes for effective decision making in several domains such as business and governance domains.

PSO2 Actionable Insights: Arrive at actionable foresight, insight, and hindsight from data for solving business and engineering problems.

PSO3 Theoretical and Practical Knowledge: Create, select, and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.

PSO4 Data Analytics and Visualization: Develop data analytics and data visualization skills, skills pertaining to knowledge acquisition, knowledge representation, and knowledge engineering, and hence be capable of coordinating complex projects.

PSO5 Fundamental Research: Able to carry out fundamental research to cater the critical needs of the society through cutting-edge technologies of AI.

Total Credits per Semester: 22

- Courses like **Mathematical Foundation, Engineering Physics, Basics of Electronics, and Programming** typically carry 4 credits each (3 L + 1 P or T).
- **Workshops, Indian Knowledge Systems, and Professional Communication Skills** generally carry 2 credits (either 2 L or 2 P).

Assessment and Exam Pattern:

1. **Internal Assessment (IA)**: This component accounts for **25 marks**. It typically includes:
   - **Attendance** and participation.
   - **Assignments** and **case studies**.
   - **Mini projects or presentations**.
2. **Mid-Semester Exam (MSE)**: A **mid-semester exam** contributes another **25 marks**. It evaluates students' understanding of the course at the halfway point.
3. **End-Semester Exam (ESE)**: The **end-semester exam** is worth **100 marks**, making it the most significant portion of the assessment. It is conducted at the end of the semester and tests the entire syllabus through written exams, practical's, or project presentations.

L: *Lecture,* ***T:*** *Tutorial,* ***P:*** *Practical,* ***Cr:*** *Credits*
ETE: *End Term Exam,* ***IA:*** *Internal Assessment*

Acronym of Course Type*	Course Type
BSC	Basic Science Course
ESC	Engineering Science Course
PCC	Program Core Course
PEC	Program Elective Course
MDM	Multidisciplinary Minor
OEC	Open Elective
VSEC	Vocational & Skill Enhancement Course
HSSM	Humanities, Social Sciences & Management
ELC	Experiential Learning Courses
LLC	Liberal Learning Courses
VSC	Vocational Skill Course
VEC	Value Education Course
MDM	Multidisciplinary Minor
SEC	Skill Enhancement Course
AEC	Ability Enhancement Course
VEC	Value Education Course
VAC	Value Added Course
CC	Co-Curricular Activities
IKS	Indian Knowledge System
OJT	On Job Training
RP	Research Project
CEP	Community Engagement Program
RM	Research Methodology
OE	Open Elective

Acronym of courses as per NEP Vertivcals and AICTE Domains

Acronym of Course Type*	Scheme Type
TH	Theory
PR	Practical
OR	Oral
TW	Term Work
SL	Self-Learning
CIE	Cumulative Internal Examination
SEE	Semester End Examination
L	Lecture
T	Tutorial
P	Practical

I YEAR SEMESTER-1

Level: 6.0														
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – I														
BSC	BTDS-101-BSC	Mathematical Foundation - I	3	1	-	50	50	-	-	-	3	1	-	4
BSC	BTDS-102-BSC	Engineering Physics	3	1	-	50	50	-	-	-	3	1	-	4
ESC	BTDS-103-ESC	Basics of Electronics	3	1	-	50	50	-	-	-	3	1	-	4
ESC	BTDS-104-ESC	Fundamentals of Programming Language	3	-	2	50	50	-	-	-	3	-	1	4
VSEC	BTDS-105-VSEC	Professional Communication Skills-I	2	-	-	25	25	-	-	-	2	-	-	2
AEC	BTDS-106-AEC	IT Workshop	-	-	4	-	-	-	25	25	-	-	2	2
Co-curricular	BTDS-107-CC	NCC/State/National/International Achievement/MOOC Certification			2				25	25			2	2
Total Hours/Week, Marks, Credits			**14**	**3**	**8**	**225**	**225**	**0**	**50**	**50**	**14**	**3**	**5**	**22**
TOTAL			**25 Hours/Week**			**550 Marks**					**22 Credits**			

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I YEAR SEMESTER-2

Level: 6.0														
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – II														
BSC	BTDS-111-BSC	Mathematical Foundation – II	3	1	-	50	50		-	-	3	1	-	4
BSC	BTDS-112-BSC	Applied Statistics*	3	-	2	50	50		-	-	3	-	1	4
BSC	BTDS-113-BSC	Computational Biology	2	-	-	25	25		-	-	2	-	-	2
ESC	BTDS-114-ESC	Discrete Mathematics	3	1	-	50	50		-	-	3	1	-	4
PCC	BTDS-115-PCC	Programming for Problem Solving *	-	-	4	-	-		25	25	-	-	2	2
AEC	BTDS-116-AEC	Indian Knowledge System	2	-	-	25	25		-	-	2	-	-	2
VSEC	BTDS-117-VSEC	Professional Communication Skills-II	2		-	25	25		-	-	2			2
Co-curricular	BTDS-118-CC	NCC/State/National/International Achievement/MOOC Certification			2				25	25			2	2
Total Hours/Week, Marks, Credits			**15**	**2**	**8**	**225**	**225**	**0**	**50**	**50**	**15**	**2**	**5**	**22**
TOTAL			**25 Hours/Week**			**550 Marks**					**22 Credits**			

**L: Lecture, T: Tutorial, P: Practical, CCE: Comprehensive Continuous Evaluation, ESE: End Sem Exam, TW: Term Work, O: Oral*

II YEAR SEMESTER-3

Level: 6.0														
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – III														
PCC	BTDS-201-PCC	Data Structures	3	-	2	50	50		-	-	3	-	1	4
PCC	BTDS-202-PCC	Object Oriented Programming	3	-	2	50	50		-	-	3	-	1	4
MDM	BTDS-203-MDM	Basket 1/Basket 2/ Basket 3	2	-	-	25	25		-	-	2	-	-	2
VEC	BTDS-204-VEC	Ethics in Data Science	2	-	-	25	25		-	-	2	-	-	2
HSSM	BTDS-205-HSSM	Business Economics & Financial Analysis	2	-	-	25	25		-	-	2	-	-	2
PCC	BTDS-206-PCC	Data Visualization	0	-	4	-	-	-	25	25	-	-	2	2
ELC	BTDS-207-ELC	Real-time Research Project/Field Based Research Project including Seminar	0		4	-	-	-	25	25	-	-	2	2
OE	BTDS-208-OE	Open Elective - I	3	-	2	50	50	-	-	-	3	-	1	4
Total Hours/Week, Marks, Credits			**15**	**0**	**14**	**225**	**225**	**0**	**50**	**50**	**15**	**0**	**7**	**22**
TOTAL			**29 Hours/Week**			**550 Marks**					**22 Credits**			

**L: Lecture, T: Tutorial, P: Practical, CCE: Comprehensive Continuous Evaluation, ESE: End Sem Exam, TW: Term Work, O: Oral*

MDM:

Sr. No	Course Code	Course Title
1	BTDS-203-MDM(A)	Introduction to Blockchain Technology
2	BTDS-203-MDM(B)	Environmental Science and Sustainability
3	BTDS-203-MDM(C)	Introduction to Imag Processing

Open Elective – I

Sr. No	Course Code	Course Title
1	BTDS-208-OE(A)	Foreign Language
2	BTDS-208-OE(B)	Cost Management of Enigineering Projects
3	BTDS-208-OE(C)	Sensors

II YEAR SEMESTER-4

Level: 6.0															
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – IV														
PCC	BTDS-211-PCC	Database Management Systems	3	-	2	50	50	-	-		3	-	1	4
PCC	BTDS-212-PCC	Operating Systems	2	-	4	50	50	-	-		2	-	2	4
MDM	BTDS-213-MDM	Basket 1/Basket 2/Basket 3	2	-	-	25	25	-	-		2	-	-	2
HSSM	BTDS-214-HSSM	History of Science and Technology in India	2	-	-	25	25	-	-		2	-	-	2
AEC	BTDS-215-AEC	Entrepreneurship Development	2	-	-	25	25	-	-		2	-	-	2
VSEC	BTDS-216-VSEC	Web Programming	0	-	4	-	-	-	25	25	-	-	2	2
PCC	BTDS-217-PCC	Introduction to Data Analytics*	0		4	-	-	-	25	25	-	-	2	2
VEC	BTDS-218-VEC	Spirituality and Mindfulness			4				25	25	-		2	2
OE	BTDS-219-OE	Open Elective - II	2	-	-	25	25	-	-		2			2
Total Hours/Week, Marks, Credits			**13**	**0**	**18**	**200**	**200**	**0**	**75**	**75**	**13**	**0**	**9**	**22**
TOTAL			**31 Hours/Week**			**550 Marks**					**22 Credits**			

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Sr. No	Course Code	Course Title
1	BTDS-213-MDM(A)	Smart Contracts and Decentralized Applications (DApps)
2	BTDS-213-MDM(B)	Fundamentals of Cybersecurity
3	BTDS-213-MDM(C)	Computer Vision

Open Elective – II

Sr. No	Course Code	Course Title
1	BTDS-219-OE(A)	Design Thinking
2	BTDS-219-OE(B)	Data Science In Biotechnology
3	BTDS-219-OE(C)	Concepts in Mobile Robots

III YEAR SEMESTER-5

Level: 6.0														
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – V														
PCC	BTDS-301-PCC	Machine Learning*	3		2	50	50		-		3	-	1	4
PCC	BTDS-302-PCC	Descriptive Analytics*	3		2	50	50		-		3	-	1	4
PEC	BTDS-303-PEC	Professional Elective-I	3		2	50	50		-		3	-	1	4
PCC	BTDS-304-PCC	Computer Organization and Architecture	3	1	-	50	50		-		3	1	-	4
MDM	BTDS-305-MDM	Basket 1/ Basket 2/ Basket 3	3		2	50	50		-		3	-	1	4
OE	BTDS-306-OE	Open Elective - III	-	-	4	-	-	-	25	25		-	2	2
Total Hours/Week, Marks, Credits			**15**	**1**	**12**	**250**	**250**	**0**	25	**25**	**15**	**1**	**6**	**22**
TOTAL			**28 Hours/Week**			**550 Marks**					**22 Credits**			

**L: Lecture, T: Tutorial, P: Practical, CCE: Comprehensive Continuous Evaluation, ESE: End Sem Exam, TW: Term Work, O: Oral.*

MDM:

Sr. No	Course Code	Course Title
1	BTDS-305-MDM(A)	Blockchain Platforms and Tools
2	BTDS-305-MDM(B)	Ethical Hacking and Penetration Testing
3	BTDS-305-MDM(C)	Digital Image Enhancement

Professional Elective-I

Sr. No	Course Code	Course Title
1	BTDS-303-PEC(A)	Graph Theory
2	BTDS-303-PEC(B)	Advanced Computer Architecture
3	BTDS-303-PEC(C)	ETL-Kafka/Talend
4	BTDS-303-PEC(D)	Image Processing
5	BTDS-303-PEC(E)	Computer Graphics
6	BTDS-303-PEC(F)	Web Framework
7	BTDS-303-PEC(G)	R Programming

Open Elective – III

Sr. No	Course Code	Course Title
1	BTDS-306-OE(A)	Introduction to Natural Language Processing
2	BTDS-306-OE(B)	Drone Technology
3	BTDS-306-OE(C)	Wearable Devices
4	BTDS-306-OE(D)	Biotechnology in Health Care

III YEAR SEMESTER-6

Level: 6.0														
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – VI														
PCC	BTDS-311-PCC	Design and Analysis of Algorithms	3	1	2	50	50		-		3	1	-	4
PCC	BTDS-312-PCC	Predictive Analytics*	3	-	2	50	50		-	25	3	-	1	4
PEC	BTDS-313-PEC	Professional Elective-II	3	-	2	50	50		-	25	3	-	1	4
PEC	BTD-DS-314-PEC	Professional Elective-III	3	-	2	50	50		-	-	3	-	1	4
MDM	BTDS-315-MDM	Basket 1/ Basket 2/ Basket 3	2	-	-	25	25		-	-	2	-	-	2
PCC	BTDS-316-PCC	Embedded Systems and IOT	-	-	4	-	-	-	25		-	-	2	2
VSEC	BTDS-317-VSEC	UI Design-Flutter	-	-	4	-	-	-	25		-	-	2	2
Total Hours/Week, Marks, Credits			**14**	**1**	**16**	**225**	**225**	**0**	**50**	**50**	**14**	**1**	**7**	**22**
TOTAL			**31 Hours/Week**			**550Marks**					**22 Credits**			

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Professional Elective – II

Sr. No	Course Code	Course Title
1	BTDS-313-PEC(A)	Software Testing Methodologies
2	BTDS-313-PEC(B)	Information Retrieval Systems
3	BTDS-313-PEC(C)	Computer Vision and Robotics
4	BTDS-313-PEC(D)	Data Warehousing and Business
5	BTDS-313-PEC(E)	Node JS/ React JS
6	BTDS-313-PEC(F)	Data Security in Data Science
7	BTDS-313-PEC(G)	Professional Practice, Law & Ethics

Professional Elective – III

Sr. No	Course Code	Course Title
1	BTDS-314-PEC(A)	Internet of Things
2	BTDS-314-PEC(B)	Scripting Languages
3	BTDS-314-PEC(C)	Mobile Application Development
4	BTDS-314-PEC(D)	Computer Networks

MDM:

Sr. No	Course Code	Course Title
1	BTDS-315-MDM(A)	Blockchain for Data Privacy and Security
2	BTDS-315-MDM(B)	Cryptography and Information Security
3	BTDS-315-MDM(C)	Deep Learning for Image Processing

IV YEAR SEMESTER-7

Level: 6.0														
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – VII														
PCC	BTDS-401-PCC	Big Data Analytics*	3	-	2	50	50	-	-	-	3	-	1	4
PCC	BTDS-402-PCC	Web and Social Media Analytics	2	-	-	25	25	-	-	-	2	-	-	2
PEC	BTDS-403-PEC	Professional Elective-IV	2	-	-	25	25	-	-	-	2	-	-	2
MDM	BTDS-404-MDM	Basket 1/ Basket 2/ Basket 3	2	-	-	25	25	-	-	-	2	-	-	2
ELC	BTDS-405-ELC	Internship / OJT	-	-	24	-	-	50	200	50	-	-	12	12
Total Hours/Week, Marks, Credits			**9**	**0**	**26**	**125**	**125**	**50**	**200**	**50**	**9**	**-**	**13**	**22**
TOTAL			**35 Hours/Week**			**550 Marks**					**22 Credits**			

**L: Lecture, T: Tutorial, P: Practical, CCE: Comprehensive Continuous Evaluation, ESE: End Sem Exam, TW: Term Work, O: Oral.*

Professional Elective – IV

Sr. No	Course Code	Course Title
1	BTDS-403-PEC(A)	Quantum computing
2	BTDS-403-PEC(B)	Expert Systems
3	BTDS-403-PEC(C)	Cloud Computing
4	BTDS-403-PEC(D)	Game Theory
5	BTDS-403-PEC(E)	Knowledge Representation and Reasoning
6	BTDS-403-PEC(F)	Speech and video Processing
7	BTDS-403-PEC(G)	Internet of Thing Lab
8	BTDS-403-PEC(H)	Scripting Langauge Lab
9	BTDS-403-PEC(I)	Cryptography and Network Security Lab

MDM:

Sr. No	Course Code	Course Title
1	BTDS-404-MDM(A)	Blockchain Scalability and Optimization
2	BTDS-404-MDM(B)	Cybersecurity in cloud and IoT Systems
3	BTDS-404-MDM(C)	Image Processing for Remote Sensing Application

IV YEAR SEMESTER-8

Level: 6.0														
Course Type	Course Code	Course Title	Teaching Scheme (Hours/Week)			Evaluation Schemes and Marks					Credits			
			L	T	P	CCE	ESE	TW	P	O	L	T	P	Total
Semester – VIII														
PCC	BTDS-411-PCC	Deep Learning	2	2	-	37	38	-	-	-	3	-	-	3
PCC	BTDS-412-PCC	Pattern Recognition	3	-	-	37	38	-	-	-	3	-	-	3
PEC	BTDS-413-PEC	Professional Elective-V	3	-	-	37	38	-	-	-	3	-	-	3
PEC	BTDS-414-PEC	Professional Elective-VI	3	-	-	37	38	-	-	-	3	-	-	3
MDM	BTDS-415-MDM	Basket 1/ Basket 2/ Basket 3	2	-	-	25	25	-	-	-	2	-		2
ELC	BTDS-416-ELC	Project Stage – II including Seminar*	-	-	16	-	-	50	100	50	-	-	8	8
Total Hours/Week, Marks, Credits			**13**	**2**	**16**	**173**	**177**	**50**	**100**	**50**	**14**	**0**	**8**	**22**
TOTAL			**31 Hours/Week**			**550 Marks**					**22 Credits**			

**L: Lecture, T: Tutorial, P: Practical, CCE: Comprehensive Continuous Evaluation, ESE: End Sem Exam, TW: Term Work, O: Oral.*

Professional Elective – V

Sr. No	Course Code	Course Title
1	BTDS-413-PEC(A)	Social Network Analysis
2	BTDS-413-PEC(B)	Federated Machine Learning
3	BTDS-413-PEC(C)	Augmented Reality & Virtual Reality
4	BTDS-413-PEC(D)	Web Security
5	BTDS-413-PEC(E)	Ad-hoc & Sensor Networks
6	BTDS-413-PEC(F)	Data Mining

Professional Elective – VI

Sr. No	Course Code	Course Title
1	BTDS-414-PEC(A)	Robotic Process Automation
2	BTDS-414-PEC(B)	Randomized Algorithms

3	BTDS-414-PEC(C)	Cognitive Computing
4	BTDS-414-PEC(D)	Semantic Web

Basket 1/Basket 2/Basket 3:

Sr. No	Course Code	Course Title
1	BTDS-415-MDM(A)	Capstone:Blockchain in Real-world Application
2	BTDS-415-MDM(B)	Capstone: Building Secure Systems
3	BTDS-415-MDM(C)	Capstone: Image Processing in Real-world Application

Multidisciplinary Minor Subject: 14 Credits

General Overview

- **Total Credits**: 14
- **Duration**: Spanning **6 semesters** (Semester 3 to Semester 8).
- **Structure**:
  - Semester 3 and 4: Foundational courses.
  - Semester 5 and 6: Intermediate courses.
  - Semester 7 and 8: Advanced and Capstone projects.

Basket 1: Blockchain

Semester 3 (Foundation)

1. **Introduction to Blockchain Technology** (2 credits)
   - Basics of blockchain, cryptography, distributed ledgers, and consensus mechanisms.
   - Lab: Set up a private blockchain.

Semester 4 (Foundation)

2. **Smart Contracts and Decentralized Applications (DApps)** (2 credits)
   - Writing smart contracts using Solidity and deploying DApps.
   - Lab: Develop a simple DApp.

Semester 5 (Intermediate)

3. **Blockchain Platforms and Tools** (4 credits)
   - Study platforms like Hyperledger Fabric and Ethereum.
   - Lab: Build a basic blockchain network.

Semester 6 (Advanced)

4. **Blockchain for Data Privacy and Security** (2 credits)
   - Blockchain's role in secure data management and privacy.
   - Lab: Implement a secure data-sharing application.

Semester 7 (Advanced)

5. **Blockchain Scalability and Optimization** (2 credits)

- Topics:
  - Techniques for optimizing blockchain scalability (e.g., sharding, layer-2 solutions).
  - Case studies on real-world scalability challenges.
- Lab: Optimize a blockchain-based application.

Semester 8 (Capstone)

6. **Capstone: Blockchain in Real-World Applications** (2 credits)
   - Example: Blockchain-based supply chain or carbon credit tracking system.

Basket 2: Cybersecurity

Semester 3 (Foundation)

1. **Environmental Science and Sustainability** (2 credits)
   - Topics:
     - Green computing and energy-efficient cryptographic algorithms.
     - IT systems and carbon neutrality.
     - Environmental policies and their impact on cybersecurity.
   - Case Studies:
     - Carbon-neutral data centers.
     - Sustainable IT systems.

Semester 4 (Foundation)

2. **Fundamentals of Cybersecurity** (2 credits)
   - Threats, vulnerabilities, and secure protocols.
   - Lab: Use tools like Wireshark for network analysis.

Semester 5 (Intermediate)

3. **Ethical Hacking and Penetration Testing** (4 credits)
   - Techniques for identifying and mitigating vulnerabilities.
   - Lab: Simulate penetration testing on virtual environments.

Semester 6 (Advanced)

4. **Cryptography and Information Security** (2 credits)
   - Study symmetric/asymmetric cryptography and hashing algorithms.
   - Lab: Implement encryption techniques like AES and RSA.

Semester 7 (Advanced)

5. **Cybersecurity in Cloud and IoT Systems** (2 credits)
    - Topics:
        - Securing cloud APIs and IoT infrastructures.
        - Case studies: Cyberattacks on IoT and cloud-based systems.
    - Lab: Implement security protocols for an IoT system.

Semester 8 (Capstone)

6. **Capstone: Building Secure Systems** (2 credits)
    - Example: Secure IoT-based home automation or a cybersecurity incident response system.

Basket 3: Image Processing

Semester 3 (Foundation)

1. **Introduction to Image Processing** (2 credits)
    - Basics of image representation, transformations, and filtering.
    - Lab: Image enhancement using OpenCV.

Semester 4 (Foundation)

2. **Computer Vision** (2 credits)
    - Object detection, tracking, and recognition.
    - Lab: Real-time face detection system.

Semester 5 (Intermediate)

3. **Digital Image Enhancement** (4 credits)
    - Techniques like histogram equalization and Fourier transforms.
    - Lab: Enhance noisy images.

Semester 6 (Advanced)

4. **Deep Learning for Image Processing** (2 credits)
    - Using CNNs for classification and segmentation.
    - Lab: Train a CNN on medical images or natural scenes.

Semester 7 (Advanced)

5. **Image Processing for Remote Sensing Applications** (2 credits)
    - Topics:
        - Applications in environmental monitoring, disaster management, and resource mapping.
        - Case studies: Remote sensing for deforestation tracking.
    - Lab: Analyze satellite images for a specific use case.

Semester 8 (Capstone)

6. **Capstone: Image Processing in Real-World Applications** (2 credits)
    - Example: Automated medical diagnosis or satellite image analysis.

MDM Credit Distribution by Semester

Semester	Credits	Focus
3	2	Foundation courses.
4	2	Foundation courses continue.
5	4	Intermediate courses for deeper learning.
6	2	Advanced intermediate courses to specialize in chosen baskets.
7	2	Advanced specialization or optimization courses.
8	2	Capstone project for real-world applications of chosen basket.

Professional Elective-I

BTDS-303-PEC(A)	Graph Theory
BTDS-303-PEC(B)	Advanced Computer Architecture
BTDS-303-PEC(C)	ETL-Kafka/Talend
BTDS-303-PEC(D)	Image Processing
BTDS-303-PEC(E)	Computer Graphics
BTDS-303-PEC(F)	Web Framework
BTDS-303-PEC(G)	R Programming

Professional Elective – II

BTDS-313-PEC(A)	Software Testing Methodologies
BTDS-313-PEC(B)	Information Retrieval Systems
BTDS-313-PEC(C)	Computer Vision and Robotics
BTDS-313-PEC(D)	Data Warehousing and Business Intelligence
BTDS-313-PEC(E)	Node JS/ React JS
BTDS-313-PEC(F)	Data Security in Data Science
BTDS-313-PEC(G)	Professional Practice, Law & Ethics

Professional Elective - III

BTDS-314-PEC(A)	Internet of Things
BTDS-314-PEC(B)	Scripting Languages
BTDS-314-PEC(C)	Mobile Application Development
BTDS-314-PEC(D)	Computer Networks

Professional Elective -IV

BTDS-403-PEC(A)	Quantum Computing
BTDS-403-PEC(B)	Expert Systems
BTDS-403-PEC(C)	Cloud Computing
BTDS-403-PEC(D)	Game Theory
BTDS-403-PEC(E)	Knowledge Representation and Reasoning
BTDS-403-PEC(D)	Speech and Video Processing

Professional Elective - V

BTDS-413-PEC(A)	Social Network Analysis
BTDS-413-PEC(B)	Federated Machine Learning
BTDS-413-PEC(C)	Augmented Reality & Virtual Reality
BTDS-413-PEC(D)	Web Security
BTDS-413-PEC(E)	Ad-hoc & Sensor Networks
BTDS-413-PEC(F)	Data Mining

Professional Elective – VI

BTDS-414-PEC(A)	Robotic Process Automation
BTDS-414-PEC(B)	Randomized Algorithms
BTDS-414-PEC(C)	Cognitive Computing
BTDS-414-PEC(D)	Semantic Web

Open Elective I:

1. BTDS-208-OE(A): Foreign Language
2. BTDS-208-OE(B): Cost Management of Engineering Projects
3. BTDS-208-OE(C): Sensors

Open Elective II:

1. BTDS-219-OE(A): Design Thinking
2. BTDS-219-OE(B): Data Science In Biotechnology
3. BTDS-219-OE(C): Concepts in Mobile Robots

Open Elective III:

1. BTDS-306-OE(A): Introduction to Natural Language Processing
2. BTDS-306-OE(B): Drone Technologies
3. BTDS-306-OE(C): Wearable Devices
4. BTDS-306-OE(D): Biotechnology in Health Care

Mapping of Course Outcome and Programme Outcome																						
Year	Semester	Course Name	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PEO1	PEO2	PEO3	PEO4	PEO5
I	1	BTDS 101	3	2	3	1	2	0	0	0	1	2	1	2	3	2	3	3	2	1	2	1
		BTDS 102	3	3	2	3	2	1	0	1	2	3	2	2	3	3	2	3	2	3	2	2
		BTDS 103	3	3	3	2	3	1	0	0	2	3	1	2	3	2	3	3	2	2	3	2
		BTDS 104	2	1	2	0	1	1	0	3	3	3	2	2	1	1	1	1	1	2	1	1
		BTDS 105	3	2	3	3	3	1	1	1	3	3	2	3	3	3	2	3	2	3	3	2
	2	BTDS 111	3	3	2	2	2	0	0	0	0	0	0	0	3	3	2	3	3	3	3	2
		BTDS 112	3	3	3	2	3	0	0	0	0	0	0	0	3	3	3	3	3	3	3	2
		BTDS 113	3	3	2	2	2	0	0	0	0	0	0	0	3	3	3	3	3	3	3	2
		BTDS 114	3	3	2	3	2	0	0	0	0	0	0	0	3	3	2	3	3	3	3	2
		BTDS 115	3	3	3	2	2	0	0	0	0	0	0	0	3	3	3	3	3	3	3	2
II	3	BTDS201	3	3	2	3	2	2	1	-	-	-	-	1	3	2	2	2	2	2	3	3
		BTDS202	3	3	2	2	3	2	1	-	-	-	-	1	3	3	2	2	2	2	3	3
		BTDS203	3	3	2	3	3	2	1	-	-	-	-	1	3	2	3	2	2	2	3	3
		BTDS204	3	3	2	2	2	2	1	-	-	-	-	1	3	3	2	2	2	2	3	3
		BTDS205	3	3	2	3	2	2	1	-	-	-	-	1	3	2	2	2	2	2	3	3
	4	BTDS211	3	3	3	2	2	2	2	3	2	2	3	2	3	3	2	3	2	3	2	2

		BTD S212	3	3	3	2	2	3	2	2	3	2	2	3	3	2	3	2	3	3	2	2
		BTD S213	3	3	3	2	3	2	3	2	3	2	3	2	3	3	3	2	3	3	2	
		BTD S214	3	3	3	2	2	2	3	2	3	2	2	3	3	3	2	3	3	2	3	2
		BTD S215	3	3	3	2	2	2	3	2	3	2	2	3	3	3	2	3	3	2	3	2
III	5	**BTD S301**	3	3	3	2	2	-	-	-	-	-	-	-	3	3	2	-	-	-	-	-
		BTD S302	3	3	3	2	2	-	-	-	-	-	-	-	3	3	2	-	-	-	-	-
		BTD S303	3	3	3	2	2	-	-	-	-	-	-	-	3	3	2	-	-	-	-	-
		BTD S304	3	3	3	2	2	-	-	-	-	-	-	-	3	3	2	-	-	-	-	-
		BTD S305	3	3	3	2	2	-	-	-	-	-	-	-	3	3	2	-	-	-	-	-
	6	**BTD S311**	3	3	2	2	3	-	-	-	-	-	-	-	3	3	3	3	3	3	3	3
		BTD S312	3	3	1	2	3	-	-	-	-	-	-	-	3	3	3	3	3	3	3	3
		BTD S313	3	3	2	2	3	-	-	-	-	-	-	-	3	3	3	3	3	3	3	3
		BTD S314	3	3	1	3	3	-	-	-	-	-	-	-	3	3	3	3	3	3	3	3
		BTD S315	3	3	2	2	3	-	-	-	-	-	-	-	3	3	3	3	3	3	3	3
IV	7	**BTD S401**	3	3	3	2	2	2	2	3	2	2	3	2	3	3	2	3	2	3	2	2
		BTD S402	3	3	3	2	2	3	2	2	3	2	2	3	3	2	3	2	3	3	2	2
		BTD S403	3	3	3	2	3	2	3	2	3	2	3	2	3	3	3	2	3	3	2	
		BTD S404	3	3	3	2	2	2	3	2	3	2	2	3	3	3	2	3	3	2	3	2
		BTD S405	3	3	3	2	2	2	2	3	2	2	3	2	3	3	2	3	2	3	2	2
	8	**BTD S411**	3	3	3	2	2	3	2	2	3	2	2	3	3	2	3	2	3	3	2	2
		BTD S412	3	3	3	2	3	2	3	2	3	2	3	2	3	3	3	2	3	3	2	
		BTD S413	3	3	3	2	2	2	3	2	3	2	2	3	3	3	2	3	3	2	3	2
		BTD S414	3	3	3	2	3	2	3	2	3	2	3	2	3	3	3	2	3	3	2	

BTDS-101-BSC: MATHEMATICAL FOUNDATION - I
B.Tech. I Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 04	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of high school mathematics (algebra, geometry, trigonometry).

Course Objective:

1. To develop the use of matrix algebra techniques that are needed by engineers for practical applications.
2. To familiarize the students with differential calculus.
3. To familiarize the student with functions of several variables. This is needed in many branches of engineering.
4. To make the students understand various techniques of integration.

Learning Course Outcome: At the end of the course the students will be able to

CO1: Use the matrix algebra methods for solving practical problems.

CO2: Apply differential calculus tools in solving various application problems.

CO3: Able to use differential calculus ideas on several variable functions.

CO4: Apply different methods of integration in solving practical problems.

Course Contents

UNIT-I	Matrices	08 Hours
Eigenvalues and Eigenvectors of a real matrix, Characteristic equation, Properties of Eigenvalues and Eigenvectors, Cayley – Hamilton.		
UNIT-II	**Algebraic Equations**	**08 Hours**
Solving Algebraic Linear Equations, Crammer's rule, Idea and Solution of Non-Linear Algebraic Equations, Gauss Elimination & Seidel. Bisection method, Newton-Raphson Method, Secant Method, False Position (Regula Falsi) Method.		
UNIT-III	**Differential Calculus**	**08 Hours**
Representation of functions - Limit of a function - Continuity - Derivatives - Differentiation rules (sum, product, quotient, chain rules) - Implicit differentiation - Logarithmic differentiation - Applications: Maxima and Minima of functions of one variable.		
UNIT-IV	**Functions of Several Variables**	**08 Hours**
Partial differentiation, Homogeneous functions and Euler's theorem, Total derivative,		

Change of variables, Jacobians, Applications: Maxima and minima of functions of two variable.		
UNIT-V	**Integral Calculus**	**08 Hours**
Definite and Indefinite integrals, Substitution rule, Techniques of Integration: Integration by parts, Trigonometric integrals, Trigonometric substitutions, Integration of rational functions by partial fraction.		

TEXTBOOKS:

1. Kreyszig.E, "Advanced Engineering Mathematics", John Wiley and Sons, 10th Edition, New Delhi, 2016.
2. Grewal.B.S., "Higher Engineering Mathematics", Khanna Publishers, New Delhi, 44th Edition , 2018.
3. James Stewart, " Calculus: Early Transcendentals ", Cengage Learning, 8th Edition, New Delhi, 2015.

REFERENCES:

1. Anton. H, Bivens. Davis and I. S, " Calculus ", Wiley, 10th Edition, 2016
2. Bali. N., Goyal. M. and Watkins. C., "Advanced Engineering Mathematics", Firewall Media (An imprint of Lakshmi Publications Pvt., Ltd.,), New Delhi, 7th Edition, 2009.
3. Jain. R.K.and Iyengar. S.R.K., "Advanced Engineering Mathematics",Narosa Publications, New Delhi, 5th Edition, 2016.
4. Narayanan. S. and Manicavachagom Pillai. T. K., "Calculus" Volume I and II, S. Viswanathan Publishers Pvt. Ltd., Chennai, 2009.
5. Ramana. B.V., "Higher Engineering Mathematics", McGraw Hill Education Pvt. Ltd, New Delhi, 2016.
6. Srimantha Pal and Bhunia. S.C, "Engineering Mathematics" Oxford University Press, 2015.
7. Thomas. G. B., Hass. J, and Weir. M.D, "Thomas Calculus ", 14th Edition, Pearson India,2018.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	1	0	0	0	0	2	0	2	3	-	-	-
CO2	3	3	1	1	0	0	0	0	2	0	2	3	-	-	-
CO3	3	3	1	1	0	0	0	0	2	0	2	3	-	-	-
CO4	3	3	1	1	0	0	0	0	2	0	2	3	-	-	-
CO5	3	3	1	1	0	0	0	0	2	0	2	3	-	-	-
Avg	3	3	1	1	0	0	0	0	2	0	2	3	-	-	-

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-102-BSC: ENGINEERING PHYSICS
B.Tech. I Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 04	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 + 2 Physics

Course Objective:

1. To make the students effectively achieve an understanding of mechanics.
2. To enable the students to gain knowledge of electromagnetic waves and its applications.
3. To introduce the basics of oscillations, optics and lasers.
4. Equipping the students to successfully understand the importance of quantum physics.
5. To motivate the students towards the applications of quantum mechanics.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the importance of mechanics.
CO2: Express their knowledge in electromagnetic waves.
CO3: Demonstrate a strong foundational knowledge in oscillations, optics and lasers.
CO4: Understand the importance of quantum physics.
CO5: Comprehend and apply quantum mechanical principles towards the formation of energy bands.

Course Contents

UNIT-I	Wave Optics	07 Hours
Newton's Rings, Michelson's Interferometer, Fraunhofer Diffraction from a Single Slit. Diffraction grating: Construction, theory and spectrum, Resolving power and Rayleigh criterion for limit of resolution, Resolving power of diffraction grating, X-Ray diffraction and Bragg's Law.		
UNIT-II	**Quantum Mechanics**	**08 Hours**
Introduction to quantum Mechanics, Wave-particle duality, Matter waves, Wave function and basic postulates, Time dependent and time independent Schrodinger's Wave Equation, Physical interpretation of wave function and its properties, Applications of the Schrodinger's Equation: Particle in one dimensional and three dimensional boxes.		
UNIT-III	**Coherence and Optical Fibers**	**08 Hours**

Spatial and temporal coherence: Coherence length; Coherence time and 'Q' factor for light, Visibility as a measure of Coherence and spectral purity, Optical fiber as optical wave guide, Numerical aperture; Maximum angle of acceptance and applications of optical fiber.		
UNIT-IV	**Laser**	**07 Hours**
Einstein's Theory of laser action; Einstein's coefficients; Properties of Laser beam, Amplification of light by population inversion, Components of laser, Construction and working of He-Ne and semiconductor lasers, Applications of Lasers in Science, engineering and medicine.		
UNIT-V	**Material Science & Semiconductor Physics**	**08 Hours**
Bonding in solids: covalent and metallic bonding, Energy bands in solids: Classification of solids as Insulators, Semiconductors and Conductors, Intrinsic and extrinsic semiconductors, Fermi dirac distribution function and Fermi energy, Conductivity in semiconductors, Hall Effect: Theory, Hall Coefficient and applications.		
UNIT-V	**Introduction to Electromagnetism**	**08 Hours**
Divergence and curl of electrostatic field, Laplace's and Poisson's equations for electrostatic potential, Bio-Savart law, Divergence and curl of static magnetic field, Faraday's law, Displacement current and magnetic field arising from time dependent electric field, Maxwell's equations, Flow of energy and Poynting vector.		
Lab Exercises: 1. To determine the wave length of monochromatic light with the help of Michelson's interferometer. 2. To determine the wave length of sodium light by Newton's Ring. 3. To determine the wave length of prominent lines of mercury by plane diffraction grating with the help of spectrometer. 4. Determination of band gap using a P-N junction diode. 5. To determine the height of given object with the help of sextant. 6. To determine the dispersive power of material of a prism with the help of spectrometer. 7. To study the charge and discharge of a condenser and hence determine the same constant (both current and voltage graphs are to be plotted. 8. To determine the coherence length and coherence time of laser using He – Ne laser. 9. To measure the numerical aperture of an optical fibre.		

10. To study the Hall Effect and determine the Hall Voltage and Hall coefficients.

REFERENCES:

1. R.Wolfson. Essential University Physics. Volume 1 & 2. Pearson Education (Indian Edition),2009.
2. Paul A. Tipler, Physic – Volume 1 & 2, CBS, (Indian Edition), 2004.
3. K.Thyagarajan and A.Ghatak. Lasers: Fundamentals and Applications, Laxmi Publications,(Indian Edition), 2019.
4. D.Halliday, R.Resnick and J.Walker. Principles of Physics, Wiley (Indian Edition), 2015.
5. N.Garcia, A.Damask and S.Schwarz. Physics for Computer Science Students. Springer-Verlag, 2012.

TEXT BOOKS:

1. D.Kleppner and R.Kolenkow. An Introduction to Mechanics. McGraw Hill Education (IndianEdition), 2017.
2. E.M.Purcell and D.J.Morin, Electricity and Magnetism, Cambridge Univ.Press, 2013.
3. Arthur Beiser, Shobhit Mahajan, S. Rai Choudhury, Concepts of Modern Physics, McGraw-Hill (Indian Edition), 2017.

CO's-PO's & PSO's MAPPING

CO	PO	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	2	1	1	1	-	-	-	-	-	-	-	-	-
2	3	3	2	1	2	1	-	-	-	-	-	-	-	-	-
3	3	3	2	2	2	1	-	-	-	-	-	1	-	-	-
4	3	3	1	1	2	1	-	-	-	-	-	-	-	-	
5	3	3	1	1	2	1	-	-	-	-	-	-	-	-	-
AV	3	3	1.6	1.2	1.8	1	-	-	-	-	-	1	-	-	-

1 - Low, 2 - medium, 3 - high, '-' - no correlation

BTDS-103-ESC: BASICS OF ELECTRONICS
B.Tech. I Year I Sem.

Teaching Scheme: TH: - 3 Hours/Week	Credit TH: 04	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of programming concepts.

Course Objective:

1. To introduce components such as diodes, BJTs and FETs.
2. To know the applications of devices.
3. To know the switching characteristics of devices.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Acquire the knowledge of various electronic devices and their use on real life.
2. Know the applications of various devices.

Acquire the knowledge about the role of special purpose devices and their applications.

Course Contents

UNIT-I	Diodes	07 Hours
Diode - Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch- switching times.		
UNIT-II	**Diode Applications**	**08 Hours**
Rectifier - Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier, Rectifiers with Capacitive and Inductive Filters, Clippers-Clipping at two independent levels, Clamper-Clamping Circuit Theorem, Clamping Operation, Types of Clampers.		
UNIT-III	**Bipolar Junction Transistor (BJT)**	**08 Hours**
Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch, switching times.		
UNIT-IV	**Junction Field Effect Transistor (FET)**	**07 Hours**
Construction, Principle of Operation, Pinch-Off Voltage, Volt- Ampere Characteristic, Comparison of BJT and FET, FET as Voltage Variable Resistor, MOSFET, MOSTET as a capacitor.		
UNIT-V	**Special Purpose Devices**	**08 Hours**

Zener Diode - Characteristics, Zener diode as Voltage Regulator, Principle of Operation - SCR, Tunnel diode, UJT, Varactor Diode, Photo diode, Solar cell, LED, Schottky diode.

Lab Exercise:

1. Basic safety precautions. Introduction and use of measuring instruments – voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors and inductors.

2. Transformers: Observation of the no-load current waveform on an oscilloscope. Loading of a transformer: measurement of primary and secondary voltages and currents, and power.

3. Three-phase transformers: Star and Delta connections. Voltage and Current relationships (line-line voltage, phase-to-neutral voltage, line and phase currents).Phase-shifts between the primary and secondary side.

4. Demonstration of cut-out sections of machines: dc machine (commutator-brush arrangement), induction machine (squirrel cage rotor), synchronous machine (field winging - slip ring arrangement) and single-phase induction machine.

5. Torque Speed Characteristic of separately excited dc motor.

6. Study of electronic devices viz. PN Junction diode, Zener diode, LED, photo diode, Bipolar Junction Transistors, MOSFET, Photo transistor, UJT, SCR.

7. Demonstration of (a) dc-dc converters (b) dc-ac converters – PWM waveform (c) Study of Half wave and Full wave Rectifiers.

TEXT BOOKS:

1. Jacob Millman - Electronic Devices and Circuits, McGraw Hill Education
2. Robert L. Boylestead, Louis Nashelsky- Electronic Devices and Circuits theory, 11th Edition, 2009, Pearson.

REFERENCE BOOKS:

1. Horowitz -Electronic Devices and Circuits, David A. Bell – 5th Edition, Oxford.
2. Chinmoy Saha, Arindam Halder, Debaati Ganguly - Basic Electronics-Principles and Applications, Cambridge, 2018.

CO's-PO's & PSO's MAPPING

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	-	-	1	1	-	-	-	-	1
CO2	3	2	3	-	-	2	1	-	-	-	-	1
CO3	3	3	3	-	-	2	1	-	-	-	-	1

1 - Low, 2 - medium, 3 - high, '-' - no correlation

BTDS-104-ESC: FUNDAMENTALS OF PROGRAMMING LANGUAGE
B.Tech. I Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of programming concepts.

Course Objective:

1. To learn the fundamentals of computers.
2. To understand the various steps in program development.
3. To learn the syntax and semantics of the C programming language.
4. To learn the usage of structured programming approaches in solving problems.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. To write algorithms and to draw flowcharts for solving problems.
2. To convert the algorithms/flowcharts to C programs.
3. To code and test a given logic in the C programming language.
4. To decompose a problem into functions and to develop modular reusable code.
5. To use arrays, pointers, strings and structures to write C programs.
6. Searching and sorting problems.

Course Contents

UNIT-I	Introduction to Programming	07 Hours

Introduction to C Programming Language: Variables (with data types and space requirements), Syntax and Logical Errors in compilation, object and executable code, Operators, expressions and precedence, Expression evaluation, Storage classes (auto, extern, static and register), type conversion, The main method and command line arguments Bitwise operations: Bitwise AND, OR, XOR and NOT operators

Conditional Branching and Loops: Writing and evaluation of conditionals and consequent branching with if, if-else, switch-case, ternary operator, goto, Iteration with for, while, do- while loops. I/O: Simple input and output with scanf and printf, formatted I/O, Introduction to stdin, stdout and stderr. Command line arguments.

UNIT-II	Arrays, Strings, Structures	08 Hours

	and Pointers	
Arrays: one and two dimensional arrays, creating, accessing and manipulating elements of arrays Strings: Introduction to strings, handling strings as array of characters, basic string functions available in C (strlen, strcat, strcpy, strstr etc.), arrays of strings Structures: Defining structures, initializing structures, unions, Array of structures Pointers: Idea of pointers, Defining pointers, Pointers to Arrays and Structures, Use of Pointers in self- referential structures, usage of self-referential structures in linked list (no implementation) Enumeration data type.		
UNIT-III	**Preprocessor and File handling in C**	**08 Hours**
Preprocessor: Commonly used Preprocessor commands like include, define, undef, if, ifdef, ifndef Files: Text and Binary files, Creating and Reading and writing text and binary files, Appending data to existing files, Writing and reading structures using binary files, Random access using fseek, ftell and rewind functions.		
UNIT-IV	**Function and Dynamic Memory Allocation**	**07 Hours**
Functions: Designing structured programs, Declaring a function, Signature of a function, Parameters and return type of a function, passing parameters to functions, call by value, Passing arrays to functions, passing pointers to functions, idea of call by reference, Some C standard functions and libraries. Simple programs, such as Finding Factorial, Fibonacci series etc., Limitations of Recursive functions Dynamic memory allocation: Allocating and freeing memory, Allocating memory for arrays of different data types.		
UNIT-V	**Searching and Sorting**	**08 Hours**
Basic searching in an array of elements (linear and binary search techniques), Basic algorithms to sort array of elements (Bubble, Insertion and Selection sort algorithms),		

TEXT BOOKS:

1. Jeri R. Hanly and Elliot B.Koffman, Problem solving and Program Design in C 7th Edition, Pearson
2. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)

REFERENCE BOOKS:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
2. E. Balagurusamy, Computer fundamentals and C, 2nd Edition, McGraw-Hill
3. Yashavant Kanetkar, Let Us C, 18th Edition, BPB
4. R.G. Dromey, How to solve it by Computer, Pearson (16th Impression)
5. Programming in C, Stephen G. Kochan, Fourth Edition, Pearson Education.
6. Herbert Schildt, C: The Complete Reference, Mc Graw Hill, 4th Edition
7. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill

CO's-PO's & PSO's MAPPING

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	-	-	1	1	-	-	-	-	1
CO2	3	2	3	-	-	2	1	-	-	-	-	1
CO3	3	3	3	-	-	2	1	-	-	-	-	1

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS104L: FUNDAMENTALS OF PROGRAMMING LANGUAGE LAB
B.Tech. I Year II Sem.

Teaching Scheme: P: - 2 Hours/Week	Credit P: 01	Examination Scheme: Total : 30 Marks

Course Prerequisites: Basic Knowledge of computer

Course Objective:

1. To work with an IDE to create, edit, compile, run and debug programs
2. To analyze the various steps in program development.
3. To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
4. To develop modular, reusable and readable C Programs using the concepts like functions, arrays etc.
5. To Write programs using the Dynamic Memory Allocation concept.
6. To create, read from and write to text and binary files

Learning Course Outcome:
After successful completion of the course, students will able to:

1. formulate the algorithms for simple problems
2. translate given algorithms to a working and correct program
3. correct syntax errors as reported by the compilers
4. identify and correct logical errors encountered during execution
5. represent and manipulate data with arrays, strings and structures
6. use pointers of different types
7. create, read and write to and from simple text and binary files
8. modularize the code with functions so that they can be reused

Course Contents

List of Experiments	60 Hours

Note: The examples suggested in each experiment are only indicative. The lab instructor is expected to design other problems on similar lines. The Examination shall not be restricted to the sample experiments listed here.

Practice sessions:

a. Write a simple program that prints the results of all the operators available in C (including pre/ post increment , bitwise and/or/not , etc.). Read required operand values from standard input.
b. Write a simple program that converts one given data type to another using auto conversion and casting. Take the values from standard input.

Simple numeric problems:

a. Write a program for finding the max and min from the three numbers.
b. Write the program for the simple, compound interest.
c. Write a program that declares Class awarded for a given percentage of marks, where mark <40%= Failed, 40% to <60% = Second class, 60% to <70%=First class, >= 70% = Distinction. Read percentage from standard input.
d. Write a program that prints a multiplication table for a given number and the number of rows in the table. For example, for a number 5 and rows = 3, the output should be:
e. 5 x 1 = 5
f. 5 x 2 = 10
g. 5 x 3 = 15
h. Write a program that shows the binary equivalent of a given positive number between 0 to 255.

Expression Evaluation:

a. A building has 10 floors with a floor height of 3 meters each. A ball is dropped from the top of the building. Find the time taken by the ball to reach each floor. (Use the formula s = ut+(1/2)at^2 where u and a are the initial velocity in m/sec (= 0) and acceleration in m/sec^2 (= 9.8 m/s^2)).
b. Write a C program, which takes two integer operands and one operator from the user, performs the operation and then prints the result. (Consider the operators +,-,*, /, % and use Switch Statement)
c. Write a program that finds if a given number is a prime number
d. Write a C program to find the sum of individual digits of a positive integer and test given number is palindrome.
e. A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.
f. Write a C program to generate all the prime numbers between 1 and n, where n is a value supplied by the user.
g. Write a C program to find the roots of a Quadratic equation.
h. Write a C program to calculate the following, where x is a fractional value.
i. 1-x/2 +x^2/4-x^3/6
j. Write a C program to read in two numbers, x and n, and then compute the sum of this geometric progression: 1+x+x^2+x^3+..... +x^n. For example: if n is 3 and x is 5, then the program computes 1+5+25+125.

Arrays, Pointers and Functions:

a. Write a C program to find the minimum, maximum and average in an array of integers.
b. Write a function to compute mean, variance, Standard Deviation, sorting of n elements in a single dimension array.
c. Write a C program that uses functions to perform the following:
d. Addition of Two Matrices
e. Multiplication of Two Matrices
f. Transpose of a matrix with memory dynamically allocated for the new matrix as row and column counts may not be the same.
g. Write C programs that use both recursive and non-recursive functions
h. To find the factorial of a given integer.
i. To find the GCD (greatest common divisor) of two given integers.
j. To find x^n
k. Write a program for reading elements using a pointer into an array and display the values using the array.
l. Write a program for display values reverse order from an array using a pointer.

m. Write a program through a pointer variable to sum of n elements from an array.

Files:

a. Write a C program to display the contents of a file to standard output device.
b. Write a C program which copies one file to another, replacing all lowercase characters with their uppercase equivalents.
c. Write a C program to count the number of times a character occurs in a text file. The file name and the character are supplied as command line arguments.
d. Write a C program that does the following:
It should first create a binary file and store 10 integers, where the file name and 10 values are given in the command line. (hint: convert the strings using atoi function)
Now the program asks for an index and a value from the user and the value at that index should be changed to the new value in the file. (hint: use fseek function)
The program should then read all 10 values and print them back.
e. Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third file).

Strings:

a. Write a C program to convert a Roman numeral ranging from I to L to its decimal equivalent.
b. Write a C program that converts a number ranging from 1 to 50 to Roman equivalent
c. Write a C program that uses functions to perform the following operations:
d. To insert a sub-string into a given main string from a given position.
e. To delete n Characters from a given position in a given string.
f. Write a C program to determine if the given string is a palindrome or not (Spelled same in both directions with or without a meaning like madam, civic, noon, abcba, etc.)
g. Write a C program that displays the position of a character ch in the string S or – 1 if S doesn‘t contain ch.
h. Write a C program to count the lines, words and characters in a given text.

Miscellaneous:

a. Write a menu driven C program that allows a user to enter n numbers and then choose between finding the smallest, largest, sum, or average. The menu and all the choices are to be functions. Use a switch statement to determine what action to take. Display an error message if an invalid choice is entered.

b. Write a C program to construct a pyramid of numbers as follows:

```
1          *          1          1          *
1 2        * *        2 3        2 2        * *
1 2 3      * * *      4 5 6      3 3 3      * * *
                                 4 4 4 4    * *
                                            *
```

Sorting and Searching:

a. Write a C program that uses non recursive function to search for a Key value in a given
b. list of integers using linear search method.
c. Write a C program that uses non recursive function to search for a Key value in a given
d. sorted list of integers using binary search method.
e. Write a C program that implements the Bubble sort method to sort a given list of
f. integers in ascending order.
g. Write a C program that sorts the given array of integers using selection sort in descending order

h. Write a C program that sorts the given array of integers using insertion sort in ascending order
i. Write a C program that sorts a given array of names

TEXT BOOKS:

1. Jeri R. Hanly and Elliot B.Koffman, Problem solving and Program Design in C 7th Edition, Pearson
2. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)

REFERENCE BOOKS:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, PHI
2. E. Balagurusamy, Computer fundamentals and C, 2nd Edition, McGraw-Hill
3. Yashavant Kanetkar, Let Us C, 18th Edition, BPB
4. R.G. Dromey, How to solve it by Computer, Pearson (16th Impression)
5. Programming in C, Stephen G. Kochan, Fourth Edition, Pearson Education.
6. Herbert Schildt, C: The Complete Reference, Mc Graw Hill, 4th Edition
7. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	3	2	-	-	-	-	-	2	2	3	3	-
2	3	3	3	3	2	-	-	-	-	-	2	2	3	-	-
3	3	3	3	3	2	-	-	-	-	-	2	-	3	-	-
4	2	2	-	2	2	-	-	-	-	-	1	-	3	-	-
5	1	2	-	-	1	-	-	-	-	-	1	-	2	-	-
6	2	2	-	-	2	-	-	-	-	-	1	-	2	-	
AVg.	**2**	**3**	**3**	**3**	**2**	-	-	-	-	-	**2**	**2**	**3**	**3**	-

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-105-VSEC: PROFESSIONAL COMMUNICATION SKILLS-I
B.Tech. I Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Prerequisites: English Grammar basic knowledge

Course Objective:

1. To improve the communicative competence of learners
2. To learn to use basic grammatic structures in suitable contexts
3. To acquire lexical competence and use them appropriately in a sentence and understand their meaning in a text
4. To help learners use language effectively in professional contexts
5. To develop learners' ability to read and write complex texts, summaries, articles, blogs, definitions, essays and user manuals.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Understand the importance of vocabulary and sentence structures.
2. Choose appropriate vocabulary and sentence structures for their oral and written communication.
3. Demonstrate their understanding of the rules of functional grammar.
4. Develop comprehension skills from the known and unknown passages.
5. Take an active part in drafting paragraphs, letters, essays, abstracts, précis and reports in various contexts.
6. Acquire basic proficiency in reading and writing modules of English.

Course Contents

UNIT-I	INTRODUCTION TO EFFECTIVE COMMUNICATION	07 Hours

What is effective communication? (Explain using activities) Why is communication critical for excellence during study, research and work? What are the seven C's of effective communication? What are key language skills? What is effective listening? What does it involve? What is effective speaking? What does it mean to be an excellent reader? What should you be able to do? What is effective writing? How does one develop language and communication skills? What does the course focus on? How are communication and language skills going to be enhanced during this course? What do

you as a learner need to do to enhance your English language and communication skills to get the best out of this course? INTRODUCTION TO FUNDAMENTALS OF COMMUNICATION Reading - Reading brochures (technical context), telephone messages / social media messages relevant to technical contexts and emails. Writing - Writing emails / letters introducing oneself. Grammar - Present Tense (simple and progressive); Question types: Wh/ Yes or No/ and Tags. Vocabulary - Synonyms; One word substitution; Abbreviations & Acronyms (as used in technical contexts).		
UNIT-II	**NARRATION AND SUMMATION**	**08 Hours**
Reading - Reading biographies, travelogues, newspaper reports, Excerpts from literature, and travel & technical blogs. Writing - Guided writing-- Paragraph writing Short Report on an event (field trip etc.) Grammar –Past tense (simple); Subject-Verb Agreement; and Prepositions. Vocabulary - Word forms (prefixes& suffixes); Synonyms and Antonyms. Phrasal verbs.		
UNIT-III	**DESCRIPTION OF A PROCESS / PRODUCT**	**08 Hours**
Reading – Reading advertisements, gadget reviews; user manuals. Writing - Writing definitions; instructions; and Product /Process description. Grammar - Imperatives; Adjectives; Degrees of comparison; Present & Past Perfect Tenses. Vocabulary - Compound Nouns, Homonyms; and Homophones, discourse markers (connectives & sequence words).		
UNIT-IV	**CLASSIFICATION AND RECOMMENDATIONS**	**07 Hours**
Reading – Newspaper articles; Journal reports –and Non Verbal Communication (tables, pie charts etc,.). Writing – Note-making / Note-taking (*Study skills to be taught, not tested); Writing recommendations; Transferring information from nonverbal (chart , graph etc, to verbal mode) Grammar – Articles; Pronouns - Possessive & Relative pronouns. Vocabulary - Collocations; Fixed / Semi fixed expressions.		
UNIT-V	**EXPRESSION**	**08 Hours**
Reading – Reading editorials; and Opinion Blogs; Writing – Essay Writing (Descriptive or narrative). Grammar – Future Tenses, Punctuation; Negation (Statements & Questions); and Simple, Compound & Complex Sentences. Vocabulary - Cause &		

Effect Expressions – Content vs Function words.

TEXT BOOKS:

1. English for Engineers & Technologists Orient Blackswan Private Ltd. Department of English,Anna University, (2020 edition)
2. English for Science & Technology Cambridge University Press, 2021. Authored by Dr. Veena Selvam, Dr. Sujatha Priyadarshini, Dr. Deepa Mary Francis, Dr. KN.Shoba, and Dr. Lourdes Joevani, Department of English, Anna University.

REFERENCES:

2. Technical Communication – Principles And Practices By Meenakshi Raman & SangeetaSharma, Oxford Univ. Press, 2016, New Delhi.
3. A Course Book On Technical English By Lakshminarayanan, Scitech Publications (India) Pvt.Ltd.
4. English For Technical Communication (With CD) By Aysha Viswamohan, Mcgraw Hill Education,ISBN : 0070264244.
5. Effective Communication Skill, Kulbhusan Kumar, RS Salaria, Khanna Publishing House.
6. Learning to Communicate – Dr. V. Chellammal, Allied Publishing House, New Delhi,2003.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	1	1	1	1	1	3	3	3	1	3	-	3	-	-	-
2	1	1	1	1	1	3	3	3	1	3	-	3	-	-	-
3	2	3	2	3	2	3	3	3	2	3	3	3	-	-	-
4	2	3	2	3	2	3	3	3	2	3	3	3	-	-	-
5	2	3	3	3	-	3	3	3	2	3	-	3	-	-	-
AVg.	**1.6**	**2.2**	**1.8**	**2.2**	**1.5**	**3**	**3**	**3**	**1.6**	**3**	**3**	**3**	**-**	**-**	**-**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-106-AEC: IT WORKSHOP
B.Tech. I Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 50 Marks

Course Prerequisites: Basic Knowledge of computer

Course Objective:

The IT Workshop for engineers is a training lab course spread over 60 hours. The modules include training on PC Hardware, Internet & World Wide Web and Productivity tools including Word, Excel, PowerPoint and Publisher.

Learning Course Outcome:

After successful completion of the course, students will able to:

1. Perform Hardware troubleshooting
2. Understand Hardware components and inter dependencies
3. Safeguard computer systems from viruses/worms
4. Document/ Presentation preparation
5. Perform calculations using spreadsheets

Course Contents

PC Hardware	07 Hours

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content.

Task 3: Every student should individually install MS windows on the personal computer. Lab instructor should verify the installation and follow it up with a Viva.

Task 4: Every student should install Linux on the computer. This computer should have

windows installed. The system should be configured as dual boot with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva

Internet & World Wide Web

Task1: **Orientation & Connectivity Boot Camp:** Students should get connected to their Local Area Network and access the Internet. In the process they configure the TCP/IP setting. Finally students should demonstrate, to the instructor, how to access the websites and email. If there is no internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop up blockers. Also, plug-ins like Macromedia Flash and JRE for applets should be configured.

Task 3: **Search Engines & Netiquette:** Students should know what search engines are and how to use the search engines. A few topics would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task 4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms.

LaTeX and WORD

Task 1 – Word Orientation: The mentor needs to give an overview of LaTeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of LaTeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using LaTeX and word — Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2: Using LaTeX and Word to create a project certificate. Features to be covered:- Formatting Fonts in word, Drop Cap in word, Applying Text effects, Using Character Spacing, Borders and Colors, Inserting Header and Footer, Using Date and Time option in both LaTeX and Word.

Task 3: Creating project abstract Features to be covered:-Formatting Styles, Inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered:- Table of Content, Newspaper columns, Images from files and clipart, Drawing toolbar and Word Art, Formatting Images, Textboxes, Paragraphs and Mail Merge in word.

Excel

Excel Orientation: The mentor needs to tell the importance of MS office or equivalent (FOSS) tool Excel as a Spreadsheet tool, give the details of the four tasks and features that would be covered in each. Using Excel — Accessing, overview of toolbars, saving excel files, Using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text

Task 2 : Calculating GPA - .Features to be covered:- Cell Referencing, Formulae in excel – average, std. deviation, Charts, Renaming and Inserting worksheets, Hyper linking, Count function, LOOKUP/VLOOKUP

Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, Conditional formatting

Powerpoint

Task 1: Students will be working on basic power point utilities and tools which help them create basic powerpoint presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2: Interactive presentations - Hyperlinks, Inserting –Images, Clip Art, Audio, Video, Objects, Tables and Charts.

Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc), and Inserting – Background, textures, Design Templates, Hidden slides.

CSS

Task 1: Introduction & Setup
Students should install a code editor (VS Code) and create their first HTML & CSS file.
Students will link CSS (inline, internal, external) and demonstrate simple styling like changing text color, font size, and text alignment.
They should use element, class, and ID selectors to style different parts of the webpage.

Task 2: Box Model & Layout
Students should apply margin, padding, and border to various elements and observe their effect.
Students will add background colors and images to sections of a webpage.
Students should experiment with display types (block, inline, inline-block) and use positioning (relative, absolute, fixed) to arrange elements on the page.

Task 3: Flexbox & Page Design
Students should create a simple navigation bar using Flexbox.
Students will align items horizontally and vertically, distribute space evenly, and center content.
Students should combine Flexbox with previous knowledge to build a basic webpage layout with a header, content area, and footer.

Task 4: Styling & Responsiveness
Students should style forms (input fields, buttons) with hover effects and style tables with borders and alternating row colors.
Students will create a simple responsive webpage using media queries and demonstrate how the layout adjusts for desktop and mobile screen sizes.

HTML

Task 1: Introduction & Basic Structure
Students should learn the basic structure of an HTML document (<!DOCTYPE html>, <html>, <head>, <body>).
Students will create a simple webpage with headings, paragraphs, and line breaks.
Students should add links (<a>), images (<img>), and lists (ordered & unordered) to the page.

Task 2: Page Layout & Semantic Tags
Students should use , <section>, <article>, <header>, <footer>, and <main> to organize content.

Students will create a simple table (<table>, <tr>, <th>, <td>) and include a caption.
Students should embed multimedia (audio/video) and use iframes to display external content.

Task 3: Forms & User Input
Students should create a form with different input types (text, number, checkbox, radio, password, submit).
Students will add <label> tags for accessibility and group related inputs using <fieldset> and <legend>.
Students should add buttons and demonstrate how a form collects data.

Task 4: Best Practices & Validation
Students should use semantic tags properly for a clean structure and add alt attributes for images.
Students will demonstrate proper use of headings (<h1>–<h6>) for accessibility.
Students should validate their HTML code using an online validator and fix any errors or warnings.

REFERENCE BOOKS:

1. Comdex Information Technology course tool kit Vikas Gupta, *WILEY Dreamtech*
2. The Complete Computer upgrade and repair book, 3rd edition Cheryl A Schmidt, *WILEY Dreamtech*
3. Introduction to Information Technology, ITL Education Solutions limited, *Pearson Education.*
4. PC Hardware - A Handbook – Kate J. Chase *PHI* (Microsoft)
5. LaTeX Companion – Leslie Lamport, *PHI/Pearson.*
6. IT Essentials PC Hardware and Software Companion Guide Third Edition by David Anfinson and Ken Quamme. — *CISCO Press, Pearson Education*.
7. IT Essentials PC Hardware and Software Labs and Study Guide Third Edition by Patrick Regan — CISCO Press, *Pearson Education.*

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3

3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-111-BSC: MATHEMATICAL FOUNDATION - II
B.Tech. I Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 04	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of calculus, linear algebra, and probability theory.

Course Objective:

1. Methods of solving the differential equations of first and higher order.
2. Concept, properties of Laplace transforms
3. Solving ordinary differential equations using Laplace transforms techniques.
4. The physical quantities involved in engineering field related to vector valued functions
5. The basic properties of vector valued functions and their applications to line, surface and volume integrals

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Identify whether the given differential equation of first order is exact or not
2. Solve higher differential equation and apply the concept of differential equation to real world problems.
3. Use the Laplace transforms techniques for solving ODE's.
4. Evaluate the line, surface and volume integrals and converting them from one to another

Course Contents

UNIT-I	First Order ODE	08 Hours
Exact differential equations, Equations reducible to exact differential equations, linear and Bernoulli's equations. Application: Orthogonal Trajectories		
UNIT-II	**Ordinary Differential Equations of Higher Order**	**08 Hours**
Second order linear differential equations with constant coefficients: Non-Homogeneous terms of the type $e$ax, sin ax , cos ax, polynomials in x, $e$ax$V(x)$ and $x\ V(x)$, method of variation of parameters, Legendre's equation, Cauchy-Euler equation.		
UNIT-III	**Laplace Transforms**	**08 Hours**

Laplace Transforms: Laplace Transform of standard functions, First shifting theorem, Second shifting theorem, Unit step function, Dirac delta function, Laplace transforms of functions when they are multiplied and divided by 't', Laplace transforms of derivatives and integrals of function, Evaluation of integrals by Laplace transforms, Laplace transform of periodic functions, Inverse Laplace transform by different methods. Applications: solving Initial value problems by Laplace Transform method.		
UNIT-IV	**Vector Differentiation**	**08 Hours**
Vector point functions and scalar point functions, Gradient, Divergence and Curl, Directional derivatives, Tangent plane and normal line. Partial differentiation of implicit functions – Taylor's series for functions of two variables		
UNIT-V	**Vector Integration**	**08 Hours**
Line, Surface and Volume Integrals, Theorems of Green, Gauss and Stokes and their applications.		

TEXT BOOKS:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010
2. R.K. Jain and S.R.K. Iyengar, Advanced Engineering Mathematics, Narosa Publications, 5th Edition, 2016.

REFERENCE BOOKS:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
3. H. K. Dass and Er. Rajnish Verma, Higher Engineering Mathematics, S Chand and Company Limited, New Delhi.
4. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	3	2.4	2.4	2	2	3	2.6	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-112-BSC: APPLIED STATISTICS
B.Tech. I Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of calculus, linear algebra, and probability theory.

Course Objective:

1. To understand the fundamental concepts and principles of applied statistics.
2. To learn how to collect, organize, and summarize data effectively.
3. To develop proficiency in probability theory and distributions.
4. To master techniques of inferential statistics for hypothesis testing and estimation.
5. To apply regression analysis and other statistical methods for predictive modeling.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Explain and apply the basic concepts of descriptive and inferential statistics.
2. Collect, organize, and summarize data using appropriate statistical methods.
3. Use probability theory to model and analyze random phenomena.
4. Conduct hypothesis testing and construct confidence intervals.
5. Perform and interpret regression analysis and other predictive modeling techniques.

Course Contents

UNIT-I	Introduction to Applied Statistics	08 Hours

1. Definition and scope of Statistics 2. Types of Characteristics, Different types of scales: nominal, ordinal, interval, and ratio. 3. Collection of Primary data: concept of a questionnaire and a schedule, Secondary data 4. Types of data: Qualitative and quantitative data; Time series data and cross section data, discrete and continuous data. 5. Tabulation. 6. Univariate frequency distribution of discrete and continuous variables. Cumulative frequency distribution. 7. Graphical representation of frequency distribution by Histogram, frequency polygon, Cumulative frequency curve.

UNIT-II	Measures of Variability I	07 Hours

i. Measures of Central Tendency Definitions and Applications of Arithmetic mean. Geometric mean, Harmonic mean, relation between Geometric mean, Harmonic mean. Arithmetic mean, Positional Averages (Median, quartiles, deciles, and percentiles), and mode, with simple examples and no any derivation. Merits and demerits of central tendency

ii. Discrete Random Variables, The Binomial Distribution, The Poisson Distribution, Continuous Random Variables, The Normal Distribution, The Standard Normal Distribution, Finding

Areas under a Normal Curve.		
UNIT-III	**Probability Distributions**	**07 Hours**
Discrete Random Variables, The Binomial Distribution, The Poisson Distribution, Continuous Random Variables, The Normal Distribution, The Standard Normal Distribution, Finding Areas under a Normal Curve.		
UNIT-IV	**Testing Hypothesis**	**07 Hours**
Hypothesis Testing, Errors arising in Hypothesis Testing, steps to Perform a Test of Hypothesis using the p-value approach, The Chi-Square Distribution, The F – Distribution		
UNIT-IV	**Examining Relationship**	**07 Hours**
Pearson Correlation Coefficient, Hypothesis Test for a Correlation, Correlation Analysis, ANOVA Overview Simple Linear Regression Analysis, Error of Prediction, Limitations of Dummy & Interaction Terms.		
UNIT-VI	**Advanced Techniques**	**07 Hours**
Non-Parametric Tests - Chi-Squared Goodness-of-Fit Test, Chi-Square Test of Independence, The Sign Test, Mann-Whitney Test, Kruskal-Wallis H-Test. Structural Equation Modeling (SEM), Cluster Analysis Background, Cluster Analysis, Hierarchical Clustering, Non-Hierarchical Method, Factor Analysis (FA), Centroid Method, Principal Components Method, Multidimensional Scaling (MDS). Making evidence-based recommendations for policy improvements.		

REFERENCE BOOKS:

1. "Numerical Methods for Engineers" by Steven C. Chapra and Raymond P. Canale
2. "Applied Numerical Methods with MATLAB" by Steven C. Chapra
3. "Elements of Information Theory" by Thomas M. Cover and Joy A. Thomas
4. "Information Theory, Inference, and Learning Algorithms" by David J.C. MacKay

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	3	2.4	2.4	2	2	3	2.6	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-112-L: APPLIED STATISTICS LAB
B.Tech. I Year II Sem.

Practical Scheme: P: - 4 Hours/Week	Credit:01	Examination Scheme: Total : 30 Marks

Course Prerequisites: Basic knowledge of Python programming

Course Objective:

1. Introduce students to essential Python packages used in data analytics.
2. Provide hands-on experience in using these packages for data manipulation, analysis, and visualization.
3. Develop the ability to implement data analytics workflows using Python.
4. Enhance problem-solving skills using Python for real-world data analytics problems.
5. Understand the integration of Python packages for comprehensive data science solutions.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Understand and set up the Python environment for data analytics. Know the applications of various devices.

CO2: Demonstrate proficiency in using NumPy and Pandas for data manipulation and preprocessing.

CO3: Create and customize data visualizations using Matplotlib and Seaborn.

CO4: Perform statistical analysis and hypothesis testing using SciPy and Statsmodels.

CO5: Implement machine learning algorithms using Scikit-Learn for both supervised and unsupervised learning tasks.

CO6: Integrate multiple Python packages to develop comprehensive data analytics workflows.

CO7: Explore advanced Python packages for deep learning and apply best practices in data analytics projects.

List of Experiments

Lab Exercises:

Unit 1: Introduction to Applied Statistics

Lab 1: Understanding Types of Data and Levels of Measurement

- Objective: Learn to classify different types of data (nominal, ordinal, interval, ratio) and their levels of measurement.
- Tools: Python (Pandas), R
- Activities: Identify types of data and levels of measurement in a given dataset.

Unit 2: Descriptive Statistics

Lab 2: Data Collection Methods

- Objective: Explore various methods of data collection.
- Tools: Python, R
- Activities: Simulate data collection using surveys, experiments, and observational studies.

Lab 3: Data Organization and Presentation

- Objective: Organize and present data using frequency distributions and graphical representations.
- Tools: Python (Pandas, Matplotlib), R
- Activities: Create frequency distributions, histograms, pie charts, and box plots.

Lab 4: Measures of Central Tendency

- Objective: Calculate mean, median, and mode.
- Tools: Python (Pandas), R
- Activities: Compute and interpret mean, median, and mode for a dataset.

Lab 5: Measures of Dispersion

- Objective: Calculate range, variance, and standard deviation.
- Tools: Python (Pandas), R
- Activities: Compute and interpret range, variance, and standard deviation for a dataset.

Lab 6: Measures of Shape and Position

- Objective: Analyze skewness, kurtosis, percentiles, and quartiles.
- Tools: Python (SciPy, Pandas), R
- Activities: Calculate and interpret skewness, kurtosis, percentiles, and quartiles.

Unit 3: Probability Theory

Lab 7: Basics of Probability

- Objective: Understand basic probability concepts.
- Tools: Python, R
- Activities: Solve probability problems using Python or R.

Lab 8: Probability Distributions

- Objective: Work with discrete and continuous probability distributions.
- Tools: Python (NumPy, SciPy), R
- Activities: Generate and analyze binomial, Poisson, normal, and exponential distributions.

Lab 9: Conditional Probability and Independence

- Objective: Explore conditional probability and independence.
- Tools: Python, R
- Activities: Calculate conditional probabilities and test for independence in datasets.

Unit 4: Inferential Statistics

Lab 10: Sampling and Sampling Distributions

- Objective: Understand sampling techniques and sampling distributions.
- Tools: Python (Pandas, NumPy), R
- Activities: Perform random sampling and create sampling distributions.

Lab 11: Point Estimation and Interval Estimation

- Objective: Perform point and interval estimation.

- Tools: Python (SciPy), R
- Activities: Estimate population parameters using point and interval estimation.

CO's-PO's & PSO's MAPPING

(CO)	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	-	-	-	3	-	3
CO2	3	3	-	-	3	3	3	3
CO3	3	-	3	-	3	-	3	-
CO4	3	3	-	3	-	3	-	3
CO5	3	3	-	-	3	3	-	3
CO6	3	3	3	-	3	-	3	3
CO7	3	3	3	-	3	3	3	-

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-113-BSC: COMPUTATIONAL BIOLOGY
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Introduction To Data Science

Course Objective:

1. To provide a solid foundation in biological data, processes, and structures.
2. To introduce algorithms and computational techniques for analyzing biological data.
3. To apply machine learning and statistical methods to biological datasets.
4. To familiarize students with biological databases and bioinformatics tools.
5. To enable students to design and analyze experiments using computational techniques.
6. To encourage hands-on experience in solving real-world biological problems using Computational approaches.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Understand the basics of molecular biology and its computational implications.
CO2: Explore and utilize various biological databases for data retrieval and analysis.
CO3: Implement sequence alignment algorithms for biological sequences.
CO4: Apply algorithmic techniques to analyze evolutionary relationships using phylogenetic trees.
CO5: Analyze and interpret NGS data using computational tools.
CO6: Identify genetic variations using variant detection algorithms.
CO7: Predict and analyze the 3D structures of proteins.
CO8: Use computational methods for structure-based drug discovery.
CO9: Apply machine-learning techniques to biological datasets.
CO10: Perform statistical analysis and hypothesis testing on biological data.
CO11: Model biological systems and interactions using network approaches.
CO12: Simulate biological processes for systems-level understanding.

Course Contents

UNIT-I	Introduction to Computational Biology and Bioinformatics	07 Hours

Fundamentals of Molecular Biology: Basics of DNA, RNA, and proteins.
Overview of Bioinformatics: What is bioinformatics? Applications in biology and medicine.
Biological Databases: Introduction to NCBI, UniProt, and PDB.
Basic Tools for Sequence Analysis: Familiarization with tools like BLAST.

UNIT-II	Introduction to Biological Sequence Analysis	08 Hours

What is Sequence Alignment?: Concepts of pairwise and multiple sequence alignment.

Basic Tools for Alignment: Introduction to BLAST and Clustal. **Simple Algorithms**: Overview of dynamic programming for sequence alignment. **Introduction to Phylogenetic**: Basics of constructing evolutionary trees		
UNIT-III	**Introduction to Genomics and Data Visualization**	**08 Hours**
What is Genomics? Understanding genes and genomes. **Next-Generation Sequencing (NGS)**: Overview and its role in genome analysis. **Introduction to Data Visualization**: Basics of creating plots using Python libraries (e.g., Matplotlib and Seaborn).		
UNIT-IV	**Basics of Protein Structure and Drug Design**	**07 Hours**
Protein Structures: Introduction to primary, secondary, and tertiary protein structures. **Protein Structure Databases**: Familiarization with PDB. **Basics of Drug Design**: Understanding how drugs interact with proteins (no advanced modelling techniques).		
UNIT-V	**Introduction to Python for Bioinformatics**	**08 Hours**
Python Basics: Variables, loops, conditionals, and data types. **File Handling**: Reading and writing biological sequence Files. **Sequence Operations**: DNA/RNA manipulation, pattern matching, and motif Search. **Biopython**: Sequence manipulation, file parsing, and metadata extraction. **Data Analysis**: Using Pandas for tabular biological Data. **Data Visualization**: Creating static and interactive plots using Matplotlib, Seaborn, and Plotly. **NCBI Data Retrieval**: Fetching sequences and metadata using Biopython's Entrez. **Statistical Analysis**: Calculating sequence properties and basic statistics.		
UNIT-VI	**Systems Biology and Biological Networks**	**08 Hours**
What is Systems Biology? Understanding how biological systems work as a whole. **Biological Networks**: Introduction to protein-protein interaction and gene regulatory networks. **Simple Simulations**: Basic examples of modeling small biological systems using computational tools.		

REFERENCE BOOKS:

1. "Bioinformatics: Sequence and Genome Analysis" by David W. Mount
2. "Introduction to Bioinformatics" by Arthur Lesk
3. "Molecular Biology of the Cell" by Bruce Alberts
4. "Bioinformatics Algorithms: An Active Learning Approach" by Phillip Compeau and Pavel Pevzner

CO's-PO's & PSO's MAPPING

(CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	-	-	-	2	-	-	2	-	-
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	-	3	-	-	3	2	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3
CO7	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-114-ESC: DISCRETE MATHEMATICS
B.Tech. I Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 04	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 +2 discrete mathematics

Course Objective:

- Introduces elementary discrete mathematics for computer science and engineering.
- Topics include formal logic notation, methods of proof, induction, sets, relations, algebraic structures, elementary graph theory, permutations and combinations, counting principles; recurrence relations and generating functions.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand and construct precise mathematical proofs
- Apply logic and set theory to formulate precise statements
- Analyze and solve counting problems on finite and discrete structures
- Describe and manipulate sequences
- Apply graph theory in solving computing problems

Course Contents

UNIT-I	Foundations of Discrete Mathematics	07 Hours

Sets, subsets, and set operations, Logic and propositions: Propositional logic, logical connectives, truth tables, tautologies, contradictions, Predicate logic: Quantifiers, predicates, and logical equivalences, Proof techniques: Direct proof, proof by contradiction, mathematical induction

Hands-on Exercises:

- Set operations and Venn diagrams
- Constructing truth tables and logical equivalences
- Writing proofs using different proof techniques

UNIT-II	Combinatorics and Counting	08 Hours

Basic counting principles: Addition and multiplication rules, Permutations and combinations, Binomial theorem and Pascal's triangle, Pigeonhole principle, Inclusion-exclusion principle

Hands-on Exercises:

- Solving combinatorial problems with permutations and combinations
- Applying the binomial theorem to data science problems
- Using inclusion-exclusion principle in real-world scenarios

UNIT-III	Graph Theory	08 Hours

Introduction to graphs: Definitions, terminology, and types of graphs, Graph representation: Adjacency matrix, adjacency list, Graph traversal: Breadth-first search (BFS), depth-first search (DFS), Shortest path algorithms: Dijkstra's algorithm, Bellman-Ford algorithm, Graph applications in data science: Social network analysis, clustering

Hands-on Exercises:

- Implementing BFS and DFS in Python
- Solving shortest path problems with Dijkstra's and Bellman-Ford algorithms
- Analyzing social networks using graph theory

UNIT-IV	Relations and Functions	07 Hours

Relations: Properties, representation, equivalence relations, partial orderings, Functions: Definitions, types of functions, composition, and inverses, Recurrence relations: Solving linear recurrence relations, applications in algorithm analysis

Hands-on Exercises:

- Working with relations and their properties
- Implementing and analyzing functions in Python
- Solving and analyzing recurrence relations

UNIT-V	Number Theory and Cryptography	08 Hours

Divisibility and prime numbers, Congruences and modular arithmetic, Euclidean algorithm and applications, Basics of cryptography: RSA algorithm, public-key

Hands-on Exercises:

- Solving problems involving divisibility and modular arithmetic
- Implementing the Euclidean algorithm in Python
- Encrypting and decrypting messages using the RSA algorithm

UNIT-VI	Probability and Statistics	08 Hours

Basic probability theory: Sample spaces, events, probability axioms, Conditional probability and Bayes' theorem, Random variables and probability distributions, Expectation, variance, and standard deviation, Applications in data science: Hypothesis

testing, regression analysis

Hands-on Exercises:

- Calculating probabilities and applying Bayes' theorem
- Working with random variables and probability distributions
- Performing hypothesis testing and regression analysis in Python

TEXT BOOKS:

1. Discrete Mathematical Structures with Applications to Computer Science: J.P. Tremblay, R. Manohar, McGraw-Hill, 1st ed.
2. Discrete Mathematics for Computer Scientists & Mathematicians: Joe l. Mott, Abraham Kandel, Teodore P. Baker, Prentis Hall of India, 2nd ed.

REFERENCE BOOKS:

3. "Discrete Mathematics and Its Applications" by Kenneth H. Rosen
4. "Discrete Mathematics with Applications" by Susanna S. Epp
5. "Graph Theory with Applications" by Bondy and Murty
6. "Introduction to the Theory of Computation" by Michael Sipser
7. "Probability and Statistics for Engineers and Scientists" by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, and Keying Ye
8. "Cryptography and Network Security: Principles and Practice" by William Stallings

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	-	-	3	-	3	-
CO2	3	3	3	-	-	3	-	3	3
CO3	3	3	-	-	3	3	-	-	3
CO4	3	3	-	3	-	-	3	-	-
CO5	3	3	-	-	3	3	-	3	-
CO6	3	3	-	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-115-PCC: PROGRAMMING FOR PROBLEM SOLVING
B.Tech. I Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 + 2 Science

Course Objective:

1. To introduce the fundamental concepts of programming using Python.
2. To develop problem-solving skills through programming.
3. To teach the basic constructs of Python for algorithmic thinking.
4. To familiarize students with Python's data structures and standard libraries.
5. To enable students to write efficient, readable, and modular code.
6. To prepare students for advanced courses in data science, machine learning, and software development.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the basic concepts of programming and problem-solving using Python.
CO2: Apply control structures for decision making and looping in Python programs.
CO3: Develop functions and modules for modular and reusable code.
CO4: Utilize Python's built-in data structures for efficient data manipulation.
CO5: Perform file-handling operations for reading and writing data.
CO6: Implement object-oriented programming concepts in Python.
CO7: Use basic libraries for data science to manipulate and visualize data.

Course Contents

UNIT-I	Introduction to Python	08 Hours

Introduction to Programming and Problem Solving, Basics of Python: Installing Python, IDEs, Python Interpreter, Python Syntax, Keywords, Identifiers, and Variables, Basic Input/Output Operations, Data Types and Type Conversion, Comments and Documentation

UNIT-II	Control Structures	08 Hours

Decision Making: if, if-else, nested if-else, Looping: for, while, nested loops, Control Statements: break, continue, pass, List Comprehensions, Practical Applications of Control Structures.

UNIT-III	Functions and Modules	08 Hours
Defining and Calling Functions, Function Arguments and Return Values, Lambda Functions, Recursion, Modules and Packages, Built-in Functions and Standard Libraries.		
UNIT-IV	**Data Structures**	**08 Hours**
Lists: Operations, Methods, and Applications, Tuples: Operations and Applications, Dictionaries: Operations, Methods, and Applications, Sets: Operations and Methods, String Manipulation and Methods, Arrays and their Applications.		
UNIT-V	**File Handling**	**08 Hours**
Reading from and Writing to Files, File Methods and Attributes, Working with Text and Binary Files, Exception Handling in File Operations, CSV File Handling.		
UNIT-VI	**Advanced Topics**	**08 Hours**
Object-Oriented Programming in Python: Classes and Objects, Inheritance and Polymorphism, Encapsulation and Abstraction, Introduction to Libraries for Data Science: NumPy: Arrays and Matrices, Pandas: DataFrames and Series, Matplotlib: Basic Plotting Introduction to Regular Expressions		

REFERENCE BOOKS:

1. "Python Crash Course" by Eric Matthes ISBN-13: 978-1593276034
2. "Automate the Boring Stuff with Python" by Al Sweigart ISBN-13: 978-1593275990
3. "Learning Python" by Mark Lutz ISBN-13: 978-1449355739
4. "Python for Data Analysis" by Wes McKinney ISBN-13: 978-1491957660

CO's-PO's & PSO's MAPPING

(CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	-	-	2	2	2
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	2	3	-	-	3	3	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3
CO7	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-115-L: PROGRAMMING FOR PROBLEM SOLVING LABORATORY
B.Tech. I Year II Sem.

Practical Scheme: P: - 2 Hours/Week	Credit:02	Examination Scheme: Total : 50 Marks

Course Prerequisites: Familiarity with basic programming concepts and syntax, preferably in another Language, and an understanding of fundamental computer science principles.

Course Objective:

To impart the basic concepts of data structures and algorithms. To understand concepts about searching and sorting techniques. To Understand basic concepts about stacks, queues, lists, trees and graphs To understand the algorithms and develop the step by step solutions of problemswith the help of data

Learning Course Outcome:

After successful completion of the course, students will able to:

The experiments will make the student gain skills on:

CO1: Understand Basic Programming Constructs
CO2: Implement Data Structures and Algorithms
CO3: Develop Object-Oriented Programming Skills
CO4: Use Python Libraries for Data Handling and Analysis
CO5: Apply File Handling and Exception Handling Techniques
CO6: Implement and Analyze Sorting and Searching Algorithms
CO7: Develop Skills in Problem Solving and Debugging
CO8: Gain Exposure to Advanced Topics

Course Contents

Experiment 1: Introduction to Python

- Write a Python program to print "Hello, World!".
- Write a Python program to take user input and print it.
- Write a Python program to swap two numbers.

Experiment 2: Control Structures

- Write a Python program to check if a number is even or odd.
- Write a Python program to find the factorial of a number using loops.
- Write a Python program to print Fibonacci series up to n terms.

Experiment 3: Functions and Modules

- Write a Python program to find the largest number among three numbers using a function.
- Write a Python program to check if a number is prime or not using a function.
- Write a Python program to generate a random number between a given range using a module.

Experiment 4: Lists and Strings

- Write a Python program to find the sum and average of elements in a list.
- Write a Python program to count the number of vowels in a string.
- Write a Python program to reverse a string using slicing.

Experiment 5: Dictionaries and Sets

- Write a Python program to demonstrate dictionary operations (add, delete, update).
- Write a Python program to find the intersection and union of two sets.
- Write a Python program to count the frequency of elements in a list using a dictionary.

Experiment 6: File Handling

- Write a Python program to read and display the contents of a file.
- Write a Python program to write user input to a file.
- Write a Python program to count the number of words in a file.

Experiment 7: Object-Oriented Programming (OOP)

- Write a Python program to demonstrate class and object creation.
- Write a Python program to demonstrate inheritance (base and derived classes).
- Write a Python program to demonstrate method overriding.

Experiment 8: Exception Handling

- Write a Python program to handle ZeroDivisionError and IndexError exceptions.
- Write a Python program to demonstrate the use of try-except-else-finally block.
- Write a Python program to create a custom exception and raise it.

Experiment 9: Recursion

- Write a Python program to find the factorial of a number using recursion.
- Write a Python program to calculate the nth Fibonacci number using recursion.
- Write a Python program to implement binary search using recursion.

Experiment 10: Sorting and Searching Algorithms

- Write a Python program to implement bubble sort and analyze its time complexity.
- Write a Python program to implement binary search on a sorted list.
- Write a Python program to implement linear search and analyze its time complexity.

Experiment 11: Data Structures (Optional)

- Write a Python program to implement a stack using lists.
- Write a Python program to implement a queue using lists.
- Write a Python program to implement a binary search tree (BST).

Experiment 12: Introduction to NumPy and Pandas (Data Handling)

- Write a Python program to create a NumPy array and perform basic operations (addition, subtraction).
- Write a Python program to create a Pandas DataFrame and perform basic operations (selection, filtering).

REFERENCE BOOKS:

1. "Python Programming: An Introduction to Computer Science" by John Zelle
2. Official documentation of Python (python.org)
3. Official documentation of NumPy (numpy.org)
4. Official documentation of Pandas (pandas.pydata.org)

CO's-PO's & PSO's MAPPING

(CO)	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
CO1	3	3	-	-	-	3	-	3
CO2	3	3	-	-	-	3	2	3
CO3	3	3	2	-	-	2	3	3
CO4	3	3	2	2	-	3	-	3
CO5	3	3	-	2	-	2	-	3
CO6	3	3	-	-	-	2	2	3
CO7	3	3	-	-	-	3	2	3
CO8	3	2	-	-	-	2	2	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-116-AEC: INDIAN KNOWLEDGE SYSTEM
B.Tech. I Year II Sem.

Teaching Scheme: TH: - 2 Hours/Week	Credit TH: 02	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Introduction To Data Science

Course Objective:

1. **To introduce students to the Indian Knowledge Systems (IKS):** Provide an understanding of traditional knowledge across various fields such as science, technology, philosophy, and arts from ancient Indian texts and practices.
2. **To demonstrate the contributions of IKS in Science and Technology:** Explore the scientific, technological, and mathematical innovations of ancient India and their relevance today.
3. **To appreciate the ethical and philosophical foundations of IKS:** Develop insights into the ethical values, governance models, and philosophical traditions that have shaped India's cultural and intellectual history.
4. **To link ancient Indian knowledge with modern engineering and technology:** Encourage students to integrate traditional Indian knowledge with modern-day applications, promoting sustainability, innovation, and problem-solving.
5. **To instill cultural pride and awareness:** Help students gain a deeper appreciation for India's rich intellectual and cultural heritage, fostering a sense of pride and identity.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. **CO1: Understand the scope and significance of Indian Knowledge Systems**
   Explain the meaning, scope, and historical evolution of Indian Knowledge Systems and their relevance to modern education and technology.

2. **CO2: Analyze contributions of IKS in Science, Mathematics, and Technology**
   Identify and describe key contributions of ancient Indian thinkers and scientists in the fields of mathematics, astronomy, medicine, and engineering.

3. **CO3: Apply ethical and philosophical principles of IKS to modern-day scenarios**
   Analyze how ancient Indian ethical and philosophical principles (such as Dharma, Karma, and Moksha) can be integrated into modern-day personal, professional, and social life.

4. **CO4: Evaluate the sustainability aspects of ancient Indian practices**
   Assess the ecological and sustainable dimensions of Indian traditional practices, such as agriculture, water management, and architecture, and their relevance to modern environmental challenges.

5. **CO5: Explore interdisciplinary applications of IKS in contemporary technology**
   Engage in case studies or projects that integrate IKS concepts into modern engineering, architecture,

healthcare, or technology, providing innovative and sustainable solutions.

6. **CO6: Develop an appreciation for the cultural and intellectual heritage of India**
   Critically evaluate the historical and cultural significance of IKS, fostering a greater awareness and appreciation of India's intellectual traditions and their global influence.

Course Contents

UNIT-I	Introduction Science and Technology in Ancient India	07 Hours
Definition & Scope of IKS: What is IKS and its importance. Historical Context: Development of Indian Knowledge Systems through history. Sources of IKS: Vedas, Upanishads, Epics (Ramayana, Mahabharata), Puranas, classical Sanskrit texts, regional literatures, inscriptions, oral traditions. Mathematics: Contribution of Indian mathematicians like Aryabhata, Brahmagupta, and Bhaskara. Astronomy: Developments in ancient Indian astronomy (Jyotisha), including the works of Varahamihira. Medicine (Ayurveda, Siddha, and Unani): Concepts of health, body, treatment systems. Engineering and Architecture: Study of ancient Indian engineering and architecture (temple construction, water management, urban planning). Metallurgy: Ancient knowledge of metals and materials science (Iron Pillar of Delhi, Wootz steel). Textile Technology: Traditional knowledge in textiles, dyeing, and fabric production.		
UNIT-II	**Philosophical and Ethical Foundations of IKS Traditional Indian Knowledge in Arts and Aesthetics**	**08 Hours**
Indian Philosophical Systems: Nyaya, Vaisheshika, Samkhya, Yoga, Mimamsa, Vedanta, and their role in scientific thought. Ethics and Dharma: Concepts of ethics in Indian philosophy and their application in professional life. Value Systems in Ancient Indian Texts: Dharma, Artha, Kama, Moksha and their relevance to modern life. Music and Dance: Understanding classical Indian music and dance forms, their mathematical and artistic rigor (e.g., Natyashastra). Art and Architecture: Study of traditional Indian art forms, painting styles (Ajanta, Ellora), temple architecture. Literature: Ancient Indian literature and poetics (e.g., Kalidasa's works, Bharata's Natyashastra).		
UNIT-III	**Indian Knowledge Systems in Agriculture and Environmental Sustainability**	**08 Hours**
Traditional Farming Practices: Crop diversity, organic farming, sustainable land use in ancient India. Water Management: Traditional knowledge in irrigation systems (e.g., step wells, tanks, canals). Sustainable Living: Concepts of sustainability in Indian traditional practices, eco-friendly architecture (e.g., mud houses, organic building		

materials).		
UNIT-IV	**Law, Governance, and Economic Thought in Ancient India**	**07 Hours**
Political Thought: Study of ancient Indian governance models (e.g., Arthashastra, Dharma Shastras). Economics: Traditional economic systems, trade routes, role of guilds and markets. Legal Systems: Early legal thought, Smriti texts, justice systems.		
UNIT-V	**Relevance of Indian Knowledge Systems in Modern Science and Technology**	**08 Hours**
IKS and Modern Science: Exploring how ancient Indian scientific thought can complement modern technological advancements. Case Studies: Applications of IKS in fields like sustainable development, healthcare, technology, and innovation. Innovation and Entrepreneurship: Lessons from ancient India's knowledge and Innovation systems that can inspire modern entrepreneurship.		
UNIT-VI	**Practical Integration and Project Work**	**08 Hours**
IKS in Daily Life: Understanding how IKS principles (such as Ayurveda, Vaastu Shastra) can be applied today. Projects/Case Studies: Students can be encouraged to work on projects that merge modern engineering principles with IKS concepts, such as sustainable architecture, eco-friendly technologies, or traditional medicine. Field Visits: Visits to historically significant sites, traditional industries, or centers of IKS research.		

REFERENCE BOOKS:

1. **Texts**: Selected readings from ancient Indian literature, scientific texts (e.g., Aryabhatiya, Charaka Samhita), and modern commentaries.
2. **Research Papers**: Modern scholarly articles on IKS.

CO's-PO's & PSO's MAPPING

(CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	-	-	-	2	-	-	2	-	-
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	-	3	-	-	3	2	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-117-VSEC: PROFESSIONAL COMMUNICATION SKILLS-II
B.Tech. I Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: English Grammar basic knowledge

Course Objective:

- To engage learners in meaningful language activities to improve their reading and writing skills
- To learn various reading strategies and apply in comprehending documents in professional context.
- To help learners understand the purpose, audience, contexts of different types of writing
- To develop analytical thinking skills for problem solving in communicative contexts
- To demonstrate an understanding of job applications and interviews for internship and placements

Learning Course Outcome:
After successful completion of the course, students will able to:

- CO1: To compare and contrast products and ideas in technical texts.
- CO2: To identify and report cause and effects in events, industrial processes through technical texts
- CO3: To analyse problems in order to arrive at feasible solutions and communicate them in the written format.
- CO4: To present their ideas and opinions in a planned and logical manner
- CO5: To draft effective resumes in the context of job search.

Course Contents

UNIT-I	MAKING COMPARISONS	06 Hours
Reading - Reading advertisements, user manuals, brochures; Writing – Professional emails, Email etiquette - Compare and Contrast Essay; Grammar – Mixed Tenses, Prepositional phrases.		
UNIT-II	**EXPRESSING CAUSAL RELATIONS IN SPEAKING AND WRITING**	**06 Hours**
Reading - Reading longer technical texts– Cause and Effect Essays, and Letters / emails of complaint, Writing - Writing responses to complaints. Grammar - Active Passive Voice transformations, Infinitive and Gerunds.		
UNIT-III	**PROBLEM SOLVING**	**06 Hours**
Reading - Case Studies, excerpts from literary texts, news reports etc. Writing – Letter to the Editor, Checklists, Problem solution essay / Argumentative Essay. Grammar – Error correction; If conditional sentences.		

UNIT-VI	REPORTING OF EVENTS AND RESEARCH	06 Hours
Reading –Newspaper articles; Writing – Recommendations, Transcoding, Accident Report, Survey Report Grammar – Reported Speech, Modals Vocabulary – Conjunctions- use of prepositions.		
UNIT-V	**THE ABILITY TO PUT IDEAS OR INFORMATION COGENTLY**	**06 Hours**
Reading – Company profiles, Statement of Purpose, (SOP), an excerpt of interview with professionals; Writing – Job / Internship application – Cover letter & Resume; Grammar – Numerical adjectives, Relative Clauses.		

TEXTBOOKS:

1. English for Engineers & Technologists (2020 edition) Orient Blackswan Private Ltd.Department of English, Anna University.
2. English for Science & Technology Cambridge University Press 2021.
3. Authored by Dr. Veena Selvam, Dr. Sujatha Priyadarshini, Dr. Deepa Mary Francis, Dr. KN.Shoba, and Dr. Lourdes Joevani, Department of English, Anna University.

REFERENCES:

1. Raman. Meenakshi, Sharma. Sangeeta (2019). Professional English. Oxford university press.New Delhi.
2. Improve Your Writing ed. V.N. Arora and Laxmi Chandra, Oxford Univ. Press, 2001, New Delhi.
3. Learning to Communicate – Dr. V. Chellammal. Allied Publishers, New Delhi, 2003
4. Business Correspondence and Report Writing by Prof. R.C. Sharma & Krishna Mohan, TataMcGraw Hill & Co. Ltd., 2001, New Delhi.
5. Developing Communication Skills by Krishna Mohan, Meera Bannerji- Macmillan India Ltd.1990, Delhi.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
1	3	3	3	3	3	3	3	3	2	3	3	3	-	-	-
2	3	3	3	3	3	3	3	3	2	3	3	3	-	-	-
3	3	3	3	3	3	3	3	3	2	3	3	3	-	-	-
4	3	3	3	3	2	3	3	3	2	3	3	3	-	-	-
5	-	-	-	-	-	-	-	-	**3**	**3**	**3**	**3**	-	-	-
AVg.	**3**	**3**	**3**	**3**	**2.75**	**3**	**3**	**3**	**2.2**	**3**	**3**	**3**	-	-	-

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-201-PCC: DATA STRUCTURES
B.Tech. II Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 04	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Programming for Problem Solving

Course Objective:

- Exploring basic data structures such as stacks and queues.
- Introduces a variety of data structures such as hash tables, search trees, tries, heaps, graphs.
- Introduces sorting and pattern matching algorithms

Learning Course Outcome:

After successful completion of the course, students will able to:

- Ability to select the data structures that efficiently model the information in a problem.
- Ability to assess efficiency trade-offs among different data structure implementations or combinations.
- Implement and know the application of algorithms for sorting and pattern matching.
- Design programs using a variety of data structures, including hash tables, binary and general tree structures, search trees, tries, heaps, graphs, and AVL-trees.

Course Contents

UNIT-I	Introduction to Data	07 Hours
Introduction to Data Structures, abstract data types, Linear list – singly linked list implementation, insertion, deletion and searching operations on linear list, Stacks-Operations, array and linked representations of stacks, stack applications, Queues-operations, array and linked representations.		
UNIT-II	**Dictionaries**	**08 Hours**
linear list representation, skip list representation, operations - insertion, deletion and searching. Hash Table Representation: hash functions, collision resolution-separate chaining, open addressing- linear probing, quadratic probing, double hashing, rehashing, extendible hashing.		
UNIT-III	**Search Trees**	**08 Hours**
Binary Search Trees, Definition, Implementation, Operations- Searching, Insertion and Deletion, B- Trees, B+ Trees, AVL Trees, Definition, Height of an AVL Tree, Operations – Insertion, Deletion and Searching, Red –Black, Splay Trees.		
UNIT-IV	**Graphs**	**07 Hours**

Graph Implementation Methods. Graph Traversal Methods. Sorting: Quick Sort, Heap Sort, External Sorting- Model for external sorting, Merge Sort.		
UNIT-V	**Pattern Matching and Tries**	**08 Hours**
Pattern matching algorithms-Brute force, the Boyer –Moore algorithm, the Knuth-Morris-Pratt algorithm, Standard Tries, Compressed Tries, Suffix tries.		

TEXT BOOKS:

1. Fundamentals of Data Structures in C, 2 nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press.
2. Data Structures using C – A. S.Tanenbaum, Y. Langsam, and M.J. Augenstein, PHI/Pearson Education.

REFERENCE BOOK:

1. Data Structures: A Pseudocode Approach with C, 2 nd Edition, R. F. Gilberg and B.A.Forouzan, Cengage Learning.

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-201-L: DATA STRUCTURES LAB
B.Tech. II Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 01	In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Prerequisites: A Course on "Programming for problem solving".

Course Objective:

- It covers various concepts of C programming language
- It introduces searching and sorting algorithms
- It provides an understanding of data structures such as stacks and queues.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Ability to develop C programs for computing and real-life applications using basic elements like control statements, arrays, functions, pointers and strings, and data structures like stacks, queues and linked lists.
- Ability to Implement searching and sorting algorithms

Course Contents

List of Experiments	07 Hours

1. Write a program that uses functions to perform the following operations on singly linked list.:
   i) Creation ii) Insertion iii) Deletion iv) Traversal
2. Write a program that uses functions to perform the following operations on doubly linked list.:
   i) Creation ii) Insertion iii) Deletion iv) Traversal
3. Write a program that uses functions to perform the following operations on circular linked list.:
   i) Creation ii) Insertion iii) Deletion iv) Traversal
4. Write a program that implement stack (its operations) using
   i) Arrays ii) Pointers
5. Write a program that implement Queue (its operations) using
   i) Arrays ii) Pointers
6. Write a program that implements the following sorting methods to sort a given list of integers in ascending order
   i) Quick sort ii) Heap sort iii) Merge sort

7. Write a program to implement the tree traversal methods (Recursive and Non Recursive).
8. Write a program to implement
   i) Binary Search tree ii) B Trees iii) B+ Trees iv) AVL trees
   v) Red - Black trees
9. Write a program to implement the graph traversal methods.
10. Implement a Pattern matching algorithms using Boyer- Moore, Knuth-Morris-Pratt

TEXT BOOKS:

1. Fundamentals of Data Structures in C, 2nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press.
2. Data Structures using C – A. S. Tanenbaum, Y. Langsam, and M. J. Augenstein, PHI/Pearson Education.

REFERENCE BOOK:

1. Data Structures: A Pseudocode Approach with C, 2nd Edition, R. F. Gilberg and B. A. Forouzan, Cengage Learning.

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-202-PCC: OBJECT ORIENTED PROGRAMMING LANGUAGE
B.Tech. II Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 04	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of programming concepts.

Course Objective:

1. To introduce the fundamental concepts of object-oriented programming.
2. To provide a comprehensive understanding of the differences and similarities between Javaand C++.
3. To develop problem-solving skills using object-oriented principles.
4. To apply object-oriented concepts in software development.
5. To understand the principles of inheritance, polymorphism, encapsulation, and abstraction.
6. To familiarize students with standard libraries and frameworks in Java and C++.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand and apply the basic principles of object-oriented programming in Java and C++.
CO2: Develop and use classes and objects in Java and C++.
CO3: Implement inheritance to promote code reuse and develop a hierarchical class structure.
CO4: Use polymorphism to create flexible and reusable code.
CO5: Apply encapsulation and abstraction to design robust software.
CO6: Handle exceptions and perform file I/O operations in Java and C++.
CO7: Utilize advanced features such as templates, generics, and multithreading.

Course Contents

UNIT-I	Introduction to Object-Oriented Programming (OOP)	07 Hours
Procedural vs. Object-Oriented Programming, Basic Concepts: Classes, Objects, Methods, and Messages, Benefits of OOP. Principles of Object-Oriented Programming, Benefits of OOP, Introduction to Java and C++, Basic Syntax and Structure, Compiling and Running Programs.		
UNIT-II	**Classes and Objects**	**08 Hours**
Defining Classes and Creating Objects, Constructors and Destructors, Access Specifiers, Static Members and Methods, This Pointer in C++ / This Keyword in Java.		

UNIT-III	Inheritance	08 Hours
Concept of Inheritance, Types of Inheritance: Single, Multiple (C++), Multilevel, Hierarchical, Hybrid, Constructor and Destructor Calls in Inheritance, Method Overriding, The super Keyword in Java, Virtual Base Classes in C++		
UNIT-IV	**Polymorphism**	**07 Hours**
Function Overloading and Operator Overloading in C++, Method Overloading in Java, Virtual Functions and Abstract Classes in C++, Interfaces and Abstract Classes in Java, Runtime Polymorphism, Type Casting.		
UNIT-V	**Encapsulation and Abstraction**	**08 Hours**
Data Hiding and Encapsulation, Access Modifiers and Their Usage, Abstract Classes and Methods, Packages in Java and Namespaces in C++, Implementing Abstraction		
UNIT-VI	**Advanced Object-Oriented Concepts & Real-World Applications and Case Studies**	**08 Hours**
Exception Handling in Java and C++, File Handling: Stream Classes in C++, File I/O in Java, Templates and Generics, Standard Template Library (STL) in C++, Collections Framework in Java, Multithreading and Concurrency: Thread Management in C++, Thread Class and Runnable Interface in Java Developing a Simple Application: Step-by-Step Approach, Case Study: Banking System Application, Case Study: Library Management System, Best Practices in OOP, Debugging and Testing Object- Oriented Programs		

TEXT BOOKS

1. Jeri R. Hanly and Elliot B.Koffman, Problem solving and Program Design in C7th Edition, Pearson
2. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)
3. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
4. E. Balagurusamy, Computer fundamentals and C, 2nd Edition, McGraw-Hill
5. Yashavant Kanetkar, Let Us C, 18th Edition, BPB
6. R.G. Dromey, How to solve it by Computer, Pearson (16th Impression)
7. Programming in C, Stephen G. Kochan, Fourth Edition, Pearson Education.
8. Herbert Schildt, C: The Complete Reference, Mc Graw Hill, 4th Edition
9. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO 1	PSO 2	PSO 3	PSO 4
CO1	3	3	2	2	3	-	-	2	2	2	2
CO2	3	3	2	2	3	-	-	3	3	3	2
CO3	3	3	2	2	3	-	-	3	3	3	2
CO4	3	3	2	2	3	-	-	3	3	3	2
CO5	3	3	2	2	3	-	-	3	3	3	2
CO6	3	3	2	2	3	-	-	3	3	3	2

CO7	3	3	2	2	3	-	-	3	3	3	2	

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-202-L: OBJECT ORIENTED PROGRAMMING LANGUAGELAB
B.Tech. II Year- I Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	P: 01	In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Prerequisites: Basic Knowledge of computer

Course Objective:

1. To reinforce object-oriented programming concepts through practical implementation.
2. To provide hands-on experience with C++ and Java programming.
3. To develop problem-solving skills using object-oriented techniques.
4. To enable students to design and implement software solutions using OOP principles.
5. To familiarize students with debugging and testing in C++ and Java environments.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Apply object-oriented programming concepts to solve problems in C++ and Java.
CO2: Design and implement classes and objects using C++ and Java.
CO3: Utilize inheritance and polymorphism to create reusable and maintainable code.
CO4: Implement encapsulation and abstraction to enhance software robustness.
CO5: Handle exceptions and perform file I/O operations in C++ and Java.
CO6: Use advanced features such as templates, generics, and multithreading in softwaredevelopment.
CO7: Develop and test object-oriented applications through a mini-project.

Course Contents

List of Experiments	60 Hours

Experiment 1: Introduction to Development Environments

- Setting up development environments for C++ and Java.
- Writing and running simple programs in C++ and Java.

Experiment 2: Classes and Objects

-
- Define classes and create objects in C++ and Java.
- Implement constructors, destructors (in C++), and methods.
- Demonstrate the use of access specifiers.

Experiment 3: Inheritance

- Implement single, multiple (C++), and multilevel inheritance.
- Use constructor and destructor calls in inheritance.
- Demonstrate method overriding in Java and virtual functions in C++.

Experiment 4: Polymorphism

- Implement function overloading and operator overloading in C++.
- Implement method overloading in Java.
- Create and use abstract classes and interfaces in Java.
- Demonstrate runtime polymorphism in C++ and Java.

Experiment 5: Encapsulation and Abstraction

- Implement encapsulation using access specifiers in C++ and Java.
- Design and use abstract classes in C++ and Java.
- Implement packages in Java and namespaces in C++.

Experiment 6: Exception Handling

- Implement try, catch, and finally blocks in Java.
- Use throw and throws in Java.
- Implement exception handling in C++ using try, catch, and throw.

Experiment 7: File Handling

- Perform file read/write operations in C++ using fstream.
- Perform file read/write operations in Java using FileReader, FileWriter, BufferedReader, andBufferedWriter.

Experiment 8: Templates and Generics

- Implement function templates and class templates in C++.
- Use generics in Java to create type-safe collections.

Experiment 9: Standard Template Library (STL) in C++

- Use various components of STL such as vectors, lists, and maps.
- Perform operations using iterators and algorithms provided by STL.

Experiment 10: Collections Framework in Java

- Use various collection classes such as ArrayList, LinkedList, HashSet, and HashMap.
- Perform operations using iterators and methods provided by the collections framework.

Experiment 11: Multithreading

- Create and manage threads in C++ using the standard thread library.
- Implement multithreading in Java using the Thread class and Runnable interface.

Experiment 12: Mini Project

Develop a mini-project using C++ or Java to apply object-oriented programming concepts learned throughout the course. This could be a small application such as a student management system, a simple game, or any other real-world application.

REFERENCE BOOKS:

1. "The Complete Reference C++" by Herbert Schildt ISBN-13: 978-0071634809
2. "Effective C++: 55 Specific Ways to Improve Your Programs and Designs" by Scott Meyers ISBN-13: 978-0321334879
3. "Thinking in Java" by Bruce Eckel ISBN-13: 978-0131872486
4. "Effective Java" by Joshua Bloch ISBN-13: 978-0134685991
5. "Java: The Complete Reference" by Herbert Schildt ISBN-13: 978-1260440232

CO's-PO's & PSO's MAPPING

Course Outcomes (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	-	-	2	2	2
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	2	3	-	-	3	3	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3
CO7	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-203-MDM(A): INTRODUCTION TO BLOCKCHAIN TECHNOLOGY
(MDM Basket – 1)
B.Tech. II Year I Sem.

Theory Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 +2 Math Statistics

Course Objective:

1. To understand the fundamental concepts of blockchain technology, including cryptography and distributed ledgers.
2. To gain knowledge of various consensus mechanisms and their importance in blockchain.
3. To apply practical skills in setting up and using a private blockchain.
4. To analyze the real-world applications of blockchain technology in various domains.
5. To understand the challenges and limitations associated with blockchain adoption.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Explain the basic principles and architecture of blockchain technology.
CO2: Demonstrate knowledge of cryptographic techniques and distributed ledgers.
CO3: Differentiate between various consensus mechanisms.
CO4: Design and implement a private blockchain network.
CO5: Analyze the potential and limitations of blockchain technology in solving real-world problems.

Course Contents

UNIT-I	Introduction to Blockchain	03 Hours

History and Evolution of Blockchain
Basic concepts: Blocks, Transactions, and Chains
Types of Blockchain: Public, Private, and Consortium
Applications of Blockchain

UNIT-II	Cryptographic Foundations	04 Hours

Hash Functions and Digital Signatures
Public and Private Keys
Symmetric vs. Asymmetric Cryptography

Merkle Trees and their role in Blockchain		
UNIT-III	**Distributed Ledgers**	**04 Hours**
Fundamentals of Distributed Ledger Technology (DLT) Decentralization vs. Centralization Peer-to-Peer Networks and Node Operations Security and Fault Tolerance in DLT		
UNIT-IV	**Consensus Mechanisms**	**03 Hours**
Proof of Work (PoW) Proof of Stake (PoS) Delegated Proof of Stake (DPoS) Practical Byzantine Fault Tolerance (PBFT) Comparative Analysis of Consensus Mechanisms		
UNIT-V	**Setting up a Private Blockchain**	**03 Hours**
Installing and Configuring Blockchain Tools (e.g., Ethereum, Hyperledger Fabric) Creating and Managing Blocks and Transactions Setting Permissions and Roles Hands-on Lab: Create a Private Blockchain Network		
UNIT-VI	**Challenges and Future of Blockchain**	**04 Hours**
Scalability and Energy Consumption Issues Security and Privacy Challenges Emerging Trends: Blockchain Interoperability, Smart Contracts, and DeFi Case Studies: Blockchain in Finance, Healthcare, and Supply Chain		

Reference Books:

1. **Mastering Blockchain** by Imran Bashir, Packt Publishing
2. **Blockchain Basics: A Non-Technical Introduction in 25 Steps** by Daniel Drescher
3. **Blockchain Revolution** by Don Tapscott and Alex Tapscott
4. **Hands-On Blockchain Development with Hyperledger** by Salman A. Baset
5. Online Resources: Hyperledger Documentation, Ethereum Developer Tutorials

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	-	-	2	3	-	-	-	-	2	-	-	-
CO2	3	2	-	-	-	3	3	-	-	-	-	2	-	-	-
CO3	3	-	1	-	-	2	2	-	-	-	-	2	-	-	-
CO4	3	2	1	1	-	2	2	-	-	-	-	2	-	-	-
CO5	3	2	1	-	-	2	2	-	-	-	-	1	-	-	-
Avg.	2.8	1.8	1	1	-	2.2	2.4	-	-	-	-	1.8	-	-	-

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-203-MDM(B): ENVIRONMENTAL SCIENCES AND SUSTAINABILITY (MDM Basket – 2)
B.Tech. II Year I Sem.

Theory Scheme: TH: - 2 Hours/Week	Credit TH: 02	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 +2 Math Statistics

Course Objective:

- To introduce the basic concepts of environment, ecosystems and biodiversity and emphasizeon the biodiversity of India and its conservation.
- To impart knowledge on the causes, effects and control or prevention measures of environmental pollution and natural disasters.
- To facilitate the understanding of global and Indian scenario of renewable and nonrenewable resources, causes of their degradation and measures to preserve them.
- To familiarize the concept of sustainable development goals and appreciate the interdependence of economic and social aspects of sustainability, recognize and analyze climate changes, concept of carbon credit and the challenges of environmental management.
- To inculcate and embrace sustainability practices and develop a broader understanding on green materials, energy cycles and analyze the role of sustainable urbanization.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: To recognize and understand the functions of environment, ecosystems and biodiversity andtheir conservation.

CO2: To identify the causes, effects of environmental pollution and natural disasters and contributeto the preventive measures in the society.

CO3: To identify and apply the understanding of renewable and non-renewable resources andcontribute to the sustainable measures to preserve them for future generations.

CO4: To recognize the different goals of sustainable development and apply them for suitable technological advancement and societal development.

CO5: To demonstrate the knowledge of sustainability practices and identify green materials, energycycles and the role of sustainable urbanization.		
Course Contents		
UNIT-I	**ENVIRONMENT AND BIODIVERSITY**	**07 Hours**
Definition, scope and importance of environment – need for public awareness. Eco-system and Energy flow– ecological succession. Types of biodiversity: genetic, species and ecosystem diversity– values of biodiversity, India as a mega-diversity nation – hot-spots of biodiversity – threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – endangered and endemic species of India – conservation of biodiversity: In-situ and ex-situ.		
UNIT-II	**ENVIRONMENTAL POLLUTION**	**07 Hours**
Causes, Effects and Preventive measures of Water, Soil, Air and Noise Pollutions. Solid, Hazardous and E-Waste management. Case studies on Occupational Health and Safety Management system (OHASMS). Environmental protection, Environmental protection acts.		
UNIT-III	**RENEWABLE SOURCES OF ENERGY**	**07 Hours**
Energy management and conservation, New Energy Sources: Need of new sources. Different types new energy sources. Applications of- Hydrogen energy, Ocean energy resources, Tidal energy conversion. Concept, origin and power plants of geothermal energy.		
UNIT-IV	**SUSTAINABILITY AND MANAGEMENT**	**07 Hours**
Development , GDP ,Sustainability- concept, needs and challenges-economic, social and aspects of sustainability-from unsustainability to sustainability-millennium development goals, and protocols- Sustainable Development Goals-targets, indicators and intervention areas Climate change- Global, Regional and local environmental issues and possible solutions-case studies. Concept of Carbon Credit, Carbon Footprint. Environmental management in industry-A case study.		
UNIT-V	**SUSTAINABILITY PRACTICES**	**07 Hours**

Zero waste and R concept, Circular economy, ISO 14000 Series, Material Life cycle assessment, Environmental Impact Assessment. Sustainable habitat: Green buildings, Green materials, Energy efficiency, Sustainable transports. Sustainable energy: Non-conventional Sources, Energy Cycles- carbon cycle, emission and sequestration, Green Engineering: Sustainable urbanization- Socio- economic and technological change.

TEXT BOOKS:

1. Anubha Kaushik and C. P. Kaushik's "Perspectives in Environmental Studies", 6th Edition,New Age International Publishers ,2018.
2. Benny Joseph, 'Environmental Science and Engineering', Tata McGraw-Hill, New Delhi,2016.
3. Gilbert M.Masters, 'Introduction to Environmental Engineering and Science', 2nd edition,Pearson Education, 2004.
4. Allen, D. T. and Shonnard, D. R., Sustainability Engineering: Concepts, Design and CaseStudies, Prentice Hall.
5. Bradley. A.S; Adebayo, A.O., Maria, P. Engineering applications in sustainable design anddevelopment, Cengage learning.
6. Environment Impact Assessment Guidelines, Notification of Government of India, 2006.
7. Mackenthun, K.M., Basic Concepts in Environmental Management, Lewis Publication,London, 1998.

REFERENCES:

1. R.K. Trivedi, 'Handbook of Environmental Laws, Rules, Guidelines, Compliances andStandards', Vol. I and II, Enviro Media. 38 . edition 2010.
2. Cunningham, W.P. Cooper, T.H. Gorhani, 'Environmental Encyclopedia', Jaico Publ., House,Mumbai, 2001.
3. Dharmendra S. Sengar, 'Environmental law', Prentice hall of India PVT. LTD, New Delhi,2007.
4. Rajagopalan, R, 'Environmental Studies-From Crisis to Cure', Oxford University Press, ThirdEdition, 2015.
5. Erach Bharucha "Textbook of Environmental Studies for Undergraduate Courses" OrientBlackswan Pvt. Ltd. 2013.

CO's-PO's & PSO's MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	2	1	-	-	-	2	3	-	-	-	-	2	-	-	-
2	3	2	-	-	-	3	3	-	-	-	-	2	-	-	-
3	3	-	1	-	-	2	2	-	-	-	-	2	-	-	-

4	3	2	1	1	-	2	2	-	-	-	-	2	-	-	-
5	3	2	1	-	-	2	2	-	-	-	-	1	-	-	-
Avg.	**2.8**	**1.8**	**1**	**1**	**-**	**2.2**	**2.4**	**-**	**-**	**-**	**-**	**1.8**	**-**	**-**	**-**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-203-MDM(c): INTRODUCTION TO IMAGE PROCESSING
(MDM Basket – 3)
B.Tech. II Year I Sem.

Theory Scheme: **TH: - 2 Hours/Week**	**Credit** **TH: 02**	**Examination Scheme:** **In Sem. Evaluation: 25 Marks** **Mid Sem. Exam: 25 Marks** **End Sem. Exam: 50 Marks** **Total : 100 Marks**

Course Prerequisites: 10 +2 Math Statistics

Course Objective:

1. To introduce the fundamental concepts of image processing.
2. To develop an understanding of image representation and manipulation.
3. To apply spatial and frequency domain techniques for image enhancement and restoration.
4. To explore methods for image segmentation, feature extraction, and pattern recognition.
5. To understand the applications of image processing in real-world scenarios such as medical imaging and computer vision.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the fundamentals of image representation and various file formats.
CO2: Apply image enhancement techniques in both spatial and frequency domains.
CO3: Implement image restoration techniques for improving degraded images.
CO4: Analyze and apply segmentation and feature extraction techniques for object recognition.
CO5: Demonstrate the use of image processing techniques in real-world applications.

Course Contents

UNIT-I	**Introduction to Image Processing**	**03 Hours**

Basics of Digital Image Processing
Components of an Image Processing System
Image Representation: Pixels, Resolution, and File Formats
Applications of Image Processing in various fields

UNIT-II	**Image Enhancement in Spatial Domain**	**04 Hours**
Point Processing Techniques: Contrast Stretching, Thresholding Histogram Processing: Equalization and Matching Spatial Filtering: Smoothing and Sharpening Filters Practical Applications of Spatial Domain Techniques		
UNIT-III	**Image Enhancement in Frequency Domain**	**04 Hours**
Fourier Transform and its Properties Frequency Domain Filtering: Low-pass and High-pass Filters Fast Fourier Transform (FFT) for Image Processing Applications of Frequency Domain Techniques		
UNIT-IV	**Image Restoration**	**03 Hours**
Degradation Model and Noise Types Noise Reduction Techniques: Mean, Median, and Adaptive Filters Inverse Filtering and Wiener Filtering Restoration of Blurred and Noisy Images		
UNIT-V	**Image Segmentation and Feature Extraction**	**03 Hours**
Thresholding Techniques: Global and Adaptive Edge Detection: Gradient-based and Laplacian Methods Region-based Segmentation: Split and Merge Techniques Feature Extraction: Shape, Texture, and Color Features		
UNIT-VI	**Challenges and Future of Blockchain**	**04 Hours**
Case Studies in Medical Imaging: CT, MRI, and Ultrasound Applications in Computer Vision: Object Detection and Recognition Image Processing in Remote Sensing and Surveillance Emerging Trends: Deep Learning for Image Processing		

Reference Books:

1. **Digital Image Processing** by Rafael C. Gonzalez and Richard E. Woods, Pearson Education
2. **Fundamentals of Digital Image Processing** by Anil K. Jain, Prentice Hall
3. **Image Processing, Analysis, and Machine Vision** by Milan Sonka, Vaclav Hlavac, and Roger Boyle
4. **Computer Vision: Algorithms and Applications** by Richard Szeliski
5. Online Resources: MATLAB Image Processing Toolbox Documentation
6. Online Resources: Hyperledger Documentation, Ethereum Developer Tutorials

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	1	-	-	2	-	-	-	-	-	2	2	1	-
CO3	2	3	1	-	-	2	-	-	-	-	-	2	2	1	-
CO4	3	3	2	1	2	2	-	-	-	-	-	2	2	1	1
CO5	3	3	2	1	2	2	1	-	-	-	-	2	2	1	1
Avg.	2.8	2.8	1.2	0.4	0.8	2	0.2	-	-	-	-	2	1.8	0.8	0.4

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-204-VEC: ETHICS IN DATA SCIENCE
B.Tech. II Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 50 Marks

Course Prerequisites: Basic understanding of computer networks and operating systems, Familiarity with data science concepts, tools, and programming languages.

Course Objective:

- **To introduce students to the ethical frameworks and principles** guiding responsible data science and AI development.
- **To educate students on data privacy, fairness, and bias**, addressing how data-driven decisions impact individuals and society.
- **To raise awareness of professional ethics and social responsibility** in data science, ensuring that students understand the implications of their work on diverse populations.
- **To provide practical insights into global data laws** and regulations such as GDPR, and how they affect data collection, storage, and usage.
- **To instil critical thinking about the societal impact** of data science applications, focusing on ethical dilemmas and the need for transparency and accountability.
- **To encourage students to design and implement ethical AI and data science systems** that prioritize fairness, inclusivity, and responsible innovation.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Understand ethical principles and frameworks
Explain and apply ethical principles (e.g., utilitarianism, deontology) and frameworks in the context of data science projects.

CO2: Recognize and address data privacy and security concerns
Identify potential data privacy risks and ensure compliance with global regulations like GDPR and data protection laws.

CO3: Analyze and mitigate bias in algorithms
Assess the presence of bias in machine learning models and take measures to ensure fairness and prevent discriminatory outcomes in AI systems.

CO4: Design data science solutions with social responsibility
Develop data science and AI solutions that consider ethical impacts on vulnerable populations, ensuring inclusivity and preventing harm.

CO5: Apply professional ethics in real-world data science contexts
Integrate ethical decision-making processes in professional settings, considering conflicts of interest,

transparency, and accountability.

CO6: Understand the role of data science in influencing society
Critically evaluate how data-driven technologies shape social, economic, and political landscapes, and how responsible data use can lead to positive societal impacts.

CO7: Develop strategies for ethical AI and data governance
Propose and implement strategies for ethical AI deployment, data governance, and responsible use of data within organizations.

Course Contents

UNIT-I	Introduction	07 Hours
Ethical frameworks for data science Data privacy laws (GDPR, etc.) Bias and fairness in machine learning models Accountability and transparency in AI systems Ethical implications of big data analytics		
UNIT-II	**Professional Ethics and Social Responsibility**	**08 Hours**
Ethical conduct in professional settings Social responsibility and the role of a data scientist Impact of data science on society and individual rights Corporate governance and sustainability Case studies on ethical dilemmas in data science		
UNIT-III	**Human-Centered Design and Social Impact**	**08 Hours**
Human-cantered design principles Social impact of data-driven decisions Inclusive design in AI systems Understanding the role of culture and values in technology adoption Case studies on data science projects that promote societal well-being		
UNIT-IV	**Data and Society**	**07 Hours**
The role of data in modern society Data-driven decision-making and its impact on democracy, economy, and culture Data ethics in the context of surveillance, control, and social justice		

The digital divide and data access equity Case studies on data misuse and its social consequences		
UNIT-V	**AI for Social Good**	**08 Hours**
Applications of AI in healthcare, education, and environmental sustainability Ethical considerations for AI-driven social good initiatives Using data science for poverty alleviation, disaster management, and public health Case studies on successful AI-driven social impact projects		

TEXT BOOKS:

1. " "Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy"
2. "The Ethical Algorithm: The Science of Socially Aware Algorithm Design"
3. "Ethics of Big Data: Balancing Risk and Innovation"
4. "AI Ethics"
5. "Data Science Ethics"
6. "Privacy, Big Data, and the Public Good: Frameworks for Engagement"
7. "Algorithms of Oppression: How Search Engines Reinforce Racism"

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO6
CO1	2	2	-	-	3
CO2	2	-	3	-	-
CO3	-	2	3	-	-
CO4	-	2	-	3	-
CO5	-	2	-	-	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-205-HSSM: BUSINESS ECONOMICS AND FINANCIAL ANALYSIS
B.Tech. II Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 50 Marks

Course Prerequisites: None

Course Objective:

To learn the basic Business types, impact of the Economy on Business and Firms specifically. To analyze the Business from the Financial Perspective.

Learning Course Outcome:

After successful completion of the course, students will able to:

The students will understand the various Forms of Business and the impact of economic variables on the Business. The Demand, Supply, Production, Cost, Market Structure, Pricing aspects are learnt. The students can study the firm's financial position by analysing the Financial Statements of a Company.

Course Contents

UNIT-I	Introduction to Business and Economics	07 Hours

Business: Structure of Business Firm, Theory of Firm, Types of Business Entities, Limited Liability Companies, Sources of Capital for a Company, Non-Conventional Sources of Finance. Economics: Significance of Economics, Micro and Macro Economic Concepts, Concepts and Importance of National Income, Inflation, Money Supply in Inflation, Business Cycle, Features and Phases of Business Cycle. Nature and Scope of Business Economics, Role of Business Economist, Multidisciplinary nature of Business Economics.

UNIT-II	Demand and Supply Analysis	08 Hours

Elasticity of Demand: Elasticity, Types of Elasticity, Law of Demand, Measurement and Significance of Elasticity of Demand, Factors affecting Elasticity of Demand, Elasticity of Demand in decision making, Demand Forecasting: Characteristics of Good Demand Forecasting, Steps in Demand Forecasting, Methods of Demand Forecasting.
Supply Analysis: Determinants of Supply, Supply Function & Law of Supply.

UNIT-III	Production, Cost, Market Structures & Pricing	08 Hours

Production Analysis: Factors of Production, Production Function, Production Function with one variable input, two variable inputs, Returns to Scale, Different Types of Production Functions.

Cost analysis: Types of Costs, Short run and Long run Cost Functions.

Market Structures: Nature of Competition, Features of Perfect competition, Monopoly, Oligopoly, Monopolistic Competition.

Pricing: Types of Pricing, Product Life Cycle based Pricing, Break Even Analysis, Cost Volume Profit Analysis.

UNIT-IV	Financial Accounting	07 Hours

Accounting concepts and Conventions, Accounting Equation, Double-Entry system of Accounting, Rules for maintaining Books of Accounts, Journal, Posting to Ledger, Preparation of Trial Balance, Elements of Financial Statements, Preparation of Final Accounts.

UNIT-V	Financial Analysis through Ratios	08 Hours

Concept of Ratio Analysis, Liquidity Ratios, Turnover Ratios, Profitability Ratios, Proprietary Ratios, Solvency, Leverage Ratios (simple problems).

Introduction to Fund Flow and Cash Flow Analysis (simple problems).

TEXT BOOKS:

1. D.D. Chaturvedi, S.L. Gupta, Business Economics - Theory and Applications, International Book House Pvt. Ltd. 2013.
2. Dhanesh K Khatri, Financial Accounting, Tata McGraw Hill, 2011.
3. Geethika Ghosh, Piyali Gosh, Purba Roy Choudhury, Managerial Economics, 2e, Tata McGraw Hill Education Pvt. Ltd. 2012.

REFERENCE BOOKS:

1. Paresh Shah, Financial Accounting for Management 2e, Oxford Press, 2015.
2. S.N. Maheshwari, Sunil K Maheshwari, Sharad K Maheshwari, Financial Accounting, 5e, Vikas Publications, 2013.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-206-PCC: DATA VISUALIZATION
B.Tech. II Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 +2 Math Statistics

Course Objective:

- Effective use of Business Intelligence (BI) technology (Tableau) to apply data visualization
- To discern patterns and relationships in the data.
- To build Dashboard applications.
- To communicate the results clearly and concisely.
- To be able to work with different formats of data sets.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand How to import data into Tableau.
- Understand Tableau concepts of Dimensions and Measures.
- Develop Programs and understand how to map Visual Layouts and Graphical Properties.
- Create a Dashboard that links multiple visualizations.
- Use graphical user interfaces to create Frames for providing solutions to real world
- problems.

Course Contents

UNIT-I	Lab Problems	07 Hours

1. Understanding Data, What is data, where to find data, Foundations for building Data Visualizations, Creating Your First visualization?
2. Getting started with Tableau Software using Data file formats, connecting your Data to Tableau, creating basic charts(line, bar charts, Tree maps),Using the Show me panel.
3. Tableau Calculations, Overview of SUM, AVR, and Aggregate features, Creating custom calculations and fields.
4. Applying new data calculations to your visualizations, Formatting Visualizations, Formatting Tools and Menus, Formatting specific parts of the view.
5. Editing and Formatting Axes, Manipulating Data in Tableau data, Pivoting Tableau data.
6. Structuring your data, Sorting and filtering Tableau data, Pivoting Tableau data.
7. Advanced Visualization Tools: Using Filters, Using the Detail panel, using the Size panels, customizing filters, Using and Customizing tooltips, Formatting your data with colors.

8. Creating Dashboards & Storytelling, creating your first dashboard and Story, Design for different displays, adding interactivity to your Dashboard, Distributing & Publishing your Visualization.
9. Tableau file types, publishing to Tableau Online, Sharing your visualizations, printing, and Exporting.
10. Creating custom charts, cyclical data and circular area charts, Dual Axis charts.

REFERENCE BOOKS:

1. Microsoft Power BI cookbook, Brett Powell, 2nd edition.
2. R Programming for Data Science by Roger D. Peng (References)
3. The Art of R Programming by Norman Matloff Cengage Learning India.

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-208-OE(A): FOREIGN LANGUAGE (Open Elective – I)
B.Tech. II Year I Sem.

Theory Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 +2 Math Statistics

Course Objective:

Develop foundational skills in French grammar, vocabulary, and pronunciation.
Engage in basic conversations in French for common social situations.
Explore the culture and traditions of French-speaking countries (France, Canada, Belgium, and Switzerland).
Build confidence in speaking, listening, reading, and writing in French.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: To recognize and understand the functions of environment, ecosystems and biodiversity andtheir conservation.

CO2: To identify the causes, effects of environmental pollution and natural disasters and contributeto the preventive measures in the society.

CO3: To identify and apply the understanding of renewable and non-renewable resources andcontribute to the sustainable measures to preserve them for future generations.

CO4: To recognize the different goals of sustainable development and apply them for suitable technological advancement and societal development.

CO5: To demonstrate the knowledge of sustainability practices and identify green materials, energycycles and the role of sustainable urbanization.

Course Contents

UNIT-I	Introduction to Foreign Language and Culture	07 Hours

alphabets and pronunciation (sounds and accents), Basic greetings and introductions, Cultural overview, Numbers, days, months, seasons, Introducing yourself (name, age, nationality), Greetings and basic expressions, Numbers/days and dates

Time, age and counter

UNIT-II	Expanding Vocabulary and Conversational Skills	07 Hours

Vocabulary: Family, professions, nationalities, Days of the week, common activities

Writing and reading foreign language scripts. Introduction to Grammar & Practicing making sentences.

UNIT-III	Prepositions and Articles	07 Hours
Vocabulary: Shopping, ordering food, buying groceries Dialogues: Ordering at a restaurant or café. Cultural etiquette: Dining out Vocabulary: Common food and drink items. Dialogues: Asking prices, talking about preferences. Using polite expressions.		
UNIT-IV	**Foreign Culture and Traditions**	**07 Hours**
Introduction to foreign holidays and traditions. Formal vs. informal speech Listening and understanding simple dialogues and cultural materials (songs, news snippets) Basic reading comprehension: Short texts.Writing practice: Writing short paragraphs (e.g., introducing yourself, describing your day) Oral presentations: A foreign country or city, traditional food, or festival		

For Japanese:

- Japanese from Zero!: Proven Techniques to Learn Japanese for Students and Professionals: 1 by George Trombley, Yukari Takenaka, Learn From Zero Publication, 2020

For German:

- Textbook: "Menschen A1" (or similar beginner-level German textbook)
- Supplementary: Deutsche Welle (DW), YouTube channels for German learners, apps like Duolingo and Babbel.

For French:

- Textbook: "Alter Ego A1" (or similar beginner-level French textbook)
- Supplementary: TV5MONDE, YouTube channels for French learners, apps like Duolingo, Babbel.

CO's-PO's & PSO's MAPPING

CO	PO												PSO		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	2	1	-	-	-	2	3	-	-	-	-	2	-	-	-
2	3	2	-	-	-	3	3	-	-	-	-	2	-	-	-
3	3	-	1	-	-	2	2	-	-	-	-	2	-	-	-
4	3	2	1	1	-	2	2	-	-	-	-	2	-	-	-
5	3	2	1	-	-	2	2	-	-	-	-	1	-	-	-
Avg.	**2.8**	**1.8**	**1**	**1**	**-**	**2.2**	**2.4**	**-**	**-**	**-**	**-**	**1.8**	**-**	**-**	**-**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-208-L: FOREIGN LANGUAGE LAB (Open Elective – I)
B.Tech. II Year I Sem.

Practical Scheme: P: - 2 Hours/Week	Credit P: 01	Examination Scheme: In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks
Course Contents		
List of Experiments		**60 Hours**

1. Writing Assignments

- **Hiragana/Katakana Practice**: Assign tasks where students practice writing Hiragana and Katakana characters with correct stroke order.
- **Kanji Mastery**: Provide lists of beginner-level Kanji characters, asking students to write their meanings, pronunciations, and sample sentences.
- **Essay Writing**: Simple essays on topics like "My Family," "My Daily Routine," or "My Hobbies" in Japanese using basic grammar.

2. Listening Assignments

- **Listening Comprehension**: Share audio clips or videos (e.g., basic Japanese conversations) and provide comprehension questions.
- **Dictation Exercises**: Play simple sentences in Japanese and ask students to transcribe them using Hiragana, Katakana, or Kanji.
- **Cultural Videos**: Assign videos about Japanese culture, festivals, or history and ask students to summarize or share key learnings.

3. Speaking Assignments

- **Role Play**: Students can enact conversations like ordering food, introducing themselves, or asking for directions.
- **Pronunciation Practice**: Record themselves reading a provided script to ensure accurate pronunciation.
- **Short Speech**: Deliver a 1-2 minute speech on topics such as their favorite Japanese food, anime, or places to visit in Japan.

4. Reading Assignments

- **Simple Stories**: Assign beginner-level Japanese reading material (e.g., children's stories or graded readers) followed by comprehension questions.
- **News Headlines**: Provide Japanese newspaper headlines and ask students to analyze or interpret them.
- **Kana/Kanji Recognition**: Worksheets where students identify and match characters to their meanings or sounds.

5. Vocabulary Building Assignments

- **Thematic Word Lists**: Assign topics like colors, numbers, family members, or food, and ask students to create sentences using those words.
- **Flashcards**: Encourage students to create digital or physical flashcards for new words and share

them with classmates.

6. Cultural Exploration

- **Festivals and Customs**: Assign research-based essays or presentations on Japanese festivals (e.g., Hanami, Tanabata) or etiquette.
- **Cuisine**: Ask students to research and write about popular Japanese dishes and their cultural significance.
- **Calligraphy Assignment**: Practice writing Kanji in traditional calligraphy style and share photos of their work.

7. Multimedia Projects

- **Anime/Movie Analysis**: Watch a simple Japanese anime episode or movie and summarize it in Japanese.
- **Japanese Song**: Learn and perform a Japanese song, focusing on pronunciation and meaning.
- **Digital Tools**: Use apps like Duolingo, Memrise, or Anki for language practice and submit progress reports.

CO's-PO's & PSO's MAPPING

Course Outcomes (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	-	-	2	2	2
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	2	3	-	-	3	3	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3
CO7	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-208-OE(B): COST MANAGEMENT OF ENGINEERING PROJECTS
(Open Elective – I)
B.Tech. II Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Fundamentals of AI

Course Objective:

- Summarize the costing concepts and their role in decision making
- Infer the project management concepts and their various aspects in selection
- Interpret costing concepts with project execution
- Develop knowledge of costing techniques in service sector and various budgetary controltechniques
- Illustrate with quantitative techniques in cost management

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the costing concepts and their role in decision-making.
CO2: Understand the project management concepts and their various aspects in selection.
CO3: Interpret costing concepts with project execution.
CO4: Gain knowledge of costing techniques in service sector and various budgetary controltechniques.
CO5: Become familiar with quantitative techniques in cost management.

Course Contents

UNIT-I	INTRODUCTION TO COSTING CONCEPTS	07 Hours
Objectives of a Costing System; Cost concepts in decision-making; Relevant cost, Differential cost, Incremental cost and Opportunity cost; Creation of a Database for operational control		
UNIT-II	**INTRODUCTION TO PROJECT MANAGEMENT**	**08 Hours**
Project: meaning, Different types, why to manage, cost overruns centres, various stages of project execution: conception to commissioning. Project execution as conglomeration of technical and nontechnical activities, Detailed Engineering activities, Pre project execution main clearances and documents, Project team: Role of each member, Importance Project site: Data required with significance, Project contracts		

UNIT-III	**PROJECT EXECUTION AND COSTING CONCEPTS**	**08 Hours**
Project execution Project cost control, Bar charts and Network diagram, Project commissioning: mechanical and process, Cost Behavior and Profit Planning Marginal Costing; Distinction between Marginal Costing and Absorption Costing; Break-even Analysis, Cost-Volume-Profit Analysis, Various decision-making problems, Pricing strategies: Pareto Analysis, Target costing, Life Cycle Costing		
UNIT-IV	**COSTING OF SERVICE SECTOR AND BUDGETERY CONTROL**	**07 Hours**
Just-in-time approach, Material Requirement Planning, Enterprise Resource Planning, Activity Based Cost Management, Bench Marking; Balanced Score Card and Value-Chain Analysis, Budgetary Control: Flexible Budgets; Performance budgets; Zero-based budgets.		
UNIT-V	**QUANTITATIVE TECHNIQUES FOR COST MANAGEMENT**	**08 Hours**
Linear Programming, PERT/CPM, Transportation problems, Assignment problems, Learning Curve Theory.		

TEXT BOOKS:

1. John M. Nicholas, Herman Steyn Project Management for Engineering, Business andTechnology, Taylor & Francis, 2 August 2020, ISBN: 9781000092561.
2. Albert Lester ,Project Management, Planning and Control, Elsevier/Butterworth-Heinemann,2007, ISBN: 9780750669566, 075066956X.

REFERENCES:

1. Ashish K. Bhattacharya, Principles & Practices of Cost Accounting A. H. Wheeler publisher,1991.
2. Charles T. Horngren and George Foster, Advanced Management Accounting, 1988.
3. Charles T. Horngren et al Cost Accounting a Managerial Emphasis, Prentice Hall of India, NewDelhi, 2011.
4. Robert S Kaplan Anthony A. Alkinson, Management & Cost Accounting, 2003.
5. Vohra N.D., Quantitative Techniques in Management, Tata McGraw Hill Book Co. Ltd, 2007.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2

Avg	3	2.4	2.4	2	2	3	2.6	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-208-OE(B): COST MANAGEMENT OF ENGINEERING PROJECT LAB
(Open Elective – I)
B.Tech. II Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 01	In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Contents	
List of Experiments	60 Hours

1. Assignments on Cost Estimation

1. **Material and Labor Cost Calculation**
   - Task: Estimate the total cost of materials and labor for constructing a small civil structure (e.g., a residential building foundation).
   - Tools: MS Excel for tabulation and calculation.
   - Outcome: Understanding basic cost breakdowns.
2. **Quantity Take-Off Assignment**
   - Task: Using a given set of drawings (e.g., architectural plans), calculate quantities of materials (e.g., concrete, steel, bricks).
   - Tools: Manual calculation or software like AutoCAD/Excel.
3. **Comparative Cost Analysis**
   - Task: Compare costs of using different materials (e.g., bricks vs. blocks) or technologies for the same project.
   - Outcome: Develop analytical skills for choosing cost-effective alternatives.

2. Assignments on Budgeting

1. **Project Budget Preparation**
   - Task: Prepare a detailed project budget for an engineering project, including direct and indirect costs, contingencies, and overheads.
   - Tools: MS Excel.
   - Outcome: Learn budget preparation and presentation skills.
2. **Earned Value Management (EVM) Calculation**
   - Task: Calculate key EVM metrics (e.g., Planned Value, Actual Cost, Earned Value, Cost Variance, and Schedule Variance) for a project with sample data.
   - Tools: Excel or Primavera.
   - Outcome: Understanding project performance metrics.

3. Assignments on Scheduling and Cost Integration

1. **Project Schedule with Cost Integration**
   - Task: Create a project schedule using MS Project/Primavera and link cost components to each activity.
   - Deliverable: A Gantt chart with cost-loaded activities.
   - Outcome: Learn scheduling and resource allocation.
2. **Crash Cost Analysis**
   - Task: Analyze a project schedule and determine the costs of crashing activities to reduce the overall project duration.

- Tools: MS Excel/Primavera.
- Outcome: Understand trade-offs between time and cost.

4. Case Studies and Reports

1. **Case Study Analysis**
   - Task: Analyze a real-life engineering project (e.g., metro rail construction, highway development) and report on its cost management challenges.
   - Deliverable: Written report or presentation.
   - Outcome: Gain insights into practical challenges in cost management.
2. **Risk Analysis and Mitigation**
   - Task: Identify potential cost-related risks in an engineering project and propose mitigation strategies.
   - Outcome: Develop risk management skills.

5. Software-Based Assignments

1. **Cost Modeling Using Primavera/MS Project**
   - Task: Develop a cost model for a project, inputting cost data and generating project reports.
   - Outcome: Proficiency in industry-standard tools.
2. **Cash Flow Analysis**
   - Task: Simulate cash inflows and outflows for a project using MS Excel. Include graphical representation and breakeven analysis.
   - Outcome: Understanding of cash flow management.

6. Real-World Data Assignments

1. **Vendor Analysis**
   - Task: Collect quotations for a specific project item (e.g., steel, concrete) from different vendors, compare costs, and recommend the best option.
   - Deliverable: Vendor comparison table and justification.
2. **Market Price Fluctuation Analysis**
   - Task: Analyze how market price changes of key materials (e.g., steel, cement) affect overall project cost.
   - Tools: Online price indices or simulated data.

7. Collaborative or Group Projects

1. **Mini Project on Cost Estimation**
   - Task: Work in groups to estimate the cost of a small engineering project, including materials, labor, equipment, and contingencies.
   - Deliverable: A detailed report and presentation.
2. **Project Performance Analysis**
   - Task: Using provided project data, evaluate the project's cost and schedule performance, identifying areas for improvement.

CO's-PO's & PSO's MAPPING

Course Outcomes (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	-	-	2	2	2
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	2	3	-	-	3	3	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3
CO7	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-308-OE(c): SENSORS (Open Elective – I)
B.Tech. II Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To learn the various types of sensors, transducers, sensor output signal types, calibrationtechniques, formulation of system equation and its characteristics.
2. To understand basic working principle, construction, Application and characteristics ofdisplacement, speed and ranging sensors.
3. To understand and analyze the working principle, construction, application andcharacteristics of force, magnetic and heading sensors.
4. To learn and analyze the working principle, construction, application and characteristics ofoptical, pressure, temperature and other sensors.
5. To familiarize students with different signal conditioning circuits design and data acquisitionsystem.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Understand various sensor effects, sensor characteristics, signal types, calibration methods and obtain transfer function and empirical relation of sensors. They can also analyze the sensor response.
CO2: Analyze and select suitable sensor for displacement, proximity and range measurement. CO3: Analyze and select suitable sensor for force, magnetic field, speed, position and direction measurement.
CO4: Analyze and Select suitable sensor for light detection, pressure and temperature measurement and also familiar with other miniaturized smart sensors.
CO5: Select and design suitable signal conditioning circuit with proper compensation and linearizing element based on sensor output signal.

Course Contents

UNIT-I	SENSOR CLASSIFICATION, CHARACTERISTICS AND SIGNAL TYPES	07 Hours
Basics of Measurement – Classification of Errors – Error Analysis – Static and Dynamic Characteristics of Transducers – Performance Measures of Sensors – Classification of Sensors – Sensor Calibration Techniques – Sensor Outputs - Signal Types - Analog and Digital Signals, PWM and PPM.		
UNIT-II	DISPLACEMENT, PROXIMITY AND RANGING SENSORS	08 Hours

Displacement Sensors – Brush Encoders - Potentiometers, Resolver, Encoders – Optical, Magnetic, Inductive, Capacitive, LVDT – RVDT – Synchro – Microsyn, Accelerometer – Range Sensors - Ultrasonic Ranging - Reflective Beacons - Laser Range Sensor (LIDAR) – GPS - RF Beacons.		
UNIT-III	**FORCE, MAGNETIC AND HEADING SENSORS**	**08 Hours**
Strain Gage – Types, Working, Advantage, Limitation, and Applications: Load Measurement – Force and Torque Measurement - Magnetic Sensors – Types, Principle, Advantage, Limitation, and Applications - Magneto Resistive – Hall Effect, Eddy Current Sensor - Heading Sensors – Compass, Gyroscope and Inclinometers.		
UNIT-IV	**OPTICAL, PRESSURE, TEMPERATURE AND OTHER SENSORS**	**07 Hours**
Photo Conductive Cell, Photo Voltaic, Photo Resistive, LDR – Fiber Optic Sensors – Pressure – Diaphragm – Bellows - Piezoelectric - Piezo-resistive - Acoustic, Temperature – IC, Thermistor, RTD, Thermocouple – Non Contact Sensor - Chemical Sensors - MEMS Sensors - Smart Sensors.		
UNIT-V	**SIGNAL CONDITIONING**	**08 Hours**
Need for Signal Conditioning – Resistive, Inductive and Capacitive Bridges for Measurement - DC and AC Signal Conditioning - Voltage, Current, Power and Instrumentation Amplifiers – Filter and Isolation Circuits – Fundamentals of Data Acquisition System.		

TEXT BOOKS

1. Bolton W., “Mechatronics”, Pearson Education, 6th Edition, 2015.
2. Ramesh S Gaonkar, “Microprocessor Architecture, Programming, and Applications with the 8085”, Penram International Publishing Private Limited, 6th Edition, 2013.

REFERENCES

3. Bradley D.A., Dawson D., Buru N.C. and Loader A.J., “Mechatronics”, Chapman and Hall,1993.
4. Davis G. Alciatore and Michael B. Histand, “Introduction to Mechatronics and Measurementsystems”, McGraw Hill Education, 2011.
5. Devadas Shetty and Richard A. Kolk, “Mechatronics Systems Design”, Cengage Learning,2010.
6. Nitaigour Premchand Mahalik, “Mechatronics Principles, Concepts and Applications”,McGraw Hill Education, 2015.
7. Smaili. A and Mrad. F, “Mechatronics Integrated Technologies for Intelligent Machines”,Oxford University Press, 2007.

CO's-PO's & PSO's MAPPING

Mapping of COs with POs and PSOs															
COs/POs & PSOs	**Pos**												**PSOs**		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**
CO1	3	3	2								1	2	3	2	1
CO2	3	3	2	1	1	1					1	2	3	2	1
CO3	3	3	2	1	1	1					1	2	3	2	1

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-308(c): SENSORS LAB (Open Elective – I)
B.Tech. III Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 01	In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Contents	
List of Experiments	60 Hours

1. Sensor Characterization and Calibration

1. **Temperature Sensor Calibration**
   - **Task**: Measure the output of a temperature sensor (e.g., LM35, thermocouple, RTD) at different temperatures using a controlled heat source. Plot the calibration curve.
   - **Deliverables**: Calibration curve, sensitivity, and linearity analysis.
   - **Tools**: Multimeter, temperature-controlled oven, or water bath.
2. **Light Sensor (LDR/Photodiode) Characterization**
   - **Task**: Measure the resistance/output voltage of an LDR or photodiode under different light intensities. Plot the relationship between light intensity and sensor output.
   - **Deliverables**: Graph of intensity vs. sensor output, sensitivity calculation.
   - **Tools**: Lux meter, variable light source.

2. Sensor Interfacing

1. **Interfacing an Ultrasonic Sensor**
   - **Task**: Interface an ultrasonic sensor (e.g., HC-SR04) with a microcontroller (e.g., Arduino, Raspberry Pi) to measure distance. Validate the output with a ruler.
   - **Deliverables**: Code, distance measurements, and error analysis.
   - **Outcome**: Understanding time-of-flight principle.
2. **Motion Detection Using PIR Sensor**
   - **Task**: Interface a Passive Infrared (PIR) sensor with an Arduino to detect motion and activate an LED/buzzer.
   - **Deliverables**: Circuit diagram, code, and working demonstration video.
   - **Outcome**: Learn motion sensing and basic event-triggered control.

3. Real-Time Data Acquisition

1. **Monitoring Temperature and Humidity**
   - **Task**: Interface a DHT11/DHT22 sensor to monitor temperature and humidity. Display data on an LCD or serial monitor.
   - **Deliverables**: Code, circuit diagram, and readings.
   - **Outcome**: Learn real-time data acquisition and display.
2. **Gas Detection Using MQ Sensors**
   - **Task**: Interface an MQ-series gas sensor (e.g., MQ-2 for LPG/CO detection) to detect gas concentration and trigger an alert when the level exceeds a threshold.
   - **Deliverables**: Code, circuit diagram, and test results.
   - **Outcome**: Understand gas sensing principles and alert mechanisms.

4. Sensor Application Projects

1. **Touch Sensor-Based Switch**
   - **Task**: Use a capacitive touch sensor (e.g., TTP223) to create a touch-controlled switch for an LED or motor.
   - **Deliverables**: Circuit diagram, code, and working demonstration.
   - **Outcome**: Learn touch sensing and control circuits.
2. **Smart Street Lighting System**
   - **Task**: Use an LDR and a motion sensor to design a smart street lighting system that activates lights only when it detects motion at night.
   - **Deliverables**: Circuit design, code, and working demo.
   - **Outcome**: Combine light and motion sensing for energy efficiency.

5. Signal Conditioning and Processing

1. **Noise Reduction in Sensor Signals**
   - **Task**: Measure a noisy sensor signal, apply filtering techniques (e.g., RC filter, software-based filtering), and compare the results.
   - **Deliverables**: Filter design, raw vs. filtered signal plots.
   - **Outcome**: Understand signal conditioning.
2. **Analog to Digital Conversion**
   - **Task**: Interface an analog sensor (e.g., potentiometer) with an ADC module to digitize the sensor output and display the result on an LCD.
   - **Deliverables**: Code, circuit, and measured values.

6. Advanced Sensor Integration

1. **Bio-Signal Acquisition (Pulse Rate Monitoring)**
   - **Task**: Use an optical heart rate sensor (e.g., MAX30102) to measure pulse rate and display it in real-time.
   - **Deliverables**: Circuit design, code, and pulse rate data.
   - **Outcome**: Introduction to biomedical sensing.
2. **Acceleration Measurement Using an Accelerometer**
   - **Task**: Interface an accelerometer (e.g., ADXL345) to measure acceleration along three axes. Visualize data using software like Python/Matlab.
   - **Deliverables**: Code, data visualization, and interpretation.

7. IoT-Based Assignments

1. **IoT Temperature Monitoring**
   - **Task**: Use a temperature sensor with an ESP8266/NodeMCU to send real-time data to an IoT dashboard (e.g., ThingSpeak).

- **Deliverables**: Code, dashboard screenshots, and live data demonstration.

2. **Wireless Sensor Network**
   - **Task**: Create a simple wireless sensor network with two nodes transmitting environmental data (e.g., temperature, humidity) using Zigbee or NRF24L01 modules.
   - **Deliverables**: Node configurations, data logs, and system description.

8. Mini-Projects

1. **Fire Detection System**
   - Combine a flame sensor and a temperature sensor to create a fire detection system with an alarm and visual indicators.
2. **Smart Parking System**
   - Use an ultrasonic sensor to detect available parking spaces and display the count on an LCD or web interface.
3. **Environment Monitoring System**
   - Design a system integrating multiple sensors (e.g., temperature, humidity, gas) to monitor environmental conditions and send alerts via SMS or email.

CO's-PO's & PSO's MAPPING

Course Outcomes (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	-	-	2	2	2
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	2	3	-	-	3	3	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3
CO7	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-211-PCC: DATABASE MANAGEMENT SYSTEMS
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: A course on "Data Structures".

Course Objective:

- To understand the basic concepts and the applications of database systems.
- To master the basics of SQL and construct queries using SQL.
- Topics include data models, database design, relational model, relational algebra, transaction control, concurrency control, storage structures and access techniques.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Gain knowledge of fundamentals of DBMS, database design and normal forms
- Master the basics of SQL for retrieval and management of data.
- Be acquainted with the basics of transaction processing and concurrency control.
- Familiarity with database storage structures and access techniques

Course Contents

UNIT-I	Database System Applications	07 Hours

A Historical Perspective, File Systems versus a DBMS, the Data Model, Levels of Abstraction in a DBMS, Data Independence, Structure of a DBMS

Introduction to Database Design: Database Design and ER Diagrams, Entities, Attributes, and Entity Sets, Relationships and Relationship Sets, Additional Features of the ER Model, Conceptual Design With the ER Model

UNIT-II	Introduction to the Relational Model	08 Hours

Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical database design, introduction to views, destroying/altering tables and views. Relational Algebra, Tuple relational Calculus, Domain relational calculus.

UNIT-III	SQL	08 Hours

QUERIES, CONSTRAINTS, TRIGGERS: form of basic SQL query, UNION, INTERSECT, and EXCEPT, Nested Queries, aggregation operators, NULL values, complex integrity constraints in SQL, triggers and active databases.

Schema Refinement: Problems caused by redundancy, decompositions, problems related to decomposition, reasoning about functional dependencies, First, Second, Third normal forms, BCNF, lossless join decomposition, multivalued dependencies, Fourth normal form, Fifth normal form.		
UNIT-IV	**Transaction Property**	**07 Hours**
Transaction Concept, Transaction State, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for serializability, Lock Based Protocols, Timestamp Based Protocols, Validation- Based Protocols, Multiple Granularity, Recovery and Atomicity, Log–Based Recovery, Recovery with Concurrent Transactions.		
UNIT-V	**File Organization**	**08 Hours**
Data on External Storage, File Organization and Indexing, Cluster Indexes, Primary and Secondary Indexes, Index data Structures, Hash Based Indexing, Tree based Indexing, Comparison of File Organizations, Indexes- Intuitions for tree Indexes, Indexed Sequential Access Methods (ISAM), B+ Trees: A Dynamic Index Structure.		

TEXT BOOKS:

1. Database System Concepts, Silberschatz, Korth, McGraw hill, V edition.3rd Edition
2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill

REFERENCE BOOKS:

1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition.
2. Fundamentals of Database Systems, Elmasri Navrate, Pearson Education
3. Introduction to Database Systems, C. J. Date, Pearson Education
4. Oracle for Professionals, The X Team, S.Shah and V. Shah, SPD.
5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI.
6. Fundamentals of Database Management Systems, M. L. Gillenson, Wiley Student Edition.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-211-L: DATABASE MANAGEMENT SYSTEMS LAB
B.Tech. II Year IISem.

Practical Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 01	In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Prerequisites: "Database Management Systems".

Course Objective:

- Introduce ER data model, database design and normalization
- Learn SQL basics for data definition and data manipulation

Learning Course Outcome:
After successful completion of the course, students will able to:

- Design database schema for a given application and apply normalization
- Acquire skills in using SQL commands for data definition and data manipulation.
- Develop solutions for database applications using procedures, cursors and triggers

Course Contents

List of Experiments	07 Hours

1. Concept design with E-R Model
2. Relational Model
3. Normalization
4. Practicing DDL commands
5. Practicing DML commands
6. A. Querying (using ANY, ALL, UNION, INTERSECT, JOIN, Constraints etc.)
   B. Nested, Correlated subqueries
7. Queries using Aggregate functions, GROUP BY, HAVING and Creation and dropping of Views.
8. Triggers (Creation of insert trigger, delete trigger, update trigger)
9. Procedures
10. Usage of Cursors

TEXT BOOKS:

1. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill, 3rd Edition
2. Database System Concepts, Silberschatz, Korth, McGraw Hill, V edition.

REFERENCE BOOKS:

1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition.
2. Fundamentals of Database Systems, Elmasri Navrate, Pearson Education
3. Introduction to Database Systems, C.J. Date, Pearson Education
4. Oracle for Professionals, The X Team, S. Shah and V. Shah, SPD.
5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI.
6. Fundamentals of Database Management Systems, M. L. Gillenson, Wiley Student Edition.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-212-PCC: OPERATING SYSTEMS
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

1. A course on "Computer Programming and Data Structures".
2. A course on "Computer Organization and Architecture".

Course Objective:

- Introduce operating system concepts (i.e., processes, threads, scheduling, synchronization, deadlocks, memory management, file and I/O subsystems and protection)
- Introduce the issues to be considered in the design and development of operating system
- Introduce basic Unix commands, system call interface for process management, interprocess communication and I/O in Unix

Learning Course Outcome:
After successful completion of the course, students will able to:

- Will be able to control access to a computer and the files that may be shared
- Demonstrate the knowledge of the components of computers and their respective roles in computing.
- Ability to recognize and resolve user problems with standard operating environments.
- Gain practical knowledge of how programming languages, operating systems, and architectures interact and how to use each effectively.

Course Contents

UNIT-I	Operating System - Introduction	07 Hours
Structures - Simple Batch, Multiprogrammed, Time-shared, Personal Computer, Parallel, Distributed Systems, Real-Time Systems, System components, Operating System services, System Calls. Process - Process concepts and scheduling, Operations on processes, Cooperating Processes, Threads		
UNIT-II	**CPU Scheduling**	**08 Hours**
Scheduling Criteria, Scheduling Algorithms, Multiple -Processor Scheduling. System call interface for process management-fork, exit, wait, waitpid, exec Deadlocks - System Model, Deadlocks Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, and Recovery from Deadlock		
UNIT-III	**Process Management and Synchronization**	**08 Hours**

The Critical Section Problem, Synchronization Hardware, Semaphores, and Classical Problems of Synchronization, Critical Regions, Monitors Interprocess Communication Mechanisms: IPC between processes on a single computer system, IPC between processes on different systems, using pipes, FIFOs, message queues, shared memory.		
UNIT-IV	**Memory Management and Virtual Memory**	**07 Hours**
Logical versus Physical Address Space, Swapping, Contiguous Allocation, Paging, Segmentation, Segmentation with Paging, Demand Paging, Page Replacement, Page Replacement Algorithms.		
UNIT-V	**File System Interface and Operations**	**08 Hours**
Access methods, Directory Structure, Protection, File System Structure, Allocation methods, Free-space Management. Usage of open, create, read, write, close, lseek, stat, ioctl system calls.		

TEXT BOOKS:

1. Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley.
2. Advanced programming in the UNIX environment, W.R. Stevens, Pearson education.

REFERENCE BOOKS:

1. Operating Systems- Internals and Design Principles, William Stallings, Fifth Edition–2005, Pearson Education/PHI
2. Operating System A Design Approach- Crowley, TMH.
3. Modern Operating Systems, Andrew S. Tanenbaum 2nd edition, Pearson/PHI
4. UNIX programming environment, Kernighan and Pike, PHI/ Pearson Education
5. UNIX Internals -The New Frontiers, U. Vahalia, Pearson Education.

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-212-L: OPERATING SYSTEMS LAB
B.Tech. II Year II Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: A course on "Programming for Problem Solving", A course on "Computer Organization and Architecture".
Co-requisite: A course on "Operating Systems".

Course Objective:

- To provide an understanding of the design aspects of operating system concepts through simulation
- Introduce basic Unix commands, system call interface for process management, interprocess communication and I/O in Unix

Learning Course Outcome:
After successful completion of the course, students will able to:

- Simulate and implement operating system concepts such as scheduling, deadlock management, file management and memory management.
- Able to implement C programs using Unix system calls

Course Contents

List of Experiments

1. Write C programs to simulate the following CPU Scheduling algorithms a) FCFS b) SJF c) Round Robin d) priority
2. Write programs using the I/O system calls of UNIX/LINUX operating system (open, read, write, close, fcntl, seek, stat, opendir, readdir)
3. Write a C program to simulate Bankers Algorithm for Deadlock Avoidance and Prevention.
4. Write a C program to implement the Producer — Consumer problem using semaphores using UNIX/LINUX system calls.
5. Write C programs to illustrate the following IPC mechanisms a) Pipes b) FIFOs c) Message Queues d) Shared Memory
6. Write C programs to simulate the following memory management techniques a) Paging b) Segmentation
7. Write C programs to simulate Page replacement policies a) FCFS b) LRU c) Optimal

TEXT BOOKS:

1. Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley
2. Advanced programming in the Unix environment, W.R.Stevens, Pearson education.

REFERENCE BOOKS:

1. Operating Systems – Internals and Design Principles, William Stallings, Fifth Edition–2005, Pearson Education/PHI
2. Operating System - A Design Approach-Crowley, TMH.
3. Modern Operating Systems, Andrew S Tanenbaum, 2nd edition, Pearson/PHI
4. UNIX Programming Environment, Kernighan and Pike, PHI/Pearson Education
5. UNIX Internals: The New Frontiers, U. Vahalia, Pearson Education

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-213-MDM(A): SMART CONTRACTS AND DECENTRALIZED APPLICATIONS (DAPPS)
(MDM Basket – 1)
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Introduction To Data Science

Course Objective:

1. To understand the concept of smart contracts and their role in blockchain ecosystems.
2. To explore the architecture and development of decentralized applications (DApps).
3. To gain hands-on experience in writing and deploying smart contracts.
4. To analyze security challenges and best practices in smart contract development.
5. To study real-world use cases of DApps across various industries.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Explain the fundamentals of smart contracts and decentralized applications.
CO2: Develop and deploy smart contracts using blockchain platforms like Ethereum.
CO3: Design and implement decentralized applications using appropriate frameworks and tools.
CO4: Analyze security vulnerabilities in smart contracts and apply mitigation strategies.
CO5: Evaluate the potential of DApps in solving real-world problems across industries.

Course Contents

UNIT-I	Introduction to Smart Contracts	03 Hours

- Definition and Characteristics of Smart Contracts
- History and Evolution of Smart Contracts
- Advantages and Limitations of Smart Contracts
- Smart Contract Platforms: Ethereum, Binance Smart Chain, and Hyperledger

UNIT-II	Smart Contract Development	03 Hours

- Introduction to Solidity Programming Language
- Data Types, Functions, and Control Structures in Solidity
- Writing, Compiling, and Deploying Smart Contracts

• Tools and Frameworks: Remix, Truffle, and Hardhat		
UNIT-III	**Decentralized Application (DApp) Development**	**04 Hours**
• Architecture of DApps: Frontend, Backend, and Smart Contracts • Integration of Smart Contracts with Web3.js and ethers.js • Developing a Full-Stack DApp with React and Solidity • Hosting and Deploying DApps on Decentralized Platforms		
UNIT-IV	**Security in Smart Contracts and DApps**	**03 Hours**
• Common Vulnerabilities in Smart Contracts: Reentrancy, Overflow, and Underflow • Security Testing and Audit Tools: MythX, Slither, and Oyente • Best Practices for Secure Smart Contract Development • Case Studies of Major Smart Contract Exploits		
UNIT-V	**Use Cases of DApps**	**04 Hours**
• Financial Services: DeFi, Lending, and Payments • Supply Chain and Logistics • Healthcare and Identity Management • Gaming and Non-Fungible Tokens (NFTs)		
UNIT-VI	**Future Trends and Challenges in DApps**	**04 Hours**
• Scalability Solutions: Layer 2 and Rollups • Cross-Chain Interoperability • Governance in Decentralized Systems • Emerging Trends: Decentralized Autonomous Organizations (DAOs) and Web3.0		

Reference Books:

1. **Mastering Ethereum** by Andreas M. Antonopoulos and Gavin Wood
2. **Solidity Programming Essentials** by Ritesh Modi
3. **Building Blockchain Projects** by Narayan Prusty
4. **Blockchain Basics: A Non-Technical Introduction in 25 Steps** by Daniel Drescher
5. Online Resources: Ethereum and Solidity Documentation, OpenZeppelin Library

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	1
CO3	3	3	3	2	3	2	-	-	-	-	-	2	2	1	1
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO5	3	3	3	2	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	1.8	2.4	2	0.4	-	-	-	-	2	1.8	1.2	0.8

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-213-MDM(B): FUNDAMENTALS OF CYBERSECURITY
(MDM Basket – 2)
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Introduction To Data Science

Course Objective:

1. To introduce the fundamental principles of cybersecurity and its significance in the digital age.
2. To provide an understanding of various types of cyber threats and attack vectors.
3. To explore key concepts of cryptography, network security, and system security.
4. To develop skills to identify, analyze, and mitigate cybersecurity risks.
5. To study cybersecurity frameworks, standards, and ethical practices.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. **CO1**: Understand the fundamentals of cybersecurity and common cyber threats.
2. **CO2**: Apply cryptographic techniques to secure data.
3. **CO3**: Implement network security measures to protect systems from attacks.
4. **CO4**: Analyze and mitigate risks associated with cybersecurity vulnerabilities.
5. **CO5**: Demonstrate knowledge of cybersecurity frameworks and ethical considerations.

Course Contents

UNIT-I	Introduction to Cybersecurity	03 Hours

- Definition and Importance of Cybersecurity
- Cyber Threat Landscape: Malware, Phishing, Ransomware, and Social Engineering
- Principles of Security: Confidentiality, Integrity, and Availability (CIA Triad)
- Cybersecurity in the Modern Era: IoT, Cloud, and Mobile Security

UNIT-II	Cryptography Basics	04 Hours

- Introduction to Cryptography and its Applications
- Symmetric and Asymmetric Encryption Techniques

• Hash Functions and Digital Signatures • Public Key Infrastructure (PKI) and Certificate Management		
UNIT-III	**Network Security**	**03 Hours**
• Overview of Network Protocols and Vulnerabilities • Firewalls, Intrusion Detection, and Prevention Systems • Virtual Private Networks (VPNs) and Secure Communication • Wireless Network Security and Best Practices		
UNIT-IV	**System Security**	**03 Hours**
• Operating System Security Features and Mechanisms • Endpoint Security: Antivirus, Anti-malware, and Patching • Securing Web Applications and Databases • Case Studies of Major System Breaches		
UNIT-V	**Risk Management and Compliance**	**04 Hours**
• Risk Assessment Methodologies and Tools • Cybersecurity Frameworks: NIST, ISO 27001, and COBIT • Regulatory Compliance: GDPR, HIPAA, and PCI-DSS • Incident Response and Disaster Recovery Planning		
UNIT-VI	**Ethics and Emerging Trends in Cybersecurity**	**04 Hours**
• Ethical Hacking and Penetration Testing • Cybersecurity for Emerging Technologies: AI, Blockchain, and Quantum Computing • Cybersecurity Careers and Certifications • Future Challenges and Opportunities in Cybersecurity		

Reference Books:

1. **Computer Security: Principles and Practice** by William Stallings and Lawrie Brown
2. **Cryptography and Network Security** by William Stallings
3. **Cybersecurity Essentials** by Charles J. Brooks, Christopher Grow, Philip Craig, and Donald Short
4. **Practical Malware Analysis** by Michael Sikorski and Andrew Honig
5. Online Resources: NIST Cybersecurity Framework, OWASP Guidelines

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	2	3	2	-	-	-	-	-	2	2	1	-
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO5	3	3	3	2	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	1.8	2.4	2	0.4	-	-	-	-	2	1.8	1.2	0.4

1 - low, 2 - medium, 3 - high, '-' - no correlation

Savitribai Phule Pune University

(Formerly University of Pune)

Department Of Technology

BTDS-213-MDM(c): COMPUTER VISION
(MDM Basket – 3)
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Introduction To Data Science

Course Objective:

1. To understand the fundamental concepts of computer vision and its applications.
2. To study image processing techniques essential for computer vision tasks.
3. To explore feature extraction, object detection, and recognition methodologies.
4. To implement and evaluate machine learning and deep learning approaches in computer vision.
5. To analyze real-world challenges and advancements in computer vision technologies.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the basics of image processing and computer vision techniques.
CO2: Apply feature extraction methods for image analysis.
CO3: Implement object detection and recognition algorithms.
CO4: Utilize machine learning and deep learning techniques for solving vision problems.
CO5: Analyze the challenges and advancements in real-world computer vision applications.

Course Contents

UNIT-I	Introduction to Computer Vision	03 Hours

- Definition and Scope of Computer Vision
- Applications of Computer Vision: Robotics, Healthcare, and Autonomous Vehicles
- Image Formation: Camera Models and Geometry
- Image Representation: Pixels, Color Spaces, and Histograms

UNIT-II	Image Processing Fundamentals	02 Hours

- Image Enhancement: Filtering, Smoothing, and Sharpening
- Edge Detection: Sobel, Canny, and Laplacian
- Morphological Operations: Erosion, Dilation, Opening, and Closing
- Image Transformations: Fourier Transform and Hough Transform

UNIT-III	**Feature Detection and Description**	**03 Hours**
• Feature Detection: Corners, Blobs, and Edges • Feature Descriptors: SIFT, SURF, and ORB • Keypoint Matching and Homography Estimation • Applications: Image Stitching and Panorama Creation		
UNIT-IV	**Object Detection and Recognition**	**02 Hours**
• Object Detection Techniques: Sliding Window, Region Proposals • Introduction to Convolutional Neural Networks (CNNs) • Object Detection Frameworks: YOLO, SSD, and Faster R-CNN • Image Classification and Object Recognition Tasks		
UNIT-V	**Machine Learning and Deep Learning for Vision**	**03 Hours**
• Supervised and Unsupervised Learning for Vision Tasks • Transfer Learning and Fine-Tuning in Vision Applications • Semantic Segmentation: U-Net, Mask R-CNN • Case Studies: Face Detection, Handwriting Recognition		
UNIT-VI	**Challenges and Future Trends in Computer Vision**	**03 Hours**
• Challenges: Occlusion, Illumination, and Scale Variations • Advances in Vision: 3D Vision, Generative Adversarial Networks (GANs) • Vision for Autonomous Systems: SLAM and Visual Odometry • Ethical Considerations in Computer Vision Applications		

Reference Books:

1. **Computer Vision: Algorithms and Applications** by Richard Szeliski
2. **Deep Learning for Computer Vision** by Rajalingappaa Shanmugamani
3. **Digital Image Processing** by Rafael C. Gonzalez and Richard E. Woods
4. **Programming Computer Vision with Python** by Jan Erik Solem
5. Online Resources: OpenCV Documentation, PyTorch and TensorFlow Tutorials.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	2	3	2	-	-	-	-	-	2	2	1	-
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO5	3	3	3	2	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	1.8	2.4	2	0.4	-	-	-	-	2	1.8	1.2	0.4

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-214-HSSM: History of Science and Technology in India
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 50 Marks

Course Prerequisites: Basic Web Programming

Course Objective:

- **Understanding Historical Context**:
- To provide students with a comprehensive understanding of the historical development of science and technology in India from ancient times to the modern era.
- **Analyzing Contributions**:
- To analyze the contributions of various civilizations in India to the fields of science and technology, including key figures, discoveries, and innovations.
- **Exploring Cultural Influence**:
- To explore the cultural, social, and political factors that influenced scientific thought and technological advancements in India throughout history.
- **Identifying Challenges and Progress**:
- To identify the challenges faced by Indian scientists and technologists in different historical contexts and how these challenges were addressed.
- **Connecting Past and Present**:
- To connect historical advancements in science and technology to contemporary issues and developments in modern India, fostering a deeper appreciation of India's scientific heritage.
- **Promoting Critical Thinking**:
- To encourage critical thinking about the implications of historical scientific practices and technological developments on society, ethics, and the environment.

Learning Course Outcome:

After successful completion of the course, students will able to:

- **Articulate Key Historical Developments**:
- Demonstrate an understanding of significant milestones in the history of science and technology in India, including key inventions and discoveries.
- **Evaluate Contributions**:
- Evaluate the contributions of notable Indian scientists and innovators, recognizing their impact on both national and global scientific communities.
- **Analyze Cultural Context**:
- Analyze how cultural, social, and political contexts influenced scientific advancements in India, and how these advancements, in turn, shaped society.
- **Assess Challenges**:

- Assess the historical challenges faced by the scientific community in India, including colonial influences, resource limitations, and societal attitudes towards science and technology.
- **Draw Connections to Modern Issues**:
- Draw connections between historical scientific practices and current technological challenges and advancements in India, fostering a deeper understanding of contemporary issues.
- **Critically Discuss Ethical Implications**:
- Engage in discussions about the ethical implications of past scientific practices and technological advancements, considering their relevance to current scientific and technological issues.
- **Demonstrate Research Skills**:
- Conduct research on specific topics related to the history of science and technology in India, utilizing primary and secondary sources effectively.
- **Communicate Effectively**:
- Communicate findings and analyses clearly and effectively through written assignments and presentations, demonstrating an ability to convey complex historical information to diverse audiences.

Course Contents

UNIT-I	CONCEPTS AND PERSPECTIVES	07 Hours
Meaning of History, Objectivity, Determinism, Relativism, Causation, Generalization in History; Moral judgment in history Extent of subjectivity, contrast with physical sciences, interpretation and speculation, causation verses evidence, concept of historical inevitability, Historical Positivism. Science and Technology-Meaning, Scope and Importance, Interaction of science, technology & society, Sources of history on science and technology in India.		
UNIT-II	**HISTORIOGRAPHY OF SCIENCE AND TECHNOLOGY IN INDIA**	**08 Hours**
Introduction to the works of D.D. Kosambi, Dharmpal, Debiprasad Chattopadhyay, Rehman, S. Irfan Habib, Deepak Kumar, Dhruv Raina, and others.		
UNIT-III	**SCIENCE AND TECHNOLOGY IN ANCIENT INDIA**	**08 Hours**
Technology in pre-historic period, Beginning of agriculture and its impact on technology Science and Technology during Vedic and Later Vedic times Science and technology from 1st century AD to C-1200.		
UNIT-IV	**SCIENCE AND TECHNOLOGY IN MEDIEVAL INDIA**	**07 Hours**
Legacy of technology in Medieval India, Interactions with Arabs. Development in medical knowledge, interaction between Unani and Ayurveda and alchemy Astronomy and Mathematics: interaction with Arabic Sciences. Science and Technology on the eve		

of British conquest		
UNIT-V	**SCIENCE AND TECHNOLOGY IN COLONIAL INDIA And IN A POST-INDEPENDENT INDIA**	**08 Hours**
Science and the Empire, Indian response to Western Science Growth of techno-scientific institutions, Science, Technology and Development discourse Shaping of the Science and Technology Policy Developments in the field of Science and Technology Science and technology in globalizing India. Social implications of new technologies like the Information Technology and Biotechnology.		

TEXT BOOKS:

1. "Science and Technology in India: A Historical Perspective", Author: S. Irfan Habib
2. "A History of Science and Technology in India", Author: K. N. Panikkar
3. "Indian Science and Technology in the 21st Century: Perspectives and Prospects", Author: C. R. Rao

REFERENCE BOOKS:

1. "The Scientific Revolution in India", Author: P. J. M. D. Silviera
2. "India's Contributions to Science and Technology: A Historical Perspective", Author: V. S. Ramachandran
3. "Technology and Society in India 1780-1980", Author: A. G. K. Menon
4. "Indian Science: A Historical Analysis", Author: A. R. M. Bhattacharya
5. "Science and Technology in India: The Early Years", Author: R. P. S. Sinha.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-215-AEC: ENTREPRENEURSHIP DEVELOPMENT
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 50 Marks

Course Prerequisites: Basic Web Programming

Course Objective:

1. Understand the importance of entrepreneurship and its role in society.
2. Analyze the entrepreneurial environment and factors affecting it.
3. Generate and evaluate business ideas and opportunities.
4. Develop a comprehensive business plan.
5. Understand funding options and manage finances.
6. Formulate marketing strategies for startups.
7. Comprehend legal aspects and regulations affecting businesses.
8. Learn strategies for scaling and growing a business.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Articulate the significance of entrepreneurship in the economic development of a society.
2. Analyze the entrepreneurial environment and identify factors that affect business opportunities.
3. Generate innovative business ideas and evaluate their feasibility through market research.
4. Create a detailed business plan that outlines strategies, financial projections, and marketing approaches.
5. Identify and utilize various funding sources for start-up businesses and manage financial operations effectively.
6. Formulate effective marketing strategies for promoting products or services in competitive markets.
7. Understand legal frameworks and ethical considerations involved in starting and operating a business.
8. Implement growth strategies and address challenges associated with scaling a business.
9. Demonstrate leadership and teamwork skills necessary for managing entrepreneurial ventures.
10. Cultivate an entrepreneurial mind-set that embraces innovation and creativity.

Course Contents

UNIT-I	**Introduction to Entrepreneurship, Entrepreneurial Environment**	**07 Hours**

Definition and importance of entrepreneurship, Characteristics and traits of successful entrepreneurs, Types of entrepreneurs and entrepreneurship models, Role of entrepreneurship in economic development, Factors affecting entrepreneurship (economic, social, legal), Government policies and support for entrepreneurs Economic, social, and cultural influences on entrepreneurship, Role of incubators and accelerators in promoting entrepreneurship.		
UNIT-II	**Idea Generation and Opportunity Recognition, Business Planning**	**08 Hours**
Techniques for idea generation (brainstorming, mind mapping), Identifying and evaluating business opportunities, Market research and feasibility analysis, Understanding customer needs and market trends, Components of a business plan (executive summary, market analysis, organizational structure, etc.), Writing a business plan: objectives, strategies, and financial projections, Importance of a business plan in securing funding Case studies of successful business plans.		
UNIT-III	**Funding and Financial Management**	**08 Hours**
Sources of funding for startups (bootstrapping, venture capital, angel investors, crowdfunding), Understanding financial statements (income statement, balance sheet, cash flow statement), Basics of budgeting and financial forecasting, Managing cash flow and working capital.		
UNIT-IV	**Marketing Strategies for Startups, Legal and Ethical Aspects of Entrepreneurship**	**07 Hours**
Developing a marketing plan, Branding and positioning strategies, Digital marketing strategies and social media, Customer relationship management (CRM) , Understanding the legal structure of businesses (sole proprietorship, partnership, corporation), Intellectual property rights and patenting, Ethical considerations in entrepreneurship Regulatory requirements and compliance.		
UNIT-V	**Growth and Scaling of Enterprises, Entrepreneurial Leadership and Team Building**	**08 Hours**
Strategies for scaling a business, Managing growth challenges, Expansion strategies (franchising, joint ventures, etc.), Exit strategies (mergers, acquisitions, IPOs), Leadership styles and their impact on entrepreneurship, Building and managing effective teams Conflict resolution and negotiation skills, Importance of networking and mentorship.		

UNIT-VI	Entrepreneurial Mindset and Innovation	08 Hours
Cultivating an entrepreneurial mindset, Importance of innovation in entrepreneurship Techniques for fostering creativity and innovation, Case studies of innovative entrepreneurs.		

TEXT BOOKS:

1. "Entrepreneurship: Successfully Launching New Ventures", Authors: Bruce R. Barringer & R. Duane Ireland
2. "The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses" Author: Eric Ries
3. "Business Model Generation: A Handbook for Visionaries, Game Changers, and Challengers" Authors: Alexander Osterwalder & Yves Pigneur
4. "Entrepreneurship: Theory, Process, Practice" Author: Donald F. Kuratko
5. "Innovation and Entrepreneurship" Author: Peter F. Drucker
6. "The Art of Start: The Time-Tested, Battle-Hardened Guide for Anyone Starting Anything" Author: Guy Kawasaki
7. "Developing the Entrepreneurial Spirit" Author: Joseph A. Schilling & Wendy Adams
8. "The E-Myth Revisited: Why Most Small Businesses Don't Work and What to Do About It" Author: Michael E. Gerber
9. "Entrepreneurship: A Real-World Approach" Authors: David A. Kirby & Chris T. Williams
10. "Business Planning: A Comprehensive Framework" Author: Jeffrey W. Timmons.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-216-VSEC: WEB PROGRAMMING
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 50 Marks

Course Prerequisites: Basic Web Programming

Course Objective:

- Understand the technologies used in Web Programming.
- Know the importance of object-oriented aspects of Scripting.
- Understand creating database connectivity using JDBC.
- Learn the concepts of web-based application using sockets.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Design web pages.
- Use technologies of Web Programming.
- Apply object-oriented aspects to Scripting.
- Create databases with connectivity using JDBC.
- Build web-based application using sockets.

Course Contents

UNIT-I	Client-side Programming	07 Hours

HTML- Basic Tags- List, Tables, Images, Forms, Frames, CSS

JAVA Script -Web page Designing using HTML, Scripting basics- Client side and server side scripting. Java ScriptObject, names, literals, operators and expressions- statements and features- events - windows - documents - frames - data types - built-in functions- Browser object model - Verifying forms.-HTML5- CSS3- HTML 5 canvas - Web site creation using tools.

UNIT-II	JAVA	08 Hours

Introduction to object-oriented programming-Features of Java – Data types, variables and arrays – Operators – Control statements – Classes and Methods – Inheritance. Packages and Interfaces – Exception Handling – Multithreaded Programming – Input/Output – Files – Utility Classes – String Handling.

UNIT-III	JDBC	08 Hours

JDBC Overview – JDBC implementation – Connection class – Statements - Catching Database Results, handling database Queries. Networking– InetAddress class – URL class- TCP sockets – UDP sockets, Java Beans –RMI.

UNIT-IV	APPLETS	07 Hours

Java applets- Life cycle of an applet – Adding images to an applet – Adding sound to an applet. Passing parameters to an applet. Event Handling. Introducing AWT: Working with Windows Graphics and Text. Using AWT Controls, Layout Managers and Menus. Servlet – life cycle of a servlet. The Servlet API, Handling HTTP Request and Response, using Cookies, Session Tracking. Introduction to JSP.

UNIT-V	XML AND WEB SERVICES	08 Hours

Xml – Introduction-Form Navigation-XML Documents- XSL – XSLT- Web services-UDDI-WSDL-Java web services – Web resources.

TEXT BOOKS:

1. Harvey Deitel, Abbey Deitel, Internet and World Wide Web: How To Program 5th Edition.
2. Herbert Schildt, Java - The Complete Reference, 7th Edition. Tata McGraw- Hill Edition.
3. Michael Morrison XML Unleashed Tech media SAMS.

REFERENCE BOOKS:

1. John Pollock, Javascript - A Beginners Guide, 3rd Edition –- Tata McGraw-Hill Edition.
2. Keyur Shah, Gateway to Java Programmer Sun Certification, Tata McGraw Hill, 2002.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-216-L: WEB FRAMEWORK LAB
B.Tech. II Year II Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Programming Lab Practice

Course Objective:

1. Understand the Django Framework.
2. Model and Database Management.
3. Dynamic Web Development.
4. URL Routing and Navigation.
5. Form Handling and Validation.
6. User Authentication and Authorization.
7. Deployment of Django Applications.
8. Testing and Quality Assurance.
9. Integration with Third-Party Libraries and APIs.
10. Implementing Security Measures.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Set up and configure the Django development environment.
CO2: Create and manage Django models and perform CRUD operations using Django ORM.
CO3: Develop dynamic web pages using Django views and templates.
CO4: Implement URL routing and navigation in Django applications.
CO5: Handle forms and perform validation in Django applications.
CO6: Implement user authentication and authorization in Django.
CO7: Deploy Django applications to live servers.
CO8: Write and run tests for Django applications to ensure quality and reliability.
CO9: Integrate third-party libraries and APIs with Django.
CO10: Implement security measures in Django applications.

Course Contents

Experiment Of List

Lab Assignment 1: Setting Up Django Environment

- **Objective**: Set up a Django development environment.
- **Tasks**:
  - Install Python and Django.
  - Create a new Django project.
  - Understand the project structure.
  - Run the development server.
- **Expected Outcome**: Successfully set up and run a Django project on a local server.

Lab Assignment 2: Creating Django Models

- **Objective**: Understand and create Django models.
- **Tasks**:
  - Define models for a sample application (e.g., Blog, E-commerce).
  - Migrate models to the database.
  - Use Django Admin to manage models.
- **Expected Outcome**: Successfully create and manage Django models.

Lab Assignment 3: Django ORM and Queries

- **Objective**: Perform CRUD operations using Django ORM.
- **Tasks**:
  - Insert, update, delete, and retrieve data using Django ORM.
  - Implement QuerySets and filtering.
  - Create relationships between models.
- **Expected Outcome**: Perform various database operations using Django ORM.

Lab Assignment 4: Building Views and Templates

- **Objective**: Develop views and templates for dynamic web pages.
- **Tasks**:
  - Create function-based views and class-based views.
  - Pass data to templates and render it.
  - Use template inheritance and filters.
- **Expected Outcome**: Develop dynamic web pages using Django views and templates.

Lab Assignment 5: URL Routing and Navigation

- **Objective**: Understand and implement URL routing in Django.
- **Tasks**:
  - Define URL patterns for different views.

  - Use named URL patterns and reverse resolution.
  - Implement navigation links in templates.
- **Expected Outcome**: Set up URL routing and navigation for a Django application.

Lab Assignment 6: Form Handling and Validation

- **Objective**: Handle forms and perform validation in Django.
- **Tasks**:
  - Create and process forms.
  - Implement form validation.
  - Use ModelForm for database operations.
- **Expected Outcome**: Successfully handle and validate forms in a Django application.

Lab Assignment 7: User Authentication and Authorization

- **Objective**: Implement user authentication and authorization.
- **Tasks**:
  - Set up user registration, login, and logout.
  - Implement password hashing and authentication.
  - Manage user permissions and access control.
- **Expected Outcome**: Implement a complete user authentication system in Django.

Lab Assignment 8: Integrating with Databases

- **Objective**: Integrate Django with various databases.
- **Tasks**:
  - Configure different databases (SQLite, PostgreSQL, MySQL).
  - Perform database migrations.
  - Use database-specific features in Django.
- **Expected Outcome**: Integrate and work with multiple databases in Django.

Lab Assignment 9: Deploying Django Applications

- **Objective**: Deploy a Django application to a live server.
- **Tasks**:
  - Prepare the application for deployment.
  - Set up a production environment (using WSGI, Gunicorn).
  - Deploy on cloud platforms (Heroku, AWS, etc.).
- **Expected Outcome**: Successfully deploy a Django application to a live server.

Lab Assignment 10: Testing Django Applications

- **Objective**: Write and run tests for Django applications.
- **Tasks**:
  - Write unit tests for models, views, and forms.
  - Use Django's testing framework.
  - Run tests and interpret results.
- **Expected Outcome**: Write and execute tests to ensure the quality of Django applications.

Lab Assignment 11: Integrating Third-Party Libraries

- **Objective**: Use third-party libraries and APIs in Django.
- **Tasks**:
  - Install and configure third-party libraries (e.g., Django Rest Framework, Celery).
  - Integrate external APIs.
  - Implement additional functionalities (e.g., social authentication, email services).
- **Expected Outcome**: Extend Django applications using third-party libraries and APIs.

Lab Assignment 12: Implementing Security Measures

- **Objective**: Implement security best practices in Django.
- **Tasks**:
  - Use Django's built-in security features (e.g., CSRF, XSS protection).
  - Implement SSL/TLS for secure communication.
  - Manage user sessions securely.
- **Expected Outcome**: Implement basic security measures to protect Django applications.

REFERENCE BOOK:

1. **"Django for Beginners" by William S. Vincent**: This book is an excellent starting point for beginners. It covers the basics of Django and takes you through building your first project.
   - ISBN-13: 978-1735467207
2. **"Two Scoops of Django 3.x: Best Practices for the Django Web Framework" by Daniel Roy Greenfeld and Audrey Roy Greenfeld**: This book provides best practices and tips for Django development, making it a valuable resource for intermediate to advanced users.
   - ISBN-13: 978-0692915721
3. **"Django Unleashed" by Andrew Pinkham**: A comprehensive guide that delves into more advanced topics in Django, including testing, deployment, and scaling applications.
   - ISBN-13: 978-0321985071
4. **"Django for APIs: Build web APIs with Python and Django" by William S. Vincent**: Focuses on building APIs with Django, which is essential for integrating with third-party services and creating backend systems.
   - ISBN-13: 978-1735467221

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO1	3	-	-	-	3	-	-	2	-	-	-
CO2	3	3	2	-	3	-	-	3	3	3	2
CO3	3	3	2	-	3	-	-	3	2	3	2
CO4	3	2	2	-	3	-	-	2	2	2	2
CO5	3	3	2	-	3	-	-	3	2	3	2
CO6	3	3	2	-	3	-	-	3	3	3	3
CO7	3	3	2	-	3	-	-	3	3	3	2
CO8	3	3	2	3	3	-	-	3	3	3	2
CO9	3	2	2	-	3	-	-	3	3	3	2
CO10	3	3	2	-	3	-	-	3	3	3	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-217-PCC: INTRODUCTION TO DATA ANALYTICS
B.Tech. II Year II Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Software Flow Basic

Course Objective:

- Introduce learners to the essential concepts, terminologies, and methodologies in data analytics, data science, and business analytics.
- Provide a comprehensive overview of the data analytics lifecycle, spanning data preparation, exploration, analysis, visualization, and decision-making processes.
- Emphasize the significance of data in various domains, industries, and modern business decision-making.
- Develop foundational skills in data analysis, interpretation, and the creation of actionable insights.
- Address the ethical considerations, biases, and responsibilities associated with handling and analyzing data.
- Expose students to popular tools, techniques, and platforms in the realms of data analytics and business intelligence.
- Explore real-world, industry-relevant use cases to bridge the gap between theory and application.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Define and articulate core concepts such as data analytics, data science, business analytics, big data, machine learning, and business intelligence.

CO2:Navigate the entire data analytics lifecycle, from data acquisition to actionable insights presentation.

CO3:Design and conceptualize basic data warehouses, appreciating the intricacies of multidimensional data representation.

CO4:Utilize tools like Tableau, Excel, Python, and R for data visualization, analysis, and representation.

CO5:Identify patterns, correlations, and trends in datasets, using both descriptive and inferential statistical methods

Course Contents

UNIT-I	Introduction to Excel for Data Analysis	07 Hours

- Overview of Excel interface
- Basics of navigating and working with sheets
- Introduction to cells, rows, columns, and ranges
- Understanding basic functions (SUM, AVERAGE, COUNT)
- Working with mathematical and statistical functions
- Introduction to text functions for data manipulation
- Working with logical functions (IF, AND, OR)
- Exploring lookup functions (VLOOKUP, HLOOKUP, INDEX, MATCH)
- Introduction to array formulas
- Identifying and handling missing data
- Removing duplicates and dealing with errors
- Text-to-columns and data-splitting techniques
- Formatting data for analysis
- Creating basic charts and graphs
- Tips for effective data presentation
- Introduction to PivotTables for dynamic data analysis
- Creating Pivot Charts for visual insights
- Customizing and formatting PivotTables and Pivot Charts
- Time-saving shortcuts and productivity hacks
- Excel with AI

UNIT-II	Introduction to Python for Data Analysis	08 Hours

- Overview of Python programming language
- Introduction to Jupyter Notebooks for data analysis
- Variables, data types, and basic operations
- Lists, tuples, and dictionaries
- Inbuilt functions

Data Manipulation with Python

- Conditional statements and loops
- User defined functions
- Functions such as map, filter, lambda

UNIT-III	Exploring Data with Pandas & Matplotlib	08 Hours

Data Manipulation with Pandas

- Overview of Pandas Library
- Reading and writing data along with basic operations with Pandas

Data Cleaning and Preprocessing with Pandas

- Handling missing data
- Removing duplicates and dealing with outliers
- Cleaning and adjustments in data

3.5.1 Incorporating analytics insights into decision-making

3.5.2 Decision-making frameworks and models

UNIT-IV	EDA & Data Visualization	07 Hours
Exploratory Data Analysis (EDA) with Pandas • Descriptive statistics and data summarization • Grouping and aggregating data • SQL like operation in data **Data Visualization with Matplotlib** • Creating basic plots (line plots, scatter plots, histograms) • Customizing and styling visualizations		
UNIT-V	**Real-time Python**	**08 Hours**
6.1 Unpacking Ethical Dimensions of Data Analysis 6.1.1 Privacy concerns and data anonymization 6.1.2 Ethical challenges in handling sensitive data 6.2 Addressing and Mitigating Biases in Data 6.2.1 Identifying and mitigating bias in algorithms and datasets 6.2.2 Ethical considerations in machine learning model development 6.3 Code of Ethics: Responsibilities of Data Practitioners 6.3.1 Ethical guidelines for data practitioners 6.3.2 Professional standards and certifications 7.1 Overview of Data Analytics Tools and Platforms 7.1.1 Comparative analysis of popular tools: Tableau, Excel, Python, R 7.1.2 Selection criteria for tools based on use cases 7.2 Hands-On Experience with Tableau, Excel, Python, and R 7.2.1 Practical exercises for each tool 7.2.2 Integrating tools for comprehensive analysis 7.3 Integration of Tools for Comprehensive Analysis 7.3.1 Developing end-to-end workflows for data analysis 7.3.2 Collaborative analytics: combining outputs from different tools		
UNIT-VI	**Real-World Use Cases**	**08 Hours**
Advanced Data Analysis with Numpy • Introduction to Numpy for numerical operations • Working with arrays and matrices **Advanced Data Visualization with Seaborn** • Creating informative and aesthetically pleasing visualizations • Pair plots, heatmaps, and advanced plotting technique		

8.3.1 Problem-solving methodologies in analytics

8.3.2 Evaluating the success of implemented solutions.

UNIT VII	Power BI for Real-Time Analytics and Advanced Features	08 Hours

Real-Time Dashboards

- Setting up real-time data streaming in Power BI
- Creating dashboards for live data monitoring

Advanced Features and Custom Visuals

- Exploring custom visuals and visuals from the marketplace
- Leveraging advanced features like forecasting and clustering
- Case Studies and Discussion
- Reviewing case studies of effective Power BI usage

Q&A and discussions on best practices in storytelling with data

REFERENCE BOOKS:

1. "Data Science for Business" by Foster Provost and Tom Fawcett
2. "Python for Data Analysis" by Wes McKinney
3. "Data Analytics Made Accessible" by Anil Maheshwari
4. "Data Science for Dummies" by Lillian Pierson
5. "Introduction to Statistical Learning" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani
6. "Storytelling with Data" by Cole Nussbaumer Knaflic
7. "Data Mining: Practical Machine Learning Tools and Techniques" by Ian H. Witten, Eibe Frank, and Mark A. Hall
8. "R for Data Science" by Hadley Wickham and Garrett Grolemund
9. "Big Data: A Revolution That Will Transform How We Live, Work, and Think" by Viktor Mayer-Schönberger and Kenneth Cukier
10. "Data Science from Scratch: First Principles with Python" by Joel Grus

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-218-VEC: SPIRITUALITY AND MINDFULNESS
B.Tech. II Year II Sem.

Practical Scheme: P: - 4 Hours/Week	Credit P: 02	Examination Scheme: In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Prerequisites: 10 +2 discrete mathematics

Course Objective:

1. To enhance self-awareness, emotional intelligence, and personal growth through mindfulness practices.
2. To understand the concept of spirituality beyond religion, focusing on inner peace, purpose, and ethical living.
3. To develop skills in stress management, mental clarity, and emotional regulation through meditation and mindfulness techniques.
4. • To promote mental and physical well-being through integration of spiritual practices and contemplative traditions.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Use MS Word to create quality documents, by structuring and organizing content for their day to day technical and academic requirements

CO2: Use MS EXCEL to perform data operations and analytics, record, retrieve data as per requirements and visualize data for ease of understanding

CO3: Use MS PowerPoint to create high quality academic presentations by including common tables, charts, graphs, interlinking other elements, and using media objects.

Course Contents		
UNIT-I	**Introduction to Spirituality**	**01 Hours**
Definition and Scope of Spirituality: Understanding spirituality beyond religious practices. Concepts of personal growth, purpose, and ethical living. Spirituality in Different Cultures: Spiritual traditions across the world (Eastern and Western philosophies). Connection between spirituality and human well-being.		
UNIT-II	**Understanding Mindfulness**	**02 Hours**
Concept of Mindfulness:What is mindfulness? Origins in Buddhism and its secular applications. Mindfulness in daily life: Presence and awareness in routine activities. **Science of Mindfulness**: Neuroscience behind mindfulness practices. Benefits for mental health, cognitive functions, and emotional regulation.		
UNIT-III	**Meditation Techniques**	**04 Hours**
Basic Meditation Practices: Introduction to breath-focused meditation. Body scan meditation and progressive muscle relaxation. Concentration and Insight Meditation (Vipassana): Techniques for observing thoughts, emotions, and sensations non-judgmentally. Developing deeper insights into the nature of the mind. Guided Meditations: Practice of loving-kindness meditation (Metta) and compassion meditation.		
UNIT-IV	**Yoga and Pranayama (Breathing Techniques) Stress Management and Emotional Intelligence**	**04 Hours**
Introduction to Yoga: Basic postures (asanas) for body-mind alignment. Yoga as a holistic practice connecting body, mind, and spirit. Pranayama (Breathing Exercises): Techniques for controlling and channeling breath (Anulom Vilom, Bhastrika). Importance of breath in mindfulness and emotional regulation. Mindfulness-Based Stress Reduction (MBSR): Techniques for managing stress through mindfulness. Real-life applications: dealing with anxiety, fear, and pressure. Developing Emotional Intelligence: How mindfulness helps in recognizing, understanding, and managing emotions. Enhancing interpersonal relationships through self-awareness and empathy.		
UNIT-IV	**Contemplative Practices and Self-Reflection, Mindfulness in Action**	**02 Hours**

Journaling and Self-Inquiry: Importance of reflection on personal growth and values. Guided exercises for journaling and self-reflection. Contemplative Practices: Silence retreats, walking meditation, and nature connection. Role of solitude and contemplation in spiritual growth. Mindfulness in Daily Activities: Cultivating mindfulness in work, relationships, and decision-making. Role of mindfulness in ethical and value-based leadership. Mindful Communication: Techniques for active listening and responding with compassion. Non-violent communication practices.

UNIT-IV	Integrating Spirituality and Mindfulness	02 Hours

Spiritual Well-Being and Life Purpose: Aligning personal goals with spiritual values. Cultivating a sense of purpose, meaning, and inner peace. Spirituality in Modern Life: Balancing material and spiritual aspirations. Applying mindfulness and spirituality in the context of career and social life.

References:

Books:

- *"The Miracle of Mindfulness"* by Thich Nhat Hanh.
- *"Wherever You Go, There You Are"* by Jon Kabat-Zinn.
- *"The Power of Now"* by Eckhart Tolle.

Research Papers:

- Studies on the effects of mindfulness on brain function, stress reduction, and well-being.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	-	-	3	-	3	-
CO2	3	3	3	-	-	3	-	3	3
CO3	3	3	-	-	3	3	-	-	3
CO4	3	3	-	3	-	-	3	-	-
CO5	3	3	-	-	3	3	-	3	-
CO6	3	3	-	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-219-OE(A) DESIGN THINKING
B.Tech. II Year II Sem.
(Open Elective – II)

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

- A course on "Computer Programming and Data Structures".
- A course on "Advanced Data Structures".

Course Objective:

- To introduce the principles and methodologies of design thinking in the context of data science and engineering.
- To foster innovation by focusing on user-centered approaches to data problem solving.
- To equip students with skills for ideation, prototyping, and testing in data-driven projects.
- To integrate creativity with data science techniques for solving real-world challenges.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Apply design-thinking principles to solve data science problems.
- Conduct user research and translate insights into data-driven solutions.
- Prototype and test data science models and solutions.
- Effectively communicate data insights with stakeholders.
- Collaborate in interdisciplinary teams to integrate data science into broader engineering contexts.

Course Contents

UNIT-I	Introduction	07 Hours
Overview of Design Thinking: Origins, evolution, and relevance in engineering and data science. Five Phases of Design Thinking: Empathize: Understanding the user and their needs. Define clearly articulating the problem. Ideate brainstorming innovative solutions. Prototype: Developing preliminary versions of the solution. Test: Iterating based on feedback. Case Studies: Applications of design thinking in data-driven environments (e.g., healthcare, finance).		
UNIT-II	**Empathy and User-Centered Design**	**08 Hours**
User Research Techniques: Interviews, observation, surveys, and data collection to understand user behavior. Journey Mapping: Identifying pain points and opportunities through customer journey maps. Contextual Inquiry: Gathering real-		

world insights that can be translated into data-driven solutions. Data Science Application: Example - Gathering user feedback to design a recommendation engine or predictive model.		
UNIT-III	**Problem Definition**	**08 Hours**
Synthesizing User Insights: From raw data to actionable insights. Problem Framing Techniques: "How Might We" questions and framing problems in user-centric ways. Defining Scope in Data Science Projects: Identifying core problems for data analytics and machine learning solutions. Data Science Application: Example - Defining the problem for optimizing business operations using data analysis.		
UNIT-IV	**Ideation and Brainstorming**	**07 Hours**
Divergent and Convergent Thinking: Encouraging creativity followed by selection of feasible solutions. Techniques for Innovation: Mind mapping, brainstorming, and collaborative idea generation. Incorporating Data: How data informs and shapes ideas during ideation. Data Science Application: Example - Ideating models for improving prediction accuracy or designing a new data visualization tool.		
UNIT-V	**Prototyping and Experimentation**	**08 Hours**
Rapid Prototyping: Building quick and low-cost models (mock-ups) to explore solution feasibility. Designing Data Experiments: Structuring data science experiments to test models or algorithms. Tools for Prototyping: Simulation tools, visualization software, and wireframing. Data Science Application: Example - Prototyping an AI chatbot for customer service.		
UNIT-VI	**Testing and Iteration, Implementation and Communication**	**08 Hours**
User Testing: Techniques for validating prototypes with real users. Data Testing: Approaches for testing algorithms and data science models. Iterating Based on Feedback: Refining solutions based on both user input and data metrics. Data Science Application: Example - Testing a machine learning model and iterating based on performance feedback. Moving from Prototype to Final Product: Planning for full-scale implementation. Communicating Data Insights: Storytelling with data and effectively communicating findings. Cross-Disciplinary Collaboration: Working with other teams to integrate data science into broader engineering projects. Data Science Application: Example - Deploying a data solution in production and presenting results to stakeholders.		

Textbooks:

1. "Design Thinking: Integrating Innovation, Customer Experience, and Brand Value"
2. "Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation"
3. "The Design of Everyday Things"
4. "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking"

Reference Books:

1. "Sprint: How to Solve Big Problems and Test New Ideas in Just Five Days"
   - Authors**:** Jake Knapp, John Zeratsky, Braden Kowitz
   - Publisher**:** Simon & Schuster
   - Edition**:** 1st Edition (2016)
2. "Design Thinking for Strategic Innovation: What They Can't Teach You at Business or Design School"
   - Author**:** Idris Mootee
   - Publisher**:** Wiley
   - Edition: 1st Edition (2013)
3. "Lean Analytics: Use Data to Build a Better Startup Faster"
   - Authors**:** Alistair Croll, Benjamin Yoskovitz
   - Publisher**:** O'Reilly Media
   - Edition**:** 1st Edition (2013)
4. "Creative Confidence: Unleashing the Creative Potential Within Us All"
   - Authors**:** Tom Kelley, David Kelley
   - Publisher**:** Crown Business
   - Edition**:** 1st Edition (2013)
5. "Design Thinking Research: Making Distinctions: Collaboration versus Cooperation"
   - Editors**:** Hasso Plattner, Christoph Meinel, Larry Leifer
   - Publisher**:** Springer
   - Edition**:** 1st Edition (2012)

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	1	1	2	1	1	1	1	1	3	2	2
CO2	2	3	2	2	2	1	2	2	1	2	1	2	3	2	2
CO3	2	2	3	2	2	1	2	1	2	2	1	2	2	3	2
CO4	2	2	3	3	3	2	2	1	2	2	1	2	2	3	3
CO5	2	1	2	1	2	1	1	1	3	3	2	2	2	2	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-219-OE(B): DATA SCIENCE IN BIOTECHNOLOGY
(Open Elective – II)
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: 10 +2 discrete mathematics

Course Objective:

To be proficient in important Microsoft Office tools: MS WORD, EXCEL, POWERPOINT.

1. **Understanding Biological Data**: To provide students with an understanding of the types of biological data (genomic, proteomic, clinical, etc.) and the challenges associated with its analysis.
2. **Data Analysis Skills**: To equip students with data analysis skills using programming languages such as Python and R, focusing on libraries and tools relevant to biological research.
3. **Machine Learning Applications**: To introduce students to machine learning concepts and their application in predicting outcomes in biotechnology, including drug discovery and disease diagnosis.
4. **Hands-on Experience**: To provide hands-on experience with real-world biological datasets and case studies, allowing students to apply theoretical concepts in practical scenarios.
5. **Ethical Considerations**: To discuss ethical issues related to the use of data in biotechnology, including data privacy, bioethics, and the implications of personalized medicine.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Analyze Biological Data:

- Demonstrate proficiency in collecting, cleaning, and analyzing various types of biological data using data science techniques.

CO2: Apply Machine Learning Algorithms:

- Implement and evaluate machine learning algorithms to solve specific problems in biotechnology, such as drug discovery or genomic data analysis.

CO3: Interpret Results:

- Effectively interpret and communicate the results of data analyses in the context of biological research and biotechnology applications.

CO4: Utilize Bioinformatics Tools:

- Utilize bioinformatics software and tools for tasks such as genome sequencing analysis, protein structure prediction, and metabolic pathway modeling.

CO5: Design Experiments:

- Design and conduct experiments that integrate data science methodologies with biological research objectives, including the use of high-throughput data.

CO6: Understand Ethical Implications:

- Recognize and discuss the ethical implications of data usage in biotechnology, including issues related to data privacy, genetic data handling, and the impact of personalized medicine on society.

CO7: Collaborate in Interdisciplinary Teams:

- Work effectively in interdisciplinary teams, collaborating with biologists, clinicians, and data scientists to address complex challenges in biotechnology.

CO8: Critical Thinking and Problem Solving:

- Develop critical thinking and problem-solving skills to identify, analyze, and address data-related challenges in the biotechnology field.

Course Contents

UNIT-I	Bioinformatics	10 Hours
Genome Sequencing: Analyzing genomic data to identify genetic variations and their association with diseases. Protein Structure Prediction: Using machine learning algorithms to predict the 3D structure of proteins based on amino acid sequences. Gene Expression Analysis: Studying patterns of gene expression across different conditions using data analytics to identify key biomarkers.		
UNIT-II	**Drug Discovery and Development**	**10 Hours**
High-Throughput Screening (HTS): Analyzing massive datasets from experiments to identify potential drug candidates. Clinical Trials Data Analysis: Using data science to evaluate drug efficacy and patient outcomes across large clinical trials. Pharmacogenomics: Analyzing how genetic variations affect an individual's response to drugs using machine learning.		
UNIT-III	**Healthcare Biotechnology**	**10 Hours**
Personalized Medicine: Applying data science to tailor treatments to individual patients based on their genetic makeup. Predictive Modeling: Predicting disease outbreaks or patient responses to treatments using large datasets and predictive algorithms. Medical Imaging: Applying computer vision and data analysis techniques to detect patterns in medical imaging for early diagnosis.		
UNIT-IV	**Systems Biology**	**10 Hours**
Metabolic Pathway Analysis: Using data science to model and simulate complex biological pathways, allowing for a deeper understanding of metabolic functions. Synthetic Biology: Data-driven approaches for designing and engineering new biological systems and organisms.		

UNIT-V	Agricultural Biotechnology	10 Hours
Genetic Modification: Using data science to identify traits in plants that can be enhanced for better yield, disease resistance, or drought tolerance. Precision Agriculture: Leveraging data from sensors, drones, and satellites to optimize farming practices and improve crop productivity.		
UNIT-VI	**Epidemiology and Public Health**	**10 Hours**
Data Analytics for Disease Surveillance: Using big data to track and predict the spread of diseases in populations. Omics Data Analysis: Integrating genomics, proteomics, and metabolomics data for a holistic view of biological processes and diseases.		

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	-	-	3	-	3	-
CO2	3	3	3	-	-	3	-	3	3
CO3	3	3	-	-	3	3	-	-	3
CO4	3	3	-	3	-	-	3	-	-
CO5	3	3	-	-	3	3	-	3	-
CO6	3	3	-	3	-	-	3	3	3
CO7	3	3	-	-	3	3	-	3	-
CO8	3	3	-	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-219-OE(c): CONCEPTS IN MOBILE ROBOTS (Open Elective – II)
B.Tech. II Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Knowledge on algorithms

Course Objective:

1. To introduce mobile robotic technology and its types in detail.
2. To learn the kinematics of wheeled and legged robot.
3. To familiarize the intelligence into the mobile robots using various sensors.
4. To acquaint the localization strategies and mapping technique for mobile robot.
5. To aware the collaborative mobile robotics in task planning, navigation and intelligence.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Evaluate the appropriate mobile robots for the desired application.
CO2: Create the kinematics for given wheeled and legged robot.
CO3: Analyze the sensors for the intelligence of mobile robotics.
CO4: Create the localization strategies and mapping technique for mobile robot.
CO5: Create the collaborative mobile robotics for planning, navigation and intelligence for desiredapplications.

Course Contents

UNIT-I	INTRODUCTION TO MOBILE ROBOTICS	07 Hours
Introduction – Locomotion of the Robots – Key Issues on Locomotion – Legged Mobile Roots – Configurations and Stability – Wheeled Mobile Robots – Design Space and Mobility Issues – Unmanned Aerial and Underwater Vehicles		

UNIT-II	KINEMATICS	08 Hours
Kinematic Models – Representation of Robot – Forward Kinematics – Wheel and Robot Constraints – Degree of Mobility and Steerability – Manoeuvrability – Workspace – Degrees of Freedom – Path and Trajectory Considerations – Motion Controls - Holonomic Robots		

UNIT-III	PERCEPTION	08 Hours
Sensor for Mobile Robots – Classification and Performance Characterization – Wheel/Motor Sensors – Heading Sensors - Ground-Based Beacons - Active Ranging - Motion/Speed Sensors– Camera - Visual Appearance based Feature Extraction.		
UNIT-IV	**LOCALIZATION**	**07 Hours**
Localization Based Navigation Versus Programmed Solutions - Map Representation - Continuous Representations - Decomposition Strategies - Probabilistic Map-Based Localization - Landmark-Based Navigation - Globally Unique Localization - Positioning Beacon Systems - Route-Based Localization - Autonomous Map Building - Simultaneous Localization and Mapping (SLAM).		
UNIT-V	**PLANNING, NAVIGATION AND COLLABORATIVE ROBOTS**	**08 Hours**
Introduction - Competences for Navigation: Planning and Reacting - Path Planning - Obstacle Avoidance - Navigation Architectures - Control Localization - Techniques for Decomposition - Case Studies – Collaborative Robots – Swarm Robots.		

TEXTBOOK

1. Roland Siegwart and IllahR.Nourbakish, “Introduction to Autonomous Mobile Robots” MITPress, Cambridge, 2004.

REFERENCES:

1. Dragomir N. Nenchev, Atsushi Konno, TeppeiTsujita, “Humanoid Robots: Modelling andControl”, Butterworth-Heinemann, 2018
2. MohantaJagadish Chandra, “Introduction to Mobile Robots Navigation”, LAP LambertAcademic Publishing, 2015.
3. Peter Corke, “Robotics, Vision and Control”, Springer, 2017.
4. Ulrich Nehmzow, “Mobile Robotics: A Practical Introduction”, Springer, 2003.
5. Xiao Qi Chen, Y.Q. Chen and J.G. Chase, “Mobile Robots - State of the Art in Land, Sea,Air, and Collaborative Missions”, Intec Press, 2009.
6. Alonzo Kelly, Mobile Robotics: Mathematics, Models, and Methods, Cambridge UniversityPress, 2013, ISBN: 978-1107031159.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-301-PCC: MACHINE LEARNING
B.Tech. III Year I Sem.

Teaching Scheme: TH: - 3 Hours/Week	Credit TH: 03	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Knowledge on Data Structures.

Course Objective:

1. Understand the core principles and theories behind machine learning algorithms.
2. Gain practical experience in implementing and applying supervised and unsupervised learning techniques.
3. Explore advanced topics such as deep learning and reinforcement learning.
4. Learn how to evaluate and optimize machine learning models for performance and accuracy.
5. Develop critical thinking and problem-solving skills in the context of data-driven decision making.
6. Explore ethical considerations and societal implications of machine learning technologies.
7. Build a foundation for further study or career advancement in the field of machine learning and artificial intelligence.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Understand the core principles and theories behind machine learning algorithms.
- Gain practical experience in implementing and applying supervised and unsupervised learning techniques.
- Explore advanced topics such as deep learning and reinforcement learning.
- Learn how to evaluate and optimize machine learning models for performance and accuracy.
- Develop critical thinking and problem-solving skills in the context of data-driven decision making.
- Explore ethical considerations and societal implications of machine learning technologies.
- Build a foundation for further study or career advancement in the field of machine learning and artificial intelligence.

Course Contents

UNIT-I	Introduction to Machine Learning	07 Hours

- **Introduction and Motivation**
  - Definition of Machine Learning
  - Motivation for Machine Learning
  - Historical Context of Machine Learning Techniques
- **Applications of Machine Learning**
  - Broad Application Areas
  - Specific Use Cases
  - Ethical Considerations and Social Implications
- **The Origin and Evolution of Machine Learning**
  - Early Foundations
  - Developments in Statistical Learning Theory
  - Rise of Neural Networks
  - Contemporary Trends and Future Directions
- **Foundational Concepts**
  - Abstraction in Machine Learning
  - Knowledge Representation
  - Generalization in Machine Learning
  - Model Complexity and Interpretability
- **Assessing Learning Success**
  - Evaluation Metrics and Performance Measures
  - Cross-Validation Techniques
  - Model Selection and Hyperparameter Tuning
  - Interpretability and Explainability
- **Steps to Apply Machine Learning**
  - Data Preprocessing and Cleaning
  - Model Training and Evaluation Pipeline
  - Deployment and Monitoring
- **Overview of Machine Learning Methods**
  - Supervised Learning
  - Unsupervised Learning
  - Reinforcement Learning

UNIT-II	Supervised Learning	08 Hours

- **Introduction to Supervised Learning**
  - Overview of Supervised Learning
  - Components of Supervised Learning
  - Importance and Applications
- **Regression**
  - Overview of Regression Analysis
  - Simple Linear Regression

○ Multiple Regression ○ Nonlinear Regression • **Classification** ○ Overview of Classification ○ Decision Trees ○ Support Vector Machines (SVMs) ○ Logistic Regression ○ Performance Evaluation • **Advanced Techniques in Classification** ○ Ensemble Learning Methods ○ Neural Networks for Classification.		
UNIT-III	**Unsupervised Learning**	**08 Hours**
• **Introduction to Unsupervised Learning** ○ Overview of Unsupervised Learning ○ Importance and Applications • **Clustering Techniques** ○ Overview of Clustering ○ Major Clustering Approaches ○ Evaluating Cluster Quality • **Dimensionality Reduction** ○ Overview of Dimensionality Reduction ○ Linear Techniques ○ Nonlinear Techniques • **Advanced Clustering Methods** ○ Soft Clustering vs. Hard Clustering ○ Fuzzy C-Means Clustering ○ Gaussian Mixture Models (GMM) ○ Spectral Clustering		
UNIT-IV	**Advanced Topics in Machine Learning**	**07 Hours**
• **Ensemble Learning Methods** ○ Introduction to Ensemble Learning ○ Bagging and Boosting • **Deep Learning** ○ Introduction to Deep Learning ○ Neural Networks, CNNs, RNNs • **Reinforcement Learning** ○ Introduction to Reinforcement Learning ○ Q-Learning, DQNs, Policy Gradient Methods • **Advanced Techniques in Model Interpretability** ○ Model-Agnostic Interpretability Techniques		

○ Model-Specific Interpretability Techniques		
UNIT-V	**Model Evaluation and Selection**	**08 Hours**
• **Model Evaluation Metrics** ○ Introduction to Evaluation Metrics ○ Classification Evaluation Metrics ○ Regression Evaluation Metrics • **Cluster Validation Measures** ○ Introduction to Cluster Validation ○ External and Internal Cluster Validation Measures ○ Relative Validity Measures • **Regression Model Assessment Techniques** ○ Overview of Regression Model Assessment ○ Hypothesis Testing in Regression Models ○ Model Selection Techniques ○ Cross-Validation Techniques • **Hyperparameter Tuning and Optimization** ○ Introduction to Hyperparameter Tuning ○ Grid Search and Random Search ○ Bayesian Optimization ○ Automated Hyperparameter Tuning Frameworks		
UNIT-VI	**Advanced Concepts and Applications**	**08 Hours**
• **Transfer Learning and Domain Adaptation** ○ Introduction to Transfer Learning ○ Techniques in Transfer Learning ○ Applications of Transfer Learning • **Meta-Learning and Learning to Learn** ○ Overview of Meta-Learning ○ Techniques in Meta-Learning ○ Applications of Meta-Learning • **Adversarial Machine Learning and Robustness** ○ Understanding Adversarial Machine Learning ○ Defenses Against Adversarial Attacks ○ Applications and Implications • **Fairness, Accountability, and Transparency** ○ Fairness in Machine Learning ○ Accountability in Machine Learning ○ Ethical Considerations and Social Implications		

Reference Books:

1. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron
2. "Pattern Recognition and Machine Learning" by Christopher M. Bishop
3. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
4. "Python Machine Learning" by Sebastian Raschka and Vahid Mirjalili
5. "Machine Learning: A Probabilistic Perspective" by Kevin P. Murphy
6. "Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller and Sarah Guido
7. "Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-301-L: MACHINE LEARNING LAB
B.Tech. III Year I Sem.

Practical Scheme: P: - 2 Hours/Week	Credit TH: 01	Examination Scheme: Total: 30 Marks
Course Prerequisites: Programming Lab Practice		
Experiment Of List		

Exercise 1: Introduction to Scikit-learn for Supervised Learning

Objective: To gain hands-on experience with implementing supervised learning algorithms using the Scikit-learn library in Python.

Exercise 2: Exploring Unsupervised Learning with K-Means Clustering

Objective: To explore the concepts of unsupervised learning and clustering using the K-Means algorithm.

Exercise 3: Implementing Linear Regression from Scratch

Objective: To gain a deeper understanding of linear regression by implementing it from scratch using Python.

Exercise 4: Binary Classification with Logistic Regression

Objective: To implement logistic regression for binary classification tasks and understand its application in real-world scenarios.

Exercise 5: Decision Tree Classifier for Multiclass Classification

Objective: To understand the working of decision tree classifiers and their application in multiclass classification problems.

Exercise 6: Support Vector Machine (SVM) for Classification

Objective: To implement support vector machine (SVM) classifiers for binary and multiclass classification tasks and understand their kernel trick.

Exercise 7: Ensemble Learning with Random Forests

Objective: To implement random forests, an ensemble learning method, and understand their advantages in improving model performance and generalization.

Exercise 8: Hierarchical Clustering

Objective: To understand hierarchical clustering algorithms and their application in grouping data points based on their similarity.

Exercise 9: Density-Based Clustering (DBSCAN)

Objective: To implement the DBSCAN algorithm and understand its effectiveness in identifying clusters of arbitrary shapes in noisy datasets.

Exercise 10: Principal Component Analysis (PCA)

Objective: To understand the concept of dimensionality reduction using PCA and apply it to high-dimensional datasets for visualization and feature extraction.

Exercise 11: Convolutional Neural Network (CNN) for Image Classification

Objective: To implement a Convolutional Neural Network (CNN) using Python and

TensorFlow/Keras library for image classification tasks.

Exercise 12: Recurrent Neural Network (RNN) for Text Generation

Objective: To implement a Recurrent Neural Network (RNN) using Python and TensorFlow/Keras library for text generation tasks.

Exercise 13: Reinforcement Learning with OpenAI Gym

Objective: To implement reinforcement learning algorithms using Python and OpenAI Gym library for training agents in various environments.

Exercise 14: Evaluation Metrics for Classification Models

Objective: To understand and implement various evaluation metrics for classification models using Python and Scikit-learn library.

Exercise 15: Regression Model Evaluation Techniques

Objective: To explore and apply various evaluation techniques for regression models using Python and Scikit-learn library.

Exercise 16: Cluster Validation Measures

Objective: To understand and apply cluster validation measures to assess the quality of clustering solutions using Python and Scikit-learn library.

Exercise 17: Model Selection Techniques

Objective: To explore and apply model selection techniques for choosing the best machine learning models using Python and Scikit-learn library.

Exercise 18: Hyperparameter Tuning and Optimization

Objective: To understand and apply hyperparameter tuning and optimization techniques for improving machine learning model performance using Python and Scikit-learn library.

Exercise 19: Transfer Learning in Image Classification

Objective: To apply transfer learning techniques using pre-trained convolutional neural network (CNN) models for image classification tasks using TensorFlow/Keras.

Exercise 20: Meta-Learning for Few-Shot Learning

Objective: To implement meta-learning algorithms for few-shot learning tasks using Python and TensorFlow/Keras.

Exercise 21: Adversarial Attacks and Defenses

Objective: To understand adversarial attacks and defenses in deep learning models and evaluate their robustness using Python and TensorFlow.

BTDS-302-PCC: DESCRIPTIVE ANALYTICS
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: The introduction of the Advanced English Communication

Course Objective:

- To understand the role and importance of descriptive analytics and business intelligence in decision-making.
- To familiarize students with business intelligence architecture and decision support science.
- To learn methods for data collection, data cleaning, and preparation for analysis.
- To develop skills in exploratory data analysis and data visualization.
- To teach students how to design and develop reports, dashboards, and scorecards for data visualization.
- To apply statistical techniques and advanced data mining approaches for summarizing and interpreting data.
- To enable students to build and manage BI projects effectively.
- To enhance problem-solving skills by analyzing real-world datasets.
- To communicate findings effectively through data storytelling and visualization.
- To equip students with the skills necessary to measure BI success and value.
- To prepare students for careers in business intelligence, analytics, and decision support systems.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Explain the principles and significance of descriptive analytics and business intelligence.
- Understand the architecture and functional areas of business intelligence tools.
- Collect, clean, and prepare data for analysis.
- Perform exploratory data analysis to identify patterns and trends.
- Use statistical methods and advanced data mining techniques to describe and summarize data.
- Create effective data visualizations, reports, dashboards, and scorecards to communicate insights.
- Utilize analytics tools such as Excel, Python, R, Power BI, and Tableau.
- Design and implement BI projects using industry-standard methodologies and tools.
- Analyze real-world datasets and derive actionable insights.
- Evaluate the effectiveness and value of business intelligence initiatives.
- Present data findings clearly and persuasively to stakeholders.
- Apply decision support techniques to analyze and explore data for decision-making.

Course Contents

UNIT-I	**Introduction to Descriptive Analytics and Business Intelligence, Business Intelligence Architecture and Decision Support Science**	**07 Hours**

1.1 Overview of Descriptive Analytics
- 1.1.1 Definition and Scope of Descriptive Analytics
  - Understanding Descriptive Analytics
  - Differentiating Descriptive Analytics from Predictive and Prescriptive Analytics
  - The role of Descriptive Analytics in the Analytics Spectrum
- 1.1.2 Key Concepts and Terminology
  - Data, Information, and Knowledge
  - Metrics and Measures
  - Key Performance Indicators (KPIs)
- 1.1.3 Historical Development of Descriptive Analytics
  - Evolution from Traditional Reporting to Modern Analytics
  - Major Milestones in the Development of Descriptive Analytics
- 1.1.4 Tools and Technologies
  - Overview of Popular Descriptive Analytics Tools (Excel, Tableau, Power BI)
  - Introduction to Programming Languages Used in Descriptive Analytics (Python, R)

1.2 Importance of Business Intelligence
- 1.2.1 Definition and Scope of Business Intelligence (BI)
  - Understanding Business Intelligence
  - Components of BI Systems
- 1.2.2 Benefits of Business Intelligence
  - Enhancing Decision-Making
  - Improving Operational Efficiency
  - Identifying Business Opportunities
  - Supporting Strategic Initiatives
- 1.2.3 BI in Various Industries
  - Case Studies: Use of BI in Healthcare, Retail, Finance, and Manufacturing
  - Industry-Specific Applications of BI
- 1.2.4 Integration of Descriptive Analytics in BI
  - How Descriptive Analytics Fits within the BI Framework
  - The Role of Descriptive Analytics in BI Workflows

1.3 Role in Decision-Making Processes
- 1.3.1 Types of Decisions Supported by BI
  - Strategic Decisions
  - Tactical Decisions
  - Operational Decisions
- 1.3.2 Decision-Making Models and Theories
  - Rational Decision-Making Model
  - Data-Driven Decision-Making
  - The Role of Intuition in Decision-Making
- 1.3.3 Data-Driven Decision-Making Process
  - Steps in the Data-Driven Decision-Making Process
  - Collecting and Preparing Data

Analyzing Data and Generating Insights
Making Decisions Based on Data Insights
1.3.4 Case Studies and Examples
Real-World Examples of BI Supporting Decision-Making
Success Stories from Various Organizations
1.3.5 Challenges and Limitations
Common Challenges in Implementing BI and Descriptive Analytics
Addressing Data Quality Issues
Overcoming Resistance to Data-Driven Decision-Making
2.1 Components of BI Architecture
2.1.1 Introduction to BI Architecture
Definition and Purpose of BI Architecture
Overview of BI Systems and Their Importance
2.1.2 Data Sources and Data Integration
Types of Data Sources (Structured, Unstructured, Semi-Structured)
ETL (Extract, Transform, Load) Processes
Data Warehousing
Characteristics of Data Warehouses
Differences Between Data Warehouses and Databases
2.1.3 Data Storage and Management
Data Marts
Data Lakes
Data Models (Star Schema, Snowflake Schema)
2.1.4 Analytical Processing
Online Analytical Processing (OLAP)
OLAP Operations (Slice, Dice, Drill Down, Roll Up)
OLAP Types (MOLAP, ROLAP, HOLAP)
Real-Time Analytics and Streaming Data
2.1.5 BI Tools and Technologies
Overview of BI Tools (Reporting Tools, Dashboards, Scorecards)
Advanced BI Tools (AI-Powered BI, Predictive Analytics Integration)
2.1.6 Security and Governance in BI
Data Security Best Practices
Data Governance Frameworks
Compliance and Regulatory Considerations
2.2 Decision Support Systems
2.2.1 Overview of Decision Support Systems (DSS)
Definition and Importance of DSS
Historical Development of DSS
2.2.2 Components of DSS
Database Management System (DBMS)
Model-Based Management System (MBMS)

User Interface
2.2.3 Types of Decision Support Systems
Data-Driven DSS
Model-Driven DSS
Knowledge-Driven DSS
Document-Driven DSS
Communication-Driven DSS
2.2.4 Decision Support Techniques
Simulation and Modeling
What-If Analysis
Sensitivity Analysis
2.2.5 Integration of DSS with BI
How DSS Enhances BI Capabilities
Case Studies of DSS Applications in BI
2.2.6 Trends and Innovations in DSS
Emerging Technologies in DSS (AI, Machine Learning)
Future Directions and Innovations
2.3 BI Project Lifecycle
2.3.1 Introduction to BI Project Lifecycle
Importance of Project Management in BI
Overview of BI Project Stages
2.3.2 Planning and Requirements Gathering
Defining Project Objectives and Scope
Stakeholder Analysis and Involvement
Requirements Elicitation Techniques
2.3.3 Designing BI Solutions
BI System Design Principles
Architecture Design
Data Modeling and Schema Design
2.3.4 Implementation and Development
Setting Up Data Integration and ETL Processes
Development of BI Reports and Dashboards
User Interface and Experience Design
2.3.5 Testing and Quality Assurance
Types of Testing (Unit Testing, Integration Testing, System Testing)
Ensuring Data Quality and Accuracy
User Acceptance Testing (UAT)
2.3.6 Deployment and Maintenance
Deployment Strategies (Phased, Parallel, Big Bang)
Training and Support for Users
Ongoing Maintenance and Optimization
2.3.7 Evaluating BI Project Success

	Key Performance Indicators (KPIs) for Project Evaluation Measuring Return on Investment (ROI) Continuous Improvement and Feedback Loops.	
UNIT-II	**Data Collection and Preparation and Exploratory Data Analysis (EDA)**	**08 Hours**
3.1 Data Sources and Types 3.1.1 Introduction to Data Sources Definition and Importance of Data Sources Overview of Data Sources in Business Intelligence 3.1.2 Structured Data Characteristics and Examples of Structured Data Sources of Structured Data (Databases, Spreadsheets) Use Cases in Business Analytics 3.1.3 Unstructured Data Characteristics and Examples of Unstructured Data Sources of Unstructured Data (Text, Images, Videos) Challenges and Opportunities with Unstructured Data 3.1.4 Semi-Structured Data Characteristics and Examples of Semi-Structured Data Sources of Semi-Structured Data (XML, JSON) Use Cases and Integration with Structured Data 3.1.5 Internal vs. External Data Sources Internal Data Sources (Enterprise Systems, CRM, ERP) External Data Sources (Social Media, Public Data, Market Data) Combining Internal and External Data for Comprehensive Analysis 3.2 Data Collection Methods 3.2.1 Overview of Data Collection Importance of Effective Data Collection Ethical Considerations in Data Collection 3.2.2 Manual Data Collection Techniques for Manual Data Collection (Surveys, Interviews) Pros and Cons of Manual Data Collection 3.2.3 Automated Data Collection Techniques for Automated Data Collection (Web Scraping, APIs) Tools and Technologies for Automated Data Collection Benefits and Challenges of Automation 3.2.4 Real-Time Data Collection Introduction to Real-Time Data Collection Methods for Collecting Real-Time Data (Sensors, IoT Devices) Applications and Use Cases for Real-Time Data 3.2.5 Data Collection in Business Intelligence		

Best Practices for Data Collection in BI Projects
Ensuring Data Quality and Integrity During Collection
Legal and Compliance Issues in Data Collection
3.3 Data Cleaning Techniques
3.3.1 Introduction to Data Cleaning
Importance of Data Cleaning in Data Preparation
Common Issues Encountered in Raw Data
3.3.2 Identifying and Correcting Data Errors
Types of Data Errors (Typos, Inconsistencies, Duplicates)
Methods for Identifying Errors (Automated Tools, Manual Review)
Techniques for Correcting Errors (Standardization, Validation)
3.3.3 Data Transformation
Normalization and Standardization of Data
Data Formatting and Conversion
Aggregation and Summarization of Data
3.3.4 Dealing with Inconsistent Data
Identifying Inconsistent Data (Date Formats, Units of Measure)
Methods for Resolving Inconsistencies (Standard Rules, Algorithms)
Tools for Data Cleaning and Transformation (Python, R, Excel)
3.4 Handling Missing Data and Outliers
3.4.1 Introduction to Missing Data
Causes and Types of Missing Data (MCAR, MAR, MNAR)
Impact of Missing Data on Analysis
3.4.2 Techniques for Handling Missing Data
Deletion Methods (Listwise, Pairwise)
Imputation Methods (Mean/Median Imputation, Regression Imputation, Multiple Imputation)
Advanced Techniques (KNN Imputation, Machine Learning-Based Imputation)
3.4.3 Introduction to Outliers
Definition and Identification of Outliers
Causes and Impact of Outliers on Data Analysis
3.4.4 Techniques for Handling Outliers
Detection Methods (Z-Score, IQR, Visualization Techniques)
Treatment Methods (Capping, Transformation, Removal)
Considerations for Keeping or Removing Outliers Based on Context
3.4.5 Tools and Software for Data Cleaning and Preparation
Overview of Popular Data Cleaning Tools (OpenRefine, Trifacta, DataWrangler)
Using Python for Data Cleaning (Pandas, NumPy)
Using R for Data Cleaning (dplyr, tidyr)
4.1 Introduction to EDA
4.1.1 Definition and Importance of EDA
What is Exploratory Data Analysis?

The Role of EDA in the Data Analysis Process
Benefits of Conducting EDA
4.1.2 Objectives of EDA
Uncovering Patterns
Spotting Anomalies
Testing Hypotheses
Checking Assumptions
4.1.3 Steps in the EDA Process
Initial Data Inspection
Data Cleaning and Preparation
Data Visualization and Summary Statistics
Identifying Patterns, Trends, and Relationships
4.1.4 Tools for EDA
Software and Tools Used for EDA (Python, R, Excel, Tableau)
Introduction to Python Libraries (Pandas, Matplotlib, Seaborn)
Introduction to R Libraries (ggplot2, dplyr)
4.2 Descriptive Statistics
4.2.1 Overview of Descriptive Statistics
Definition and Importance
Types of Descriptive Statistics
4.2.2 Measures of Central Tendency
Mean
Definition and Calculation
Advantages and Disadvantages
Applications in Business Analytics
Median
Definition and Calculation
Advantages and Disadvantages
Applications in Business Analytics
Mode
Definition and Calculation
Advantages and Disadvantages
Applications in Business Analytics
4.2.3 Measures of Dispersion
Range
Definition and Calculation
Interpretation
Variance
Definition and Calculation
Interpretation
Standard Deviation
Definition and Calculation

- Interpretation
- Applications of Dispersion Measures in Business Analytics
- 4.2.4 Measures of Shape
  - Skewness
    - Definition and Calculation
    - Interpretation and Applications
  - Kurtosis
    - Definition and Calculation
    - Interpretation and Applications
- 4.2.5 Visualizing Descriptive Statistics
  - Histograms
    - Construction and Interpretation
    - Use Cases
  - Boxplots
    - Construction and Interpretation
    - Use Cases
  - Bar Charts
    - Construction and Interpretation
    - Use Cases
  - Best Practices for Visualizing Summary Statistics
- 4.3 Identifying Patterns and Trends
  - 4.3.1 Time Series Analysis
    - Introduction to Time Series Data
      - Definition and Importance
      - Examples of Time Series Data in Business
    - Identifying Trends
      - Techniques for Identifying Trends
      - Visualization Techniques (Line Charts, Moving Averages)
    - Seasonality and Cycles
      - Identifying Seasonal Patterns
      - Techniques for Seasonality Analysis
      - Visualization Techniques (Seasonal Decomposition)
    - Applications of Time Series Analysis in Business
  - 4.3.2 Pattern Recognition Techniques
    - Clustering
      - K-Means Clustering
        - Algorithm and Steps
        - Advantages and Disadvantages
        - Practical Applications
      - Hierarchical Clustering
        - Algorithm and Steps
        - Advantages and Disadvantages

- Practical Applications
- Classification
  - Decision Trees
    - Algorithm and Steps
    - Advantages and Disadvantages
    - Practical Applications
  - Support Vector Machines (SVM)
    - Algorithm and Steps
    - Advantages and Disadvantages
    - Practical Applications
- Identifying Patterns in Data
  - Using Visualization Techniques (Heatmaps, Scatter Plots)
  - Applications of Pattern Recognition in Business
- 4.3.3 Anomaly Detection
  - Definition and Importance of Anomaly Detection
  - Methods for Detecting Anomalies
    - Z-Score Method
      - Calculation and Interpretation
      - Practical Applications
    - Interquartile Range (IQR) Method
      - Calculation and Interpretation
      - Practical Applications
    - Machine Learning Techniques for Anomaly Detection
      - Overview of Algorithms (Isolation Forest, LOF)
      - Practical Applications
  - Applications of Anomaly Detection in Business (Fraud Detection, Quality Control)

4.4 Correlation Analysis

- 4.4.1 Introduction to Correlation
  - Definition and Importance of Correlation
  - Types of Correlation
    - Positive Correlation
    - Negative Correlation
    - No Correlation
- 4.4.2 Calculating Correlation Coefficients
  - Pearson Correlation Coefficient
    - Definition and Calculation
    - Interpretation and Applications
  - Spearman Rank Correlation
    - Definition and Calculation
    - Interpretation and Applications
  - Kendall Tau Correlation
    - Definition and Calculation

Interpretation and Applications 4.4.3 Interpreting Correlation Results Strength and Direction of Correlation Interpretation of Correlation Coefficients Practical Applications in Business Correlation vs. Causation Understanding the Difference Common Misconceptions Practical Applications of Correlation Analysis Identifying Relationships Between Variables Use Cases in Business Analytics 4.4.4 Visualizing Correlations Correlation Matrices Construction and Interpretation Practical Applications Scatter Plot Matrices Construction and Interpretation Practical Applications Heatmaps for Correlation Construction and Interpretation Practical Applications		
UNIT-III	**Statistical Methods for Data Summarization and Advanced Data Mining Techniques**	**08 Hours**
5.1 Measures of Central Tendency 5.1.1 Overview of Central Tendency Definition and Importance in Data Analysis Applications in Business Intelligence 5.1.2 Mean (Arithmetic Mean) Definition and Formula Calculation Methods Advantages and Disadvantages Practical Examples and Applications 5.1.3 Median Definition and Formula Calculation Methods Advantages and Disadvantages Practical Examples and Applications 5.1.4 Mode Definition and Formula Calculation Methods Advantages and Disadvantages		

Practical Examples and Applications
5.1.5 Comparing Measures of Central Tendency
When to Use Mean, Median, and Mode
Impact of Outliers on Central Tendency
Case Studies and Practical Applications
5.2 Measures of Dispersion and Variability
5.2.1 Overview of Dispersion and Variability
Definition and Importance in Data Analysis
Applications in Business Intelligence
5.2.2 Range
Definition and Formula
Calculation Methods
Advantages and Disadvantages
Practical Examples and Applications
5.2.3 Variance
Definition and Formula
Calculation Methods
Advantages and Disadvantages
Practical Examples and Applications
5.2.4 Standard Deviation
Definition and Formula
Calculation Methods
Advantages and Disadvantages
Practical Examples and Applications
5.2.5 Interquartile Range (IQR)
Definition and Formula
Calculation Methods
Advantages and Disadvantages
Practical Examples and Applications
5.2.6 Comparing Measures of Dispersion
When to Use Range, Variance, Standard Deviation, and IQR
Impact of Outliers on Dispersion Measures
Case Studies and Practical Applications
5.3 Probability Distributions
5.3.1 Introduction to Probability Distributions
Definition and Importance in Data Analysis
Types of Probability Distributions
5.3.2 Discrete Probability Distributions
Binomial Distribution
Definition, Formula, and Properties
Practical Examples and Applications
Poisson Distribution

Definition, Formula, and Properties
Practical Examples and Applications
5.3.3 Continuous Probability Distributions
Normal Distribution
Definition, Formula, and Properties
Practical Examples and Applications
Exponential Distribution
Definition, Formula, and Properties
Practical Examples and Applications
5.3.4 Understanding and Using Probability Distributions
Fitting Data to Distributions
Using Probability Distributions for Predictions
Case Studies and Practical Applications
5.4 Hypothesis Testing
5.4.1 Introduction to Hypothesis Testing
Definition and Importance in Data Analysis
Types of Hypotheses: Null and Alternative
5.4.2 Steps in Hypothesis Testing
Formulating Hypotheses
Choosing the Appropriate Test
Setting the Significance Level (Alpha)
Calculating Test Statistic and P-Value
Making a Decision: Reject or Fail to Reject the Null Hypothesis
5.4.3 Types of Hypothesis Tests
Z-Test
Definition, Formula, and When to Use
Practical Examples and Applications
T-Test
One-Sample T-Test
Definition, Formula, and When to Use
Practical Examples and Applications
Two-Sample T-Test (Independent and Paired)
Definition, Formula, and When to Use
Practical Examples and Applications
Chi-Square Test
Definition, Formula, and When to Use
Practical Examples and Applications
ANOVA (Analysis of Variance)
Definition, Formula, and When to Use
Practical Examples and Applications
5.4.4 Interpreting Hypothesis Test Results
Understanding P-Values and Confidence Intervals

Making Business Decisions Based on Hypothesis Test Results
Case Studies and Practical Applications
6.1 Introduction to Data Mining
6.1.1 Overview of Data Mining
Definition and Importance
History and Evolution of Data Mining
Applications in Various Industries
6.1.2 Data Mining Process
Steps in the Data Mining Process: Data Selection, Data Preprocessing, Data Transformation, Data Mining, Interpretation/Evaluation
Data Mining Methodologies: CRISP-DM, SEMMA, KDD
6.1.3 Key Concepts in Data Mining
Data Warehousing and Data Marts
Data Mining vs. Machine Learning
Ethical Considerations in Data Mining
6.2 Clustering and Classification
6.2.1 Clustering Techniques
Definition and Importance
Types of Clustering: Hard vs. Soft Clustering
Common Clustering Algorithms
K-Means Clustering
Algorithm Overview
Choosing the Number of Clusters (K)
Practical Examples and Applications
Hierarchical Clustering
Algorithm Overview
Agglomerative vs. Divisive Methods
Dendrogram Analysis
DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
Algorithm Overview
Identifying Core Points and Clusters
Handling Noise in Data
6.2.2 Classification Techniques
Definition and Importance
Types of Classification: Binary, Multiclass, Multilabel
Common Classification Algorithms
Decision Trees
Algorithm Overview
Splitting Criteria (Gini, Entropy)
Pruning Techniques
Practical Examples and Applications
Random Forest

- Ensemble Method Overview
- Building and Interpreting Random Forest Models
- Support Vector Machines (SVM)
  - Algorithm Overview
  - Kernel Functions
  - Practical Examples and Applications
- Naive Bayes
  - Algorithm Overview
  - Assumptions and Practical Use Cases
- Neural Networks for Classification
  - Overview of Perceptrons and Multilayer Networks
  - Training Neural Networks for Classification Tasks

6.2.3 Model Evaluation and Validation
- Confusion Matrix and Performance Metrics (Accuracy, Precision, Recall, F1-Score)
- Cross-Validation Techniques
- Handling Imbalanced Data

6.3 Association Rule Mining

6.3.1 Introduction to Association Rule Mining
- Definition and Importance
- Applications in Market Basket Analysis, Recommender Systems

6.3.2 Key Concepts in Association Rule Mining
- Support, Confidence, and Lift
- Apriori Algorithm
  - Algorithm Overview
  - Finding Frequent Itemsets
  - Generating Association Rules
  - Practical Examples and Applications
- Eclat Algorithm
  - Algorithm Overview
  - Depth-First Search Approach
  - Practical Examples and Applications
- FP-Growth (Frequent Pattern Growth)
  - Algorithm Overview
  - Building and Using FP-Trees
  - Practical Examples and Applications

6.3.3 Advanced Topics in Association Rule Mining
- Mining Multidimensional Association Rules
- Handling Categorical and Continuous Data
- Pruning and Improving Rule Quality

6.4 Anomaly Detection

6.4.1 Introduction to Anomaly Detection
- Definition and Importance

Applications in Fraud Detection, Network Security, Fault Detection 6.4.2 Techniques for Anomaly Detection Statistical Methods Z-Score and Thresholding Box Plot Analysis Practical Examples and Applications Machine Learning Methods Clustering-Based Methods (e.g., K-Means for Anomaly Detection) Classification-Based Methods (e.g., One-Class SVM, Isolation Forest) Neural Networks for Anomaly Detection Autoencoders LSTM for Sequence Anomalies 6.4.3 Evaluation of Anomaly Detection Methods Precision, Recall, and F1-Score for Anomaly Detection ROC and AUC Analysis Handling Imbalanced Data and Rare Events		
UNIT-IV	**Data Visualization and Business Intelligence Tools**	**07 Hours**
7.1 Principles of Data Visualization 7.1.1 Introduction to Data Visualization Definition and Importance History and Evolution of Data Visualization Applications in Various Industries 7.1.2 Key Principles of Effective Data Visualization Clarity and Simplicity Accuracy and Integrity Efficiency and Speed Aesthetics and Design 7.1.3 Understanding the Audience Identifying Audience Needs Tailoring Visualizations to Different Stakeholders Ensuring Accessibility and Inclusivity 7.1.4 Data Visualization Techniques Comparison of Different Visualization Techniques Choosing the Right Visualization for Different Data Types 7.2 Creating Effective Charts and Graphs 7.2.1 Types of Charts and Graphs Bar Charts Vertical and Horizontal Bar Charts Stacked and Grouped Bar Charts Line Charts Simple Line Charts Multiple Line Charts		

- Pie Charts
  - When and How to Use Pie Charts
  - Alternatives to Pie Charts
- Scatter Plots
  - Correlation and Trend Analysis
  - Bubble Charts
- Histograms and Box Plots
  - Distribution and Variability Analysis
- Heatmaps
  - Visualizing Data Density
  - Using Color Scales Effectively

7.2.2 Best Practices for Creating Charts and Graphs
- Choosing Appropriate Scales and Axes
- Labeling and Annotating Effectively
- Using Color and Style Consistently
- Avoiding Common Pitfalls (e.g., Misleading Scales, Overcomplication)

7.2.3 Advanced Charting Techniques
- Dual-Axis Charts
- Combination Charts
- Small Multiples
- Interactive and Dynamic Charts

7.3 Dashboards and Scorecards

7.3.1 Introduction to Dashboards
- Definition and Purpose
- Types of Dashboards (Strategic, Analytical, Operational)

7.3.2 Designing Effective Dashboards
- Key Elements of a Dashboard
- Principles of Good Dashboard Design
  - Layout and Composition
  - Visual Hierarchy
  - Interactivity and User Experience

7.3.3 Tools for Creating Dashboards
- Overview of Popular Tools (Tableau, Power BI, Excel, Google Data Studio)
- Features and Capabilities of Each Tool
- Choosing the Right Tool for the Job

7.3.4 Building and Implementing Dashboards
- Step-by-Step Guide to Building a Dashboard
- Connecting to Data Sources
- Creating Interactive Elements (Filters, Drill-Downs)
- Testing and Validating Dashboards
- Deploying and Sharing Dashboards

7.3.5 Scorecards for Performance Monitoring

Definition and Purpose
Key Metrics and KPIs
Designing Effective Scorecards
Integrating Scorecards with Dashboards
7.4 Data Storytelling
7.4.1 The Art of Data Storytelling
Definition and Importance
Elements of a Good Data Story
The Role of Narrative in Data Visualization
7.4.2 Structuring a Data Story
Identifying the Core Message
Building the Narrative Arc (Introduction, Rising Action, Climax, Conclusion)
Supporting the Story with Data
7.4.3 Techniques for Effective Data Storytelling
Using Visuals to Enhance the Narrative
Combining Text and Visuals
Engaging the Audience Emotionally and Intellectually
7.4.4 Case Studies in Data Storytelling
Analyzing Successful Data Stories
Lessons Learned from Real-World Examples
7.4.5 Tools and Technologies for Data Storytelling
Overview of Tools (e.g., PowerPoint, Tableau Story Points, Flourish)
Integrating Different Media (Text, Visuals, Audio, Video)
Tips for Using Tools Effectively
8.1 Excel for Data Analysis
8.1.1 Introduction to Excel for BI
Overview of Excel's Capabilities in Data Analysis
Differences between Excel and other BI Tools
8.1.2 Data Management in Excel
Data Entry and Data Cleaning
Using Functions and Formulas
Data Validation and Conditional Formatting
8.1.3 Descriptive Statistics and Analysis
Applying Basic Statistical Functions (SUM, AVERAGE, COUNT, etc.)
Using Advanced Statistical Functions (STDEV, VAR, etc.)
8.1.4 PivotTables and PivotCharts
Creating and Customizing PivotTables
Using PivotCharts for Data Visualization
Filtering and Sorting Data in PivotTables
8.1.5 Advanced Excel Techniques
VLOOKUP, HLOOKUP, and INDEX-MATCH Functions
Using Excel Tables and Data Models

Introduction to Power Query and Power Pivot
8.1.6 Visualization and Reporting
Creating Charts and Graphs
Customizing Chart Elements
Building Interactive Dashboards in Excel
8.2 Python (pandas, matplotlib, seaborn)
8.2.1 Introduction to Python for Data Analysis
Overview of Python's Role in Data Analysis
Setting Up the Python Environment (Anaconda, Jupyter Notebooks)
8.2.2 Data Manipulation with pandas
Introduction to pandas Data Structures (Series, DataFrame)
Data Importing and Exporting
Data Cleaning and Preparation (Handling Missing Values, Duplicates)
Data Transformation (Filtering, Sorting, Grouping, Merging)
8.2.3 Data Visualization with matplotlib
Basics of matplotlib
Creating Basic Plots (Line, Bar, Scatter)
Customizing Plots (Labels, Titles, Legends)
8.2.4 Advanced Visualization with seaborn
Introduction to seaborn and Its Features
Creating Statistical Graphics (Histograms, KDE Plots, Pair Plots)
Customizing Seaborn Plots
8.2.5 Integrating pandas with matplotlib and seaborn
Combining Data Manipulation and Visualization
Creating Complex Visualizations for EDA
8.2.6 Real-World Data Analysis Projects
Hands-On Projects Using Real Datasets
Best Practices for Code and Data Management
8.3 R for Statistical Computing
8.3.1 Introduction to R and RStudio
Overview of R's Capabilities for Statistical Analysis
Setting Up R and RStudio
8.3.2 Data Management in R
Importing and Exporting Data
Data Cleaning and Preparation
Data Transformation with dplyr
8.3.3 Statistical Analysis with R
Descriptive Statistics (Summary, Mean, Median, Mode)
Inferential Statistics (t-tests, ANOVA)
Correlation and Regression Analysis
8.3.4 Data Visualization with ggplot2
Introduction to ggplot2

	Creating Basic Plots (Histograms, Scatter Plots, Box Plots) Customizing Plots with Themes and Labels 8.3.5 Advanced Visualization Techniques Faceting and Grouping Data in Plots Creating Interactive Visualizations with Shiny 8.3.6 Real-World Data Analysis Projects Hands-On Projects Using R Applying R to Business Intelligence Scenarios 8.4 Power BI and Tableau 8.4.1 Introduction to Power BI and Tableau Overview of Power BI and Tableau Capabilities Differences and Use Cases for Each Tool 8.4.2 Data Import and Preparation Connecting to Data Sources (Excel, SQL, Web, etc.) Data Cleaning and Transformation Creating Data Models 8.4.3 Building Visualizations in Power BI Creating Basic Visuals (Charts, Tables, Cards) Customizing Visuals with Power BI Features Using DAX for Advanced Calculations 8.4.4 Building Visualizations in Tableau Creating Basic Visuals (Bar, Line, Pie Charts) Customizing Visuals with Tableau Features Using Calculated Fields and Table Calculations 8.4.5 Dashboards and Reports Designing Interactive Dashboards Best Practices for Dashboard Layout and Design Sharing and Publishing Dashboards 8.4.6 Advanced Analytics and Insights Using Power BI and Tableau for Advanced Data Analysis Applying Machine Learning Models in BI Tools Real-World Business Intelligence Projects	
UNIT-V	**Building and Managing BI Projects and Real-World Applications and Case Studies**	**08 Hours**

9.1 BI Project Planning and Execution
- 9.1.1 Introduction to BI Projects
  - Understanding the Scope and Objectives of BI Projects
  - Key Elements of a Successful BI Project
- 9.1.2 Project Planning and Management
  - Setting Project Goals and Milestones
  - Developing a Project Timeline
  - Allocating Resources and Responsibilities
  - Risk Management and Mitigation Strategies
- 9.1.3 Data Requirements and Integration
  - Identifying Data Sources and Requirements
  - Data Integration Techniques and Best Practices
  - Ensuring Data Quality and Consistency
- 9.1.4 Agile Methodology in BI Projects
  - Overview of Agile Methodology
  - Applying Agile Practices to BI Projects
  - Sprints, Backlogs, and Iterative Development
- 9.1.5 Collaboration and Communication
  - Building Effective Project Teams
  - Stakeholder Engagement and Management
  - Communication Strategies and Tools

9.2 Developing Reports and Dashboards
- 9.2.1 Introduction to Reporting in BI
  - Purpose and Importance of BI Reports
  - Types of BI Reports (Operational, Analytical, Strategic)
- 9.2.2 Designing Effective Reports
  - Principles of Effective Report Design
  - Structuring Reports for Clarity and Impact
  - Using KPIs and Metrics in Reports
- 9.2.3 Dashboard Design and Development
  - Introduction to Dashboards
  - Best Practices for Dashboard Design
  - Interactive Dashboards and User Experience
- 9.2.4 Tools for Report and Dashboard Development
  - Overview of BI Reporting Tools (Power BI, Tableau, etc.)
  - Step-by-Step Guide to Creating Reports and Dashboards
  - Advanced Features and Customization
- 9.2.5 Real-World Reporting and Dashboard Projects
  - Case Studies and Examples of Effective BI Reports
  - Hands-On Projects: Creating Reports and Dashboards
  - Peer Reviews and Feedback

9.3 Case Studies of BI Projects

9.3.1 Overview of Case Studies
Importance of Learning from Real-World Examples
Selection Criteria for Case Studies
9.3.2 Successful BI Project Case Studies
Case Study 1: Implementing a Sales Dashboard
Project Background and Objectives
Data Integration and Analysis
Challenges and Solutions
Outcomes and Impact
Case Study 2: Enhancing Customer Insights with BI
Project Background and Objectives
Data Collection and Cleaning
Analysis Techniques and Tools
Outcomes and Impact
9.3.3 Lessons Learned from BI Projects
Common Challenges in BI Projects
Best Practices and Key Takeaways
Applying Lessons to Future Projects
9.4 Measuring BI Success and Value
9.4.1 Importance of Measuring BI Success
Why Measuring Success is Crucial
Key Metrics for Evaluating BI Projects
9.4.2 Developing a BI Success Framework
Defining Success Criteria
Establishing KPIs and Metrics
Creating a Measurement Plan
9.4.3 Quantitative Measures of BI Success
Financial Metrics (ROI, Cost Savings)
Performance Metrics (Efficiency, Accuracy)
User Adoption and Satisfaction
9.4.4 Qualitative Measures of BI Success
User Feedback and Surveys
Case Study Reviews and Success Stories
Impact on Decision-Making and Strategy
9.4.5 Continuous Improvement in BI
Monitoring and Reporting on BI Performance
Identifying Areas for Improvement
Implementing Changes and Measuring Impact
10.1 Industry Applications of Descriptive Analytics
10.1.1 Introduction to Industry Applications
Overview of Descriptive Analytics Across Industries
Importance of Domain Knowledge in Analytics

10.1.2 Descriptive Analytics in Retail
Sales Analysis and Customer Segmentation
Inventory Management and Demand Forecasting
Case Study: Analyzing Sales Data for a Retail Chain
10.1.3 Descriptive Analytics in Healthcare
Patient Data Analysis and Healthcare Outcomes
Resource Allocation and Utilization
Case Study: Improving Patient Care Through Data Analysis
10.1.4 Descriptive Analytics in Finance
Fraud Detection and Risk Management
Portfolio Performance Analysis
Case Study: Risk Assessment and Mitigation in Banking
10.1.5 Descriptive Analytics in Marketing
Campaign Performance and Customer Insights
Market Basket Analysis and Customer Loyalty
Case Study: Enhancing Marketing Strategies with Analytics
10.1.6 Descriptive Analytics in Manufacturing
Quality Control and Process Improvement
Predictive Maintenance and Operational Efficiency
Case Study: Optimizing Production Processes Through Data
10.2 Hands-on Projects with Real-World Datasets
10.2.1 Project Planning and Setup
Defining Project Goals and Objectives
Selecting Appropriate Datasets
Setting Up Analytical Tools and Environment
10.2.2 Data Collection and Cleaning
Gathering Data from Multiple Sources
Data Preprocessing Techniques
Ensuring Data Quality and Integrity
10.2.3 Exploratory Data Analysis (EDA)
Applying Descriptive Statistics
Visualizing Data to Identify Trends and Patterns
Using EDA to Formulate Hypotheses
10.2.4 Advanced Data Analysis Techniques
Applying Statistical Methods and Data Mining Techniques
Clustering and Classification for Segmentation
Association Rule Mining and Anomaly Detection
10.2.5 Data Visualization and Reporting
Creating Charts, Graphs, and Dashboards
Developing Reports to Communicate Findings
Best Practices for Effective Data Storytelling
10.2.6 Presenting Project Results

Preparing a Presentation Outline Communicating Insights to Stakeholders Receiving and Incorporating Feedback 10.3 Analyzing and Interpreting Data-Driven Insights 10.3.1 Principles of Data Interpretation Understanding Data Context and Relevance Differentiating Between Correlation and Causation 10.3.2 Data-Driven Decision-Making Translating Data Insights into Business Strategies Case Study: Using Data to Drive Business Decisions 10.3.3 Communicating Insights to Non-Technical Audiences Simplifying Complex Data for Stakeholder Understanding Techniques for Effective Data Storytelling 10.3.4 Evaluating the Impact of Data Insights Measuring the Effectiveness of Data-Driven Strategies Case Study: Assessing the Impact of Data Insights on Business Performance 10.3.5 Continuous Improvement Through Data Analysis Using Feedback Loops to Enhance Analytical Models Adapting Strategies Based on New Data Insights Case Study: Iterative Improvement in Business Analytics		
UNIT-VI	**Communication and Data Storytelling, Ethics and Best Practices in Business Intelligence**	**07 Hours**
11.1 Techniques for Effective Data Storytelling 11.1.1 Introduction to Data Storytelling Definition and Importance of Data Storytelling Components of a Data Story: Data, Narrative, and Visuals 11.1.2 Crafting a Narrative with Data Identifying the Main Message and Key Points Structuring the Story: Beginning, Middle, and End Techniques for Engaging the Audience 11.1.3 Using Visuals to Enhance the Narrative Choosing the Right Visualization for Your Data Best Practices for Designing Clear and Informative Visuals Integrating Visuals Seamlessly into the Story 11.1.4 Techniques for Simplifying Complex Data Avoiding Information Overload Highlighting Key Insights and Trends Using Analogies and Metaphors to Explain Data 11.1.5 Case Studies in Data Storytelling Examples of Effective Data Stories from Various Industries Analysis of What Makes These Stories Compelling 11.2 Creating Compelling Reports and Presentations		

11.2.1 Principles of Report Writing Structure and Organization of Reports
Clarity and Conciseness in Writing Incorporating Data Visualizations into Reports
11.2.2 Designing Effective Presentations
Principles of Good Presentation Design
Balancing Text, Images, and Data Visualizations
Using Slide Transitions and Animations Effectively
11.2.3 Tools for Creating Reports and Presentations
Overview of Tools: Microsoft PowerPoint, Google Slides, Tableau, Power BI
Tips and Tricks for Using Each Tool Efficiently
Integrating Interactive Elements in Presentations
11.2.4 Tailoring Reports and Presentations to the Audience
Understanding the Needs and Interests of Your Audience
Customizing Content for Different Stakeholder Groups
Using Language and Examples that Resonate with the Audience
11.2.5 Reviewing and Refining Reports and Presentations
Techniques for Self-Review and Peer Review
Iterative Improvement: Incorporating Feedback
Ensuring Consistency and Professionalism
11.3 Presenting Findings to Stakeholders
11.3.1 Preparation for Stakeholder Presentations
Understanding Stakeholder Objectives and Concerns
Planning and Structuring the Presentation
Rehearsing and Refining Your Delivery
11.3.2 Techniques for Effective Oral Communication
Public Speaking Tips and Strategies
Using Body Language and Voice Modulation
Managing Nervousness and Building Confidence
11.3.3 Engaging Stakeholders During the Presentation
Techniques for Capturing and Maintaining Attention
Encouraging Questions and Interactive Discussion
Handling Difficult Questions and Feedback Gracefully
11.3.4 Follow-Up After the Presentation
Summarizing Key Takeaways and Next Steps
Providing Supplementary Materials and Detailed Reports
Gathering Feedback for Continuous Improvement
11.3.5 Case Studies of Successful Stakeholder Presentations
Analysis of Real-World Examples
Identifying Best Practices and Lessons Learned
12.1 Ethical Considerations in Data Analysis
12.2 Ensuring Data Privacy and Security
12.3 Best Practices for Sustainable BI Practices.

Books Recommended:

1. **"Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking"**
   - Authors: Foster Provost and Tom Fawcett
   - Publisher: O'Reilly Media
   - Description: This book provides a solid foundation in data science principles, focusing on data mining techniques and their application in business contexts. It is a practical guide for understanding data-analytic thinking.
2. **"Business Intelligence: A Managerial Approach"**
   - Authors: Efraim Turban, Ramesh Sharda, Dursun Delen, and David King
   - Publisher: Pearson
   - Description: This book covers the comprehensive architecture of business intelligence, including methodologies, tools, and applications. It is an excellent resource for understanding BI from a managerial perspective.
3. **"The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling"**
   - Authors: Ralph Kimball and Margy Ross
   - Publisher: Wiley
   - Description: A classic book on data warehousing and dimensional modeling, this text is essential for students looking to understand the structure and design of data warehouses, which are crucial for BI.
4. **"Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython"**
   - Author: Wes McKinney
   - Publisher: O'Reilly Media
   - Description: This book provides a thorough introduction to data analysis in Python, focusing on libraries like pandas and NumPy. It is ideal for hands-on learning and practical application of data analysis techniques.
5. **"Storytelling with Data: A Data Visualization Guide for Business Professionals"**
   - Author: Cole Nussbaumer Knaflic
   - Publisher: Wiley
   - Description: This book emphasizes the importance of data visualization and effective communication. It provides practical tips and strategies for creating compelling data stories.
6. **"Data Mining: Concepts and Techniques"**
   - Authors: Jiawei Han, Micheline Kamber, and Jian Pei
   - Publisher: Morgan Kaufmann
   - Description: This comprehensive guide to data mining covers the theoretical foundations and practical applications of various data mining techniques. It is suitable for both beginners and advanced learners.
7. **"Practical Business Intelligence"**
   - Authors: Ahmed Sherif, R. Kelly Rainer Jr., and Brad Prince
   - Publisher: McGraw Hill Education
   - Description: This book focuses on the practical aspects of implementing business intelligence solutions, including project management, data integration, and real-world case studies. It is designed for students and professionals looking to apply BI concepts in business settings.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-302-L: DESCRIPTIVE ANALYTICS LAB
B.Tech. III Year I Sem.

Practical Scheme: P: - 4 Hours/Week	Credit P: 01	Examination Scheme: Total : 30 Marks

Course Contents

Exercises	07 Hours

Lab 1: Introduction to R and Python
Objective: Familiarize students with basic operations and functions in R and Python.
Tools: R, Python
1. Install and set up R and Python.
2. Write simple scripts to perform basic arithmetic operations.
3. Load and explore datasets using R and Python.

Lab 2: Data Collection and Importing Data
Objective: Learn methods to collect and import data from various sources.
Tools: R, Python, Excel
1. Import data from CSV, Excel, and web sources.
2. Clean and preprocess the imported data.
3. Handle missing values and remove duplicates.

Lab 3: Data Cleaning and Preparation
Objective: Perform data cleaning and preparation tasks.
Tools: R, Python
1. Identify and handle missing values.
2. Detect and remove outliers.
3. Standardize and normalize data.

Lab 4: Exploratory Data Analysis (EDA)
Objective: Conduct exploratory data analysis to understand data patterns.
Tools: R, Python
1. Compute summary statistics (mean, median, mode).
2. Create histograms, box plots, and scatter plots.
3. Identify patterns and correlations in the data.

Lab 5: Descriptive Statistics
Objective: Apply descriptive statistical methods to summarize data.
Tools: R, Python
1. Calculate measures of central tendency (mean, median, mode).
2. Calculate measures of dispersion (range, variance, standard deviation).
3. Visualize the results using appropriate charts.

Lab 6: Probability Distributions
Objective: Understand and apply different probability distributions.
Tools: R, Python

1. Generate and plot normal, binomial, and Poisson distributions.
2. Fit data to these distributions.
3. Analyze real-world data using these distributions.

Lab 7: Hypothesis Testing
Objective: Conduct hypothesis testing on datasets.
Tools: R, Python

1. Perform t-tests and chi-square tests.
2. Interpret p-values and confidence intervals.
3. Apply these tests to real-world data.

Lab 8: Data Visualization with Excel
Objective: Create basic data visualizations using Excel.
Tools: Excel

1. Create bar charts, line charts, and pie charts.
2. Use pivot tables to summarize data.
3. Customize charts for better readability.

Lab 9: Advanced Data Visualization with Python
Objective: Use Python libraries for advanced data visualization.
Tools: Python (matplotlib, seaborn)

1. Create complex plots like pair plots and heatmaps.
2. Customize visualizations with colors and themes.
3. Use subplots to compare multiple plots.

Lab 10: Data Visualization with Tableau
Objective: Create interactive data visualizations using Tableau.
Tools: Tableau

1. Import data into Tableau.
2. Create dashboards and stories.
3. Add interactivity to visualizations with filters and actions.

Lab 11: Data Visualization with Power BI
Objective: Develop interactive reports and dashboards using Power BI.
Tools: Power BI

1. Import and transform data in Power BI.
2. Create reports using different visualizations.
3. Develop interactive dashboards with slicers and filters.

Lab 12: Clustering Analysis
Objective: Perform clustering analysis to group similar data points.
Tools: R, Python (scikit-learn)

1. Implement K-means clustering.
2. Visualize the clusters.
3. Evaluate clustering results using silhouette score.

Lab 13: Classification Analysis
Objective: Apply classification techniques to labeled data.
Tools: R, Python (scikit-learn)

1. Implement decision trees and logistic regression.
2. Evaluate models using confusion matrix and accuracy.
3. Visualize decision boundaries.

Lab 14: Association Rule Mining
Objective: Discover associations and relationships in large datasets.
Tools: R, Python

1. Apply the Apriori algorithm.
2. Extract frequent itemsets and association rules.
3. Visualize association rules.

Lab 15: Anomaly Detection
Objective: Identify outliers and anomalies in data.
Tools: R, Python

1. Implement isolation forest and DBSCAN algorithms.
2. Visualize anomalies in the dataset.
3. Analyze and interpret the results.

Lab 16: Creating Reports and Dashboards in Tableau
Objective: Design professional reports and dashboards.
Tools: Tableau

1. Create comprehensive reports with multiple visualizations.
2. Develop interactive dashboards.
3. Share reports and dashboards with stakeholders.

Lab 17: Creating Reports and Dashboards in Power BI
Objective: Design and implement interactive reports and dashboards.
Tools: Power BI

1. Develop detailed reports with various visualizations.
2. Implement interactive elements like slicers.
3. Publish and share dashboards with stakeholders.

Lab 18: Data Storytelling with Visualizations
Objective: Learn techniques for effective data storytelling.
Tools: Tableau, Power BI

1. Create a narrative using visualizations.
2. Develop a cohesive story through a series of visualizations.
3. Present the story to a simulated audience.

Lab 19: Real-World Data Analysis Project 1
Objective: Apply learned skills to analyze a real-world dataset.
Tools: R, Python, Tableau, Power BI

1. Collect and clean a real-world dataset.
2. Perform exploratory data analysis.
3. Present findings through visualizations and reports.

Lab 20: Real-World Data Analysis Project 2
Objective: Conduct a comprehensive analysis on a different dataset.
Tools: R, Python, Tableau, Power BI

1. Prepare and preprocess the dataset.
2. Apply statistical and data mining techniques.
3. Develop a dashboard to present insights.

Lab 21: Advanced Data Mining Techniques
Objective: Use advanced data mining methods to uncover insights.
Tools: R, Python

1. Implement clustering, classification, and regression.
2. Analyze complex datasets.
3. Interpret the results and derive insights.

Lab 22: BI Project Management
Objective: Manage and execute a BI project from start to finish.
Tools: Excel, Power BI

1. Plan a BI project.
2. Develop and implement the project.
3. Measure the project's success and value.

Lab 23: Evaluating BI Success and Value
Objective: Measure the effectiveness of BI initiatives.
Tools: Excel, Power BI

1. Define metrics for success.
2. Collect and analyze data related to BI initiatives.
3. Present evaluation results.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

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BTDS-303-PEC(A): GRAPH THEORY (Professional Elective – I)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Understanding graphs, trees, connected paths, applications of trees and graphs.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Know some important classes of graph theoretic problems;
- Prove central theorems about trees, matching, connectivity, coloring and planar graphs;
- Describe and apply some basic algorithms for graphs;
- Use graph theory as a modeling tool.

Course Contents

UNIT-I	Introduction	07 Hours
Discovery of graphs, Definitions, Subgraphs, Isomorphic graphs, Matrix representations of graphs, Degree of a vertex, Directed walks, paths and cycles, Connectivity in digraphs, Eulerian and Hamilton digraphs, Eulerian digraphs, Hamilton digraphs, Special graphs, Complements, Larger graphs from smaller graphs, Union, Sum, Cartesian Product, Composition, Graphic sequences, Graph theoretic model of the LAN problem, Havel-Hakimi criterion, Realization of a graphic sequence.		
UNIT-II	**Connected graphs and shortest paths**	**08 Hours**
Walks, trails, paths, cycles, Connected graphs, Distance, Cut-vertices and cut-edges, Blocks, Connectivity, Weighted graphs and shortest paths, Weighted graphs, Dijkstra‟s shortest path algorithm, Floyd-Warshall shortest path algorithm.		
UNIT-III	**Trees**	**08 Hours**
Definitions and characterizations, Number of trees, Cayley‟s formula, Kircho↵-matrix-tree theorem, Minimum spanning trees, Kruskal‟s algorithm, Prim‟s algorithm, Special classes of graphs, Bipartite Graphs, Line Graphs, Chordal Graphs, Eulerian Graphs, Fleury‟s algorithm, Chinese Postman problem, Hamilton Graphs, Introduction, Necessary conditions and sufficient conditions.		

UNIT-IV	Independent sets coverings and matchings	07 Hours
Introduction, Independent sets and coverings: basic equations, Matchings in bipartite graphs, Hall‟s Theorem, K¨onig‟s Theorem, Perfect matchings in graphs, Greedy and approximation algorithms.		
UNIT-V	**Vertex Colorings**	**08 Hours**
Basic definitions, Cliques and chromatic number, Mycielski‟s theorem, Greedy coloring algorithm, Coloring of chordal graphs, Brooks theorem, Edge Colorings, Introduction and Basics, Gupta-Vizing theorem, Class-1 and Class-2 graphs, Edge-coloring of bipartite graphs, Class-2 graphs, Hajos union and Class-2 graphs, A scheduling problem and equitable edge-coloring.		

TEXT BOOKS:

1. J. A. Bondy and U. S. R. Murty. Graph Theory, volume 244 of Graduate Texts in Mathematics. Springer, 1st edition, 2008.
2. J. A. Bondy and U. S. R. Murty. Graph Theory with Applications.

REFERENCE BOOKS:

1. Lecture Videos: http://nptel.ac.in/courses/111106050/13
2. Introduction to Graph Theory, Douglas B. West, Pearson.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-303-PEC(B): ADVANCED COMPUTER ARCHITECTURE
(Professional Elective – I)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Computer Organization

Course Objective:

- Impart the concepts and principles of parallel and advanced computer architectures.
- Develop the design techniques of Scalable and multithreaded Architectures.
- Apply the concepts and techniques of parallel and advanced computer architectures to design modern computer systems.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Computational models and Computer Architectures.
- Concepts of parallel computer models.
- Scalable Architectures, Pipelining, Superscalar processors

Course Contents

UNIT-I	SIMD/ PRAM/ VLSI models	07 Hours

Theory of Parallelism, Parallel computer models, The State of Computing, Multiprocessors and Multicomputers, Multivector and SIMD Computers, PRAM and VLSI models, Architectural development tracks, Program and network properties, Conditions of parallelism, Program partitioning and Scheduling, Program flow Mechanisms, System interconnect Architectures.

UNIT-II	Principles of Scalable performance	08 Hours

Performance metrics and measures, Parallel Processing applications, Speed up performance laws, Scalability Analysis and Approaches, Hardware Technologies, Processes and Memory Hierarchy, Advanced Processor Technology, Superscalar and Vector Processors.

UNIT-III	Shared-Memory Organizations	08 Hours

Sequential and weak consistency models, Pipelining and superscalar techniques, Linear Pipeline Processors, Non-Linear Pipeline Processors, Instruction Pipeline design, Arithmetic pipeline design, superscalar pipeline design.

UNIT-IV	Parallel and Scalable Architectures	07 Hours
Multiprocessors and Multicomputers, Multiprocessor system interconnects, cache coherence and synchronization mechanism, Three Generations of Multicomputers, Message-passing Mechanisms, Multivetor and SIMD computers.		
UNIT-V	**Vector Processing Principles**	**08 Hours**
Multivector Multiprocessors, Compound Vector processing, SIMD computer Organizations, The connection machine CM-5.		

TEXT BOOK:

1. Advanced Computer Architecture, Kai Hwang, 2nd Edition, Tata McGraw Hill Publishers.

REFERENCE BOOKS:

1. Computer Architecture, J.L. Hennessy and D.A. Patterson, 4th Edition, ELSEVIER.
2. Advanced Computer Architectures, S.G.Shiva, Special Indian edition, CRC, Taylor &Francis.
3. Introduction to High Performance Computing for Scientists and Engineers, G. Hager and G. Wellein, CRC Press, Taylor & Francis Group.
4. Advanced Computer Architecture, D. Sima, T. Fountain, P. Kacsuk, Pearson education.
5. Computer Architecture, B. Parhami, Oxford Univ. Press.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-303-PEC(c): ETL- KAFKA/TALEND
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Develop a comprehensive understanding of Extract, Transform, Load (ETL) processes using Apache Kafka and Talend.
- Understand how to scale Kafka clusters seamlessly to handle growing data volumes, ensuring optimal performance for ETL operations.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Learn to design and deploy fault-tolerant Kafka clusters, ensuring data integrity and availability in real-world scenarios.
2. Gain practical experience in cluster management, topic creation, and basic operations such as producing and consuming messages.

Course Contents

LIST OF EXPERIMENTS	07 Hours

1. Install Apache Kafka on a single node.
2. Demonstrate setting up a single-node, single-broker Kafka cluster and show basic operations such as creating topics and producing/consuming messages.
3. Extend the cluster to multiple brokers on a single node.
4. Write a simple Java program to create a Kafka producer and Produce messages to a topic.
5. Implement sending messages both synchronously and asynchronously in the producer.
6. Develop a Java program to create a Kafka consumer and subscribe to a topic and consume messages.
7. Write a script to create a topic with specific partition and replication factor settings.
8. Simulate fault tolerance by shutting down one broker and observing the cluster behavior.
9. Implement operations such as listing topics, modifying configurations, and deleting topics.
10. Introduce Kafka Connect and demonstrate how to use connectors to integrate with external systems.
11. Implement a simple word count stream processing application using Kafka Stream
12. Implement Kafka integration with the Hadoop ecosystem.

TEXT BOOK:

1. Neha Narkhede, Gwen Shapira, Todd Palino, Kafka – The Definitive Guide: Real-time data and stream processing at scale, O′Reilly

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-303-PEC(D): IMAGE PROCESSING (Professional Elective – I)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Students are expected to have knowledge in linear signals and systems, Fourier Transform, basic linear algebra, basic probability theory and basic programming techniques; knowledge of digital signal processing is desirable.

- A course on "Computational Mathematics"
- A course on "Computer Oriented Statistical Methods"

Course Objective:

- Provide a theoretical and mathematical foundation of fundamental Digital Image Processing concepts.
- The topics include image acquisition; sampling and quantization; preprocessing; enhancement; restoration; segmentation; and compression.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Demonstrate the knowledge of the basic concepts of two-dimensional signal acquisition, sampling, and quantization.
- Demonstrate the knowledge of filtering techniques.
- Demonstrate the knowledge of 2D transformation techniques.
- Demonstrate the knowledge of image enhancement, segmentation, restoration and compression techniques.

Course Contents

UNIT-I	Digital Image Fundamentals	07 Hours
Digital Image through Scanner, Digital Camera. Concept of Gray Levels.Gray Level to Binary Image Conversion. Sampling and Quantization. Relationship between Pixels.Imaging Geometry. 2D Transformations-DFT, DCT, KLT and SVD.		
UNIT-II	**Image Processing**	**08 Hours**
Image Enhancement in Spatial Domain Point Processing, Histogram Processing, Spatial Filtering, Enhancement in Frequency Domain, Image Smoothing, Image Sharpening.		
UNIT-III	**Image Restoration Degradation Model**	**08 Hours**

Algebraic Approach to Restoration, Inverse Filtering, Least Mean Square Filters, Constrained Least Squares Restoration, Interactive Restoration.		
UNIT-IV	**Image Segmentation Detection of Discontinuities**	**07 Hours**
Edge Linking and Boundary Detection, Thresholding, Region Oriented Segmentation.		
UNIT-V	**Image Compression**	**08 Hours**
Image Compression Redundancies and their Removal Methods, Fidelity Criteria, Image Compression Models, Source Encoder and Decoder, Error Free Compression, Lossy Compression.		

TEXT BOOK:

1. Digital Image Processing: R.C. Gonzalez & R. E. Woods, Addison Wesley/ Pearson Education,2nd Ed, 2004.

REFERENCE BOOKS:

1. Fundamentals of Digital Image Processing: A. K. Jain, PHI.
2. Digital Image Processing using MAT LAB: Rafael C. Gonzalez, Richard E. Woods, Steven L.Eddins: Pearson Education India, 2004.
3. Digital Image Processing: William K. Pratt, John Wilely, 3rd Edition, 2004.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-303-PEC(E): COMPUTER GRAPHICS (Professional Elective – I)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Programming for problem solving and Data Structures

Course Objective:

- Provide the basics of graphics systems including Points and lines, line drawing algorithms, 2D, 3D objective transformations

Learning Course Outcome:
After successful completion of the course, students will able to:

- Explore applications of computer graphics
- Understand 2D, 3D geometric transformations and clipping algorithms
- Understand 3D object representations, curves, surfaces, polygon rendering methods, color models
- Analyze animation sequence and visible surface detection methods

Course Contents

UNIT-I	Introduction	07 Hours

Application areas of Computer Graphics, overview of graphics systems, video-display devices, raster-scan systems, random-scan systems, graphics monitors and work stations and input devices

Output primitives: Points and lines, line drawing algorithms (DDA and Bresenham's Algorithm) circle- generating algorithms and ellipse - generating algorithms

Polygon Filling: Scan-line algorithm, boundary-fill and flood-fill algorithms

UNIT-II	transformations	08 Hours

2-D geometric transformations: Translation, scaling, rotation, reflection and shear transformations, matrix representations and homogeneous coordinates, composite transforms, transformations between coordinate systems

2-D viewing: The viewing pipeline, viewing coordinate reference frame, window to view-port coordinate transformation, viewing functions, clipping operations, point clipping, Line clipping-Cohen Sutherland algorithms, Polygon clipping-Sutherland Hodgeman polygon clipping algorithm.

UNIT-III	object representation	08 Hours

3-D object representation: Polygon surfaces, quadric surfaces, spline representation, Hermite curve, Bezier curve and B-Spline curves, Bezier and B-Spline surfaces, Polygon rendering methods, color models and color applications.

UNIT-IV	Geometric transformations	07 Hours

3-D Geometric transformations: Translation, rotation, scaling, reflection and shear transformations, composite transformations.

3-D viewing: Viewing pipeline, viewing coordinates, projections, view volume and general projection transforms and clipping.

UNIT-V	Computer animation	08 Hours

Design of animation sequence, general computer animation functions, raster animations, computer animation languages, key frame systems, motion specifications.

Visible surface detection methods: Classification, back-face detection, depth-buffer method, BSP- tree method, area sub-division method and octree method.

TEXT BOOKS:

1. "Computer Graphics C version", Donald Hearn and M. Pauline Baker, Pearson Education

REFERENCE BOOKS:

1. Procedural elements for Computer Graphics, David F Rogers, Tata Mc Graw hill, 2nd edition.
2. Principles of Interactive Computer Graphics", Neuman and Sproul, TMH.
3. Principles of Computer Graphics, Shalini Govil, Pai, 2005, Springer.
4. "Computer Graphics Principles & practice", second edition in C, Foley, Van Dam, Feiner and Hughes, Pearson Education.
5. Computer Graphics, Steven Harrington, TMH.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-303-PEC(F): WEB FRAMEWORK (Professional Elective – I)
B.Tech. III Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of Python programming.

Course Objective:

1. To understand the basics of Django and its applications in web development.
2. To learn to build dynamic web applications using the Django framework.
3. To integrate databases with Django applications.
4. To implement user authentication and authorization in Django.
5. To develop skills for testing and deploying Django applications.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the architecture and components of the Django framework.
CO2: Develop dynamic web applications using Django.
CO3: Integrate and manage databases with Django ORM.
CO4: Implement user authentication and authorization.
CO5: Test and deploy Django applications.

Course Contents

UNIT-I	Overview of Django	07 Hours
History and evolution of Django, Features and benefits of using Django, Django architecture: MTV (Model-Template-View) pattern, setting up Django: Installation and configuration, Creating a Django project, Overview of Django project structure.		

UNIT-II	Django Models and ORM	08 Hours
Introduction to Django Models: Creating and defining models, Model fields and field types, Database setup and configuration, Django ORM: QuerySets and querying the database, CRUD (Create, Read, Update, Delete) operations, Relationships between models (One-to-One, One-to-Many, Many-to-Many)		

UNIT-III	Django Views and Templates	08 Hours
Django Views: Function-based views (FBV), Class-based views (CBV), Handling forms and form validation, Django Templates: Template language and syntax, Template inheritance and reusability, Context and context processors.		

UNIT-IV	Django URLs and Routing	07 Hours
URL Configuration: URL patterns and routing, Including and referencing URLs from other apps, Named URL patterns and reverse resolution, Dynamic URLs: Capturing URL parameters, URL dispatching and redirection.		
UNIT-V	**User Authentication and Authorization**	**08 Hours**
Authentication System: User model and user management, Authentication views and forms, Password hashing and security, Authorization: Permissions and groups, Role-based access control, Custom user models and extending user profiles.		
UNIT-VI	**Advanced Django Topics**	**08 Hours**
Django Forms: Form classes and form handling, Model forms and form sets, Custom form fields and validation Django Admin: Customizing the admin interface, Adding models to the admin site, Custom admin actions and filters Testing and Deployment: Writing and running tests, Django testing framework, Deployment strategies (using WSGI, deploying on cloud platforms), Configuring static and media files		

REFERENCE BOOKS:

1. "Django for Beginners: Build websites with Python and Django" by William S. Vincent
2. "Two Scoops of Django 3.x: Best Practices for the Django Web Framework" by Audrey Roy Greenfeld and Daniel Roy Greenfeld
3. "Django 3 By Example: Build powerful and reliable Python web applications from scratch" by Antonio Mele
4. Official Django documentation: https://docs.djangoproject.com

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO1	3	2	-	-	3	-	-	2	-	2	-
CO2	3	3	2	-	3	-	-	3	2	3	-
CO3	3	3	2	-	3	-	-	3	3	3	2
CO4	3	3	-	2	3	-	-	2	3	3	3
CO5	3	3	2	3	3	-	-	3	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-303-PEC(G): R PROGRAMMING (Professional Elective – I)
B.Tech. III Year I Sem.

Teaching Scheme: TH: - 4 Hours/Week	Credit TH: 03	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Introduction To Data Science

Course Objective:

1. To introduce students to the R programming environment.
2. To teach data manipulation, visualization, and analysis using R.
3. To cover statistical modeling and machine learning techniques in R.
4. To enable students to perform reproducible research using R.
5. To familiarize students with advanced topics such as parallel computing and package development in R.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the basic concepts and environment of R programming.
CO2: Perform data manipulation and cleaning using R packages.
CO3: Create and customize various types of data visualizations.
CO4: Conduct statistical analysis and interpret the results.
CO5: Apply machine-learning algorithms to real-world datasets.
CO6: Utilize advanced R programming techniques including parallel computing and package development.
CO7: Conduct reproducible research and document analysis using RMarkdown.

Course Contents

UNIT-I	Introduction to R	07 Hours
Overview of R and RStudio, Installing R and RStudio, Basic R Syntax, Variables and Data Types, Vectors, Lists, Matrices, Arrays, and Data Frames, Basic Operations in R		
UNIT-II	**Data Manipulation in R**	**08 Hours**
Importing and Exporting Data, Data Cleaning and Preparation, dplyr and tidyr for Data Manipulation, String Manipulation with stringr, Handling Missing Values		
UNIT-III	**Data Visualization in R**	**08 Hours**
Introduction to Data Visualization, Basic Plotting with Base R, Advanced Plotting with ggplot2, Customizing Plots, Creating Interactive Visualizations with plotly and shiny		

UNIT-IV	Statistical Analysis in R	07 Hours
Descriptive Statistics, Probability Distributions, Hypothesis Testing, ANOVA and Regression Analysis, Time Series Analysis.		
UNIT-V	**Machine Learning in R**	**08 Hours**
Introduction to Machine Learning, Supervised Learning: Regression and Classification, Unsupervised Learning: Clustering and Association Rules, Model Evaluation and Tuning, Using caret and mlr Packages for Machine Learning.		
UNIT-V	**Advanced Topics in R**	**08 Hours**
Introduction to Parallel Computing in R, Writing Functions and Scripts, Package Development in R, Version Control with Git and GitHub, Reproducible Research with RMarkdown		

REFERENCE BOOKS:

1. "R for Data Science" by Hadley Wickham and Garrett Grolemund ISBN-13: 978-1491910399
2. "Advanced R" by Hadley Wickham ISBN-13: 978-1498759809
3. "The Art of R Programming" by Norman Matloff ISBN-13: 978-1593273842
4. "R in Action" by Robert I. Kabacoff ISBN-13: 978-1617291388

CO's-PO's & PSO's MAPPING

(CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	-	-	-	2	-	-	2	-	-
CO2	3	3	2	2	3	-	-	3	3	3
CO3	3	3	2	-	3	-	-	3	2	3
CO4	3	3	2	2	3	-	-	3	3	3
CO5	3	3	2	2	3	-	-	3	3	3
CO6	3	3	2	2	3	-	-	3	3	3
CO7	3	3	2	2	3	-	-	3	3	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS303-L: R PROGRAMMING LAB
B.Tech. III Year I Sem.

Practical Scheme: P: - 2 Hours/Week	Credit:01	Examination Scheme: Total : 30 Marks

Course Prerequisites: Any programming language.

Course Objective:

- Familiarize with R basic programming concepts, various data structures for handling datasets, various graph representations and Exploratory Data Analysis concepts.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Setup R programming environment.
- Understand and use R – Data types and R – Data Structures.
- Develop programming logic using R – Packages.
- Analyze data sets using R – programming capabilities

Course Contents

LIST OF EXPERIMENTS	07 Hours

1. Download and install R-Programming environment and install basic packages using install.packages() command in R.

2. Learn all the basics of R-Programming (Data types, Variables, Operators etc,.)

3. Write R command to
i) Illustrate summation, subtraction, multiplication, and division operations on vectors using vectors.
ii) Enumerate multiplication and division operations between matrices and vectors in R console

4. Write R command to
i) Illustrates the usage of Vector subsetting and Matrix subsetting
ii) Write a program to create an array of 3×3 matrices with 3 rows and 3 columns.

5. Write an R program to draw i) Pie chart ii) 3D Pie Chart, iii) Bar Chart along with chart legend by considering suitable CSV file

6. Create a CSV file having Speed and Distance attributes with 1000 records. Write R program to draw
i) Box plots
ii) Histogram
iii) Line Graph

iv) Multiple line graphs
v) Scatter plot
to demonstrate the relation between the cars speed and the distance.

7. Implement different data structures in R (Vectors, Lists, Data Frames)

8. Write an R program to read a csv file and analyze the data in the file using EDA (Explorative Data Analysis) techniques.

9. Write an R program to illustrate Linear Regression and Multi linear Regression considering suitable CSV file.

TEXT BOOKS:

1. R Programming for Data Science by Roger D. Peng
2. The Art of R Programming by Norman Matloff Cengage Learning India.

REFERENCE BOOKS:

1. Hadley Wickham, Garrett Grolemund, R for Data Science: Import, Tidy, Transform, Visualize, and Model Data 1st Edition, O'Reilly
2. Tilman M. Davies, The book of R a first course in programming and statistics, no starch press

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-304-PCC: COMPUTER ORGANIZATION AND ARCHITECTURE
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: A Course on "Digital Electronics".

Course Objective:

- The purpose of the course is to introduce principles of computer organization and the basic architectural concepts.
- It begins with basic organization, design, and programming of a simple digital computer and introduces simple register transfer language to specify various computer operations.
- Topics include computer arithmetic, instruction set design, microprogrammed control unit, pipelining and vector processing, memory organization and I/O systems, and multiprocessors

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the basics of instruction sets and their impact on processor design.
- Demonstrate an understanding of the design of the functional units of a digital computer system.
- Evaluate cost performance and design trade-offs in designing and constructing a computer processor including memory.
- Design a pipeline for consistent execution of instructions with minimum hazards.
- Recognize and manipulate representations of numbers stored in digital computers

Course Contents

UNIT-I	Digital Computers	07 Hours

Introduction, Block diagram of Digital Computer, Definition of Computer Organization, Computer Design and Computer Architecture.

Register Transfer Language and Micro operations: Register Transfer language, Register Transfer, Bus and memory transfers, Arithmetic Micro operations, logic micro operations, shift micro operations, Arithmetic logic shift unit.

Basic Computer Organization and Design: Instruction codes, Computer Registers Computer instructions, Timing and Control, Instruction cycle, Memory Reference Instructions, Input – Output and Interrupt.

UNIT-II	Microprogrammed Control	08 Hours
Control memory, Address sequencing, micro program example, design of control unit. Central Processing Unit: General Register Organization, Instruction Formats, Addressing modes, Data Transfer and Manipulation, Program Control.		
UNIT-III	**Data Representation**	**08 Hours**
Data types, Complements, Fixed Point Representation, Floating Point Representation. Computer Arithmetic: Addition and subtraction, multiplication Algorithms, Division Algorithms, Floating – point Arithmetic operations. Decimal Arithmetic unit, Decimal Arithmetic operations.		
UNIT-IV	**Input-Output Organization**	**07 Hours**
Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupt Direct memory Access. Memory Organization: Memory Hierarchy, Main Memory, Auxiliary memory, Associate Memory, Cache Memory.		
UNIT-V	**Reduced Instruction Set Computer**	**08 Hours**
CISC Characteristics, RISC Characteristics. Pipeline and Vector Processing: Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processor. Multi Processors: Characteristics of Multiprocessors, Interconnection Structures, Interprocessor arbitration, Interprocessor communication and synchronization, Cache Coherence.		

TEXT BOOK:

1. Computer System Architecture – M. Morris Mano, Third Edition, Pearson/PHI.

REFERENCE BOOKS:

1. Computer Organization – Carl Hamacher, Zvonks Vranesic, SafeaZaky, Vth Edition, McGraw Hill.
2. Computer Organization and Architecture – William Stallings Sixth Edition, Pearson/PHI.
3. Structured Computer Organization – Andrew S. Tanenbaum, 4 th Edition, PHI/Pearson.

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-305-MDM(A): BLOCKCHAIN PLATFORMS AND TOOLS
(MDM Basket – 1)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Data structures and compiler design

Course Objective:

- To provide an in-depth understanding of various blockchain platforms and their architectures.
- To explore tools and frameworks for developing blockchain-based applications.
- To examine the scalability, interoperability, and consensus mechanisms of blockchain platforms.
- To enable the practical implementation of smart contracts using popular blockchain tools.
- To analyze and evaluate the performance and security of blockchain applications.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the architecture and features of different blockchain platforms.
CO2: Utilize blockchain development tools and frameworks for building applications.
CO3: Implement and deploy smart contracts on blockchain networks.
CO4: Analyze the scalability and interoperability challenges in blockchain systems.
CO5: Evaluate the security and performance of blockchain applications.

Course Contents

UNIT-I	Introduction to Blockchain Platforms	03 Hours

- Overview of Blockchain Platforms: Ethereum, Hyperledger Fabric, Corda, and Polkadot
- Architecture and Components of Blockchain Platforms
- Types of Blockchain Networks: Public, Private, and Consortium
- Use Cases of Different Platforms in Industry

UNIT-II	Blockchain Development Tools	04 Hours

- Development Frameworks: Truffle, Hardhat, and Remix
- Blockchain Nodes and Clients: Geth, Parity, and Besu
- Blockchain APIs and SDKs
- Setting Up a Local Blockchain Environment

UNIT-III	Smart Contracts Development	03 Hours

- Writing Smart Contracts Using Solidity
- Testing and Debugging Smart Contracts
- Gas Optimization and Cost Estimation
- Deploying Smart Contracts on Testnet and Mainnet

UNIT-IV	Scalability and Interoperability in Blockchain	04 Hours

- Layer-2 Solutions: Sidechains, State Channels, and Rollups
- Blockchain Interoperability Protocols: Polkadot, Cosmos, and Chainlink
- Challenges and Trade-offs in Scalability and Interoperability
- Case Studies on Cross-Platform Applications

UNIT-V	Security and Performance in Blockchain	04 Hours

- Common Vulnerabilities in Smart Contracts: Reentrancy, Overflow, and Underflow
- Tools for Auditing Smart Contracts: MythX, Slither, and CertiK
- Performance Metrics: Transaction Throughput, Latency, and Finality
- Strategies for Enhancing Security and Performance

UNIT-VI	Future Trends in Blockchain Platforms	04 Hours

- Emerging Platforms: Avalanche, Solana, and Algorand
- Blockchain for Decentralized Finance (DeFi) and Non-Fungible Tokens (NFTs)
- Quantum Computing and Its Impact on Blockchain
- Ethical and Regulatory Challenges in Blockchain Adoption

Reference Books:

1. **Mastering Blockchain** by Imran Bashir
2. **Blockchain Basics** by Daniel Drescher
3. **Hands-On Blockchain Development with Hyperledger** by Salman A. Baset and Luc Desrosiers
4. **Building Ethereum DApps** by Roberto Infante
5. Online Resources: Ethereum, Hyperledger Fabric, and Polkadot Documentation

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	2	3	2	-	-	-	-	-	2	2	1	-
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO5	3	3	3	2	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	1.8	2.4	2	0.4	-	-	-	-	2	1.8	1.2	0.4

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-305-L: BLOCKCHAIN PLATFORMS AND TOOLS LAB
B.Tech. III Year I Sem.

Practical Scheme: P: - 2 Hours/Week	Credit:01	Examination Scheme: Total : 30 Marks

Course Prerequisites: Any programming language.

Course Objective:

- Familiarize with blockchain basic programming concepts, various data structures for handling datasets, various graph representations and Exploratory Data Analysis concepts.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Setup blockchain programming environment.
- Understand and use R – Data types and blockchain – Data Structures.
- Develop programming logic using blockchain – Packages.
- Analyze data sets using blockchain – programming capabilities

Course Contents

LIST OF EXPERIMENTS	07 Hours

Assignment 1: Exploring Blockchain Platforms

- Task: Set up a private Ethereum blockchain using Geth.
- Objective: Understand the architecture and functionality of a blockchain platform.
- Deliverable: Working private blockchain network and a report on its setup and use cases.

Assignment 2: Blockchain Development Tools

- Task: Use Truffle and Remix to create, compile, and deploy a simple smart contract.
- Objective: Gain hands-on experience with blockchain development frameworks.
- Deliverable: Deployed smart contract and a brief presentation on tools used.

Assignment 3: Smart Contracts Implementation

- Task: Develop a voting system using Solidity and deploy it on a local blockchain.
- Objective: Learn smart contract development and gas optimization techniques.
- Deliverable: Functional voting system and code documentation.

Assignment 4: Scalability Solutions

- Task: Implement a Layer-2 solution using a state channel or sidechain.
- Objective: Explore scalability challenges and solutions.
- Deliverable: Demonstration of the Layer-2 solution and a report on its performance.

Assignment 5: Security Analysis of Smart Contracts

- Task: Audit a smart contract for vulnerabilities using tools like MythX or Slither.
- Objective: Understand common security issues in blockchain applications.
- Deliverable: Security report highlighting vulnerabilities and mitigations.

Assignment 6: Blockchain Interoperability

- Task: Demonstrate interoperability between two blockchain platforms (e.g., Ethereum and Polkadot).
- Objective: Learn protocols and techniques for blockchain interoperability.
- Deliverable: Working demonstration and a technical report on interoperability challenges and solutions.

Assignment 7: Case Study on Emerging Blockchain Platforms

- Task: Research and present on an emerging blockchain platform (e.g., Solana or Avalanche).
- Objective: Analyze advancements and features of new platforms.
- Deliverable: Presentation slides and a detailed case study report.

CO's-PO's & PSO's MAPPING

Cos	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-305-MDM(B): ETHICAL HACKING AND PENETRATION TESTING
(MDM Basket – 2)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Data structures and compiler design

Course Objective:

- To introduce students to the fundamentals of ethical hacking and penetration testing methodologies.
- To develop skills for identifying vulnerabilities in computer systems and networks.
- To understand various tools and techniques used for penetration testing.
- To ensure ethical and legal practices in cybersecurity.
- To enable students to create secure environments by understanding and mitigating security risks.

Learning Course Outcome:
After successful completion of the course, students will able to:

- **CO1**: Understand the principles and ethics of hacking and penetration testing.
- **CO2**: Identify vulnerabilities and threats in computer systems and networks.
- **CO3**: Utilize tools and techniques for ethical hacking and penetration testing.
- **CO4**: Perform comprehensive security assessments and provide recommendations.
- **CO5**: Adhere to ethical and legal standards in cybersecurity practices.

Course Contents

UNIT-I	Introduction to Ethical Hacking	03 Hours

- Basics of Ethical Hacking: Definition, Types, and Importance
- Difference Between Ethical and Malicious Hacking
- Hacking Methodology: Reconnaissance, Scanning, Exploitation, and Post-Exploitation
- Legal and Ethical Considerations
- Case Studies on Real-World Hacking Incidents

UNIT-II	Information Gathering and Reconnaissance	04 Hours

- Footprinting Techniques: Passive and Active
- Social Engineering: Techniques and Prevention

• Open Source Intelligence (OSINT) Tools • Network Scanning and Enumeration • Tools: Nmap, Maltego, and Shodan		
UNIT-III	**Vulnerability Analysis**	**03 Hours**
• Types of Vulnerabilities: Software, Hardware, and Network • Vulnerability Assessment Tools: Nessus, OpenVAS, and Nikto • Exploiting Vulnerabilities: Exploit Frameworks and Payloads • Common Vulnerabilities and Exposure (CVE) Database		
UNIT-IV	**Exploitation and Post-Exploitation**	**04 Hours**
• Exploitation Techniques: Buffer Overflows, SQL Injection, and Cross-Site Scripting (XSS) • Privilege Escalation and Lateral Movement • Post-Exploitation: Data Exfiltration and Covering Tracks • Tools: Metasploit, Burp Suite, and OWASP ZAP		
UNIT-V	**Penetration Testing Frameworks and Methodologies**	**04 Hours**
• Penetration Testing Phases: Planning, Discovery, Exploitation, and Reporting • Types of Penetration Testing: Black Box, White Box, and Gray Box • Writing Penetration Testing Reports • Tools: Kali Linux, Parrot Security OS		
UNIT-VI	**Securing Systems and Networks**	**04 Hours**
• Mitigating Vulnerabilities: Patches, Updates, and Configuration Changes • Secure Coding Practices • Firewall and Intrusion Detection/Prevention Systems (IDS/IPS) • Security Best Practices and Frameworks • Case Studies on Incident Response and Mitigation		

Reference Books:

1. **The Basics of Hacking and Penetration Testing** by Patrick Engebretson
2. **Ethical Hacking and Penetration Testing Guide** by Rafay Baloch
3. **CEH v11: Certified Ethical Hacker Study Guide** by Ric Messier
4. **Web Application Hacker's Handbook** by Dafydd Stuttard and Marcus Pinto
5. Online Resources: OWASP, MITRE ATT&CK, and CVE Database

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	3	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	2	3	2	-	-	-	-	-	2	2	1	-
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO5	3	3	3	2	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	1.8	2.4	2.2	0.4	-	-	-	-	2	1.8	1.2	0.4

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-305-MDM(c): DIGITAL IMAGE ENHANCEMENT
(MDM Basket – 3)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Data structures and compiler design

Course Objective:

- To introduce the fundamental concepts and techniques of digital image enhancement.
- To enable students to process and enhance images for improved visual quality.
- To familiarize students with various spatial and frequency domain techniques for image enhancement.
- To explore applications of image enhancement in real-world scenarios.
- To develop problem-solving skills using modern image processing tools and frameworks.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the principles and techniques of digital image enhancement.
CO2: Apply spatial domain techniques for image processing.
CO3: Implement frequency domain methods for image enhancement.
CO4: Evaluate and compare different image enhancement algorithms.
CO5: Use image enhancement techniques in various application domains.

Course Contents

UNIT-I	Introduction to Image Enhancement	03 Hours

- Basics of Image Processing and Enhancement
- Need for Image Enhancement
- Image Quality Metrics
- Applications of Image Enhancement in Medicine, Remote Sensing, and Surveillance

UNIT-II	Spatial Domain Techniques	04 Hours

- Point Processing: Contrast Stretching, Histogram Equalization
- Image Smoothing: Mean, Median, and Gaussian Filters
- Image Sharpening: Laplacian and High-Boost Filtering
- Adaptive Enhancement Techniques

UNIT-III	Frequency Domain Techniques	03 Hours

- Fourier Transform for Image Processing
- Frequency Domain Filters: Low-Pass, High-Pass, and Band-Pass
- Image Smoothing and Sharpening in the Frequency Domain
- Discrete Cosine Transform (DCT) and Its Applications

UNIT-IV	Edge Detection and Enhancement	04 Hours

- Basics of Edge Detection
- Gradient-Based Methods: Sobel, Prewitt, and Canny
- Edge Enhancement Techniques
- Applications of Edge Detection in Object Recognition

UNIT-V	Color Image Enhancement	04 Hours

- Fundamentals of Color Models: RGB, CMY, HSV
- Histogram Equalization for Color Images
- Color Balancing and Enhancement Techniques
- Pseudo-Color and False-Color Processing

UNIT-VI	Modern Techniques and Applications	04 Hours

- Wavelet-Based Image Enhancement
- Machine Learning for Image Enhancement
- Image Enhancement for Low-Light and Underwater Images
- Case Studies: Enhancing Images in Real-World Scenarios

Reference Books:

1. **Digital Image Processing** by Rafael C. Gonzalez and Richard E. Woods
2. **Fundamentals of Digital Image Processing** by Anil K. Jain
3. **Image Processing: Principles and Applications** by Tinku Acharya and Ajoy K. Ray
4. **Computer Vision: Algorithms and Applications** by Richard Szeliski
5. Online Resources: IEEE Xplore, SpringerLink, and GitHub Repositories for Image Processing

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	2	3	2	-	-	-	-	-	2	2	1	-
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO5	3	3	3	2	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	1.8	2.4	2	0.4	-	-	-	-	2	1.8	1.2	0.4

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-306-OE(A): INTRODUCTION TO NATURAL LANGUAGE PROCESSING
(Open Elective – III)
B.Tech. III Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks End Sem. Exam: 25 Marks Total : 50 Marks

Course Prerequisites: Data structures and compiler design

Course Objective:

- Introduction to some of the problems and solutions of NLP and their relation to linguistics and statistics.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Show sensitivity to linguistic phenomena and an ability to model them with formal grammars.
- Understand and carry out proper experimental methodology for training and evaluating empirical NLP systems
- Able to manipulate probabilities, construct statistical models over strings and trees, and estimate parameters using supervised and unsupervised training methods.
- Able to design, implement, and analyze NLP algorithms; and design different language modeling Techniques.

Course Contents

UNIT-I	Finding the Structure of Words	07 Hours
Words and Their Components, Issues and Challenges, Morphological Models Finding the Structure of Documents: Introduction, Methods, Complexity of the Approaches, Performances of the Approaches, Features		
UNIT-II	**Syntax I**	**08 Hours**
Parsing Natural Language, Treebanks: A Data-Driven Approach to Syntax, Representation of Syntactic Structure, Parsing Algorithms.		
UNIT-III	**Syntax II**	**08 Hours**
Models for Ambiguity Resolution in Parsing, Multilingual Issues Semantic Parsing I: Introduction, Semantic Interpretation, System Paradigms, Word Sense.		
UNIT-IV	**Semantic Parsing II**	**07 Hours**

Predicate-Argument Structure, Meaning Representation Systems.		
UNIT-V	**Language Modeling**	**08 Hours**
Introduction, N-Gram Models, Language Model Evaluation, Bayesian parameter estimation, Language Model Adaptation, Language Models- class based, variable length, Bayesian topic based, Multilingual and Cross Lingual Language Modeling		

TEXT BOOKS:

1. Multilingual natural Language Processing Applications: From Theory to Practice – Daniel M. Bikel and Imed Zitouni, Pearson Publication

REFERENCE BOOK:

1. Speech and Natural Language Processing - Daniel Jurafsky& James H Martin, Pearson Publications.
2. Natural Language Processing and Information Retrieval: Tanvier Siddiqui, U.S. Tiwary.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-306-OE(B): DRONE TECHNOLOGIES (Open Elective – III)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To understand the basics of drone concepts
2. To learn and understand the fundaments of design, fabrication and programming of drone
3. To impart the knowledge of an flying and operation of drone
4. To know about the various applications of drone
5. To understand the safety risks and guidelines of fly safely

Course Outcome:

After successful completion of the course, students will able to:

CO1: Know about a various type of drone technology, drone fabrication and programming.
CO2: Execute the suitable operating procedures for functioning a drone
CO3: Select appropriate sensors and actuators for Drones
CO4: Develop a drone mechanism for specific applicationsCO5: Create the programs for various drones

Course Contents

UNIT-I	INTRODUCTION TO DRONE TECHNOLOGY	07 Hours
Drone Concept - Vocabulary Terminology- History of drone - Types of current generation of drones based on their method of propulsion- Drone technology impact on the businesses- Drone business through entrepreneurship- Opportunities/applications for entrepreneurship and employability.		
UNIT-II	**DRONE DESIGN, FABRICATION AND PROGRAMMING**	**08 Hours**
Classifications of the UAV -Overview of the main drone parts- Technical characteristics of the parts -Function of the component parts -Assembling a drone- The energy sources- Level of autonomy- Drones configurations -The methods of programming drone- Download program - Install program on computer- Running Programs- Multi rotor stabilization- Flight modes -Wi-Fi connection.		
UNIT-III	**DRONE FLYING AND OPERATION**	**08 Hours**
Concept of operation for drone -Flight modes- Operate a small drone in a controlled environment- Drone controls Flight operations –management tool –Sensors-Onboard storage capacity - Removable storage devices- Linked mobile devices and applications.		

UNIT-IV	DRONE COMMERCIAL APPLICATIONS	07 Hours
Choosing a drone based on the application -Drones in the insurance sector- Drones in delivering mail, parcels and other cargo- Drones in agriculture- Drones in inspection of transmission lines and power distribution -Drones in filming and panoramic picturing.		
UNIT-V	**FUTURE DRONES AND SAFETY**	**07 Hours**
The safety risks- Guidelines to fly safely -Specific aviation regulation and standardization- Drone license- Miniaturization of drones- Increasing autonomy of drones -The use of drones in swarms.		

TEXT BOOKS

1. Daniel Tal and John Altschuld, "Drone Technology in Architecture, Engineering and Construction: A Strategic Guide to Unmanned Aerial Vehicle Operation and Implementation", 2021 John Wiley & Sons, Inc.
2. Terry Kilby and Belinda Kilby, "Make:Getting Started with Drones ",Maker Media, Inc, 2016

REFERENCES

1. John Baichtal, "Building Your Own Drones: A Beginners' Guide to Drones, UAVs, and ROVs",Que Publishing, 2016
2. Zavrsnik, "Drones and Unmanned Aerial Systems: Legal and Social Implications for Securityand Surveillance", Springer, 2018.

CO's-PO's & PSO's MAPPING

CO's-PO's & PSO's MAPPING															
COs/Pos&PSOs	**POs**												**PSOs**		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**
CO1	1	2	3	1	3	2						1	2	1	3
CO2	1	2	3	1	3	2						1	2	1	3
CO3	1	2	3	1	3	2						1	2	1	3
CO4	1	2	3	1	3	2						1	2	1	3
CO5	1	2	3	1	3	2						1	2	1	3
CO/PO &PSO Average	1	2	3	1	3	2						1	2	1	3
1 - low, 2 - medium, 3 - high, '-"- no correlation															

BTDS-306-OE(c): WEARABLE DEVICES (Open Elective – III)
B.Tech. III Year I Sem.

Teaching Scheme: TH: - 4 Hours/Week	Credit P: 02	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To know the hardware requirement of wearable systems
2. To understand the communication and security aspects in the wearable devices
3. To know the applications of wearable devices in the field of medicine

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Describe the concepts of wearable system.
CO2: Explain the energy harvestings in wearable device.
CO3: Use the concepts of BAN in health care.
CO4: Illustrate the concept of smart textile
CO5: Compare the various wearable devices in healthcare system

Course Contents

UNIT-I	INTRODUCTION TO WEARABLE SYSTEMS AND SENSORS	07 Hours

Wearable Systems- Introduction, Need for Wearable Systems, Drawbacks of Conventional Systems for Wearable Monitoring, Applications of Wearable Systems, Types of Wearable Systems, Components of wearable Systems. Sensors for wearable systems-Inertia movement sensors, Respiration activity sensor, Impedance plethysmography, Wearable ground reaction force sensor.

UNIT-II	SIGNAL PROCESSING AND ENERGY HARVESTING FOR WEARABLEDEVICES	08 Hours

Wearability issues -physical shape and placement of sensor, Technical challenges - sensor design, signal acquisition, sampling frequency for reduced energy consumption, Rejection of irrelevant information. Power Requirements- Solar cell, Vibration based, Thermal based, Human body as a heat source for power generation,

Hybrid thermoelectric photovoltaic energy harvests, Thermopiles.

UNIT-III	WIRELESS HEALTH SYSTEMS	08 Hours

Need for wireless monitoring, Definition of Body area network, BAN and Healthcare, Technical Challenges- System security and reliability, BAN Architecture – Introduction, Wireless communication Techniques.

UNIT-IV	SMART TEXTILE	07 Hours

Introduction to smart textile- Passive smart textile, active smart textile. Fabrication Techniques- Conductive Fibres, Treated Conductive Fibres, Conductive Fabrics, Conductive Inks.Case study- smart fabric for monitoring biological parameters - ECG, respiration.

UNIT-V	APPLICATIONS OF WEARABLE SYSTEMS	08 Hours

Medical Diagnostics, Medical Monitoring-Patients with chronic disease, Hospital patients, Elderly patients, neural recording, Gait analysis, Sports Medicine.

TEXT BOOK:

1. Annalisa Bonfiglo and Danilo De Rossi, Wearable Monitoring Systems, Springer, 2011
2. Zhang and Yuan-Ting, Wearable Medical Sensors and Systems,Springer, 2013
3. Edward Sazonov and Micheal R Neuman, Wearable Sensors: Fundamentals,Implementation and Applications, Elsevier, 2014
4. Mehmet R. Yuce and JamilY.Khan, Wireless Body Area Networks Technology,Implementation applications,Pan Stanford Publishing Pte.Ltd, Singapore, 2012

REFERENCES

1. Sandeep K.S, Gupta, Tridib Mukherjee and Krishna Kumar Venkatasubramanian, BodyArea Networks Safety, Security, and Sustainability, Cambridge University Press, 2013.
2. Guang-Zhong Yang, Body Sensor Networks, Springer, 2006.

CO's-PO's & PSO's MAPPING

CO's	PO's												PSO's		
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
1	3	2	1	1	2			1					1		1
2	3	2	1	1	2			1					1		1
3	3	2	1	1	2			1					1		1
4	3	2	1	1	2			1					1		1
5	3	2	1	1	2			1					1		1

AVg.	3	2	1	1	2			1					1		1

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-306-OE(D): BIOTECHNOLOGY IN HEALTH CARE
(Open Elective – III)
B.Tech. III Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
P: - 4 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basics of Biotechnology

Course Objective:

- Create higher standard of knowledge on healthcare system and services
- Prioritize advanced technologies for the diagnosis and treatment of various diseases

Learning Course Outcome:
After successful completion of the course, students will able to:

- Gain the knowledge of what is AI, risks and benefits of AI, limits of AI and the ethics involved in building an AI application.
- Understand the nature of environments and the structure of agents.
- Possess the ability to select a search algorithm for a problem and characterize its time and space complexities.
- Possess the skill for representing knowledge using the appropriate technique
- Gain an understanding of the applications of AI

Course Contents

UNIT-I	PUBLIC HEALTH	07 Hours
Definition and Concept of Public Health, Historical aspects of Public Health, Changing Concepts of Public Health, Public Health versus Medical Care, Unique Features of Public Health, Determinants of Health (Social, Economic, Cultural, Environmental, Education, Genetics, Food and Nutrition). Indicators of health, Burden of disease, Role of different disciplines in Public Health.		
UNIT-II	**CLINICAL DISEASES**	**08 Hours**

Communicable diseases: Chickenpox / Shingles, COVID-19, Tuberculosis, Hepatitis B, Hepatitis C, HIV / AIDS, Influenza, Swine flu. Non-Communicable diseases: Diabetes mellitus, atherosclerosis, fatty liver, Obesity, Cancer.

UNIT-III	VACCINOLOGY	08 Hours

History of Vaccinology, conventional approaches to vaccine development, live attenuated and killed vaccines, adjuvants, quality control, preservation and monitoring of microorganisms in seed lot systems. Instruments related to monitoring of temperature, sterilization, and environment.

UNIT-IV	OUTPATIENT & IN PATIENT SERVICES	07 Hours

Radiotherapy, Nuclear medicine, surgical units, OT Medical units, G & Obs. units Pediatric, neonatal units, Critical care units, Physical medicine & Rehabilitation, Neurology, Gastroenterology, Endoscopy, Pulmonology, Cardiology.

UNIT-V	LEARNING FROM EXAMPLES	08 Hours

Diagnostic X-rays - Computer tomography – MRI – Ultrasonography – Endoscopy – Thermography – Different types of biotelemetry systems.

TEXT BOOKS

1. Joseph J.carr and John M. Brown, Introduction to Biomedical Equipment Technology, JohnWiley and sons, New York, 4th Edition, 2012.
2. Thomas M. Devlin.Textbook of Biochemistry with clinical correlations. Wiley LissPublishers
3. The Vaccine Book (2nd Ed.), Rafi Ahmed, Roy M. Anderson et. al.Editor(s): Barry R.Bloom, PaulHenri Lambert, Academic Press, 2016, Pages xxi-xxiv.

REFERENCE BOOKS

1. Suh, Sang, Gurupur, Varadraj P., Tanik, Murat M., Health Care Systems, Technology andTechniques, Springer, 1st Edition, 2011
2. Burtis & Ashwood W.B. Tietz Textbook of Clinical chemistry. Saunders Company
3. Levine, M. M. (2004). New Generation Vaccines. New York: M. Dekker

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-311-PCC DESIGN AND ANALYSIS OF ALGORITHMS
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 04	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

- A course on "Computer Programming and Data Structures".
- A course on "Advanced Data Structures".

Course Objective:

- Introduces the notations for analysis of the performance of algorithms and the data structure of disjoint sets.
- Describes major algorithmic techniques (divide-and-conquer, backtracking, dynamic programming, greedy, branch and bound methods) and mention problems for which each technique is appropriate
- Describes how to evaluate and compare different algorithms using worst-, average-, and best case analysis.
- Explains the difference between tractable and intractable problems, and introduces the problems that are P, NP and NP complete.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Analyze the performance of algorithms
- Choose appropriate data structures and algorithm design methods for a specified application
- Understand the choice of data structures and the algorithm design methods

Course Contents

	Introduction	**07 Hours**
Algorithm, Performance Analysis-Space complexity, Time complexity, Asymptotic Notations- Big oh notation, Omega notation, Theta notation and Little oh notation. Divide and conquer: General method, applications-Binary search, Quick sort, Merge sort, Strassen's matrix multiplication.		
UNIT-II	**Disjoint Sets**	**08 Hours**
Disjoint set operations, union and find algorithms, Priority Queue- Heaps, Heapsort Backtracking: General method, applications, n-queen's problem, sum of subsets problem, graph Coloring, hamitonian cycles.		
UNIT-III	**Dynamic Programming**	**08 Hours**

General method, applications- Optimal binary search tree, 0/1 knapsack problem, All pairs shortest path problem, Traveling salesperson problem, Reliability design.		
UNIT-IV	**Greedy method**	**07 Hours**
General method, applications-Job sequencing with deadlines, knapsack problem, Minimum cost spanning trees, Single source shortest path problem. Basic Traversal and Search Techniques: Techniques for Binary Trees, Techniques for Graphs, Connected components, Biconnected components.		
UNIT-V	**Branch and Bound**	**08 Hours**
General method, applications - Traveling salesperson problem, 0/1 knapsack problem - LC Branch and Bound solution, FIFO Branch and Bound solution. NP-Hard and NP-Complete problems: Basic concepts, non-deterministic algorithms, NP-Hard and NP-Complete classes, Cook's theorem.		

TEXT BOOK:

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekharan, University press, 1998.

REFERENCE BOOKS:

1. Design and Analysis of algorithms, Aho, Ullman and Hopcroft, Pearson education.
2. Introduction to Algorithms, second edition, T. H. Cormen, C.E. Leiserson, R. L. Rivest, and C. Stein, PHI Pvt. Ltd./ Pearson Education.
3. Algorithm Design: Foundations, Analysis and Internet Examples, M.T. Goodrich and R. Tamassia, John Wiley and sons.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

Savitribai Phule Pune University

(Formerly University of Pune)

Department Of Technology

BTDS-312-PCC: PREDICTIVE ANALYTICS
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Understand the fundamental concepts and importance of predictive analytics in various industries.
- Learn methods for data collection, cleaning, and preprocessing, including handling missing values and normalization.
- Gain proficiency in exploratory data analysis and feature engineering.
- Master the application of probability theory and statistical methods in predictive modeling.
- Develop skills in building and interpreting regression models, including simple and multiple linear regression.
- Explore machine learning algorithms for supervised and unsupervised learning, as well as ensemble methods.
- Understand and apply time series analysis techniques for forecasting.
- Learn model evaluation and validation techniques to ensure robust predictive performance.
- Gain practical experience in deploying, monitoring, and updating predictive models.
- Apply predictive analytics techniques to solve real-world problems in various industries, such as business analytics, financial forecasting, marketing, healthcare, and e-commerce.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Explain the significance and applications of predictive analytics in decision-making processes.
- Prepare and preprocess datasets for predictive modeling tasks.
- Apply various data mining and machine learning techniques to extract patterns and build predictive models.
- Evaluate and compare predictive models using appropriate metrics and validation techniques.
- Develop and interpret predictive models using regression analysis and advanced machine learning algorithms.
- Implement time series forecasting models and understand their applications.
- Deploy predictive models and monitor their performance in a real-world setting.
- Apply predictive analytics techniques to specific industry problems and present findings effectively.

Course Contents

UNIT-I	**Introduction to Predictive Analytics, Data Understanding and Preparation**	**07 Hours**

1.1 Overview of Predictive Analytics

Definition of Predictive Analytics: Understanding what predictive analytics is and its role in data analysis.

History and Evolution: A brief history of predictive analytics and its evolution over time.

Key Components: Introduction to the major components of predictive analytics, including data collection, data processing, model building, and model deployment.

1.2 Importance and Applications in Various Industries

Business:

Customer Retention: Using predictive analytics to predict customer churn and implement retention strategies.

Sales Forecasting: Forecasting future sales based on historical data.

Finance:

Risk Management: Predicting financial risks and defaults.

Fraud Detection: Identifying potential fraudulent activities.

Marketing:

Targeted Marketing: Personalizing marketing campaigns based on customer behavior predictions.

Market Basket Analysis: Understanding product purchase patterns.

Healthcare:

Predictive Diagnosis: Predicting disease outbreaks and patient health outcomes.

Resource Allocation: Optimizing the allocation of healthcare resources.

Others:

Manufacturing: Predicting equipment failures and maintenance needs.

Retail: Inventory management and demand forecasting.

Government: Predictive policing and public policy formulation.

1.3 Key Concepts and Terminology

Data Mining: Techniques for extracting patterns from large datasets.

Machine Learning: Algorithms that allow computers to learn from and make predictions based on data.

Big Data: Large and complex data sets that traditional data-processing software can't handle efficiently.

Predictive Model: A model that predicts future outcomes based on historical data.

Training Data: The data used to train predictive models.

Test Data: Data used to test the accuracy of predictive models.

Overfitting: A model that fits the training data too well but performs poorly on unseen data.

Underfitting: A model that is too simple to capture the underlying patterns in the data.

Feature Engineering: The process of selecting, modifying, or creating features (variables) for use in model building.

Cross-Validation: A technique for assessing how the results of a predictive model will generalize to an independent data set.

1.4 Steps in the Predictive Analytics Process

Problem Definition:

Understanding the Business Problem: Clearly defining the problem you aim to solve with predictive analytics.

Setting Objectives: Establishing what you want to achieve through predictive modeling.

Data Collection:

Sources of Data: Identifying and gathering relevant data from various sources.

Data Quality: Ensuring the accuracy, completeness, and reliability of the data collected.

Data Preparation:

Data Cleaning: Removing noise and correcting inconsistencies in the data.

Data Transformation: Normalizing and transforming data to a suitable format for analysis.

Feature Engineering: Creating new features from existing data to improve model performance.

Model Building:

Selecting Algorithms: Choosing appropriate algorithms based on the problem and data.

Training the Model: Using training data to create the predictive model.

Tuning Hyperparameters: Adjusting the model parameters to optimize performance.

Model Evaluation:

Validation Techniques: Using cross-validation and other techniques to evaluate model performance.

Performance Metrics: Assessing model accuracy, precision, recall, F1-score, ROC-AUC, etc.

Model Deployment:

Integration: Integrating the model into business processes or systems.

Monitoring: Continuously monitoring model performance and updating as needed.

Interpretation and Communication:

Result Interpretation: Understanding and interpreting the results generated by the model.

Reporting: Communicating findings and insights to stakeholders through reports and visualizations.

2.1 Data Collection Methods

Primary Data Collection:

Surveys and Questionnaires: Designing and conducting surveys for data collection.

Experiments: Collecting data through controlled experiments.

Observations: Gathering data through direct observation.

Secondary Data Collection:

Public Databases: Using data from public repositories and government databases.

Corporate Databases: Accessing internal corporate databases for historical data.

Web Scraping: Extracting data from websites and online sources.

Automated Data Collection:

Sensors and IoT Devices: Collecting data from interconnected devices.

APIs: Using Application Programming Interfaces for data retrieval from different platforms.

2.2 Data Cleaning and Preprocessing

Data Cleaning:

Handling Inconsistent Data: Correcting errors and inconsistencies in the data.

Removing Duplicates: Identifying and removing duplicate records.

Outlier Detection and Treatment: Detecting outliers and deciding whether to remove or transform them.

Data Transformation and Normalization:

Standardization: Transforming data to have a mean of zero and a standard deviation of one.

Min-Max Normalization: Scaling data to fit within a specified range, usually 0 to 1.

Log Transformation: Applying logarithmic transformation to stabilize variance and make the data more normal distribution-like.

Categorical Data Encoding: Converting categorical variables into numerical format using techniques like one-hot encoding and label encoding.

Handling Missing Values:

Identifying Missing Data: Detecting missing values in the dataset.

Imputation Techniques:

Mean/Median/Mode Imputation: Replacing missing values with the mean, median, or mode of the column.

K-Nearest Neighbors (KNN) Imputation: Using KNN algorithm to estimate and replace missing values.

Regression Imputation: Using regression models to predict and fill in missing values.

Multiple Imputation: Generating multiple imputations to account for the uncertainty of missing data.

2.3 Exploratory Data Analysis (EDA)

Descriptive Statistics:

Measures of Central Tendency: Calculating mean, median, and mode.

Measures of Dispersion: Understanding range, variance, and standard deviation.

Distribution Analysis: Analyzing the distribution of data using histograms, probability plots, and skewness/kurtosis measures.

Data Visualization Techniques:

Univariate Analysis: Visualizing single variable distributions using histograms, box plots, and density plots.

Bivariate Analysis: Exploring relationships between two variables using scatter plots, correlation matrices, and bar charts.

Multivariate Analysis: Visualizing relationships among multiple variables using pair plots, heatmaps, and parallel coordinates plots.

Time Series Visualization: Plotting time series data to identify trends and patterns over time.

2.4 Feature Engineering

Feature Selection:

Filter Methods: Using statistical tests and metrics (e.g., Chi-square test, ANOVA, correlation coefficient) to select relevant features.

Wrapper Methods: Employing algorithms like Recursive Feature Elimination (RFE) to identify the best subset of features.

Embedded Methods: Using algorithms that perform feature selection during the model training process (e.g., Lasso regression, decision trees).

Feature Extraction:

Principal Component Analysis (PCA): Reducing dimensionality by transforming features into a new set of orthogonal components.

Linear Discriminant Analysis (LDA): Reducing dimensionality while preserving class separability.

t-Distributed Stochastic Neighbor Embedding (t-SNE): Visualizing high-dimensional data in a lower-dimensional space.

Text Feature Extraction: Converting text data into numerical features using techniques like TF-IDF, word embeddings, and n-grams.

Image Feature Extraction: Using convolutional neural networks (CNNs) and other techniques to extract features from image data.

UNIT-II	Probability Theory and Statistics for Predictive Modeling, Regression Analysis, Probability Theory and Statistics for Predictive Modeling	08 Hours

3.1 Basics of Probability Theory

Foundational Concepts:

Probability Definitions: Understanding probability as a measure of uncertainty.

Sample Space: Defining the set of all possible outcomes.

Events: Identifying and describing events within a sample space.

Probability Rules:

Addition Rule: Calculating the probability of the union of two events.

Multiplication Rule: Calculating the probability of the intersection of two events.

Complementary Rule: Understanding the probability of the complement of an event.

Conditional Probability and Independence:

Conditional Probability: Calculating the probability of an event given another event has occurred.

Independence: Identifying independent events and understanding their implications.

3.2 Probability Distributions

Discrete and Continuous Distributions:

Discrete Distributions:

Binomial Distribution: Characteristics, mean, and variance.

Poisson Distribution: Characteristics, mean, and variance.

Continuous Distributions:

Uniform Distribution: Understanding the properties and applications.

Exponential Distribution: Characteristics and applications in time-between-events modeling.

Normal Distribution:

Properties of Normal Distribution: Mean, variance, skewness, and kurtosis.

Standard Normal Distribution: Z-scores and the standard normal table.

Central Limit Theorem: Understanding its significance in the context of predictive modeling.

Applications of Normal Distribution: Usage in confidence intervals, hypothesis testing, and error analysis.

3.3 Statistical Inference
- Point Estimation and Confidence Intervals:
  - Point Estimation:
    - Estimators and Estimates: Understanding the concepts and properties of estimators.
    - Methods of Estimation: Maximum likelihood estimation (MLE) and method of moments.
  - Confidence Intervals:
    - Constructing Confidence Intervals: For population mean and proportion.
    - Interpretation: Understanding the confidence level and interval bounds.
    - Margin of Error: Calculating and interpreting the margin of error.
- Hypothesis Testing:
  - Formulating Hypotheses:
    - Null Hypothesis (H0): Definition and formulation.
    - Alternative Hypothesis (H1): Definition and formulation.
  - Types of Errors:
    - Type I Error (α): Probability and implications.
    - Type II Error (β): Probability and implications.
  - Test Statistics and P-values:
    - Calculating Test Statistics: Using Z-tests, T-tests, Chi-square tests, and F-tests.
    - P-values: Interpretation and decision-making.
  - Decision Rules:
    - Significance Level: Understanding α and its role in hypothesis testing.
    - Critical Values: Using critical values to make decisions.
    - Power of the Test: Understanding and calculating the power of a test.

4.1 Simple Linear Regression
- 4.1.1 Model Assumptions
  - Linearity: Relationship between independent and dependent variables is linear.
  - Independence: Observations are independent of each other.
  - Homoscedasticity: Constant variance of errors.
  - Normality: Errors are normally distributed.
  - No Multicollinearity: For single predictor, this is not applicable.
- 4.1.2 Estimation of Parameters
  - Least Squares Method: Derivation and computation.
  - Interpretation of Coefficients: Slope and intercept.
  - Standard Error of Estimate: Calculation and interpretation.
  - Confidence Intervals for Parameters: Construction and interpretation.
- 4.1.3 Goodness of Fit: R-squared
  - Definition and Calculation: Proportion of variance explained by the model.
  - Adjusted R-squared: Adjusting for the number of predictors.
  - Interpretation: Assessing model fit.

4.2 Multiple Linear Regression
- 4.2.1 Model Building

Selecting Predictors: Criteria for inclusion.
Model Specification: Including interaction terms and polynomial terms.
Assumptions: Linearity, independence, homoscedasticity, normality, and no multicollinearity.

4.2.2 Interpretation of Coefficients
Partial Slopes: Meaning and interpretation in the context of other variables.
Standardized Coefficients: Comparison across variables.
Significance Testing: T-tests for individual coefficients.

4.2.3 Diagnostics and Residual Analysis
Residual Plots: Checking assumptions.
Influential Points: Leverage and Cook's distance.
Variance Inflation Factor (VIF): Detecting multicollinearity.
Normal Probability Plots: Checking normality of residuals.

4.3 Advanced Regression Techniques

4.3.1 Logistic Regression
Binary Outcomes: Modeling probability of binary outcomes.
Logit Function: Definition and interpretation.
Odds Ratios: Calculation and interpretation.
Model Fitting: Maximum likelihood estimation.
Goodness of Fit: Likelihood-ratio tests, Hosmer-Lemeshow test.

4.3.2 Polynomial Regression
Higher Order Terms: Including polynomial terms in regression.
Model Building: Selecting the degree of the polynomial.
Interpretation: Understanding coefficients in polynomial terms.
Overfitting: Balancing model complexity and predictive accuracy.

UNIT-III	Machine Learning Algorithms for Predictive Modeling, Time Series Analysis	08 Hours

5.1 Overview of Machine Learning
Definition and Types: Supervised, Unsupervised, Semi-supervised, and Reinforcement Learning.
Applications and Use Cases: Various domains like finance, healthcare, marketing, etc.
Key Concepts and Terminology: Features, labels, training and testing data, model training, overfitting, underfitting.
Evaluation Metrics: Accuracy, precision, recall, F1 score, ROC-AUC.

5.2 Supervised Learning Algorithms

5.2.1 Decision Trees
Structure and Components: Nodes, branches, leaves.
Splitting Criteria: Gini impurity, information gain, entropy.
Pruning: Avoiding overfitting, post-pruning and pre-pruning techniques.
Advantages and Limitations: Interpretability, complexity, overfitting issues.

5.2.2 Random Forests

Ensemble of Trees: Concept of bagging, how random forests combine multiple decision trees.

Bootstrap Aggregating: Sampling with replacement.

Feature Selection: Random selection of features for splitting.

Out-of-Bag Error: Estimating the error of the model.

Advantages and Limitations: Robustness, handling missing values, interpretability.

5.2.3 Support Vector Machines (SVM)

Hyperplane and Margin: Definition, support vectors, and maximum margin.

Kernels: Linear, polynomial, radial basis function (RBF).

Regularization Parameter (C): Controlling the trade-off between margin maximization and error minimization.

Advantages and Limitations: Effectiveness in high-dimensional spaces, computational efficiency, choice of kernel.

5.2.4 k-Nearest Neighbors (k-NN)

Algorithm Mechanics: Distance metrics (Euclidean, Manhattan).

Choosing k: Impact on model performance.

Advantages and Limitations: Simplicity, non-parametric nature, computational cost, curse of dimensionality.

5.3 Unsupervised Learning Algorithms

5.3.1 Clustering

K-means Clustering

Algorithm Steps: Initialization, assignment, update steps.

Choosing k: Elbow method, silhouette analysis.

Advantages and Limitations: Simplicity, scalability, sensitivity to initialization.

Hierarchical Clustering

Agglomerative vs Divisive: Bottom-up and top-down approaches.

Linkage Criteria: Single, complete, average.

Dendrogram: Interpretation and cutting.

Advantages and Limitations: Dendrogram interpretability, computational cost.

5.3.2 Association Rule Mining

Apriori Algorithm: Frequent itemsets, support, confidence, lift.

Applications: Market basket analysis, cross-selling strategies.

Advantages and Limitations: Rule discovery, computational complexity.

5.4 Ensemble Methods

5.4.1 Bagging and Boosting

Bagging (Bootstrap Aggregating)

Concept and Benefits: Reducing variance, improving stability.

Implementation: Bagged decision trees.

Boosting

Concept and Benefits: Reducing bias, combining weak learners.

Algorithms: AdaBoost, basic principles.

Advantages and Limitations: Improved accuracy, risk of overfitting.

5.4.2 Gradient Boosting Machines (GBM)

Concept and Mechanics: Building models sequentially, residual errors.

Algorithm Steps: Initialization, learning rate, number of trees.

Regularization Techniques: Shrinkage, subsampling.

Advantages and Limitations: High accuracy, interpretability, computational cost.

6.1 Introduction to Time Series Data

Definition and Importance: Understanding time series data and its significance in predictive analytics.

Components of Time Series: Trend, seasonality, cyclic patterns, and irregular components.

Time Series Plots: Visualization techniques for time series data.

Applications: Forecasting in finance, economics, supply chain management, meteorology, and more.

6.2 Time Series Decomposition

Additive and Multiplicative Models: Understanding the difference between additive and multiplicative decomposition.

Trend Extraction: Identifying and extracting the underlying trend component.

Seasonal Adjustment: Techniques for identifying and adjusting seasonal effects.

Residual Analysis: Examining the irregular component after trend and seasonality removal.

Practical Applications: Using decomposition to better understand and model time series data.

6.3 Smoothing Techniques

Purpose of Smoothing: Reducing noise to reveal important patterns.

6.3.1 Moving Average

Simple Moving Average (SMA): Calculation and interpretation.

Weighted Moving Average (WMA): Giving different weights to data points.

Applications: Smoothing time series data to identify trends.

6.3.2 Exponential Smoothing

Simple Exponential Smoothing: Applying smoothing parameters and exponential weights.

Holt's Linear Trend Model: Handling linear trends in time series data.

Holt-Winters Seasonal Model: Extending Holt's model to account for seasonality.

Comparison and Applications: Choosing appropriate smoothing techniques for different types of time series data.

6.4 Time Series Forecasting Models

Overview of Forecasting: Importance of accurate forecasting in various domains.

6.4.1 ARIMA (AutoRegressive Integrated Moving Average)

ARIMA Model Structure: Understanding autoregressive (AR), differencing (I), and moving average (MA) components.

Model Identification: Using ACF (autocorrelation function) and PACF (partial autocorrelation function) plots to identify appropriate parameters.

Parameter Estimation: Estimating ARIMA model parameters using statistical software.

Model Diagnostics: Checking residuals to ensure a good fit.

Applications and Examples: Real-world scenarios where ARIMA is applied.

6.4.2 Seasonal Decomposition of Time Series (STL)

STL Decomposition: Breaking down time series data into seasonal, trend, and residual components.

Seasonal Adjustment Techniques: Handling complex seasonal patterns with STL.

Model Implementation: Using STL for robust forecasting in seasonal data.

Case Studies and Examples: Practical applications of STL in various industries.

UNIT-IV	Model Evaluation and Validation, Model Deployment	07 Hours

7.1 Model Evaluation Metrics

Importance of Evaluation: Understanding why model evaluation is crucial in predictive analytics.

7.1.1 Accuracy, Precision, Recall, F1-Score

Accuracy: Definition, calculation, and limitations.

Precision: Definition, calculation, and interpretation.

Recall: Definition, calculation, and significance.

F1-Score: Combining precision and recall for a balanced evaluation metric.

Application and Examples: Practical scenarios for each metric, particularly in classification problems.

7.1.2 ROC-AUC

ROC Curve (Receiver Operating Characteristic Curve): Plotting true positive rate against false positive rate.

AUC (Area Under the Curve): Interpreting the AUC value for model performance.

Use Cases: Situations where ROC-AUC is particularly useful.

7.2 Cross-Validation Techniques

Purpose of Cross-Validation: Ensuring model generalizability and robustness.

7.2.1 k-Fold Cross-Validation

Methodology: Splitting data into k subsets, training on k-1 and validating on the remaining one.

Choosing k: Guidelines for selecting the appropriate number of folds.

Advantages and Disadvantages: Balancing bias-variance trade-off and computational efficiency.

Practical Implementation: Step-by-step guide on applying k-fold cross-validation.

7.2.2 Leave-One-Out Cross-Validation (LOOCV)

Methodology: Using each data point as a single validation case.

Comparison with k-Fold: Pros and cons of LOOCV vs. k-Fold.

When to Use LOOCV: Specific scenarios where LOOCV is beneficial.

7.3 Overfitting and Underfitting

Understanding Overfitting: Recognizing when a model is too complex and tailored to training data.

Symptoms and Consequences: How overfitting affects model performance on unseen data.

Prevention Techniques: Regularization, pruning, and other methods to mitigate overfitting.

Understanding Underfitting: Recognizing when a model is too simple to capture underlying

patterns.

Symptoms and Consequences: How underfitting affects model accuracy and generalizability.

Improvement Techniques: Increasing model complexity, feature engineering, and other strategies to reduce underfitting.

7.4 Model Selection Criteria

Importance of Model Selection: Choosing the right model for the best predictive performance.

7.4.1 AIC (Akaike Information Criterion)

Definition and Calculation: Understanding AIC and its role in model selection.

Interpreting AIC Values: How to use AIC for comparing models.

Applications: Situations where AIC is most effective.

7.4.2 BIC (Bayesian Information Criterion)

Definition and Calculation: Understanding BIC and its role in model selection.

Interpreting BIC Values: Differences between AIC and BIC, and when to use BIC.

Applications: Scenarios where BIC provides more reliable model selection.

8.1 Preparing Models for Deployment

Model Export and Serialization

Formats: Overview of common formats (e.g., PMML, ONNX, pickle).

Serialization Techniques: How to serialize models for deployment.

Best Practices: Ensuring reproducibility and integrity of models during export.

Environment Setup

Infrastructure Requirements: Understanding hardware and software needs.

Deployment Platforms: Overview of cloud services (e.g., AWS SageMaker, Google AI Platform) and on-premise options.

Dependencies Management: Handling libraries, packages, and version control.

8.2 Model Implementation

Integrating Models with Applications

APIs and Microservices: Creating and deploying REST APIs for model inference.

Frameworks: Introduction to frameworks like Flask, FastAPI, and Django for serving models.

Use Cases and Examples: Practical implementation scenarios in different industries.

Batch vs. Real-Time Processing

Batch Processing: Techniques for handling large-scale batch predictions.

Real-Time Inference: Requirements and strategies for real-time prediction serving.

Case Studies: Examples of batch and real-time deployments.

8.3 Monitoring Model Performance

Performance Metrics

Operational Metrics: Latency, throughput, and resource utilization.

Prediction Quality: Ongoing assessment of accuracy, precision, recall, and other relevant metrics.

Monitoring Tools

Logging and Alerts: Implementing logging systems and alert mechanisms for model monitoring.

Visualization Dashboards: Using tools like Grafana, Kibana, or custom dashboards to visualize performance metrics.

Drift Detection

Concept Drift: Understanding and detecting shifts in data distributions over time.

Methods for Drift Detection: Techniques like statistical tests, monitoring metrics, and control charts.

Handling Drift: Strategies to address drift in model performance.

8.4 Updating and Retraining Models

Retraining Strategies

Scheduled Retraining: Regular intervals for model retraining.

Triggered Retraining: Based on performance degradation or drift detection.

Automated Retraining Pipelines

Continuous Integration/Continuous Deployment (CI/CD): Setting up CI/CD pipelines for automated retraining and deployment.

Tools and Frameworks: Utilizing tools like Jenkins, GitLab CI, and MLflow for automation.

Version Control

Model Versioning: Managing and tracking different versions of models.

Rollback Mechanisms: Ensuring quick recovery to previous versions if new models underperform.

Post-Deployment Evaluation

A/B Testing: Implementing A/B tests to compare new models with existing ones.

Canary Deployment: Gradual rollout of new models to minimize risk.

Feedback Loops: Incorporating user feedback and new data into model improvement processes.

UNIT-V	Applications of Predictive Analytics	08 Hours

9.1 Business Analytics

Sales Forecasting

Techniques and Models: Time series analysis, regression models.

Case Studies: Real-world examples of sales forecasting.

Supply Chain Optimization

Predictive Models: Demand forecasting, inventory optimization.

Applications: Case studies from manufacturing and retail industries.

Risk Management

Fraud Detection: Techniques for identifying and predicting fraudulent activities.

Credit Scoring: Predictive models for assessing credit risk.

9.2 Financial Forecasting

Stock Market Prediction

Models: Time series analysis, machine learning techniques.

Data Sources: Financial statements, market data, news sentiment.

Portfolio Management
Risk Assessment: Predictive models for portfolio risk and return analysis.
Optimization Techniques: Techniques for optimizing investment portfolios.
Economic Forecasting
Macroeconomic Indicators: Using predictive analytics to forecast economic trends.
Case Studies: Examples of economic forecasting in government and finance sectors.
9.3 Marketing Campaigns and Customer Segmentation
Customer Segmentation
Clustering Techniques: K-means, hierarchical clustering, DBSCAN.
Applications: Segmenting customers based on purchasing behavior and demographics.
Marketing Campaign Optimization
Response Modeling: Predictive models to optimize marketing efforts and maximize ROI.
Case Studies: Examples of successful marketing campaign optimizations.
Customer Lifetime Value (CLV)
Predictive Models: Techniques for calculating and predicting CLV.
Applications: Using CLV to inform business strategies and marketing efforts.
9.4 Healthcare Analytics
Predictive Models in Healthcare
Disease Prediction: Using machine learning to predict disease onset and progression.
Patient Readmission Rates: Models to predict hospital readmissions and improve patient outcomes.
Personalized Medicine
Genomics and Predictive Analytics: Predicting patient responses to treatments based on genetic data.
Case Studies: Examples from personalized medicine and treatment plans.
Operational Efficiency
Resource Allocation: Predictive models for optimizing staffing and resource allocation in hospitals.
Case Studies: Real-world applications in healthcare management.
9.5 Web Usage Mining and E-commerce Data Analysis
Web Usage Mining
Techniques: Analyzing web logs, user behavior, and session patterns.
Applications: Improving website usability and user experience.
Recommendation Systems
Collaborative Filtering: Techniques and algorithms for recommending products.
Content-Based Filtering: Personalizing recommendations based on user preferences and item characteristics.
Customer Behavior Analysis
E-commerce Data Analysis: Analyzing purchase patterns, cart abandonment, and conversion rates.
Case Studies: Examples from leading e-commerce platforms and online retailers.
A/B Testing and Multivariate Testing

Experiment Design: Setting up and running experiments to test changes on websites.
Analysis: Interpreting results to make data-driven decisions.

TEXT BOOK:

1. Applied Predictive Analytics, Principles and Techniques for the Professional Data Analyst by Dean Abbott, 2014.

Reference Books:

1. "Data Mining: Concepts and Techniques" by Jiawei Han, Micheline Kamber, and Jian Pei
   A comprehensive guide to data mining concepts, techniques, and applications.
2. "Applied Predictive Modeling" by Max Kuhn and Kjell Johnson
   An in-depth exploration of predictive modeling techniques with practical examples.
3. "Machine Learning: A Probabilistic Perspective" by Kevin P. Murphy
   A detailed book on machine learning algorithms and their probabilistic foundations.
4. "Introduction to Statistical Learning" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani
   A beginner-friendly introduction to statistical learning and predictive modeling.
5. "Time Series Analysis and Its Applications: With R Examples" by Robert H. Shumway and David S. Stoffer
   A practical guide to time series analysis using R.
6. "Pattern Recognition and Machine Learning" by Christopher M. Bishop
   A comprehensive textbook on pattern recognition and machine learning techniques.
7. "Practical Data Science with R" by Nina Zumel and John Mount
   A hands-on guide to data science and predictive modeling using R.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-312-L: PREDICTIVE ANALYTICS LAB
B.Tech. III Year II Sem

Teaching Scheme: P: - 2 Hours/Week	Credit P: 01	Examination Scheme: Total : 30 Marks

Course Prerequisites: ML Practice

Course Objective:

- The objective of this lab is to get an overview of the various machine learning techniques and can demonstrate them using python.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand modern notions in predictive data analysis
- Select data, model selection, model complexity and identify the trends
- Understand a range of machine learning algorithms along with their strengths and weaknesses
- Build predictive models from data and analyze their performance

Course Contents

List of Experiments	07 Hours

Unit 1: Introduction to Predictive Analytics

Lab 1: Introduction to Predictive Analytics Tools

- Objective: Familiarize students with Python, R, and Tidyverse for predictive analytics.
- Tasks:
  - Install and set up Python and R environments.
  - Introduction to Jupyter Notebooks and RStudio.
  - Basic operations and syntax in Python and R.

Unit 2: Data Understanding and Preparation

Lab 2: Data Collection and Importing Data

- Objective: Understand data collection methods and how to import data.
- Tasks:
  - Import datasets from CSV, Excel, and databases into Python (using Pandas) and R (using readr and DBI packages).

Lab 3: Data Cleaning and Preprocessing

- Objective: Learn data cleaning and preprocessing techniques.
- Tasks:
  - Handle missing values using Pandas in Python and Tidyverse in R.
  - Perform data transformations and normalization.

Lab 4: Exploratory Data Analysis (EDA)

- Objective: Perform EDA to understand data distributions and relationships.

- Tasks:
  - Generate descriptive statistics in Python (Pandas) and R (dplyr).
  - Create visualizations (histograms, box plots) using Matplotlib and Seaborn in Python, and ggplot2 in R.

Lab 5: Feature Engineering

- Objective: Practice feature selection and extraction techniques.
- Tasks:
  - Use Python's Scikit-learn and R's caret package for feature selection.
  - Perform feature extraction techniques such as PCA.

Unit 3: Probability Theory and Statistics for Predictive Modeling

Lab 6: Probability Distributions

- Objective: Understand and visualize different probability distributions.
- Tasks:
  - Generate and plot discrete and continuous distributions in Python (SciPy) and R (ggplot2).

Lab 7: Statistical Inference

- Objective: Perform point estimation, confidence intervals, and hypothesis testing.
- Tasks:
  - Conduct hypothesis tests and construct confidence intervals using Python (SciPy, Statsmodels) and R (stats package).

Unit 4: Regression Analysis

Lab 8: Simple Linear Regression

- Objective: Implement and evaluate simple linear regression models.
- Tasks:
  - Build and interpret a simple linear regression model in Python (Scikit-learn) and R (lm function).

Lab 9: Multiple Linear Regression

- Objective: Develop and analyze multiple linear regression models.
- Tasks:
  - Perform multiple linear regression and interpret coefficients in Python (Scikit-learn) and R (lm function).
  - Conduct diagnostics and residual analysis.

Lab 10: Advanced Regression Techniques

- Objective: Implement logistic and polynomial regression models.
- Tasks:
  - Build logistic regression models in Python (Scikit-learn) and R (glm function).
  - Create polynomial regression models and evaluate performance.

Unit 5: Machine Learning Algorithms for Predictive Modeling

Lab 11: Decision Trees

- Objective: Implement and evaluate decision tree models.
- Tasks:

- Train decision tree models using Python (Scikit-learn) and R (rpart package).
- Visualize and interpret decision trees.

Lab 12: Random Forests

- Objective: Develop and analyze random forest models.
- Tasks:
  - Build random forest models in Python (Scikit-learn) and R (randomForest package).
  - Evaluate model performance using accuracy and feature importance.

Lab 13: Support Vector Machines (SVM)

- Objective: Implement SVM for classification tasks.
- Tasks:
  - Train SVM models in Python (Scikit-learn) and R (e1071 package).
  - Tune hyperparameters and evaluate model performance.

Lab 14: k-Nearest Neighbors (k-NN)

- Objective: Apply k-NN algorithm for classification and regression.
- Tasks:
  - Implement k-NN in Python (Scikit-learn) and R (class package).
  - Optimize the value of k and assess model accuracy.

Lab 15: Clustering Algorithms

- Objective: Perform clustering using k-means and hierarchical clustering.
- Tasks:
  - Apply k-means clustering in Python (Scikit-learn) and R (stats package).
  - Implement hierarchical clustering and visualize dendrograms.

Lab 16: Association Rule Mining

- Objective: Discover association rules in datasets.
- Tasks:
  - Use Python (mlxtend package) and R (arules package) to perform association rule mining.

Lab 17: Ensemble Methods

- Objective: Implement ensemble learning techniques.
- Tasks:
  - Apply bagging and boosting methods using Python (Scikit-learn) and R (caret package).
  - Build Gradient Boosting Machines (GBM) and evaluate performance.

Unit 6: Time Series Analysis

Lab 18: Time Series Decomposition

- Objective: Decompose time series data into trend, seasonality, and residuals.
- Tasks:
  - Perform time series decomposition in Python (Statsmodels) and R (stl function).

Lab 19: Smoothing Techniques

- Objective: Apply smoothing techniques to time series data.
- Tasks:
  - Implement moving average and exponential smoothing in Python (Pandas) and R (forecast

package).

Lab 20: Time Series Forecasting Models

- Objective: Develop time series forecasting models.
- Tasks:
  - Build ARIMA models in Python (Statsmodels) and R (forecast package).
  - Apply Seasonal Decomposition of Time Series (STL) and forecast.

Unit 7: Model Evaluation and Validation

Lab 21: Model Evaluation Metrics

- Objective: Calculate and interpret model evaluation metrics.
- Tasks:
  - Compute accuracy, precision, recall, F1-score, and ROC-AUC in Python (Scikit-learn) and R (caret package).

Lab 22: Cross-Validation Techniques

- Objective: Implement cross-validation methods.
- Tasks:
  - Perform k-fold and leave-one-out cross-validation in Python (Scikit-learn) and R (caret package).

Lab 23: Overfitting and Underfitting

- Objective: Identify and mitigate overfitting and underfitting.
- Tasks:
  - Visualize model performance to detect overfitting and underfitting.
  - Apply regularization techniques (Lasso, Ridge) in Python (Scikit-learn) and R (glmnet package).

Lab 24: Model Selection Criteria

- Objective: Use AIC and BIC for model selection.
- Tasks:
  - Calculate AIC and BIC for model comparison in Python (Statsmodels) and R (stats package).

Unit 8: Model Deployment

Lab 25: Preparing Models for Deployment

- Objective: Prepare predictive models for deployment.
- Tasks:
  - Export trained models using Python (joblib, pickle) and R (saveRDS, PMML).

Lab 26: Model Implementation

- Objective: Implement models in a production environment.
- Tasks:
  - Deploy models as web services using Flask (Python) and plumber (R).

Lab 27: Monitoring and Updating Models

- Objective: Monitor and update deployed models.
- Tasks:

- Set up monitoring for model performance metrics.
- Retrain models with new data and update deployment using Python (Scikit-learn, Flask) and R (caret, plumber).

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-313-PEC(A): SOFTWARE TESTING METHODOLOGIES
(Professional Elective – II)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Software Engineering

Course Objective:

- To provide knowledge of the concepts in software testing such as testing process, criteria, strategies, and methodologies.
- To develop skills in software test automation and management using the latest tools.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Understand purpose of testing and path testing
- Understand strategies in data flow testing and domain testing
- Develop logic-based test strategies
- Understand graph matrices and its applications
- Implement test cases using any testing automation tool

Course Contents

UNIT-I	testing	07 Hours
Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing.		
UNIT-II	**Transaction Flow Testing**	**08 Hours**
transaction flows, transaction flow testing techniques. Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing. Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability.		
UNIT-III	**Paths, Path products and Regular expressions**	**08 Hours**
path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection. Logic Based Testing: overview, decision tables, path expressions, kv charts, specifications.		
UNIT-IV	**State, State Graphs and Transition testing**	**07 Hours**

state graphs, good & bad state graphs, state testing, Testability tips.		
UNIT-V	**Graph Matrices and Application**	**08 Hours**
Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon).		

TEXT BOOKS:

1. Software Testing techniques - Baris Beizer, Dreamtech, second edition.
2. Software Testing Tools – Dr. K. V. K. K. Prasad, Dreamtech.

REFERENCE BOOKS:

1. The craft of software testing - Brian Marick, Pearson Education.
2. Software Testing Techniques – SPD(Oreille)
3. Software Testing in the Real World – Edward Kit, Pearson.
4. Effective methods of Software Testing, Perry, John Wiley.
5. Art of Software Testing – Meyers, John Wiley.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-313-PEC(B): INFORMATION RETRIEVAL SYSTEMS
(Professional Elective – II)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Data Structures

Course Objective:

- To learn the concepts and algorithms in Information Retrieval Systems
- To understand the data/file structures that are necessary to design, and implement information retrieval (IR) systems.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Ability to apply IR principles to locate relevant information large collections of data
- Ability to design different document clustering algorithms
- Implement retrieval systems for web search tasks.
- Design an Information Retrieval System for web search tasks.

Course Contents

UNIT-I	Introduction to Information Retrieval Systems	07 Hours

Definition of Information Retrieval System, Objectives of Information Retrieval Systems, Functional Overview, Relationship to Database Management Systems, Digital Libraries and Data Warehouses Information Retrieval System Capabilities: Search Capabilities, Browse Capabilities, Miscellaneous Capabilities

UNIT-II	Cataloging and Indexing	08 Hours

History and Objectives of Indexing, Indexing Process, Automatic Indexing, Information Extraction Data Structure: Introduction to Data Structure, Stemming Algorithms, Inverted File Structure, N-Gram Data Structures, PAT Data Structure, Signature File Structure, Hypertext and XML Data Structures, Hidden Markov Models.

UNIT-III	Automatic Indexing	08 Hours

Classes of Automatic Indexing, Statistical Indexing, Natural Language, Concept Indexing, Hypertext Linkages

Document and Term Clustering: Introduction to Clustering, Thesaurus Generation, Item Clustering, Hierarchy of Clusters

UNIT-IV	User Search Techniques	07 Hours
Search Statements and Binding, Similarity Measures and Ranking, Relevance Feedback, Selective Dissemination of Information Search, Weighted Searches of Boolean Systems, Searching the INTERNET and Hypertext Information Visualization: Introduction to Information Visualization, Cognition and Perception, Information Visualization Technologies		
UNIT-V	**Text Search Algorithms**	**08 Hours**
Introduction to Text Search Techniques, Software Text Search Algorithms, Hardware Text Search Systems. Multimedia Information Retrieval: Spoken Language Audio Retrieval, Non-Speech Audio Retrieval, Graph Retrieval, Imagery Retrieval, Video Retrieval		

TEXT BOOK:

1. Information Storage and Retrieval Systems – Theory and Implementation, Second Edition, Gerald J. Kowalski, Mark T. Maybury, Springer

REFERENCE BOOKS:

1. Frakes, W.B., Ricardo Baeza-Yates: Information Retrieval Data Structures and Algorithms,
2. Prentice Hall, 1992.
3. Information Storage & Retrieval By Robert Korfhage – John Wiley & Sons.
4. Modern Information Retrieval By Yates and Neto Pearson Education.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-313-PEC(c): COMPUTER VISION AND ROBOTICS
(Professional Elective – II)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Linear Algebra and Probability.

Course Objective:

1. To understand the Fundamental Concepts Related To sources, shadows and shading
2. To understand the The Geometry of Multiple Views

Learning Course Outcome:
After successful completion of the course, students will able to:

- Implement fundamental image processing techniques required for computer vision
- Implement boundary tracking techniques
- Apply chain codes and other region descriptors, Hough Transform for line, circle, and ellipse detections.
- Apply 3D vision techniques and Implement motion related techniques.
- Develop applications using computer vision techniques.

Course Contents

UNIT-I	CAMERAS: Pinhole Cameras	07 Hours

Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Interreflections: Global Shading Models Color: The Physics of Color, Human Color Perception, Representing Color, A Model for Image Color, Surface Color from Image Color.

UNIT-II	Linear Filters	08 Hours

Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates

Edge Detection: Noise, Estimating Derivatives, Detecting Edges

Texture: Representing Texture, Analysis (and Synthesis) Using Oriented Pyramids, Application: Synthesis by Sampling Local Models, Shape from Texture.

UNIT-III	The Geometry of Multiple Views: Two Views	08 Hours

Stereopsis: Reconstruction, Human Stereposis, Binocular Fusion, Using More Cameras Segmentation by Clustering: Segmentation, Human Vision: Grouping and Getstalt, Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph-Theoretic Clustering,		
UNIT-IV	**Model and Camera Calibration**	**07 Hours**
Segmentation by Fitting a Model: The Hough Transform, Fitting Lines, Fitting Curves, Fitting as a Probabilistic Inference Problem, Robustness Geometric Camera Models: Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical Photogrammetry, An Application: Mobile Robot Localization		
UNIT-V	**Introduction to Robotics**	**08 Hours**
Social Implications of Robotics, Brief history of Robotics, Attributes of hierarchical paradigm, Closed world assumption and frame problem, Representative Architectures, Attributes of Reactive Paradigm, Subsumption Architecture, Potential fields and Perception Common sensing techniques for Reactive Robots: Logical sensors, Behavioural Sensor Fusion, Pro- prioceptive sensors, Proximity Sensors, Topological Planning and Metric Path Planning		

TEXT BOOKS:

1. David A. Forsyth and Jean Ponce: Computer Vision — A Modern Approach, PHI Learning (Indian Edition), 2009.
2. Robin Murphy, Introduction to AI Robotics, MIT Press

REFERENCE BOOKS:

1. E. R. Davies: Computer and Machine Vision – Theory, Algorithms and Practicalities, Elsevier (Academic Press), 4th edition, 2013.
2. The Robotics premier, Maja J Matari, MIT Press
3. Richard Szeliski "Computer Vision: Algorithms and Applications" Springer-Verlag London Limited 2011.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-313-PEC(D): DATA WAREHOUSING AND BUSINESS INTELLIGENCE (Professional Elective – II)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- This course is concerned with extracting data from the information systems that deal with the day-to-day operations and transforming it into data that can be used by businesses to drive high-level decision making
- Students will learn how to design and create a data warehouse, and how to utilize the process of extracting, transforming, and loading (ETL) data into data warehouses.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand architecture of data warehouse and OLAP operations.
- Understand Fundamental concepts of BI
- Application of BI Key Performance indicators
- Understand Utilization of Advanced BI Tools and their Implementation.
- Implementation of BI Techniques and BI Ethics.

Course Contents

UNIT-I	Data Warehouse	07 Hours
Data Warehouse Modelling, OLAP operations, Data Qube Computation methods		
UNIT-II	**Business Intelligence Introduction**	**08 Hours**
Definition, Leveraging Data and Knowledge for BI, BI Components, BI Dimensions, Information Hierarchy, Business Intelligence and Business Analytics. BI Life Cycle. Data for BI - Data Issues and Data Quality for BI.		
UNIT-III	**BI Implementation**	**08 Hours**
Key Drivers, Key Performance Indicators and Performance Metrics, BI Architecture/Framework, Best Practices, Business Decision Making, Styles of BI-vent-Driven alerts-A cyclic process of Intelligence Creation. The value of Business Intelligence-Value driven and Information use.		
UNIT-IV	**Advanced BI**	**07 Hours**

Big Data and BI, Social Networks, Mobile BI, emerging trends, Description of different BI-Tools (Pentaho, KNIME)		
UNIT-V	**Business Intelligence and integration implementation**	**08 Hours**
connecting in BI systems- Issues of legality- Privacy and ethics- Social networking and BI.		

TEXT BOOKS:

1. Data Mining — Concepts and Techniques - JIAWEI HAN & MICHELINE KAMBER, Elsevier, 4th Edition.
2. Rajiv Sabherwal "Business Intelligence" Wiley Publications, 2012.

REFERENCE BOOKS:

1. Efraim Turban, Ramesh Sharda, Jay Aronson, David King, Decision Support and Business Intelligence Systems, 9th Edition, Pearson Education, 2009.
2. David Loshin, Business Intelligence - The Savy Manager's Guide Getting Onboard with Emerging IT, Morgan Kaufmann Publishers, 2009.
3. Philo Janus, Stacia Misner, Building Integrated Business Intelligence. Solutions with SQL Server, 2008 R2 & Office 2010, TMH, 2011.
4. Business Intelligence Data Mining and Optimization for decision making [Author: Carlo-Verellis] [Publication: (Wiley)]
5. Data Warehousing, Data Mining & OLAP- Alex Berson and Stephen J. Smith- Tata McGraw- Hill Edition, Tenth reprint 2007
6. Building the Data Warehouse- W. H. Inmon, Wiley Dreamtech India Pvt. Ltd.
7. Data Mining Introductory and Advanced topics – Margaret H Dunham, PEA.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-313-PEC(e): NODE JS/ REACT JS
B.Tech. III Year II Sem.

Practical Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Object Oriented Programming through Java, HTML Basics

Course Objective:

- To implement the static web pages using HTML and do client side validation using JavaScript.
- To design and work with databases using Java
- To develop an end to end application using java full stack.
- To introduce Node JS implementation for server side programming.
- To experiment with single page application development using React.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Build a custom website with HTML, CSS, and Bootstrap and little JavaScript.
- Demonstrate Advanced features of JavaScript and learn about JDBC
- Develop Server – side implementation using Java technologies like
- Develop the server – side implementation using Node JS.
- Design a Single Page Application using React.

Course Contents

Exercises	07 Hours

1. Build a responsive web application for shopping cart with registration, login, catalog and cart pages using CSS3 features, flex and grid.
2. Make the above web application responsive web application using Bootstrap framework.
3. Use JavaScript for doing client – side validation of the pages implemented in experiment 1 and experiment 2.
4. Explore the features of ES6 like arrow functions, callbacks, promises, async/await. Implement an application for reading the weather information from openweathermap.org and display the information in the form of a graph on the web page.
5. Develop a java stand alone application that connects with the database (Oracle / mySql) and perform the CRUD operation on the database tables.
6. Create an xml for the bookstore. Validate the same using both DTD and XSD.
7. Design a controller with servlet that provides the interaction with application developed in experiment 1 and the database created in experiment 5.
8. Maintaining the transactional history of any user is very important. Explore the

various session tracking mechanism (Cookies, HTTP Session)

9. Create a custom server using http module and explore the other modules of Node JS like OS, path, event.
10. Develop an express web application that can interact with REST API to perform CRUD operations on student data. (Use Postman)
11. For the above application create authorized end points using JWT (JSON Web Token).
12. Create a react application for the student management system having registration, login, contact, about pages and implement routing to navigate through these pages.
13. Create a service in react that fetches the weather information from openweathermap.org and the display the current and historical weather information using graphical representation using chart.js
14. Create a TODO application in react with necessary components and deploy it into github.

REFERENCE BOOKS:

1. Jon Duckett, Beginning HTML, XHTML, CSS, and JavaScript, Wrox Publications, 2010
2. Bryan Basham, Kathy Sierra and Bert Bates, Head First Servlets and JSP, O'Reilly Media, 2nd Edition, 2008.
3. Vasan Subramanian, Pro MERN Stack, Full Stack Web App Development with Mongo, Express, React, and Node, 2nd Edition, A Press.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-313-PEC(F): Data Security in Data Science
(Professional Elective-II)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Basic understanding of computer networks and operating systems, Familiarity with data science concepts, tools, and programming languages.

Course Objective:

- To understand the fundamental concepts of data security in the context of data science.
- To learn various cryptographic techniques and their applications in data science.
- To explore methods for securing data during storage, transmission, and processing in data science workflows.
- To examine security protocols and standards relevant to data science.
- To gain practical experience in implementing data security measures in data science projects.

Learning Course Outcome:

- Demonstrate an understanding of data security principles and practices in data science.
- Apply cryptographic techniques to secure data used in data science.
- Implement security measures for data storage, transmission, and processing in data science workflows.
- Evaluate and use security protocols and standards in data science projects.
- Analyze security risks in data science and develop strategies to mitigate them.

Course Contents

UNIT-I	Introduction to Data Security in Data Science	07 Hours
Overview of Data Security: Importance and Challenges in Data Science, Basic Concepts: Confidentiality, Integrity, Availability, Threats and Vulnerabilities in Data Science: Types and Examples, Risk Management and Assessment in Data Science Projects, Legal and Ethical Issues in Data Security for Data Science		
UNIT-II	**Cryptography for Data Science**	**08 Hours**
Introduction to Cryptography: Goals and Applications in Data Science, Symmetric Key Cryptography: Algorithms (AES, DES), Asymmetric Key Cryptography: Algorithms (RSA, ECC), Hash Functions and Digital Signatures, Key Management and Distribution in Data Science		
UNIT-III	**Data Security in Storage for Data Science**	**08 Hours**
Data Encryption at Rest: Techniques and Tools, Secure Data Storage Solutions: Databases, Data Lakes, and Cloud Storage, Access Controls and Authentication		

Mechanisms, Backup and Recovery Strategies for Data Science Projects, Case Studies: Implementing Data Security in Storage for Data Science.		
UNIT-IV	**Data Security in Transmission for Data Science**	**07 Hours**
Data Encryption in Transit: Techniques and Tools, Secure Communication Protocols: SSL/TLS, HTTPS, Network Security Measures: Firewalls, Intrusion Detection Systems, Virtual Private Networks (VPNs) for Data Science, Case Studies: Implementing Data Security in Transmission for Data Science		
UNIT-V	**Security Protocols and Standards for Data Science**	**08 Hours**
Overview of Security Protocols: Importance and Applications in Data Science, Authentication Protocols: Kerberos, OAuth, Secure Data Sharing Protocols: Secure Multiparty Computation, Differential Privacy, Security Standards and Frameworks: ISO/IEC 27001, NIST, GDPR, Case Studies: Implementing Security Protocols and Standards in Data Science Projects		
UNIT-VI	**Advanced Topics and Emerging Trends in Data Security for Data Science**	**08 Hours**
Advanced Encryption Techniques: Quantum Cryptography, Homomorphic Encryption, Data Privacy and Anonymization Techniques: GDPR, CCPA Compliance, Cloud Security for Data Science: Challenges and Solutions, Security in Big Data and IOT: Techniques and Best Practices, Future Trends in Data Security for Data Science		

TEXT BOOKS:

1. "Cryptography and Network Security: Principles and Practice" by William Stallings
2. "Data Security and Privacy: A Practical Guide for Protecting Data" by David Salomon
3. "Security and Privacy in Data Mining and Machine Learning" by Charu C. Aggarwal
4. "Applied Cryptography: Protocols, Algorithms, and Source Code in C" by Bruce Schneier
5. "Database Security: Concepts, Approaches, and Challenges" by Alfred Basta and Melissa Zgola

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO6
CO1	2	2	-	-	3
CO2	2	-	3	-	-
CO3	-	2	3	-	-
CO4	-	2	-	3	-
CO5	-	2	-	-	3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-313-PEC(G): PROFESSIONAL PRACTICE, LAW & ETHICS
(Professional Elective -II)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Understand the types of roles they are expected to play in the society as practitioners of the engineering profession.
- To develop some ideas of the legal and practical aspects of their profession.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Practice ethics and rule of the land in their profession
- Follow the principles and elements of legal contracts
- Able to resolve disputes pertaining to arbitration, reconciliation
- Aware of intellectual property loss

Course Contents

UNIT-I	Professional Practice and Ethics	07 Hours
Definition of Ethics, Professional Ethics - Engineering Ethics, Personal Ethics; Code of Ethics - Profession, Professionalism, Professional Responsibility, Conflict of Interest, Gift Vs Bribery, Environmental breaches, Negligence, Deficiencies in state-of-the-art; Vigil Mechanism, Whistle blowing, protected disclosures. Introduction to GST- Various Roles of Various Stake holders		
UNIT-II	**Law of Contract**	**08 Hours**
Nature of Contract and Essential elements of valid contract, Offer and Acceptance, Consideration, Capacity to contract and Free Consent, Legality of Object. Unlawful and illegal agreements, Contingent Contracts, Performance and discharge of Contracts, Remedies for breach of contract. Contracts-II: Indemnity and guarantee, Contract of Agency, Sale of goods Act -1930: General Principles, Conditions & Warranties, Performance of Contract of Sale.		
UNIT-III	**Arbitration, Conciliation and ADR (Alternative Dispute Resolution) system**	**08 Hours**

Arbitration – meaning, scope and types – distinction between laws of 1940 and 1996; UNCITRAL model law – Arbitration and expert determination; Extent of judicial intervention; International commercial arbitration.		
UNIT-IV	**Arbitration agreements**	**07 Hours**
essential and kinds, validity, reference and interim measures by court; Arbitration tribunal – appointment, challenge, jurisdiction of arbitral tribunal, powers, grounds of challenge, procedure and court assistance; Distinction between conciliation, negotiation, mediation and arbitration, confidentiality, resort to judicial proceedings, costs; Dispute Resolution Boards; Lok Adalats.		
UNIT-V	**Law relating to Intellectual property**	**08 Hours**
Introduction – meaning of intellectual property, main forms of IP, Copyright, Trademarks, Patents and Designs, Secrets; Law relating to Copyright in India including Historical evolution of Copy Rights Act, 1957, Meaning of copyright – computer programs, Ownership of copyrights and assignment, Criteria of infringement, Piracy in Internet – Remedies and procedures in India; Law relating to Patents under Patents Act, 1970		

TEXT BOOKS:

1. Professional Ethics: R. Subramanian, Oxford University Press, 2015.
2. Ravinder Kaur, Legal Aspects of Business, 4e, Cengage Learning, 2016.

REFERENCE BOOKS:

1. Wadhera (2004), Intellectual Property Rights, Universal Law Publishing Co.
2. T. Ramappa (2010), Intellectual Property Rights Law in India, Asia Law House.
3. O.P. Malhotra, Law of Industrial Disputes, N.M. Tripathi Publishers.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2

Avg	3	2.4	2.4	2	2	3	2.6	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-315-MDM(A): BLOCKCHAIN FOR DATA PRIVACY AND SECURITY
(MDM Basket – 1)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To provide an in-depth understanding of blockchain technology and its applications in data privacy and security.
2. To study cryptographic methods and mechanisms to enhance blockchain security.
3. To analyze privacy issues and challenges in blockchain systems.
4. To explore blockchain-based solutions for securing sensitive data.
5. • To design and implement privacy-preserving blockchain applications for real-world problems.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Explain blockchain technology and its role in privacy and security.
CO2: Apply cryptographic techniques to strengthen blockchain security.
CO3: Analyze privacy challenges in blockchain systems and evaluate mitigation strategies.
CO4: Design blockchain-based solutions for data privacy and secure storage.
CO5: Develop and test privacy-preserving blockchain applications.

Course Contents

UNIT-I	Fundamentals of Blockchain Technology	03 Hours

- Overview of Blockchain Technology
- Architecture: Blocks, Chains, Nodes, and Transactions
- Consensus Mechanisms: Proof of Work (PoW), Proof of Stake (PoS), Delegated Proof of Stake (DPoS), and Byzantine Fault Tolerance
- Blockchain Types: Public, Private, and Consortium
- Smart Contracts and Their Role in Security

UNIT-II	Cryptographic Foundations of Blockchain	03 Hours

• Basics of Cryptography: Symmetric and Asymmetric Encryption • Hashing Algorithms: SHA-256, MD5, and Keccak • Digital Signatures and Public Key Infrastructure (PKI) • Zero-Knowledge Proofs (ZKP) and Homomorphic Encryption • Advanced Cryptographic Protocols: zk-SNARKs, zk-STARKs		
UNIT-III	**Privacy Challenges in Blockchain Systems**	**03 Hours**
• Transparency vs. Privacy Dilemma • Anonymity and Pseudonymity in Blockchain Networks • Privacy Vulnerabilities in Public Blockchains • Data Breaches and Privacy Concerns in Blockchain Applications • Case Studies: Analysis of Privacy Failures in Blockchain Projects		
UNIT-IV	**Privacy-Preserving Blockchain Solutions**	**04 Hours**
• Privacy-Enhancing Techniques: Mixers, Ring Signatures, and CoinJoin • Off-Chain Privacy Solutions: Sidechains and State Channels • Cryptographic Privacy Protocols: Monero, Zcash, and MimbleWimble • Secure Multiparty Computation (SMPC) in Blockchain		
UNIT-V	**Blockchain for Data Security Applications**	**03 Hours**
• Blockchain-Based Identity Management Systems • Securing Medical Records with Blockchain • Supply Chain Security and Blockchain Implementation • Financial Data Security and Fraud Prevention Using Blockchain • Case Studies: Real-World Blockchain Security Applications		
UNIT-VI	**Building Blockchain Privacy and Security Solutions**	**04 Hours**
• Development Tools: Ethereum, Hyperledger Fabric, and Corda • Frameworks for Privacy-Preserving Blockchain Solutions • Designing and Testing Privacy-Centric Blockchain Applications • Future Trends: Quantum-Resistant Blockchains and AI Integration		

Reference Books:

1. **Mastering Blockchain** by Imran Bashir
2. **Blockchain Basics: A Non-Technical Introduction in 25 Steps** by Daniel Drescher
3. **Blockchain and Cryptocurrencies** by Arshdeep Bahga and Vijay Madisetti
4. **Data Privacy and Security** by David A. Ricks
5. Research Papers and Resources from IEEE Xplore, SpringerLink, and GitHub

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	3	3	2	-	-	-	-	-	2	2	2	1
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	2
CO5	3	3	3	3	3	2	1	-	-	-	-	2	2	2	2
Avg.	3	2.8	2.2	2.2	2.4	2	0.4	-	-	-	-	2	1.8	1.4	1.2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-315-MDM(B): CRYPTOGRAPHY AND INFORMATION SECURITY
(MDM Basket – 2)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To introduce the fundamental principles and techniques of cryptography.
2. To understand various cryptographic algorithms and their applications.
3. To explore key management and distribution techniques in secure communication.
4. To study advanced cryptographic techniques and protocols for secure data transmission.
5. To analyze real-world security challenges and apply cryptographic principles to mitigate them.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Explain the core principles and objectives of cryptography and information security.
CO2: Apply symmetric and asymmetric encryption algorithms for secure communication.
CO3: Implement key management and authentication techniques.
CO4: Analyze advanced cryptographic protocols and their applications.
CO5: Design and evaluate cryptographic solutions for real-world security problems.

Course Contents

UNIT-I	Introduction to Cryptography	03 Hours

- Principles of Cryptography: Confidentiality, Integrity, Availability
- Classical Cryptography: Substitution and Transposition Techniques
- Cryptanalysis Techniques: Brute Force and Statistical Methods
- Applications of Cryptography in Information Security

UNIT-II	Symmetric Encryption Algorithms	03 Hours

- Public Key Cryptography: Principles and Applications
- RSA Algorithm: Key Generation, Encryption, and Decryption
- Elliptic Curve Cryptography (ECC)
- Key Exchange Protocols: Diffie-Hellman and ElGamal

UNIT-III	Asymmetric Encryption Algorithms	03 Hours

- Transparency vs. Privacy Dilemma
- Anonymity and Pseudonymity in Blockchain Networks
- Privacy Vulnerabilities in Public Blockchains
- Data Breaches and Privacy Concerns in Blockchain Applications
- Case Studies: Analysis of Privacy Failures in Blockchain Projects

UNIT-IV	Authentication and Key Management	04 Hours

- Hash Functions: MD5, SHA Family, and Message Integrity
- Digital Signatures: Concepts and Standards (DSA, RSA)
- Authentication Protocols: Kerberos and Certificates
- Key Management Techniques: Public Key Infrastructure (PKI) and Key Lifecycle

UNIT-V	Advanced Cryptographic Protocols	03 Hours

- Zero-Knowledge Proofs and Applications
- Secure Multi-Party Computation
- Homomorphic Encryption and Applications in Cloud Security
- Blockchain Cryptography: Merkle Trees and Consensus Algorithms

UNIT-VI	Applications and Emerging Trends	04 Hours

- Cryptography in Wireless Networks and IoT Security
- Quantum Cryptography and Post-Quantum Cryptography
- Cryptographic Attacks: Side Channel and Timing Attacks
- Case Studies: SSL/TLS, End-to-End Encryption in Messaging Apps

Reference Books

1. **Cryptography and Network Security: Principles and Practice** by William Stallings
2. **Applied Cryptography: Protocols, Algorithms, and Source Code in C** by Bruce Schneier
3. **Network Security Essentials: Applications and Standards** by William Stallings
4. **Introduction to Modern Cryptography** by Jonathan Katz and Yehuda Lindell
5. **Handbook of Applied Cryptography** by Alfred J. Menezes, Paul C. van Oorschot, and Scott A. Vanstone

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	2	-	-	-	-	-	2	1	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	3	3	2	-	-	-	-	-	2	2	2	1
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	2
CO5	3	3	3	3	3	2	1	-	-	-	-	2	2	2	2
Avg.	3	2.8	2.4	2.2	2.4	2	0.4	-	-	-	-	2	1.8	1.4	1.2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-315-MDM(c): DEEP LEARNING FOR IMAGE PROCESSING
(MDM Basket – 3)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To introduce the fundamental concepts of deep learning and its applications in image processing.
2. To understand various deep learning architectures and their suitability for image processing tasks.
3. To develop skills in implementing and training deep learning models for image classification, segmentation, and enhancement.
4. To explore state-of-the-art techniques like convolutional neural networks (CNNs) and generative adversarial networks (GANs).
5. To analyze and evaluate the performance of deep learning models on real-world image datasets.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Understand the foundational principles of deep learning and their applications in image processing.

CO2: Implement convolutional neural networks (CNNs) for tasks like image classification and object detection.

CO3: Apply advanced architectures such as GANs and autoencoders for image enhancement and generation.

CO4: Evaluate deep learning models using performance metrics and optimize them for better accuracy.

CO5: Solve real-world problems using deep learning techniques on large-scale image datasets.

Course Contents

UNIT-I	Introduction to Deep Learning for Image Processing	03 Hours

- Basics of Neural Networks: Perceptrons, Activation Functions, and Loss Functions
- Deep Learning Architectures and Applications in Image Processing
- Image Datasets: MNIST, CIFAR-10, ImageNet
- Tools and Frameworks: TensorFlow, PyTorch, Keras

UNIT-II	Convolutional Neural Networks (CNNs)	03 Hours

- Introduction to CNNs: Convolution, Pooling, and Fully Connected Layers
- Popular Architectures: AlexNet, VGG, ResNet, and EfficientNet
- Transfer Learning and Fine-Tuning for Image Processing
- Case Studies: Image Classification and Object Detection

UNIT-III	Advanced Architectures and Techniques	03 Hours

- Recurrent Neural Networks (RNNs) for Sequential Image Data
- Autoencoders for Dimensionality Reduction and Image Denoising
- Variational Autoencoders (VAEs) for Image Synthesis
- Generative Adversarial Networks (GANs): Structure and Applications

UNIT-IV	Image Segmentation and Detection	04 Hours

- Image Segmentation Techniques: Semantic and Instance Segmentation
- U-Net and Mask R-CNN for Medical and General Image Segmentation
- Object Detection Algorithms: YOLO, SSD, and Faster R-CNN
- Applications of Segmentation and Detection in Autonomous Vehicles and Healthcare

UNIT-V	Optimization and Regularization in Deep Learning	03 Hours

- Optimization Techniques: Gradient Descent, Adam, and RMSProp
- Regularization Methods: Dropout, L2 Regularization, and Batch Normalization
- Data Augmentation and Preprocessing for Image Datasets
- Hyperparameter Tuning for Image Processing Models

UNIT-VI	Applications and Emerging Trends	04 Hours

- Applications of Deep Learning in Image Super-Resolution and Style Transfer
- Edge AI for Real-Time Image Processing
- Explainability and Interpretability in Deep Learning for Image Processing
- Recent Advances in Vision Transformers and Large Vision Models

Reference Books

1. **Deep Learning** by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
2. **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow** by Aurélien Géron
3. **Deep Learning for Computer Vision** by Rajalingappaa Shanmugamani
4. **Deep Learning with PyTorch** by Eli Stevens, Luca Antiga, and Thomas Viehmann
5. Research Papers and Online Resources: arXiv, IEEE Xplore, and Kaggle

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	2	1	-	-	-	-	-	2	2	-	-
CO2	3	3	2	2	3	2	-	-	-	-	-	2	2	2	-
CO3	3	3	3	3	3	2	-	-	-	-	-	2	2	2	1
CO4	3	3	3	3	3	2	-	-	-	-	-	2	2	2	1
CO5	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	2.2	2.8	1.8	0.2	-	-	-	-	2	2	1.6	0.6

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-316-PCC: EMBEDDED SYSTEMS AND IOT
B.Tech. III Year II Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

- A course on "Computer Programming and Data Structures".
- A course on "Advanced Data Structures".

Course Objective:

- To learn the internal architecture and programming of an embedded processor.
- To introduce interfacing I/O devices to the processor.
- To introduce the evolution of the Internet of Things (IoT).
- To build a small low-cost embedded and IoT system using Arduino/Raspberry Pi/ open platform.
- To apply the concept of Internet of Things in real world scenario.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Explain the architecture of embedded processors.
CO2: Write embedded C programs.
CO3: Design simple embedded applications.
CO4: Compare the communication models in IOT
CO5: Design IoT applications using Arduino/Raspberry Pi /open platform.

Course Contents

UNIT-I	8-BIT EMBEDDED PROCESSOR	07 Hours
8-Bit Microcontroller – Architecture – Instruction Set and Programming – Programming Parallel Ports. Timers and Serial Port – Interrupt Handling.		
UNIT-II	**EMBEDDED C PROGRAMMING**	**08 Hours**
Memory And I/O Devices Interfacing – Programming Embedded Systems in C – Need For RTOS – Multiple Tasks and Processes – Context Switching – Priority Based Scheduling Policies.		
UNIT-III	**IOT AND ARDUINO PROGRAMMING**	**08 Hours**
Introduction to the Concept of IoT Devices – IoT Devices Versus. Computers – IoT Configurations – Basic Components – Introduction to Arduino – Types of Arduino –		

Arduino Toolchain – Arduino Programming Structure – Sketches – Pins – Input/Output From Pins Using Sketches – Introduction to Arduino Shields – Integration of Sensors and Actuators with Arduino.		
UNIT-IV	**IOT COMMUNICATION AND OPEN PLATFORMS**	**07 Hours**
IoT Communication Models and APIs – IoT Communication Protocols – Bluetooth – WiFi – ZigBee – GPS – GSM modules – Open Platform (like Raspberry Pi) – Architecture – Programming – Interfacing – Accessing GPIO Pins – Sending and Receiving Signals Using GPIO Pins – Connecting to the Cloud.		
UNIT-V	**APPLICATIONS DEVELOPMENT**	**08 Hours**
Complete Design of Embedded Systems – Development of IoT Applications – Home Automation – Smart Agriculture – Smart Cities – Smart Healthcare.		
PRACTICAL EXERCISES		
1. Write 8051 Assembly Language experiments using simulator. 2. Test data transfer between registers and memory. 3. Perform ALU operations. 4. Write Basic and arithmetic Programs Using Embedded C. 5. Introduction to Arduino platform and programming 6. Explore different communication methods with IoT devices (Zigbee, GSM,Bluetooth) 7. Introduction to Raspberry PI platform and python programming 8. Interfacing sensors with Raspberry PI 9. Communicate between Arduino and Raspberry PI using any wireless medium 10. Setup a cloud platform to log the data 11. Log Data using Raspberry PI and upload to the cloud platform 12. Design an IOT based system		

TEXT BOOK:

1. Muhammed Ali Mazidi, Janice Gillispie Mazidi, Rolin D. McKinlay, "The 8051 Microcontroller andEmbedded Systems", Pearson Education, Second Edition, 2014
2. Robert Barton, Patrick Grossetete, David Hanes, Jerome Henry, Gonzalo Salgueiro, "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things",CISCO Press, 2017.

REFERENCES

1. Michael J. Pont, "Embedded C", Pearson Education, 2007.
2. Wayne Wolf, "Computers as Components: Principles of Embedded Computer System Design",Elsevier, 2006.
3. Andrew N Sloss, D. Symes, C. Wright, "Arm System Developer's Guide", Morgan Kauffman/Elsevier, 2006.
4. Arshdeep Bahga, Vijay Madisetti, "Internet of Things – A hands-on approach", Universities Press,2015

CO's-PO's & PSO's MAPPING

	PO's												PSO's		
CO's	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**
1	3	1	1	3	1	-	-	-	3	1	2	3	2	1	1
2	2	3	1	3	2	-	-	-	1	2	3	1	1	1	2
3	2	2	3	2	3	-	-	-	1	3	1	1	2	1	3
4	2	2	2	1	1	-	-	-	1	3	1	1	2	2	3
5	3	2	1	1	3	-	-	-	3	2	3	2	1	3	1
AVG	2.4	2	1.6	2	2	-	-	-	1.8	2.2	2	1.6	1.6	1.6	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-317-VSEC: UI DESIGN-FLUTTER
B.Tech. III Year II Sem.

Practical Scheme: P: - 4 Hours/Week	Credit: 02 P: 02	Examination Scheme: Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Learns to Implement Flutter Widgets and Layouts
- Understands Responsive UI Design and with Navigation in Flutter
- Knowledge on Widges and customize widgets for specific UI elements, Themes
- Understand to include animation apart from fetching data

Learning Course Outcome:
After successful completion of the course, students will able to:

- Implements Flutter Widgets and Layouts
- Responsive UI Design and with Navigation in Flutter
- Create custom widgets for specific UI elements and also Apply styling using themes and custom styles.
- Design a form with various input fields, along with validation and error handling
- Fetches data and write code for unit Test for UI components and also animation

Course Contents

List of Experiments	07 Hours

Students need to implement the following experiments

1. a) Install Flutter and Dart SDK.
   b) Write a simple Dart program to understand the language basics.

2. a) Explore various Flutter widgets (Text, Image, Container, etc.).
   b) Implement different layout structures using Row, Column, and Stack widgets.

3. a) Design a responsive UI that adapts to different screen sizes.
   b) Implement media queries and breakpoints for responsiveness.

4. a) Set up navigation between different screens using Navigator.
   b) Implement navigation with named routes.

5. a) Learn about stateful and stateless widgets.
   b) Implement state management using set State and Provider.

6. a) Create custom widgets for specific UI elements.

b) Apply styling using themes and custom styles.

7. a) Design a form with various input fields.
   b) Implement form validation and error handling.

8. a) Add animations to UI elements using Flutter's animation framework.
   b) Experiment with different types of animations (fade, slide, etc.).

9. a) Fetch data from a REST API.
   b) Display the fetched data in a meaningful way in the UI.

10. a) Write unit tests for UI components.
    b) Use Flutter's debugging tools to identify and fix issues.

TEXT BOOK:

1. Marco L. Napoli, Beginning Flutter: A Hands-on Guide to App Development.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-314-PEC(A): INTERNET OF THINGS (Professional Elective – III)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Computer organization, Computer Networks

Course Objective:

- To introduce the terminology, technology and its applications
- To introduce the concept of M2M (machine to machine) with necessary protocols
- To introduce the Python Scripting Language which is used in many IoT devices
- To introduce the Raspberry PI platform, that is widely used in IoT applications
- To introduce the implementation of web-based services on IoT devices

Learning Course Outcome:
After successful completion of the course, students will able to:

- Interpret the impact and challenges posed by IoT networks leading to new architectural models.
- Compare and contrast the deployment of smart objects and the technologies to connect them to network.
- Appraise the role of IoT protocols for efficient network communication.
- Identify the applications of IoT in Industry.

Course Contents

UNIT-I	Introduction to Internet of Things	07 Hours
Definition and Characteristics of IoT, Physical Design of IoT, Logical Design of IoT, IoT Enabling Technologies, IoT Levels and Deployment Templates Domain Specific IoTs – Home automation, Environment, Agriculture, Health and Lifestyle		
UNIT-II	**IoT and M2M**	**08 Hours**
M2M, Difference between IoT and M2M, SDN and NFV for IoT, IoT System Management with NETCOZF, YANG- Need for IoT system Management, Simple Network management protocol, Network operator requirements, NETCONF, YANG, IoT Systems Management with NETCONF-YANG		
UNIT-III	**IoT Systems – Logical design using Python**	**08 Hours**
Introduction to Python – Python Data types & Data structures, Control flow, Functions, Modules, Packaging, File handling, Data/Time operations, Classes, Exception, Python packages of Interest for IoT		

UNIT-IV	IoT Physical Devices and Endpoints	07 Hours
Raspberry Pi, Linux on Raspberry Pi, Raspberry Pi Interfaces, Programming Raspberry PI with Python, Other IoT devices. IoT Physical Servers and Cloud Offerings – Introduction to Cloud Storage models and communication APIs, WAMP-AutoBahn for IoT, Xively Cloud for IoT, Python web application framework –Designing a RESTful web API		
UNIT-V	**Case studies**	**08 Hours**
Home Automation, Environment-weather monitoring-weather reporting- air pollution monitoring, Agriculture.		

TEXT BOOK:

1. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547.

REFERENCE BOOK:

1. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014, ISBN: 9789350239759.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-314-PEC(B): SCRIPTING LANGUAGES (Professional Elective – III)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

1. A course on "Computer Programming and Data Structures".
2. A course on "Object Oriented Programming Concepts".

Course Objective:

- This course introduces the script programming paradigm
- Introduces scripting languages such as Perl, Ruby and TCL.
- Learning TCL

Learning Course Outcome:

After successful completion of the course, students will able to:

- Comprehend the differences between typical scripting languages and typical system and application programming languages.
- Gain knowledge of the strengths and weakness of Perl, TCL and Ruby; and select an appropriate language for solving a given problem.
- Acquire programming skills in scripting language

Course Contents

UNIT-I	Introduction	07 Hours
Ruby, Rails, The structure and Execution of Ruby Programs, Package Management with RUBYGEMS, Ruby and web: Writing CGI scripts, cookies, Choice of Webservers, SOAP and web services RubyTk – Simple Tk Application, widgets, Binding events, Canvas, scrolling		
UNIT-II	**Extending Ruby**	**08 Hours**
Ruby Objects in C, the Jukebox extension, Memory allocation, Ruby Type System, Embedding Ruby to Other Languages, Embedding a Ruby Interpreter		
UNIT-III	**Introduction to PERL and Scripting**	**08 Hours**
Scripts and Programs, Origin of Scripting, Scripting Today, Characteristics of Scripting Languages, Uses for Scripting Languages, Web Scripting, and the universe of Scripting Languages. PERL- Names and Values, Variables, Scalar Expressions, Control Structures, arrays, list, hashes, strings, pattern and regular expressions, subroutines.		
UNIT-IV	**Advanced Perl**	**07 Hours**

Finer points of looping, pack and unpack, filesystem, eval, data structures, packages, modules, objects, interfacing to the operating system, Creating Internet ware applications, Dirty Hands Internet Programming, security Issues.

UNIT-V	TCL/ Tk	08 Hours

TCL Structure, syntax, Variables and Data in TCL, Control Flow, Data Structures, input/output, procedures, strings, patterns, files, Advance TCL- eval, source, exec and uplevel commands, Name spaces, trapping errors, event driven programs, making applications internet aware, Nuts and Bolts Internet Programming, Security Issues, C Interface.

Tk. Tk-Visual Tool Kits, Fundamental Concepts of Tk, Tk by example, Events and Binding, Perl-Tk.

TEXT BOOKS:

1. The World of Scripting Languages, David Barron, Wiley Publications.
2. Ruby Programming language by David Flanagan and Yukihiro Matsumoto O'Reilly
3. "Programming Ruby" The Pramatic Progammers guide by Dabve Thomas Second edition

REFERENCE BOOKS:

1. Open Source Web Development with LAMP using Linux Apache, MySQL, Perl and PHP, J.Lee and B. Ware (Addison Wesley) Pearson Education.
2. Perl by Example, E. Quigley, Pearson Education.
3. Programming Perl, Larry Wall, T. Christiansen and J. Orwant, O'Reilly, SPD.
4. Tcl and the Tk Tool kit, Ousterhout, Pearson Education.

Perl Power, J. P. Flynt, Cengage Learning.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-314-PEC(c): MOBILE APPLICATION DEVELOPMENT
(Professional Elective – III)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

1. Acquaintance with JAVA programming
2. A Course on DBMS

Course Objective:

- To demonstrate their understanding of the fundamentals of Android operating systems
- To improves their skills of using Android software development tools
- To demonstrate their ability to develop software with reasonable complexity on mobile platform
- To demonstrate their ability to deploy software to mobile devices
- To demonstrate their ability to debug programs running on mobile devices

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the working of Android OS Practically.
- Develop Android user interfaces
- Develop, deploy and maintain the Android Applications.

Course Contents

UNIT-I	Introduction to Android Operating System	07 Hours

Android OS design and Features – Android development framework, SDK features, Installing and running applications on Android Studio, Creating AVDs, Types of Android applications, Best practices in Android programming, Android tools Android application components – Android Manifest file, Externalizing resources like values, themes, layouts, Menus etc, Resources for different devices and languages, Runtime Configuration Changes

Android Application Lifecycle – Activities, Activity lifecycle, activity states, monitoring state changes

UNIT-II	Android User Interface: Measurements	08 Hours

Device and pixel density independent measuring unit - s Layouts – Linear, Relative, Grid and Table Layouts

User Interface (UI) Components –Editable and non-editable TextViews, Buttons, Radio and Toggle Buttons, Checkboxes, Spinners, Dialog and pickers

Event Handling – Handling clicks or changes of various UI components

Fragments – Creating fragments, Lifecycle of fragments, Fragment states, Adding fragments to Activity, adding, removing and replacing fragments with fragment transactions, interfacing between fragments and Activities, Multi-screen Activities

UNIT-III	Intents and Broadcasts	08 Hours

Intent – Using intents to launch Activities, Explicitly starting new Activity, Implicit Intents, Passing data to Intents, Getting results from Activities, Native Actions, using Intent to dial a number or to send SMS

Broadcast Receivers – Using Intent filters to service implicit Intents, Resolving Intent filters, finding and using Intents received within an Activity

Notifications – Creating and Displaying notifications, Displaying Toasts

UNIT-IV	Persistent Storage	07 Hours

Files – Using application specific folders and files, creating files, reading data from files, listing contents of a directory Shared Preferences – Creating shared preferences, saving and retrieving data using Shared Preference

UNIT-V	Database	08 Hours

Introduction to SQLite database, creating and opening a database, creating tables, inserting retrieving and etindelg data, Registering Content Providers, Using content Providers (insert, delete, retrieve and update)

TEXT BOOK:

1. Professional Android 4 Application Development, Reto Meier, Wiley India, (Wrox), 2012

REFERENCE BOOKS:

1. Android Application Development for Java Programmers, James C Sheusi, Cengage Learning, 2013
2. Beginning Android 4 Application Development, Wei-Meng Lee, Wiley India (Wrox), 2013

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	3	2.4	2.4	2	2	3	2.6	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-314-PEC(D): COMPUTER NETWORKS
(Professional Elective – III)
B.Tech. III Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

1. Acquaintance with JAVA programming
2. A Course on DBMS

Course Objective:

1. To understand the concept of layering in networks.
2. To know the functions of protocols of each layer of TCP/IP protocol suite.
3. To visualize the end-to-end flow of information.
4. To learn the functions of network layer and the various routing protocols
5. To familiarize the functions and protocols of the Transport layer

Learning Course Outcome:

After successful completion of the course, students will able to:

CO 1: Explain the basic layers and its functions in computer networks.
CO 2: Understand the basics of how data flows from one node to another.
CO 3: Analyze routing algorithms.
CO 4: Describe protocols for various functions in the network.
CO 5: Analyze the working of various application layer protocols.

Course Contents

UNIT-I	INTRODUCTION AND APPLICATION LAYER	07 Hours
Data Communication - Networks – Network Types – Protocol Layering – TCP/IP Protocol suite – OSI Model – Introduction to Sockets - Application Layer protocols: HTTP – FTP – Email protocols (SMTP - POP3 - IMAP - MIME) – DNS – SNMP		
UNIT-II	**TRANSPORT LAYER**	**08 Hours**
Introduction - Transport-Layer Protocols: UDP – TCP: Connection Management – Flow control - Congestion Control - Congestion avoidance (DECbit, RED) – SCTP – Quality of Service.		
UNIT-III	**NETWORK LAYER**	**08 Hours**

Switching : Packet Switching - Internet protocol - IPV4 – IP Addressing – Subnetting - IPV6, ARP, RARP, ICMP, DHCP.

UNIT-IV	ROUTING	07 Hours

Routing and protocols: Unicast routing - Distance Vector Routing - RIP - Link State Routing – OSPF. Path-vector routing - BGP - Multicast Routing: DVMRP – PIM.

UNIT-V	DATA LINK AND PHYSICAL LAYERS	08 Hours

Data Link Layer – Framing – Flow control – Error control – Data-Link Layer Protocols – HDLC – PPP. Media Access Control – Ethernet Basics – CSMA/CD – Virtual LAN – Wireless LAN (802.11) - Physical Layer: Data and Signals - Performance – Transmission media- Switching – Circuit Switching.

PRACTICAL EXERCISES:

1. Learn to use commands like tcpdump, netstat, ifconfig, nslookup and traceroute. Capture ping and trace route PDUs using a network protocol analyzer and examine.
2. Write a HTTP web client program to download a web page using TCP sockets.
3. Applications using TCP sockets like: a) Echo client and echo server b) Chat
4. Simulation of DNS using UDP sockets.
5. Use a tool like Wireshark to capture packets and examine the packets
6. Write a code simulating ARP /RARP protocols.
7. Study of Network simulator (NS) and Simulation of Congestion Control Algorithms using NS.
8. Study of TCP/UDP performance using Simulation tool.
9. Simulation of Distance Vector/ Link State Routing algorithm.
10. Simulation of an error correction code (like CRC)

TEXT BOOK:

1. James F. Kurose, Keith W. Ross, Computer Networking, A Top-Down Approach Featuringthe Internet, Eighth Edition, Pearson Education, 2021.
2. Behrouz A. Forouzan, Data Communications and Networking with TCP/IP Protocol Suite,Sixth Edition TMH, 2022

REFERENCES

1. Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, Fifth Edition,Morgan Kaufmann Publishers Inc., 2012.
2. William Stallings, Data and Computer Communications, Tenth Edition, Pearson Education,2013.
3. Nader F. Mir, Computer and Communication Networks, Second Edition, Prentice Hall, 2014.
4. Ying-Dar Lin, Ren-Hung Hwang, Fred Baker, "Computer Networks: An Open SourceApproach", McGraw Hill, 2012.

CO's-PO's & PSO's MAPPING

	PO's												PSO's		
CO's	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**
1	3	1	2	3	-	-	-	-	1	1	3	1	3	2	1
2	3	2	1	2	2	-	-	-	2	2	2	1	3	2	3
3	2	2	3	2	1	-	-	-	3	3	1	2	1	1	3
4	1	3	1	3	1	-	-	-	1	2	1	1	1	3	1
5	3	3	1	1	2	-	-	-	2	2	2	2	2	2	2
AVG	2	2	2	2	2	-	-	-	2	2	2	1	2	2	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-401-PCC: BIG DATA ANALYTICS
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Understand the fundamental concepts and characteristics of big data and its importance in modern data-driven decision-making.
- Explore the architecture and components of big data platforms like Hadoop and Spark.
- Develop skills in data collection, storage, and processing techniques specific to big data.
- Learn about various big data storage solutions, including NoSQL databases.
- Master techniques for data cleaning, transformation, and preparation for analysis.
- Apply advanced analytical methods and machine learning techniques to big data.
- Gain proficiency in using big data tools and frameworks for real-world data analysis.
- Interpret and visualize large data sets effectively to communicate insights.
- Analyze real-world case studies and understand the application of big data analytics in different industries.
- Prepare students for advanced study and careers in big data analytics and data science.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Explain the key concepts, opportunities, and challenges of big data.
- Utilize the Hadoop ecosystem and Spark framework for big data processing.
- Implement data collection, storage, and processing workflows for large datasets.
- Employ NoSQL databases for efficient data storage and retrieval.
- Perform data cleaning, transformation, and preparation techniques on big data.
- Apply machine learning algorithms and advanced analytics techniques to big data.
- Use big data tools such as Hadoop, Spark, and NoSQL databases for practical data analysis tasks.
- Create visualizations to effectively communicate insights derived from big data.
- Implement stream processing pipelines for real-time data analysis.
- Analyze case studies to understand the application of big data analytics in various industries.
- Demonstrate the ability to manage and analyze big data projects, preparing for roles in data science and big data analytics.

Course Contents

UNIT-I	Introduction to Big Data, Big Data Technologies	07 Hours

1.1. Overview of Big Data
Definition and Concept
Evolution of Big Data
Big Data Ecosystem

1.2. Characteristics and Challenges of Big Data
- Volume, Velocity, Variety, Veracity, and Value
- Data Quality and Management Issues
- Scalability and Storage Solutions
- Security and Privacy Concerns

1.3. Big Data vs Traditional Data Analysis
- Differences in Data Processing
- Advantages of Big Data
- Limitations and Trade-offs
- Case Studies Comparing Both Approaches

1.4. Importance and Applications of Big Data Analytics
- Role in Decision-Making
- Industry-Specific Applications (Healthcare, Finance, Retail, etc.)
- Predictive and Prescriptive Analytics
- Success Stories and Real-World Examples

2.1. Introduction to Hadoop Ecosystem
- Overview of Hadoop Ecosystem Components
  - HDFS (Hadoop Distributed File System)
  - MapReduce
  - YARN (Yet Another Resource Negotiator)
  - Hadoop Common
- Role of Each Component in Big Data Processing
- Installation and Configuration of Hadoop

2.2. Hadoop Distributed File System (HDFS)
- Architecture and Design Principles
  - NameNode and DataNode
  - Block Storage Mechanism
- HDFS Operations
  - Read/Write Operations
  - Data Replication
  - Fault Tolerance and Reliability
- HDFS Shell and Command Line Interface
- Best Practices for HDFS Usage

2.3. MapReduce Programming Model
- Basic Concepts of MapReduce
  - Map and Reduce Functions
  - Data Flow and Processing
- Writing a MapReduce Program
  - Key-Value Pair Concept
  - Example Programs (Word Count, Sorting)
- Advanced MapReduce Techniques

Combiner and Partitioner Optimization and Performance Tuning Real-World Use Cases and Applications 2.4. Introduction to Spark Framework Overview of Apache Spark Components and Architecture Spark Core, Spark SQL, Spark Streaming, MLlib, GraphX Spark vs. Hadoop: A Comparative Analysis Programming with Spark RDD (Resilient Distributed Dataset) DataFrames and Datasets Writing Spark Applications Basic Operations (Transformations and Actions) Example Programs (Word Count, DataFrame Operations) 2.5. Comparison of Hadoop and Spark Performance Comparison Batch Processing vs. Real-Time Processing Speed and Efficiency Usability and Flexibility Ease of Use and Learning Curve Programming Languages Supported (Java, Scala, Python, R) Ecosystem and Integration Compatibility with Other Big Data Tools Scalability and Fault Tolerance Cost and Resource Utilization Resource Management and Optimization Cost of Deployment and Maintenance		
UNIT-II	**Data Collection and Storage**	**08 Hours**
3.1. Data Collection Techniques Overview of Data Collection Methods Structured Data Collection Semi-Structured and Unstructured Data Collection Data Sources Logs and Event Data Social Media and Web Data Sensors and IoT Devices Data Quality and Preprocessing Data Cleaning Handling Missing Values Data Normalization and Transformation		

- 3.2. Data Ingestion Tools (Apache Flume, Sqoop)
  - Introduction to Data Ingestion
    - Importance of Data Ingestion in Big Data
    - Batch vs. Real-Time Data Ingestion
  - Apache Flume
    - Architecture and Components
    - Sources, Channels, and Sinks
    - Configuring and Deploying Flume Agents
    - Example Use Cases
  - Apache Sqoop
    - Introduction and Use Cases
    - Importing Data from Relational Databases to Hadoop
    - Exporting Data from Hadoop to Relational Databases
    - Performance Tuning and Optimization
- 3.3. Data Storage Solutions
  - Overview of Big Data Storage Requirements
    - Scalability
    - Durability
    - Consistency
  - Distributed File Systems
    - HDFS (Hadoop Distributed File System)
    - Google File System (GFS)
  - Cloud-Based Storage Solutions
    - Amazon S3
    - Google Cloud Storage
    - Azure Blob Storage
- 3.4. Introduction to NoSQL Databases
  - Characteristics of NoSQL Databases
    - Schema-less Data Models
    - Horizontal Scalability
    - High Availability and Fault Tolerance
  - Types of NoSQL Databases
    - Key-Value Stores
    - Document Stores
    - Column-Family Stores
    - Graph Databases
  - Comparison with Traditional RDBMS
    - Advantages and Disadvantages
    - Use Cases and Applications
- 3.5. Key-Value Stores, Document Stores, Column-Family Stores
  - Key-Value Stores
    - Basic Concepts and Architecture

Examples: Redis, DynamoDB Use Cases and Performance Considerations Document Stores Document-Oriented Data Model Examples: MongoDB, CouchDB Querying and Indexing Documents Column-Family Stores Data Model and Architecture Examples: Apache Cassandra, HBase Column Families and Wide-Column Data Storage Querying and Performance Tuning		
UNIT-III	**Data Processing and Management**	**08 Hours**
4.1. Data Cleaning and Preparation Importance of Data Cleaning Impact on Analysis Accuracy Data Quality Issues Data Cleaning Techniques Removing Duplicates Handling Missing Values Detecting and Correcting Errors Data Standardization and Normalization Data Preparation Data Integration and Fusion Feature Engineering Splitting Data into Training and Testing Sets Dealing with Imbalanced Datasets 4.2. Data Transformation Techniques Overview of Data Transformation Objectives and Benefits ETL (Extract, Transform, Load) Process Common Data Transformation Methods Aggregation and Summarization Filtering and Subsetting Data Enrichment Advanced Transformation Techniques Data Binning Encoding Categorical Variables Dimensionality Reduction (e.g., PCA) Text and Time Series Data Transformations 4.3. Workflow Orchestration Tools (Apache Oozie, Airflow)		

- Introduction to Workflow Orchestration
  - Importance in Big Data Processing
  - Key Features of Orchestration Tools
- Apache Oozie
  - Overview and Architecture
  - Defining Workflows and Coordinators
  - Job Scheduling and Dependency Management
  - Oozie Workflow Language
- Apache Airflow
  - Overview and Architecture
  - Creating and Managing Workflows with DAGs (Directed Acyclic Graphs)
  - Task Scheduling, Execution, and Monitoring
  - Integrating Airflow with Hadoop, Spark, and Other Big Data Tools

4.4. Batch Processing with Hadoop and Spark

- Introduction to Batch Processing
  - Batch vs. Real-Time Processing
  - Use Cases for Batch Processing
- Batch Processing with Hadoop
  - MapReduce Programming Model
    - Job Configuration
    - Map and Reduce Functions
  - Workflow Execution with YARN
  - Performance Optimization Techniques
- Batch Processing with Spark
  - RDDs (Resilient Distributed Datasets) and DataFrames
  - Spark SQL and DataFrame API
  - Spark Core: Transformations and Actions
  - Optimizing Spark Jobs
  - Integrating Spark with Hive, HBase, and Other Data Sources

4.5. Real-Time Processing with Apache Storm and Kafka

- Introduction to Real-Time Processing
  - Importance and Applications
  - Challenges in Real-Time Data Processing
- Real-Time Processing with Apache Storm
  - Storm Architecture and Components (Spouts, Bolts, Topologies)
  - Building and Deploying Storm Topologies
  - Fault Tolerance and Reliability in Storm
  - Integrating Storm with Hadoop and HDFS
- Real-Time Data Streaming with Apache Kafka
  - Kafka Architecture (Producers, Consumers, Brokers, Topics)
  - Kafka Streams API for Stream Processing
  - Building Real-Time Data Pipelines with Kafka Connect

Ensuring Data Durability and Scalability with Kafka Integrating Kafka with Spark Streaming and Storm		
UNIT-IV	**Query Languages and Tools, Advanced Analytical Methods**	**07 Hours**
5.1. Introduction to HiveQL Overview of Hive Apache Hive Architecture Key Features and Use Cases HiveQL Basics Data Definition Language (DDL) CREATE, ALTER, DROP Statements Data Manipulation Language (DML) SELECT, INSERT, UPDATE, DELETE Statements Data Types and Storage Formats Primitive and Complex Data Types File Formats (Text, Parquet, ORC) 5.2. Querying Big Data with Hive Writing Hive Queries Basic Query Structure Joins and Unions Subqueries and Nested Queries Aggregation Functions and Grouping Hive Functions Built-in Functions String Functions Date Functions Mathematical Functions User-Defined Functions (UDFs) Creating and Using UDFs Hive Query Performance Optimization Partitioning and Bucketing Indexing and Query Caching Cost-Based Optimization (CBO) Integrating Hive with Other Tools Hive and HDFS Hive and Spark SQL Hive and HBase 5.3. Data Processing with Pig Introduction to Pig Pig Architecture and Execution Modes		

- Pig Latin Language Overview
- Writing Pig Scripts
  - Loading and Storing Data
  - Relational Operators (FOREACH, FILTER, JOIN, GROUP, ORDER BY)
  - User-Defined Functions (UDFs) in Pig
- Pig Data Types and Schemas
  - Scalar and Complex Data Types
  - Defining and Using Schemas
- Advanced Pig Techniques
  - Data Transformations and Aggregations
  - Error Handling and Debugging Pig Scripts
  - Optimizing Pig Performance
- Pig and Hadoop Integration
  - Running Pig on Hadoop
  - Pig and HCatalog Integration

5.4. Using JAQL for Data Analysis

- Introduction to JAQL
  - Overview and Architecture
  - Key Features and Use Cases
- JAQL Syntax and Data Model
  - Basic Syntax and Expressions
  - Handling JSON Data in JAQL
- Writing JAQL Queries
  - Data Loading and Transformation
  - Filtering, Mapping, and Reducing Data
  - Aggregations and Grouping
- Advanced JAQL Techniques
  - Joins and Nested Queries
  - User-Defined Functions in JAQL
  - Error Handling and Debugging
- Integrating JAQL with Hadoop Ecosystem
  - Running JAQL on Hadoop
  - JAQL and HDFS Integration
  - JAQL with Hive and HBase

5.5. Advanced Query Optimization Techniques

- Importance of Query Optimization
  - Impact on Performance and Resource Utilization
- Optimization Strategies in Hive
  - Cost-Based Optimization (CBO)
  - Partition Pruning
  - Predicate Pushdown
  - Vectorized Query Execution

- Optimizing Pig Scripts
  - Understanding Logical and Physical Plans
  - Combiner Optimization
  - Multi-query Execution
  - Schema and Type Optimization
- General Optimization Techniques
  - Data Partitioning and Distribution
  - Indexing and Caching Strategies
  - Efficient Join and Aggregation Techniques
- Monitoring and Tuning Query Performance
  - Analyzing Query Execution Plans
  - Resource Allocation and Management
  - Using Tools for Performance Monitoring (Ganglia, Nagios, Ambari)

6.1. Introduction to Machine Learning on Big Data
- Overview of Machine Learning Concepts
  - Supervised Learning
  - Unsupervised Learning
  - Reinforcement Learning
- Machine Learning Challenges with Big Data
  - Scalability
  - Data Quality and Preprocessing
  - Model Training and Evaluation
- Big Data Machine Learning Platforms
  - Apache Spark MLlib
  - Hadoop Mahout
  - TensorFlow on Big Data

6.2. Implementing Algorithms with Spark MLlib
- Overview of Spark MLlib
  - Key Features and Architecture
  - MLlib vs. ML Pipeline API
- Data Preparation for MLlib
  - Loading and Transforming Data
  - Feature Engineering and Selection
- Supervised Learning Algorithms
  - Linear Regression
  - Logistic Regression
  - Decision Trees and Random Forests
  - Support Vector Machines
- Unsupervised Learning Algorithms
  - K-means Clustering
  - Principal Component Analysis (PCA)

- Singular Value Decomposition (SVD)
- Model Evaluation and Tuning
  - Cross-Validation
  - Hyperparameter Tuning
  - Metrics for Model Performance

6.3. Scalable Analytics with R and Python

- Introduction to Scalable Analytics
  - Benefits and Challenges
  - Key Concepts and Approaches
- Using R for Big Data Analytics
  - R Packages for Big Data (e.g., dplyr, data.table)
  - Integrating R with Hadoop and Spark
  - Parallel Processing in R
- Using Python for Big Data Analytics
  - Python Libraries for Big Data (e.g., NumPy, Pandas, Dask)
  - Integrating Python with Hadoop and Spark
  - Parallel Processing in Python with Dask and PySpark
- Comparative Analysis
  - R vs. Python for Big Data
  - Selecting the Right Tool for the Task

6.4. Predictive Analytics and Modeling

- Overview of Predictive Analytics
  - Concepts and Applications
  - Workflow for Predictive Modeling
- Building Predictive Models
  - Data Collection and Preparation
  - Feature Engineering
  - Model Selection and Training
- Techniques in Predictive Analytics
  - Regression Analysis
  - Classification Techniques
    - Decision Trees, Random Forests, Gradient Boosting
  - Time Series Analysis and Forecasting
- Model Evaluation and Deployment
  - Evaluating Model Performance
  - Model Deployment Strategies
  - Monitoring and Updating Models

6.5. Case Studies on Advanced Analytics

- Case Study 1: Predictive Maintenance
  - Problem Definition and Data Collection
  - Model Building and Evaluation
  - Implementation and Results

Case Study 2: Customer Segmentation Data Preparation and Feature Selection Clustering Analysis Business Implications and Actionable Insights Case Study 3: Fraud Detection Identifying Fraudulent Patterns Model Training and Validation Real-time Fraud Detection System Case Study 4: Recommendation Systems Collaborative Filtering Techniques Building a Recommendation Model Evaluation and Tuning Lessons Learned and Best Practices Key Takeaways from Case Studies Common Pitfalls and How to Avoid Them Future Trends in Big Data Analytics		
UNIT-V	**Data Visualization and Interpretation, Stream Processing and Real-Time Analytics**	**08 Hours**
7.1. Principles of Data Visualization Basics of Data Visualization Definition and Importance History and Evolution of Data Visualization Key Principles Clarity and Simplicity Accuracy and Integrity Context and Relevance Types of Data Visualizations Charts: Bar, Line, Pie, Scatter Plots Graphs and Networks Maps and Geospatial Visualizations Dashboards and Interactive Visualizations Design Best Practices Choosing the Right Visualization Type Color Theory and Usage Layout and Design Principles Avoiding Common Pitfalls 7.2. Tools for Big Data Visualization (Tableau, D3.js) Overview of Visualization Tools Criteria for Selecting Visualization Tools Tableau Introduction to Tableau		

- Key Features and Capabilities
- Data Connection and Preparation
- Creating Basic Visualizations
- Advanced Tableau Features (Calculated Fields, Parameters, Tableau Prep)

D3.js
- Introduction to D3.js
- Key Features and Capabilities
- Basic D3.js Syntax and Structure
- Creating Simple Visualizations with D3.js
- Advanced D3.js Techniques (Transitions, Animations, Interactivity)

Comparison and Integration
- Integrating Tableau and D3.js with Big Data Platforms
- Strengths and Weaknesses of Each Tool

7.3. Creating Effective Visualizations

Visualization Process
- Data Preparation and Cleaning
- Identifying Key Insights and Messages
- Selecting Appropriate Visualization Techniques

Design and Implementation
- Designing for Your Audience
- Using Visual Elements to Enhance Understanding
- Interactive vs. Static Visualizations

Enhancing Visual Appeal
- Using Color and Fonts Effectively
- Incorporating Branding Elements
- Creating Storyboards and Prototypes

Testing and Refinement
- User Testing and Feedback
- Iterative Design and Improvement
- Ensuring Accessibility and Inclusivity

7.4. Interpreting and Communicating Insights

Interpreting Data Visualizations
- Analyzing Patterns, Trends, and Outliers
- Drawing Conclusions from Visual Data
- Identifying Misleading Visualizations

Storytelling with Data
- Crafting a Narrative
- Structuring the Presentation of Insights
- Using Visuals to Support Your Story

Effective Communication Strategies
- Tailoring Communication to Different Audiences
- Combining Visuals with Verbal and Written Explanations

Persuasive Presentation Techniques
Documentation and Reporting
Creating Comprehensive Reports
Including Visualizations in Reports and Presentations
Ensuring Reproducibility and Transparency
7.5. Real-World Visualization Examples
Case Study 1: Business Analytics
Visualizing Sales and Revenue Data
Creating Dashboards for Business Intelligence
Interactive Visualizations for Decision Support
Case Study 2: Healthcare Analytics
Visualizing Patient Data and Health Outcomes
Mapping Disease Spread and Epidemiological Data
Using Visualizations for Public Health Communication
Case Study 3: Social Media Analytics
Analyzing Social Media Trends and Sentiments
Network Visualizations of Social Connections
Interactive Visuals for Social Media Engagement
Case Study 4: Environmental Data
Visualizing Climate Data and Environmental Changes
Geospatial Visualizations of Environmental Impact
Using Visualizations for Environmental Policy and Advocacy

8.1. Introduction to Stream Processing
Concepts and Definitions
Definition of Stream Processing
Differences Between Stream Processing and Batch Processing
Key Terminologies: Streams, Events, Windows
Importance of Stream Processing
Applications and Benefits
Challenges in Stream Processing
Stream Processing Models
Event-driven Architecture
Data Flow Models
Windowing and Aggregation Techniques
Stream Processing vs. Real-Time Analytics
Understanding the Need for Real-Time Data
Use Cases in Various Industries (Finance, Healthcare, IoT)
8.2. Tools for Real-Time Data Processing (Apache Flume, Kafka)
Apache Flume
Overview and Architecture
Key Components: Sources, Channels, Sinks

- Data Flow in Flume
- Configuring and Deploying Flume Agents
- Use Cases and Examples
- Apache Kafka
  - Overview and Architecture
  - Key Concepts: Producers, Consumers, Topics, Partitions, Brokers
  - Kafka Streams API
  - Setting Up a Kafka Cluster
  - Data Ingestion and Processing with Kafka
  - Use Cases and Examples
- Comparison of Flume and Kafka
  - Strengths and Weaknesses
  - Use Case Scenarios for Each Tool

8.3. Building Stream Processing Pipelines

- Pipeline Architecture
  - Designing a Stream Processing Pipeline
  - Key Components and Their Roles
  - Integrating Various Tools and Technologies
- Data Ingestion
  - Techniques for Capturing Real-Time Data
  - Integrating Data Sources with Flume and Kafka
  - Handling Data Streams at Scale
- Stream Processing Frameworks
  - Introduction to Apache Storm
    - Architecture and Key Concepts
    - Setting Up and Configuring Storm Clusters
    - Developing Storm Topologies
  - Introduction to Apache Samza
    - Architecture and Key Concepts
    - Setting Up and Configuring Samza Jobs
    - Developing Samza Applications
  - Introduction to Apache Flink
    - Architecture and Key Concepts
    - Setting Up and Configuring Flink Clusters
    - Developing Flink Applications
- Data Transformation and Enrichment
  - Applying Transformations on Streaming Data
  - Joining and Aggregating Data Streams
  - Handling Out-of-Order and Late Arriving Data
- Monitoring and Managing Pipelines
  - Tools for Monitoring Stream Processing
  - Troubleshooting and Debugging Pipelines

Scaling Stream Processing Applications 8.4. Real-Time Analytics Use Cases Financial Services Fraud Detection and Prevention Real-Time Trading Analytics Customer Behavior Analysis Healthcare Real-Time Patient Monitoring Predictive Analytics for Emergency Response Disease Outbreak Tracking E-Commerce and Retail Real-Time Inventory Management Personalized Recommendations Customer Sentiment Analysis Telecommunications Network Performance Monitoring Real-Time Customer Support Usage Pattern Analysis Internet of Things (IoT) Smart Cities and Infrastructure Real-Time Environmental Monitoring Predictive Maintenance Case Studies and Examples Detailed Analysis of Successful Real-Time Analytics Implementations Lessons Learned and Best Practices Future Trends in Real-Time Analytics		
UNIT-VI	**Case Studies and Applications**	**08 Hours**
9.1. Big Data Analytics in Business Overview of Big Data in Business Importance and Benefits of Big Data Analytics in Business Common Challenges and Solutions Customer Insights and Personalization Analyzing Customer Behavior and Preferences Personalized Marketing and Recommendation Systems Case Study: Netflix's Recommendation Algorithm Operational Efficiency and Optimization Supply Chain and Logistics Optimization Predictive Maintenance and Asset Management Case Study: UPS's ORION Routing System Financial Analytics		

- Risk Management and Fraud Detection
- Real-Time Financial Analytics and Trading
- Case Study: Credit Scoring Models in Banking
- Human Resources and Talent Management
  - People Analytics and Employee Retention Strategies
  - Workforce Planning and Optimization
  - Case Study: Google's People Operations

9.2. Big Data in Healthcare

- Overview of Big Data in Healthcare
  - Importance and Impact on Healthcare
  - Challenges and Ethical Considerations
- Electronic Health Records (EHR)
  - Data Integration and Management
  - Predictive Analytics for Patient Outcomes
  - Case Study: Predictive Analytics in Hospital Readmissions
- Personalized Medicine and Genomics
  - Analyzing Genomic Data for Personalized Treatments
  - Drug Discovery and Development
  - Case Study: IBM Watson in Oncology
- Public Health and Epidemiology
  - Tracking and Predicting Disease Outbreaks
  - Health Surveillance and Population Health Management
  - Case Study: Google Flu Trends
- Remote Monitoring and Telemedicine
  - Real-Time Patient Monitoring Systems
  - Big Data in Telehealth Services
  - Case Study: Remote Monitoring for Chronic Disease Management

9.3. Big Data for Social Media Analysis

- Overview of Big Data in Social Media
  - Importance and Applications in Social Media
  - Data Privacy and Ethical Issues
- Sentiment Analysis and Opinion Mining
  - Techniques for Analyzing Social Media Sentiments
  - Tools and Frameworks for Sentiment Analysis
  - Case Study: Sentiment Analysis for Brand Monitoring
- Influencer Identification and Network Analysis
  - Identifying Key Influencers and Opinion Leaders
  - Social Network Analysis Techniques
  - Case Study: Influencer Marketing Strategies
- Trend Analysis and Event Detection
  - Real-Time Trend Analysis on Social Media Platforms
  - Event Detection and Analysis

Case Study: Twitter Data Analysis for Event Detection
Campaign Performance and Social Media ROI
Measuring the Impact of Social Media Campaigns
Techniques for Calculating Social Media ROI
Case Study: Campaign Analysis for a Major Brand
9.4. Big Data in E-commerce
Overview of Big Data in E-commerce
Importance and Benefits for E-commerce Businesses
Key Challenges and Considerations
Customer Behavior and Personalization
Analyzing Browsing and Purchase Behavior
Personalized Recommendations and Marketing
Case Study: Amazon's Recommendation Engine
Inventory Management and Demand Forecasting
Optimizing Inventory Levels with Predictive Analytics
Demand Forecasting Techniques
Case Study: Walmart's Inventory Management System
Fraud Detection and Prevention
Techniques for Detecting and Preventing Fraudulent Activities
Real-Time Fraud Detection Systems
Case Study: PayPal's Fraud Detection Algorithm
Customer Retention and Churn Analysis
Analyzing Customer Churn and Retention Strategies
Implementing Loyalty Programs Based on Data Insights
Case Study: Customer Retention Strategies in E-commerce
9.5. Industry-Specific Case Studies
Big Data in Manufacturing
Predictive Maintenance and Quality Control
Supply Chain Optimization
Case Study: GE's Big Data Strategy for Predictive Maintenance
Big Data in Energy and Utilities
Smart Grids and Energy Management
Predictive Analytics for Equipment Maintenance
Case Study: Big Data for Energy Consumption Optimization
Big Data in Transportation and Logistics
Route Optimization and Fleet Management
Real-Time Traffic Analysis and Prediction
Case Study: DHL's Logistics Optimization with Big Data
Big Data in Telecommunications
Network Performance Monitoring and Optimization
Customer Experience and Churn Management
Case Study: Telecoms Using Big Data for Customer Retention

Big Data in Retail
Personalized Shopping Experiences and Targeted Promotions
Inventory and Supply Chain Management
Case Study: Big Data Analytics in a Major Retail Chain

TEXT BOOKS:

1. Big Data Analytics, Seema Acharya, Subhashini Chellappan, Wiley 2015.
2. R programming for beginners, sandhya arora, latesh malik, university press.

REFERENCE BOOKS:

1. chandramouli subramanian, Asha A Geroge, C R Rene Robin, big data analytics, University press.
2. Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Business, Michael Minelli, Michehe Chambers, 1st Edition, Ambiga Dhiraj, Wiley CIO Series, 2013.
3. Hadoop: The Definitive Guide, Tom White, 3rd Edition, O'Reilly Media, 2012.
4. Big Data Analytics: Disruptive Technologies for Changing the Game, Arvind Sathi, 1st Edition, IBM Corporation, 2012.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-401-L: BIG DATA ANALYTICS LAB
B.Tech. IV Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
P: - 2 Hours/Week	P: 01	Total : 30 Marks

Course Prerequisites: None

Course Objective:

- Provide knowledge of Big data Analytics principles and techniques.
- Designed to give an exposure of the frontiers of Big data Analytics

Learning Course Outcome:
After successful completion of the course, students will able to:

- Use Excel as an Analytical tool and visualization tool.
- Ability to program using HADOOP and Map reduce
- Ability to perform data analytics using ML in R.
- Use MongoDB to perform data analytics

Course Contents

List of Experiments	07 Hours

Introduction to Big Data
Lab 1. Exploring Big Data Sets
- Objective: Introduction to large datasets.
- Tools: Python, Pandas, Jupyter Notebook.
- Description: Load and explore a large dataset, summarize its key characteristics, and visualize basic statistics.

Big Data Technologies
Lab 2. Setting Up Hadoop Environment
- Objective: Install and configure Hadoop.
- Tools: Hadoop.
- Description: Set up a single-node Hadoop cluster and explore its basic functionalities.

Lab 3. Working with HDFS
- Objective: Understand HDFS.
- Tools: Hadoop HDFS.
- Description: Perform file operations on HDFS (upload, download, delete files).

Lab 4. MapReduce Programming
- Objective: Write a simple MapReduce program.
- Tools: Hadoop MapReduce, Java.
- Description: Implement a word count example using MapReduce.

Lab 5. Introduction to Spark
- Objective: Get started with Apache Spark.

- Tools: Apache Spark, PySpark.
- Description: Perform basic operations and transformations using Spark.

Lab 6. Comparison of Hadoop and Spark
- Objective: Compare performance between Hadoop and Spark.
- Tools: Hadoop, Spark.
- Description: Implement a simple data processing task and compare execution times.

Data Collection and Storage

Lab 7. Data Collection with Apache Flume
- Objective: Use Flume to collect data.
- Tools: Apache Flume.
- Description: Set up a Flume agent to collect and transport log data.

Lab 8. Data Ingestion with Sqoop
- Objective: Import data from SQL to Hadoop.
- Tools: Apache Sqoop.
- Description: Use Sqoop to import data from MySQL to HDFS.

Lab 9. Setting Up NoSQL Database (MongoDB)
- Objective: Install and use a NoSQL database.
- Tools: MongoDB.
- Description: Install MongoDB, create collections, and perform CRUD operations.

Lab 10. Working with Key-Value Stores
- Objective: Understand key-value stores.
- Tools: Redis.
- Description: Set up Redis, store and retrieve key-value pairs.

Data Processing and Management

Lab 11. Data Cleaning with Pandas
- Objective: Clean and preprocess data.
- Tools: Python, Pandas.
- Description: Perform data cleaning operations such as handling missing values and data normalization.

Lab 12. Data Transformation with Spark
- Objective: Transform data using Spark.
- Tools: Apache Spark, PySpark.
- Description: Perform various data transformation tasks using Spark.

Lab 13. Workflow Orchestration with Apache Airflow
- Objective: Automate data workflows.
- Tools: Apache Airflow.
- Description: Create and schedule a data processing pipeline using Airflow.

Lab 14. Batch Processing with Hadoop
- Objective: Implement batch processing.
- Tools: Hadoop, MapReduce.
- Description: Perform a batch processing task using MapReduce.

Lab 15. Real-Time Processing with Kafka

- Objective: Implement real-time data processing.
- Tools: Apache Kafka, Apache Storm.
- Description: Set up Kafka and Storm to process real-time data streams.

Query Languages and Tools

Lab 16. Querying Big Data with Hive
- Objective: Use Hive for data queries.
- Tools: Apache Hive.
- Description: Write and execute HiveQL queries.

Lab 17. Data Processing with Pig
- Objective: Use Pig for data processing.
- Tools: Apache Pig.
- Description: Write Pig scripts to process data.

Lab 18. Data Analysis with JAQL
- Objective: Analyze data with JAQL.
- Tools: JAQL.
- Description: Write JAQL queries for data analysis.

Lab 19. Advanced Query Optimization in Hive
- Objective: Optimize Hive queries.
- Tools: Apache Hive.
- Description: Implement and test various query optimization techniques in Hive.

Advanced Analytical Methods

Lab 20. Machine Learning with Spark MLlib
- Objective: Implement ML algorithms.
- Tools: Apache Spark MLlib.
- Description: Build and evaluate a machine learning model using Spark MLlib.

Lab 21. Scalable Analytics with R
- Objective: Perform analytics using R.
- Tools: R, SparkR.
- Description: Use SparkR for scalable data analysis.

Lab 22. Predictive Analytics with Python
- Objective: Implement predictive models.
- Tools: Python, scikit-learn.
- Description: Build and evaluate predictive models on large datasets.

Lab 23. Case Study Analysis
- Objective: Analyze real-world data.
- Tools: Various.
- Description: Perform a comprehensive analysis of a case study involving advanced analytics.

Data Visualization and Interpretation

Lab 24. Data Visualization with Tableau
- Objective: Create visualizations.
- Tools: Tableau.
- Description: Use Tableau to create and interpret data visualizations.

Lab 25. Data Visualization with D3.js
- Objective: Visualize data using D3.js.
- Tools: D3.js.
- Description: Create interactive data visualizations using D3.js.

Lab 26. Interpreting Data Insights
- Objective: Communicate insights effectively.
- Tools: Various visualization tools.
- Description: Create a report or presentation to communicate data insights.

Stream Processing and Real-Time Analytics

Lab 27. Building Stream Processing Pipelines
- Objective: Implement a stream processing pipeline.
- Tools: Apache Kafka, Apache Flume.
- Description: Build a stream processing pipeline to process real-time data.

TEXT BOOKS:

1. Big Data Analytics, Seema Acharya, Subhashini Chellappan, Wiley 2015.
2. Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Business, Michael Minelli, Michehe Chambers, 1st Edition, Ambiga Dhiraj, Wiely CIO Series, 2013.
3. Hadoop: The Definitive Guide, Tom White, 3rd Edition, O"Reilly Media, 2012.
4. Big Data Analytics: Disruptive Technologies for Changing the Game, Arvind Sathi, 1st Edition, IBM Corporation, 2012.

REFERENCE BOOKS:

1. Big Data and Business Analytics, Jay Liebowitz, Auerbach Publications, CRC press (2013)
2. Using R to Unlock the Value of Big Data: Big Data Analytics with Oracle R Enterprise and Oracle R Connector for Hadoop, Tom Plunkett, Mark Hornick, McGraw-Hill/Osborne Media (2013), Oracle press.
3. Professional Hadoop Solutions, Boris lublinsky, Kevin t. Smith, Alexey Yakubovich, Wiley, ISBN: 9788126551071, 2015.
4. Understanding Big data, Chris Eaton, Dirk deroos et al, McGraw Hill, 2012.
5. Intelligent Data Analysis, Michael Berthold, David J. Hand, Springer, 2007.
6. Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics, Bill Franks, 1st Edition, Wiley and SAS Business Series, 2012.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-402-PCC: WEB AND SOCIAL MEDIA ANALYTICS
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Exposure to various web and social media analytic techniques.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Knowledge on decision support systems
- Apply natural language processing concepts on text analytics
- Understand sentiment analysis
- Knowledge on search engine optimization and web analytics

Course Contents

UNIT-I	An Overview of Business Intelligence, Analytics, and Decision Support	07 Hours
Analytics to Manage a Vaccine Supply Chain Effectively and Safely, Changing Business Environments and Computerized Decision Support, Information Systems Support for Decision Making, The Concept of Decision Support Systems (DSS), Business Analytics Overview, Brief Introduction to Big Data Analytics		
UNIT-II	**Text Analytics and Text Mining**	**08 Hours**
Machine Versus Men on Jeopardy: The Story of Watson, Text Analytics and Text Mining Concepts and Definitions, Natural Language Processing, Text Mining Applications, Text Mining Process, Text Mining Tools		
UNIT-III	**Sentiment Analysis**	**08 Hours**
Sentiment Analysis Overview, Sentiment Analysis Applications, Sentiment Analysis Process, Sentiment Analysis and Speech Analytics		
UNIT-IV	**Web Analytics, Web Mining**	**07 Hours**
Security First Insurance Deepens Connection with Policyholders, Web Mining Overview, Web Content and Web Structure Mining, Search Engines, Search Engine Optimization, Web Usage Mining (Web Analytics), Web Analytics Maturity Model and Web Analytics Tools		

UNIT-V	Social Analytics and Social Network Analysis	08 Hours

Social Analytics and Social Network Analysis, Social Media Definitions and Concepts, Social Media Analytics, Prescriptive Analytics - Optimization and Multi-Criteria Systems: Multiple Goals, Sensitivity Analysis, What-If Analysis, and Goal Seeking

TEXT BOOK:

1. Ramesh Sharda, Dursun Delen, Efraim Turban, Business Intelligence and Analytics: Systems for Decision Support, Pearson Education

REFERENCE BOOKS:

1. Rajiv Sabherwal, Irma Becerra- Fernandez," Business Intelligence– Practice, Technologies and Management", John Wiley 2011.
2. Lariss T. Moss, Shaku Atre, "Business Intelligence Roadmap", Addison-Wesley It Service.
3. Yuli Vasiliev, "Oracle Business Intelligence: The Condensed Guide to Analysis and Reporting", SPD Shroff, 2012.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-404-MDM(A): BLOCKCHAIN SCALABILITY AND OPTIMIZATION
(MDM Basket – 1)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To understand the challenges of blockchain scalability and the impact of increasing transaction loads.
2. To explore consensus mechanisms and optimization techniques for enhancing blockchain performance.
3. To introduce scalability solutions like sharding, sidechains, and layer-2 protocols.
4. To develop skills for designing and implementing scalable blockchain systems.
5. To analyze real-world use cases and trends in blockchain scalability research and development.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the fundamental scalability issues and challenges in blockchain systems.
CO2: Analyze and compare various consensus mechanisms and their scalability impacts.
CO3: Apply scalability solutions like sharding, sidechains, and layer-2 protocols to optimize blockchain systems.
CO4: Design scalable architectures for blockchain networks considering performance trade-offs.
CO5: Evaluate the performance of blockchain systems using metrics like throughput, latency, and energy efficiency.

Course Contents

UNIT-I	Introduction to Blockchain Scalability	07 Hours

- Overview of Blockchain Architecture and Challenges
- Scalability Metrics: Throughput, Latency, and Transaction Per Second (TPS)
- Trade-offs in Blockchain Systems: Scalability, Security, and Decentralization (Blockchain Trilemma)
- Case Studies of Scalability Challenges in Popular Blockchains (Bitcoin, Ethereum, etc.)

UNIT-II	Consensus Mechanisms and Their Scalability	08 Hours

- Proof-of-Work (PoW) and Proof-of-Stake (PoS): Limitations and Scalability
- Delegated Proof-of-Stake (DPoS) and Byzantine Fault Tolerance (BFT) Mechanisms
- Energy-Efficient Consensus Protocols: Proof-of-Authority (PoA), Proof-of-History (PoH)
- Comparative Analysis of Consensus Mechanisms and Their Optimization Techniques

UNIT-III	Sharding and Partitioning	08 Hours
• Concepts of Sharding: State Sharding, Network Sharding, and Transaction Sharding • Design Challenges in Sharded Blockchains • Cross-Shard Communication Protocols and Their Implementation • Case Study: Ethereum 2.0 and Zilliqa		
UNIT-IV	**Layer-2 Solutions**	**07 Hours**
• Introduction to Layer-2 Scaling Solutions • Payment Channels and State Channels: Lightning Network and Raiden Network • Rollups: Optimistic Rollups and zk-Rollups • Sidechains: Architecture, Benefits, and Examples (Polygon, Liquid Network)		
UNIT-V	**Blockchain Optimization Techniques**	**08 Hours**
• Block Compression Techniques and Data Pruning • Adaptive Block Sizes and Transaction Fee Mechanisms • Gossip Protocol Optimization for Network Scalability • Optimizing Storage Requirements with Directed Acyclic Graphs (DAGs)		
UNIT-V	**Trends and Emerging Solutions**	**03 Hours**
• Cross-Chain Interoperability and Scalability • Decentralized Autonomous Organizations (DAOs) and Their Scalability Needs • Emerging Solutions: Hydra, Avalanche Subnets, and Blockchain 4.0 • Research Directions in Blockchain Scalability and Optimization		

Reference Books

1. **Mastering Blockchain** by Imran Bashir
2. **Blockchain Basics: A Non-Technical Introduction in 25 Steps** by Daniel Drescher
3. **Blockchain Scalability and Its Foundations in Distributed Systems** by Joseph Bonneau et al.
4. **Building Scalable Blockchain Applications** by Kassim Dandashly
5. Research Articles and Online Resources: IEEE Xplore, arXiv, SpringerLink

CO's-PO's & PSO's MAPPING

CO PO	CO	CO P	CO PC	CO PO	CO PO1	CO PO 1 PC	CO PO1 PO2	CO PO1
CO1 3	**CO1**	**CO1** 3	**CO1** 3	CO1 3	**CO1** 3	**CO1** 3 2	**CO1** 3 2	**CO1** 3
CO2 3	**CO2**	**CO2** 3	**CO2** 3	**CO2** 3	**CO2** 3	CO2 3 3	CO2 3 3	**CO2** 3
CO3 3	CO3	CO3 3	CO3 3	**CO3** 3	CO3 3	**CO3** 3 3	**CO3** 3 3	**CO3** 3
CO4 3	CO4	CO4 3	**CO4** 3	**CO4** 3	CO4 3	**CO4** 3 3	**CO4** 3 3	**CO4** 3
CO5 3	CO5	**CO5** 3	CO5 3	CO5 3	**CO5** 3	**CO5** 3 3	CO5 3 3	**CO5** 3
Avg. 3	Avg.	Avg. 3	**Avg.** 3	Avg. 3	Avg. 3	Avg. 3 2.8	Avg. 3 2.8	**Avg.** 3

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-404-MDM(B): CYBERSECURITY IN CLOUD AND IOT SYSTEMS
(MDM Basket – 2)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To understand the unique security challenges in cloud computing and IoT systems.
2. To study and apply various security protocols for cloud and IoT environments.
3. To analyze the risks, threats, and vulnerabilities specific to cloud services and IoT networks.
4. To explore best practices and tools for ensuring data privacy and secure communication in distributed systems.
5. To evaluate case studies and design secure architectures for cloud and IoT systems.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand the fundamentals of cybersecurity and their relevance to cloud and IoT systems.
CO2: Identify and analyze potential threats and vulnerabilities in cloud and IoT environments.
CO3: Implement security protocols to ensure secure data storage, processing, and transmission.
CO4: Design and evaluate secure architectures for cloud and IoT systems.
CO5: Apply best practices, tools, and frameworks for real-world cybersecurity challenges in cloud and IoT systems.

Course Contents

UNIT-I	Introduction to Cybersecurity in Cloud and IoT	07 Hours

- Basics of Cybersecurity: CIA Triad (Confidentiality, Integrity, Availability)
- Overview of Cloud Computing and IoT Systems
- Security Challenges and Opportunities in Cloud and IoT Environments
- Case Studies of Major Security Breaches

UNIT-II	Cloud Security Fundamentals	08 Hours

- Cloud Security Architecture and Models (IaaS, PaaS, SaaS)
- Identity and Access Management (IAM) in Cloud
- Data Security in Cloud: Encryption, Masking, and Tokenization
- Security in Virtualization and Multi-Tenancy Environments
- Cloud Security Alliance (CSA) Best Practices

UNIT-III	IoT Security Fundamentals	08 Hours

- IoT Network Architecture and Protocols (MQTT, CoAP, Zigbee)
- IoT Device Authentication and Identity Management
- Securing IoT Gateways and Edge Devices
- Privacy Concerns and Solutions in IoT Networks
- Lightweight Cryptographic Techniques for IoT

UNIT-IV	Threats, Vulnerabilities, and Risk Management	07 Hours

- Common Attacks in Cloud and IoT: DDoS, Man-in-the-Middle, Ransomware
- Vulnerabilities in IoT Devices and Cloud Systems
- Threat Modeling and Risk Assessment Frameworks
- Incident Detection, Response, and Recovery

UNIT-V	Security Protocols and Tools	08 Hours

- Public Key Infrastructure (PKI) for Cloud and IoT
- Blockchain-Based Security Solutions for IoT Networks
- Intrusion Detection and Prevention Systems (IDPS)
- Tools for Cloud Security: AWS Shield, Azure Security Center
- IoT Security Tools: OWASP IoT Framework, Shodan

UNIT-V	Emerging Trends and Case Studies	03 Hours

- AI and Machine Learning for Threat Detection
- Zero Trust Security in Cloud and IoT Systems
- 5G Security and Its Implications for IoT
- Case Studies: Securing Smart Cities, Industrial IoT, and Healthcare Systems
- Research Directions in Cloud and IoT Security

Reference Books

1. **Cybersecurity and Privacy in IoT** by Pethuru Raj and Anupama C. Raman
2. **Cloud Security: A Comprehensive Guide** by Ronald L. Krutz and Russell Dean Vines
3. **IoT Security: Advances in Authentication** by Shancang Li and Li Da Xu
4. **Cloud and IoT-Based Vehicular Ad Hoc Networks** by Rohit Sharma and Sheetal Kalra
5. Research Articles and Online Resources: IEEE Xplore, NIST Cybersecurity Framework, and OWASP

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	2	-	-	-	-	-	-	2	2	-	-
CO2	3	3	3	2	3	2	-	-	-	-	-	2	2	1	-
CO3	3	3	3	3	3	2	-	-	-	-	-	2	2	2	1
CO4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO5	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.4	2.2	2.8	1.6	0.4	-	-	-	-	2	2	1.4	0.6

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-404-MDM(c): IMAGE PROCESSING FOR REMOTE SENSING APPLICATIONS
(MDM Basket – 3)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To understand the fundamental principles of image processing and their application in remote sensing.
2. To familiarize students with remote sensing data acquisition, interpretation, and processing techniques.
3. To explore the use of various image enhancement, classification, and feature extraction methods for remote sensing applications.
4. To develop skills in applying image processing techniques for environmental monitoring, urban planning, and natural resource management.
5. To analyze real-world remote sensing datasets using modern tools and algorithms.

Learning Course Outcome:

After successful completion of the course, students will able to:

CO1: Understand the basics of remote sensing and image processing.
CO2: Preprocess remote sensing images for analysis and interpretation.
CO3: Apply image enhancement and classification techniques to remote sensing datasets.
CO4: Extract meaningful features and interpret spatial patterns for various applications.
CO5: Use advanced algorithms and tools to process and analyze remote sensing data.

Course Contents

UNIT-I	Fundamentals of Remote Sensing and Image Processing	03 Hours

- Principles of Remote Sensing: Electromagnetic Spectrum, Platforms, and Sensors
- Types of Remote Sensing Data: Multispectral, Hyperspectral, Radar, and LiDAR
- Basics of Digital Image Processing
- Applications of Remote Sensing in Environmental and Resource Management

UNIT-II	Image Preprocessing for Remote Sensing	04 Hours

- Radiometric and Geometric Corrections
- Noise Reduction Techniques
- Atmospheric Correction Methods
- Image Resampling and Reprojection

UNIT-III	Image Enhancement and Transformation	03 Hours

- Contrast Stretching and Histogram Equalization
- Spatial Filtering: Smoothing and Sharpening Filters
- Principal Component Analysis (PCA) for Dimensionality Reduction
- Vegetation Indices (NDVI, EVI) and Other Spectral Indices

UNIT-IV	Image Classification and Feature Extraction	03 Hours

- Supervised and Unsupervised Classification Techniques
- Support Vector Machines (SVM) and Random Forest for Classification
- Object-Based Image Analysis (OBIA)
- Feature Selection and Extraction Techniques for Remote Sensing

UNIT-V	Security Protocols and Tools	03 Hours

- Techniques for Change Detection: Post-Classification Comparison, Image Differencing
- Applications in Monitoring Urban Growth, Deforestation, and Disaster Management
- Accuracy Assessment and Validation Methods
- Case Studies of Change Detection in Remote Sensing

UNIT-VI	Advanced Tools and Applications	03 Hours

- Google Earth Engine for Remote Sensing Analysis
- Cloud Computing for Big Remote Sensing Data
- AI and Machine Learning for Remote Sensing Applications
- Case Studies: Agriculture, Climate Change, and Urban Planning

Reference Books

1. **Remote Sensing and Image Interpretation** by Thomas Lillesand, Ralph W. Kiefer, and Jonathan Chipman
2. **Introduction to Remote Sensing** by James B. Campbell and Randolph H. Wynne
3. **Digital Image Processing** by Rafael C. Gonzalez and Richard E. Woods
4. **Remote Sensing Digital Image Analysis** by John A. Richards
5. **Hyperspectral Remote Sensing: Principles and Applications** by Marcus Borengasser, William S. Hungate, and Russell Watkins

CO's-PO's & PSO's MAPPING

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 1	3	2	-	-	-	-	-	-	-	-	-	2	2	-	-
CO 2	3	3	2	2	3	-	-	-	-	-	-	2	2	1	-
CO 3	3	3	3	3	3	2	-	-	-	-	-	2	2	2	1
CO 4	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
CO 5	3	3	3	3	3	2	1	-	-	-	-	2	2	2	1
Avg.	3	2.8	2.2	2.2	2.4	1.2	0.4	-	-	-	-	2	2	1.4	0.6

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(A): QUANTUM COMPUTING
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. To introduce the fundamentals of quantum computing
2. The problem-solving approach using finite dimensional mathematics

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand basics of quantum computing
- Understand physical implementation of Qubit
- Understand Quantum algorithms and their implementation
- Understand The Impact of Quantum Computing on Cryptography

Course Contents

UNIT-I	History of Quantum Computing	07 Hours
Importance of Mathematics, Physics and Biology. Introduction to Quantum Computing: Bits Vs Qubits, Classical Vs Quantum logical operations		
UNIT-II	**Background Mathematics**	**08 Hours**
Basics of Linear Algebra, Hilbert space, Probabilities and measurements. Background Physics: Paul's exclusion Principle, Superposition, Entanglement and super-symmetry, density operators and correlation, basics of quantum mechanics, Measurements in bases other than computational basis. Background Biology: Basic concepts of Genomics and Proteomics (Central Dogma)		
UNIT-III	**Qubit**	**08 Hours**
Physical implementations of Qubit. Qubit as a quantum unit of information. The Bloch sphere Quantum Circuits: single qubit gates, multiple qubit gates, designing the quantum circuits. Bell states.		
UNIT-IV	**Quantum Algorithms**	**07 Hours**
Classical computation on quantum computers. Relationship between quantum and classical complexity classes. Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor's factorization		

algorithm, Grover's search algorithm.		
UNIT-V	**Noise and error correction**	**08 Hours**
Graph states and codes, Quantum error correction, fault-tolerant computation. Quantum Information and Cryptography: Comparison between classical and quantum information theory. Quantum Cryptography, Quantum teleportation		

TEXT BOOK:

1. Nielsen M. A., Quantum Computation and Quantum Information, Cambridge.

REFERENCE BOOKS:

1. Quantum Computing for Computer Scientists by Noson S. Yanofsky and Mirco A. Mannucci
2. Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol. I: Basic Concepts, Vol II.
3. Basic Tools and Special Topics, World Scientific. Pittenger A. O., An Introduction to Quantum Computing Algorithms.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(B): EXPERT SYSTEMS
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Understand the basic techniques of artificial intelligence.
- Understand the Non-monotonic reasoning and statistical reasoning

Learning Course Outcome:
After successful completion of the course, students will able to:

- Apply the basic techniques of artificial intelligence.
- Discuss the architecture of an expert system and its tools.
- Understand the importance of building an expert systems
- Understand various problems with an expert systems

Course Contents

UNIT-I	Introduction to AI programming languages	07 Hours
Blind search strategies, Breadth-first – Depth-first – Heuristic search techniques Hill Climbing – Best first – A Algorithms AO* algorithm – game trees, Min- max algorithms, game playing – Alpha-beta pruning.		
UNIT-II	**Knowledge representation issues predicate logic**	**08 Hours**
Logic programming Semantic nets- frames and inheritance, constraint propagation; Representing Knowledge using rules, Rules-based deduction systems.		
UNIT-III	**Introduction to Expert Systems**	**08 Hours**
Architecture of expert systems, Representation and organization of knowledge, Basics characteristics, and types of problems handled by expert systems.		
UNIT-IV	**Expert System Tools**	**07 Hours**
Techniques of knowledge representations in expert systems, knowledge engineering, system-building aids, support facilities, stages in the development of expert systems.		
UNIT-V	**Expert System**	**08 Hours**

Building an Expert System: Expert system development, Selection of the tool, Acquiring Knowledge, Building process.

Problems with Expert Systems: Difficulties, common pitfalls in planning, dealing with domain experts, difficulties during development.

TEXT BOOKS:

1. Elain Rich and Kevin Knight, "Artificial Intelligence", Tata McGraw-Hill, New Delhi,
2. Waterman D.A., "A Guide to Expert Systems", Addison Wesley Longman,

REFERENCE BOOKS:

1. Stuart Russel and other Peter Norvig, "Artificial Intelligence – A Modern Approach", Prentice- Hall.
2. Patrick Henry Winston, "Artificial Intelligence", Addison Wesley.
3. Patterson, Artificial Intelligence & Expert System, Prentice Hall India, 1999.
4. Hayes-Roth, Lenat, and Waterman: Building Expert Systems, Addison Wesley, Weiss S. M. and Kulikowski C.A., "A Practical Guide to Designing Expert Systems", Rowman & Allanheld, New Jersey.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(c): CLOUD COMPUTING
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

1. A course on "Computer Networks".
2. A course on "Operating System".

Course Objective:

- This course provides an insight into cloud computing
- Topics covered include- Cloud Computing Architecture, Deployment Models, Service Models, Technological Drivers for Cloud Computing, Networking for Cloud Computing and Security in Cloud Computing

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand different computing paradigms and potential of the paradigms and specifically cloud computing
- Understand cloud service types, cloud deployment models and technologies supporting and driving the cloud
- Acquire the knowledge of programming models for cloud and development of software application that runs the cloud and various services available from major cloud providers
- Understand the security concerns and issues in cloud computing
- Acquire the knowledge of advances in cloud computing.

Course Contents

UNIT-I	Cloud Computing Fundamentals	07 Hours
Computing Paradigms, Cloud Computing Fundamentals, Cloud Computing Architecture and Management		
UNIT-II	**Models**	**08 Hours**
Cloud Deployment Models, Cloud Service Models, Technological Drivers for Cloud Computing: SOA and Cloud, Multicore Technology, Web 2.0 and Web 3.0, Pervasive Computing, Operating System, Application Environment		
UNIT-III	**Virtualization**	**08 Hours**

Programming Models for Cloud Computing: MapReduce, Cloud Haskell, Software Development in Cloud		
UNIT-IV	**Networking for Cloud Computing**	**07 Hours**
Introduction, Overview of Data Center Environment, Networking Issues in Data Centers, Transport Layer Issues in DCNs, Cloud Service Providers		
UNIT-V	**Security in Cloud Computing**	**08 Hours**
Security in Cloud Computing, and Advanced Concepts in Cloud Computing		

TEXT BOOK:

1. Chandrasekaran, K. Essentials of cloud computing. CRC Press, 2014.

REFERENCE BOOKS:

1. Cloud Computing: Principles and Paradigms, Editors: Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Wiley, 2011
2. Enterprise Cloud Computing - Technology, Architecture, Applications, Gautam Shroff, Cambridge University Press, 2010
3. Cloud Computing Bible, Barrie Sosinsky, Wiley-India, 2010.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(D): GAME THEORY
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:
The course will explain in depth the standard equilibrium concepts (such as Nash equilibrium, Subgame-Perfect Nash Equilibrium, and others) in Game Theory.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the basic concepts of game theory and solutions
- Understand different types of equilibrium interpretations
- Understand and analyze knowledge and solution concepts
- Analyze extensive games with perfect information

Course Contents

UNIT-I	Introduction	07 Hours

Game Theory, Games and Solutions, Game Theory and the Theory of Competitive Equilibrium, Rational Behavior, The Steady State and Deductive Interpretations, Bounded Rationality Terminology and Notation

Nash Equilibrium- Strategic Games, Nash Equilibrium, Examples, Existence of a Nash Equilibrium, Strictly Competitive Games, Bayesian Games: Strategic Games with Imperfect Information

UNIT-II	Features	08 Hours

Mixed, Correlated, and Evolutionary Equilibrium -Mixed Strategy Nash Equilibrium, Interpretations of Mixed Strategy Nash Equilibrium, Correlated Equilibrium, Evolutionary Equilibrium. Rationalizability and Iterated Elimination of Dominated Actions- Rationalizability Iterated Elimination of Strictly Dominated Actions, Iterated Elimination of Weakly Dominated Actions

UNIT-III	Knowledge and Equilibrium	08 Hours

A Model of Knowledge Common Knowledge, Can People Agree to Disagree? Knowledge and Solution Concepts, The Electronic Mail Game

UNIT-IV	Extensive Games with Perfect Information	07 Hours
Extensive Games with Perfect Information, Subgame Perfect Equilibrium, Two Extensions of the Definition of a Game, The Interpretation of a Strategy, Two Notable Finite Horizon Games, Iterated Elimination of Weakly Dominated Strategies Bargaining Games -Bargaining and Game Theory, A Bargaining Game of Alternating Offers, Subgame Perfect Equilibrium, Variations and Extensions		
UNIT-V	**Repeated Games**	**08 Hours**
The Basic Idea Infinitely Repeated Games vs. Finitely Repeated Games, Infinitely Repeated Games: Definitions, Strategies as Machines, Trigger Strategies: Nash Folk Theorems, Punishing for a Limited Length of Time: A Perfect Folk Theorem for the Limit of Means Criterion, Punishing the Punisher: A Perfect Folk Theorem for the Overtaking Criterion, Rewarding Players Who Punish: A Perfect Folk Theorem for the Discounting Criterion, The Structure of Subgame Perfect Equilibria Under the Discounting Criterion, Finitely Repeated Game		

TEXT BOOKS:

1. A course in Game Theory, M. J. Osborne and A. Rubinstein, MIT Press.

REFERENCE BOOKS:

1. Game Theory, Roger Myerson, Harvard University Press.
2. Game Theory, D. Fudenberg and J. Tirole, MIT Press.
3. Theory of Games and Economic Behavior, J. von Neumann and O. Morgenstern, New York: John Wiley and Sons.
4. Games and Decisions, R.D. Luce and H. Raiffa, New York: John Wiley and Sons.
5. Game Theory, G. Owen, 2nd Edition, New York: Academic Press.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(e): KNOWLEDGE REPRESENTATION AND REASONING
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Verbal Communication

Course Objective:

- To investigate the key concepts of Knowledge Representation (KR) techniques and different notations.
- To integrate the KR view as a knowledge engineering approach to model organizational knowledge.
- To introduce the study of ontologies as a KR paradigm and applications of ontologies.
- To understand various KR techniques and process, knowledge acquisition and sharing of ontology.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Analyze and design knowledge-based systems intended for computer implementation.
- Acquire theoretical knowledge about principles for logic-based representation and reasoning.
- Ability to understand knowledge-engineering process
- Ability to implement production systems, frames, inheritance systems and approaches to handle uncertain or incomplete knowledge.

Course Contents

UNIT-I	The Key Concepts	07 Hours
Knowledge, Representation, Reasoning, Why knowledge representation and reasoning, Role of logic. Logic: Historical background, Representing knowledge in logic, Varieties of logic, Name, Type, Measures, Unity Amidst diversity		
UNIT-II	**Ontology**	**08 Hours**
Ontological categories, Philosophical background, Top-level categories, Describing physical entities, Defining abstractions, Sets, Collections, Types and Categories, Space and Time		
UNIT-III	**Knowledge Representations**	**08 Hours**

Knowledge Engineering, Representing structure in frames, Rules and data, Object-oriented systems, Natural language Semantics, Levels of representation		
UNIT-IV	**Processes**	**07 Hours**
Times, Events and Situations, Classification of processes, Procedures, Processes and Histories, Concurrent processes, Computation, Constraint satisfaction, Change Contexts: Syntax of contexts, Semantics of contexts, First-order reasoning in contexts, Modal reasoning in contexts, Encapsulating objects in contexts.		
UNIT-V	**Knowledge Soup**	**08 Hours**
Vagueness, Uncertainty, Randomness and Ignorance, Limitations of logic, Fuzzy logic, Nonmonotonic Logic, Theories, Models and the world, Semiotics Knowledge Acquisition and Sharing: Sharing Ontologies, Conceptual schema, Accommodating multiple paradigms, Relating different knowledge representations, Language patterns, Tools for knowledge acquisition		

TEXT BOOKS:

1. Knowledge Representation logical, Philosophical, and Computational Foundations by John F. Sowa, Thomson Learning.
2. Knowledge Representation and Reasoning by Ronald J. Brachman, Hector J. Levesque, Elsevier.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(F): SPEECH AND VIDEO PROCESSING
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- To make students understand speech and video processing techniques

Learning Course Outcome:

After successful completion of the course, students will able to:

- Describe the mechanisms of human speech production systems and methods for speech feature extraction.
- Understand basic algorithms of speech analysis and speech recognition.
- Explain basic techniques in digital video processing, including imaging characteristics and sensors.
- Apply motion estimation and object tracking algorithms on video.

Course Contents

UNIT-I	Speech processing concepts	07 Hours

The speech production mechanism, Discrete time speech signals, Pole-Zero modeling of speech, relevant properties of the fast Fourier transform for speech recognition, convolution, linear and nonlinear filter banks, spectral estimation of speech using DFT. Linear Prediction analysis of speech.

UNIT-II	Speech recognition	08 Hours

Feature extraction for speech, static and dynamic feature for speech recognition, MFCC, LPCC, Distance measures, vector quantization models, Gaussian Mixture model, HMM.

UNIT-III	Multi-Dimensional Signals and Systems	08 Hours

Multi-Dimensional Signals, Multi-Dimensional Transforms, Multi-Dimensional Systems, Multi- Dimensional Sampling Theory, Sampling Structure Conversion Digital Images and Video: Human Visual System and Color, Digital Video		
UNIT-IV	**Motion Estimation**	**07 Hours**
Image Formation, Motion Models, 2D Apparent-Motion Estimation, Differential Methods, Matching Methods, Nonlinear Optimization Methods, Transform-Domain Methods, 3D Motion and Structure Estimation.		
UNIT-V	**Video Segmentation and Tracking**	**08 Hours**
Image Segmentation, Change Detection, Motion Segmentation, Motion Tracking, Image and Video Matting, Performance Evaluation.		

TEXT BOOKS:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series
2. Digital Video processing, A Murat Tekalp, 2nd edition, Prentice Hall.

REFERENCE BOOKS:

1. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.
2. Video Processing and Communications, Yao Wang, J. Osternann and Qin Zhang, Pearson Education.
3. "Speech and Audio Signal Processing", B.Gold and N. Morgan, Wiley.
4. "Digital image sequence processing, Compression, and analysis", Todd R. Reed, CRC Press.
5. "Handbook of Image and Video processing", Al Bovik, Academic press, second Edition.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(G): INTERNET of THINGS LAB
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Web Browser Basic knowledge

Course Objective:

- To introduce the raspberry PI platform, that is widely used in IoT applications
- To introduce the implementation of distance sensor on IoT devices.

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Ability to introduce the concept of M2M (machine to machine) with necessary protocols and get awareness in implementation of distance sensor
2. Get the skill to program using python scripting language which is used in many IoT devices.

Course Contents

List of Experiments

1. Using Raspberry pi
   a. Calculate the distance using a distance sensor.
   b. Interface an LED and switch with Raspberry pi.
   c. Interface an LDR with Raspberrry Pi.
2. Using Arduino
   a. Calculate the distance using a distance sensor.
   b. Interface an LED and switch with Aurdino.
   c. Interface an LDR with Aurdino
   d. Calculate temperature using a temperature sensor.
3. Using Node MCU
   a. Calculate the distance using a distance sensor.
   b. Interface an LED and switch with Raspberry pi.
   c. Interface an LDR with Node MCU
   d. Calculate temperature using a temperature sensor.
4. Installing OS on Raspberry Pi
   a) Installation using PiImager
   b) Installation using image file
      - Downloading an Image
      - Writing the image to an SD card

- using Linux
- using Windows
- Booting up Follow the instructions given in the URL https://www.raspberrypi.com/documentation/computers/getting-started.html

5. Accessing GPIO pins using Python
   a) Installing GPIO Zero library.
      update your repositories list:
      install the package for Python 3:
   b) Blinking an LED connected to one of the GPIO pin
   c) Adjusting the brightness of an LED Adjust the brightness of an LED (0 to 100, where 100 means maximum brightness) using the in-built PWM wavelength.
6. Create a DJANGO project and an app.
7. Create a DJANGO view for weather station REST API
8. Create DJANGO template
9. Configure MYSQL with DJANGO framework

TEXT BOOKS:

1. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547.
2. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014, ISBN: 9789350239759.

REFERENCE BOOKS:

1. Bernd Scholz-Reiter, Florian Michahelles, "Architecting the Internet of Things", ISBN 978-3- 642-19156-5 e-ISBN 978-3-642-19157-2, Springer, 2016
2. N. Ida, Sensors, Actuators and Their Interfaces, Scitech Publishers, 2014.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(H): SCRIPTING LANGUAGES LAB
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Any High level programming language (C, C++)

Course Objective:

- To Understand the concepts of scripting languages for developing web based projects
- To understand the applications the of Ruby, TCL, Perl scripting languages

Learning Course Outcome:

After successful completion of the course, students will able to:

1. Ability to understand the differences between Scripting languages and programming languages
2. Gain some fluency programming in Ruby, Perl, TCL.

Course Contents

LIST OF EXPERIMENTS

1. Write a Ruby script to create a new string which is n copies of a given string where n is a non-negative integer
2. Write a Ruby script which accept the radius of a circle from the user and compute the parameter and area.
3. Write a Ruby script which accept the users first and last name and print them in reverse order with a space between them
4. Write a Ruby script to accept a filename from the user print the extension of that
5. Write a Ruby script to find the greatest of three numbers
6. Write a Ruby script to print odd numbers from 10 to 1
7. Write a Ruby script to check two integers and return true if one of them is 20 otherwise return their sum
8. Write a Ruby script to check two temperatures and return true if one is less than 0 and the other is greater than 100
9. Write a Ruby script to print the elements of a given array
10. Write a Ruby program to retrieve the total marks where subject name and marks of a student stored in a hash
11. Write a TCL script to find the factorial of a number
12. Write a TCL script that multiplies the numbers from 1 to 10
13. Write a TCL script for sorting a list using a comparison function

14. Write a TCL script to (i) create a list (ii) append elements to the list (iii) Traverse the list (iv) Concatenate the list
15. Write a TCL script to comparing the file modified times.
16. Write a TCL script to Copy a file and translate to native format.
17. a) Write a Perl script to find the largest number among three numbers.
    b) Write a Perl script to print the multiplication tables from 1-10 using subroutines.
18. Write a Perl program to implement the following list of manipulating functions
    a) Shift
    b) Unshift
    c) Push
19. a) Write a Perl script to substitute a word, with another word in a string.
    b) Write a Perl script to validate IP address and email address.
20. Write a Perl script to print the file in reverse order using command line arguments

TEXT BOOKS:

1. The World of Scripting Languages, David Barron,Wiley Publications.
2. Ruby Programming language by David Flanagan and Yukihiro Matsumoto O'Reilly
3. "Programming Ruby" The Pramatic Progammers guide by Dabve Thomas Second edition

REFERENCE BOOKS:

1. Open Source Web Development with LAMP using Linux Apache, MySQL, Perl and PHP, J.Lee and B. Ware (Addison Wesley) Pearson Education.
2. Perl by Example, E. Quigley, Pearson Education.
3. Programming Perl, Larry Wall, T. Christiansen and J. Orwant, O'Reilly, SPD.
4. Tcl and the Tk Tool kit, Ousterhout, Pearson Education.
5. Perl Power, J. P. Flynt, Cengage Learning.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	3	2.4	2.4	2	2	3	2.6	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-403-PEC(I): CRYPTOGRAPHY AND NETWORK SECURITY LAB
(Professional Elective – IV)
B.Tech. IV Year I Sem.

Practical Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 02	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Explain the objectives of information security
- Explain the importance and application of each of confidentiality, integrity, authentication and availability
- Understand various cryptographic algorithms.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand basic cryptographic algorithms, message and web authentication and security issues.
- Identify information system requirements for both of them such as client and server.
- Understand the current legal issues towards information security.

Course Contents

List of Experiments

1. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should XOR each character in this string with 0 and display the result.
2. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should AND or and XOR each character in this string with 127 and display the result.
3. Write a Java program to perform encryption and decryption using the following algorithms
   a. Ceaser cipher b. Substitution cipher c. Hill Cipher
4. Write a C/JAVA program to implement the DES algorithm logic.
5. Write a C/JAVA program to implement the Blowfish algorithm logic.
6. Write a C/JAVA program to implement the Rijndael algorithm logic.
7. Write the RC4 logic in Java Using Java cryptography; encrypt the text "Hello world" using Blowfish. Create your own key using Java key tool.
8. Write a Java program to implement the RSA algorithm.
9. Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript.
10. Calculate the message digest of a text using the SHA-1 algorithm in JAVA.
11. Calculate the message digest of a text using the MD5 algorithm in JAVA.

TEXT BOOKS:

1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6th Edition
2. Cryptography and Network Security: Atul Kahate, McGraw Hill, 3rd Edition

REFERENCE BOOKS:

1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.
2. Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 3rd Edition
3. Information Security, Principles, and Practice: Mark Stamp, Wiley India.
4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH
5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning
6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-411-PCC: DEEP LEARNING
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 2 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Machine Learning

Course Objective:

1. **Understand the Fundamentals**: To provide a strong foundation in the principles of neural networks and deep learning.
2. **Explore Deep Architectures**: To explore various deep learning models such as CNNs, RNNs, and advanced architectures like GANs and Transformers.
3. **Apply Deep Learning to Real-world Problems**: To equip students with the skills to apply deep learning algorithms in areas such as computer vision, natural language processing, and other real-world domains.
4. **Develop Practical Skills**: To enable students to implement deep learning models using frameworks like TensorFlow and PyTorch.
5. **Promote Research and Innovation**: To encourage students to engage with cutting-edge research in the field of deep learning and AI, fostering innovation and new applications.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Understand Neural Network Concepts
Explain the concepts of artificial neural networks, activation functions, and the process of training models using backpropagation and gradient descent.

CO2: Build and Train Deep Neural Networks
Design and implement deep neural networks, optimize their performance using various techniques, and apply regularization methods to improve model generalization.

CO3: Apply Convolutional Neural Networks
Develop convolutional neural networks (CNNs) and use them for image processing tasks such as classification, object detection, and segmentation.

CO4: Implement Recurrent Neural Networks
Apply recurrent neural networks (RNNs), including LSTMs and GRUs, for sequence prediction tasks such as time-series forecasting and natural language processing.

CO5: Work with Advanced Architectures
Understand and implement advanced deep learning models like GANs, Autoencoders, and Transformers for tasks like image generation, unsupervised learning, and language modeling.

CO6: Utilize Deep Learning Frameworks

Demonstrate proficiency in using deep learning frameworks such as TensorFlow and PyTorch to build and train models efficiently.

CO7: Apply Deep Learning in Real-world Scenarios
Analyze real-world problems, design appropriate deep learning solutions, and critically assess the performance and ethical implications of AI systems.

CO8: Engage with Deep Learning Research
Read, understand, and present state-of-the-art research papers in deep learning, identifying trends and areas of innovation in the field.

Course Contents

UNIT-I	Introduction to Deep Learning	07 Hours
Overview of Machine Learning & AI, Introduction to Neural Networks, Perceptron, Linear Separability, Activation Functions (ReLU, Sigmoid, Tanh), Multilayer Perceptrons (MLP) Forward and Backward Propagation, Loss Functions (MSE, Cross Entropy), Gradient Descent, Stochastic Gradient Descent, Overfitting, Regularization Techniques (L1/L2 Regularization, Dropout)		
UNIT-II	**Deep Neural Networks**	**08 Hours**
Deep Networks and Training Deep Models, Introduction to TensorFlow and PyTorch Optimizers (Adam, RMSProp), Batch Normalization, Vanishing and Exploding Gradient Problems, Xavier and He Initialization		
UNIT-III	**Convolutional Neural Networks (CNNs)**	**08 Hours**
Introduction to CNNs, Convolution Operations, Pooling Layers, Applications of CNNs (Image Classification, Object Detection), Architectures: LeNet, AlexNet, VGG, GoogLeNet, ResNet Transfer Learning in CNNs		
UNIT- IV	**Recurrent Neural Networks (RNNs)**	**07 Hours**
Introduction to RNNs, Sequence Models and Time Series Data, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Applications of RNNs in NLP, Word Embeddings (Word2Vec, GloVe), Seq2Seq Models		
UNIT- V	**Advanced Topics**	**08 Hours**
Generative Adversarial Networks (GANs), Autoencoders, Variational Autoencoders (VAE), Attention Mechanism, Transformers (Self-Attention, Multi-Head Attention), BERT and GPT Models		
UNIT- VI	**Reinforcement Learning**	**08 Hours**

Introduction to Reinforcement Learning, Q-learning, Deep Q Networks, Policy Gradients, Case Studies: Computer Vision, Natural Language Processing, Speech Recognition, Applications in Autonomous Systems, Healthcare, and Finance, Research Paper Presentations

Real-world Applications and Ethics in AI

Assignments/Projects:

- **Assignment 1:** Implementation of MLP for a Classification Problem
- **Assignment 2:** CNN-based Image Classifier (e.g., MNIST, CIFAR-10)
- **Assignment 3:** RNN/LSTM for Time Series or Text Prediction

Project: Build a Deep Learning Model for a Real-world Application (e.g., sentiment analysis, image classification, chatbot)

Textbooks/Resources:

- **Primary Textbook:**
  - Ian Goodfellow, Yoshua Bengio, Aaron Courville, "Deep Learning," MIT Press.
- **Supplementary Readings:**
  - François Chollet, "Deep Learning with Python"
  - Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow"

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-412-PCC: PATTERN RECOGNITION
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Programming for problem solving

- Computer Oriented Statistical Methods

Course Objective:

- Introducing fundamental concepts, theories, and algorithms for pattern recognition and machine learning.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the importance of pattern recognition and its representation
- Analyza the variants of NN algorithm
- Understand the necessity of Hidden markov models, decision tree and SVM for classification
- Understand different types of clustering algorithms

Course Contents

UNIT-I	Introduction: Pattern Recognition	07 Hours

Data Sets for Pattern Recognition, Different Paradigms for Pattern Recognition. Representation: Data Structures for Pattern Representation, Representation of Clusters, Proximity Measures, Size of Patterns, Abstractions of the Data Set, Feature Extraction, Feature Selection, Evaluation of Classifier, Evaluation of Clustering.

UNIT-II	Nearest Neighbor Based Classifier	08 Hours

Nearest Neighbor Algorithm, Variants of the NN Algorithm, use of the Nearest Neighbor Algorithm for Transaction Databases, Efficient Algorithms, Data Reduction, Prototype Selection. Bayes Classifier: Bayes Theorem, Minimum Error Rate Classifier, Estimation of Probabilities, Comparison with the NNC, Naïve Bayes Classifier, Bayesian Belief Network.

UNIT-III	Hidden Markov Models	08 Hours

Markov Models for Classification, Hidden Morkov Models, Classification using HMMs. Decision Trees: Introduction, Decision Tree for Pattern Classification, Construction of Decision Trees, Splitting at the Nodes, Overfitting and Pruning, Examples of Decision Tree Induction.

UNIT-IV	Support Vector Machines	07 Hours
Introduction, Learning the Linear Discriminant Functions, Neural Networks, SVM for Classification. Combination of Classifiers: Introduction, Methods for Constructing Ensembles of Classifiers, Methods for Combining Classifiers.		
UNIT-V	**Clustering**	**08 Hours**
Importance of clustering, Hierarchical Algorithms, Partitional Clustering, Clustering Large Data Sets. An Application-Hand Written Digit Recognition: Description of the Digit Data, Preprocessing of Data, Classification Algorithms, Selection of Representative Patterns, Results.		

TEXT BOOK:

1. Pattern Recognition: An Algorithmic Approach: Murty, M. Narasimha, Devi, V. Susheela, Spinger Pub, 1st Ed.

REFERENCE BOOKS:

1. Machine Learning - Mc Graw Hill, Tom M. Mitchell.
2. Fundamentals Of Speech Recognition: Lawrence Rabiner and Biing- Hwang Juang. Prentice Hall Pub.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-415-MDM(A): CAPSTONE: BLOCKCHAIN IN REAL-WORLD APPLICATIONS
(MDM Basket – 1)
B.Tech. IV Year II Sem.

Teaching Scheme: P:02	Credit P: 02	Examination Scheme: Total : 50 Marks

Course Prerequisites:
Knowledge of Blockchain

Course Objective:

1. To enable students to design and implement blockchain solutions for practical challenges.
2. To explore use cases in domains like supply chain management, finance, healthcare, and sustainability.
3. To develop expertise in blockchain platforms, tools, and consensus mechanisms.
4. To ensure security, scalability, and efficiency in blockchain-based solutions.
5. To foster innovation through real-world problem-solving and teamwork.

Learning Course Outcome:
After successful completion of the course, students will able to:

CO1: Analyze and define the requirements of blockchain-based applications.
CO2: Design and implement blockchain solutions tailored to specific industries.
CO3: Integrate security and privacy mechanisms into blockchain systems.
CO4: Test, evaluate, and optimize blockchain solutions for scalability and efficiency.
CO5: Demonstrate a blockchain-based system to solve real-world problems, such as supply chain tracking or carbon credit systems.

Course Contents

UNIT-I	Problem Identification and Requirement Analysis	03 Hours

- Identifying potential use cases for blockchain in real-world domains.
- Evaluating technical, legal, and social implications of blockchain adoption.

UNIT-II	Solution Design and Architecture	03 Hours

- Designing blockchain workflows and smart contracts.
- Selecting suitable platforms and consensus mechanisms (e.g., Ethereum, Hyperledger, Polygon).

UNIT-III	Implementation	04 Hours

- Setting up blockchain environments and smart contract coding.
- Deployment and integration of blockchain solutions with existing systems.

UNIT- IV	Testing and Optimization	04 Hours

- Performance evaluation: Throughput, latency, and fault tolerance.
- Scalability and interoperability testing.

UNIT- V	Case Studies and Documentation	04 Hours

- Examples: Blockchain for supply chain tracking, carbon credit management, or digital identity systems.
- Report writing, system documentation, and project presentation.

Reference Books:

1. **Mastering Blockchain** by Imran Bashir
2. **Blockchain Basics** by Daniel Drescher
3. **Blockchain for Dummies** by Tiana Laurence
4. IEEE and ACM papers on blockchain applications.

CO's-PO's & PSO's MAPPING

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 1	3	3	2	2	3	2	1	-	-	-	2	2	3	2	1
CO 2	3	3	3	3	3	3	2	-	-	-	2	2	3	3	2
CO 3	3	3	3	3	3	3	3	-	-	-	2	2	3	3	2
CO 4	3	3	3	3	3	3	2	-	-	-	2	3	3	3	2
CO 5	3	3	3	3	3	3	3	-	-	2	2	3	3	3	3
Avg.	3	3	2.8	2.8	3	2.8	2.2	-	-	0.4	2	2.4	3	2.8	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-415-MDM(B): CAPSTONE: BUILDING SECURE SYSTEMS
(MDM Basket – 2)
B.Tech. IV Year II Sem.

Teaching Scheme: P:02	Credit P: 02	Examination Scheme: Total : 50 Marks

Course Prerequisites:
Knowledge of Cyber Security

Course Objective:
1. To design and implement secure systems for IoT and cybersecurity applications.
2. To explore security challenges in IoT and develop solutions for system hardening.
3. To apply incident response frameworks for real-time threat mitigation.
4. To integrate security policies and tools for end-to-end system protection.
5. To build a deployable secure system as a capstone project.

Learning Course Outcome:
After successful completion of the course, students will able to:
CO1: Analyze security requirements for IoT systems and networks.
CO2: Design secure system architectures using industry best practices.
CO3: Implement and test robust security measures, including encryption and authentication.
CO4: Develop incident response plans for real-world scenarios.
CO5: Demonstrate and document a secure system prototype.

Course Contents

UNIT-I	Problem Identification and Security Analysis	03 Hours

- Identifying security gaps in IoT-based systems.
- Threat modeling and risk assessment.

UNIT-II	System Design and Architecture	03 Hours

- Designing secure IoT systems with layered security models.
- Integrating encryption, authentication, and secure communication.

UNIT-III	Implementation and Integration	04 Hours

- Securing IoT protocols (e.g., MQTT, CoAP).
- Testing and validating security controls.

UNIT- IV	Incident Response and Mitigation	04 Hours
• Developing real-time incident response systems. • Case studies: Handling DDoS attacks and malware threats.		
UNIT- V	**Documentation and Presentation**	**04 Hours**
• Example: Secure IoT-based home automation or incident response system. • Report and final project demonstration.		

Reference Books:

1. **Cybersecurity for IoT** by Shancang Li and Li Da Xu
2. **IoT Security: Advances in Authentication** by Abhik Chaudhuri
3. NIST Cybersecurity Framework and related publications.

CO's-PO's & PSO's MAPPING

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 1	3	3	2	2	3	2	1	-	-	-	2	2	3	2	1
CO 2	3	3	3	3	3	3	2	-	-	-	2	2	3	3	2
CO 3	3	3	3	3	3	3	3	-	-	-	2	2	3	3	2
CO 4	3	3	3	3	3	3	2	-	-	-	2	3	3	3	2
CO 5	3	3	3	3	3	3	3	-	-	2	2	3	3	3	3
Avg.	3	3	2.8	2.8	3	2.8	2.2	-	-	0.4	2	2.4	3	2.8	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-415-MDM(c): CAPSTONE: IMAGE PROCESSING IN REAL-WORLD APPLICATIONS
(MDM Basket – 3)
B.Tech. IV Year II Sem.

Teaching Scheme: P:02	Credit P: 02	Examination Scheme: Total: 50 Marks

Course Prerequisites:
Knowledge of Image Processing

Course Objective:
1. To implement image processing algorithms for solving practical challenges.
2. To explore applications in medical imaging, satellite image analysis, and automation.
3. To develop expertise in feature extraction and object detection.
4. To integrate machine learning and deep learning for image-based tasks.
5. To build a functional image processing system for a specific use case.

Learning Course Outcome:
After successful completion of the course, students will able to:
CO1: Understand real-world requirements for image processing applications.
CO2: Apply advanced image processing techniques for problem-solving.
CO3: Integrate deep learning models for object detection and classification.
CO4: Optimize systems for speed, accuracy, and scalability.
CO5: Deliver a project prototype addressing a specific application.

Course Contents

UNIT-I	Problem Analysis and Dataset Preparation	03 Hours

- Identifying potential use cases for blockchain in real-world domains.
- Evaluating technical, legal, and social implications of blockchain adoption.
- Identifying application areas: Medical imaging, satellite data, etc.
- Acquiring and preprocessing image datasets.

UNIT-II	Algorithm Design and Implementation	03 Hours

- Applying enhancement, segmentation, and feature extraction methods.
- Integrating deep learning frameworks like TensorFlow and PyTorch.

UNIT-III	Optimization and Testing	04 Hours

- Evaluating accuracy using performance metrics.
- Real-world constraints and optimizations.

UNIT- IV	Case Studies and Presentation	04 Hours
• Examples: Automated medical diagnosis or satellite image analysis. • Final project demonstration and documentation.		

Reference Books:

1. **Digital Image Processing** by Gonzalez and Woods
2. **Deep Learning for Computer Vision** by Rajalingappaa Shanmugamani
3. Research papers and online repositories like Kaggle datasets and GitHub.

CO's-PO's & PSO's MAPPING

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO 1	3	3	2	2	3	2	1	-	-	-	2	2	3	2	1
CO 2	3	3	3	3	3	3	2	-	-	-	2	2	3	3	2
CO 3	3	3	3	3	3	3	3	-	-	-	2	2	3	3	2
CO 4	3	3	3	3	3	3	2	-	-	-	2	3	3	3	2
CO 5	3	3	3	3	3	3	3	-	-	2	2	3	3	3	3
Avg.	3	3	2.8	2.8	3	2.8	2.2	-	-	0.4	2	2.4	3	2.8	2

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-413-PEC(A): SOCIAL NETWORK ANALYSIS
(Professional Elective – V)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

- Web Technologies
- Computer Networks
- Data Warehousing and Data Mining

Course Objective:

1. Understand the concepts of social media
2. Learn the mechanisms for social network analysis
3. Analysis of widely used services such as email, Wikis, Twitter, flickr, YouTube, etc.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Ability to construct social network maps easily
- Gain skills in tracking the content flow through the social media
- Understand NodeXL use to perform social network analysis

Course Contents

UNIT-I	Introduction	07 Hours
Social Media and Social Networks, Social Media: New Technologies of Collaboration Social Network Analysis: Measuring, Mapping, and Modelling collections of Connections.		
UNIT-II	**Preparing Data**	**08 Hours**
NodeXL, Layout, Visual Design, and Labelling, Calculating and Visualising Network Metrics, Preparing Data and Filtering, Clustering and Grouping.		
UNIT-III	**CASE STUDIES**	**08 Hours**
Email: The lifeblood of Modern Communication. Thread Networks: Mapping Message Boards and Email Lists, Twitter: Conversation, Entertainment and Information		
UNIT-IV	**CASE STUDIES**	**07 Hours**

Visualizing and Interpreting Facebook Networks, WWW Hyperlink Networks		
UNIT-V	**CASE STUDIES**	**08 Hours**
You Tube: Contrasting Patterns of Content Interaction, and Prominence. Wiki Networks: Connections of Creativity and Collaboration		

TEXT BOOK:

1. Hansen, Derek, Ben Sheiderman, Marc Smith, Analyzing Social Media Networks with NodeXL: Insights from a Connected World, Morgan Kaufmann, 2011.

REFERENCE BOOKS:

1. Avinash Kaushik, Web Analytics 2.0: The Art of Online Accountability, Sybex, 2009.
2. Marshall Sponder, Social Media Analytics: Effective Tools for Building, Interpreting and Using Metrics, 1st Edition, MGH, 2011.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-413-PEC(B): FEDERATED MACHINE LEARNING
(Professional Elective – V)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

- The prerequisite knowledge for this course includes machine learning, basic computer systems and basic programming skills.

Course Objective:

1. Understand the key concepts and issues behind Federated Learning
2. Get familiar with key theoretical results of Federated Learning

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the basics on privacy-preserving ML
- Analyze the key concepts of Distributed ML and FL
- Understand the key concepts and applications of Horizontal FL and Vertical FL
- Motivates the intensive mechanism design for FL
- Analyze the concepts of federated reinforcement learning

Course Contents

UNIT-I	Introduction	07 Hours
Motivation, Federated Learning as a Solution, The Definition of Federated Learning, Categories of Federated Learning, Current Development in Federated Learning, Research Issues in Federated Learning, Open-Source Projects, Standardization Efforts, The Federated AI Ecosystem Background: Privacy-Preserving Machine Learning, PPML and Secure ML, Threat and Security Models, Privacy Threat Models, Adversary and Security Models, Privacy Preservation Techniques, Secure Multi-Party Computation, Homomorphic Encryption, Differential Privacy		
UNIT-II	**Distributed Machine Learning**	**08 Hours**
Introduction to DML, The Definition of DML, DML Platforms, Scalability- Motivated DML, Large-Scale Machine Learning, Scalability-Oriented DML Schemes, Privacy-Motivated DML, Privacy-Preserving Decision Trees, Privacy-Preserving Techniques, Privacy-Preserving DML Schemes, Privacy-Preserving Gradient Descent, Vanilla Federated Learning, Privacy-Preserving Methods		
UNIT-III	**Horizontal Federated Learning**	**08 Hours**

The Definition of HFL, Architecture of HFL, The Client- Server Architecture, The Peer-to-Peer Architecture, Global Model Evaluation, The Federated Averaging Algorithm, Federated Optimization, The FedAvg Algorithm, The Secured FedAvg Algorithm, Improvement of the FedAvg Algorithm, Communication Efficiency, Client Selection Vertical Federated Learning: The Definition of VFL, Architecture of VFL, Algorithms of VFL, Secure Federated Linear Regression, Secure Federated Tree-Boosting		
UNIT-IV	**Federated Transfer Learning**	**07 Hours**
Heterogeneous Federated Learning, Federated Transfer Learning, The FTL Framework, Additively Homomorphic Encryption, The FTL Training Process, The FTL Prediction Process, Security Analysis, Secret Sharing-Based FTL Incentive Mechanism Design for Federated Learning: Paying for Contributions, Profit- Sharing Games, Reverse Auctions, A Fairness-Aware Profit Sharing Framework, Modeling Contribution, Modeling Cost, Modeling Regret, Modeling Temporal Regret, The Policy Orchestrator, Computing Payoff Weightage		
UNIT-V	**Federated Learning**	**08 Hours**
Vision, Language, and Recommendation: Federated Learning for Computer Vision, Federated CV, Federated Learning for NLP, Federated NLP, Federated Learning for Recommendation Systems, Recommendation Model, Federated Recommendation System Federated Reinforcement Learning: Introduction to Reinforcement Learning, Policy, Reward, Value Function, Model of the Environment, RL Background Example, Reinforcement Learning Algorithms, Distributed Reinforcement Learning, Asynchronous Distributed Reinforcement Learning, Synchronous Distributed Reinforcement Learning, Federated Reinforcement Learning, Background and Categorization		
TEXT BOOK: 1. Federated Learning, Qiang Yang, Yang Liu, Yong Cheng, Yan Kang, Tianjian Chen, and Han Yu Synthesis Lectures on Artificial Intelligence and Machine Learning 2019.		

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-413-PEC(c): AUGMENTED REALITY & VIRTUAL REALITY
(Professional Elective –V)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Provide a foundation to the fast-growing field of AR and make the students aware of the various AR concepts.
- To give historical and modern overviews and perspectives on virtual reality. It describes the fundamentals of sensation, perception, technical and engineering aspects of virtual reality systems.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Describe how AR systems work and list the applications of AR.
- Understand the software architectures of AR.
- Understand the Visual perception and rendering in VR
- Understand the interaction, auditory perception and rendering in VR

Course Contents

UNIT-I	Introduction to Augmented Reality	07 Hours
Augmented Reality - Defining augmented reality, history of augmented reality, Examples, Related fields. Displays: Multimodal Displays, Visual Perception, Requirements and Characteristics, Spatial Display Model, Visual Displays. Tracking: Tracking, Calibration, and Registration, Coordinate Systems, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors		
UNIT-II	**Computer Vision for Augmented Reality**	**08 Hours**
Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Outdoor Tracking. Interaction: Output Modalities, Input Modalities, Tangible Interfaces, Virtual User Interfaces on Real Surfaces, Augmented Paper, Multi-view Interfaces, Haptic Interaction Software Architectures: AR Application Requirements, Software Engineering Requirements, Distributed Object Systems, Dataflow, Scene Graphs		
UNIT-III	**Introduction to Virtual Reality**	**08 Hours**

Defining Virtual Reality, History of VR, Human Physiology and Perception

The Geometry of Virtual Worlds: Geometric Models, Axis-Angle Representations of Rotation, Viewing Transformations. Light and Optics: Basic Behavior of Light, Lenses, Optical Aberrations, The Human Eye, Cameras, Displays

UNIT-IV	The Physiology of Human Vision	07 Hours

From the Cornea to Photoreceptors, From Photoreceptors to the Visual Cortex, Eye Movements, Implications for VR Visual Perception: Visual Perception - Perception of Depth, Perception of Motion, Perception of Color Visual Rendering: Visual Rendering -Ray Tracing and Shading Models, Rasterization, Correcting Optical Distortions, Improving Latency and Frame Rates, Immersive Photos and Videos.

UNIT-V	Motion in Real and Virtual Worlds	08 Hours

Velocities and Accelerations, The Vestibular System, Physics in the Virtual World, Mismatched Motion and Vection

Interaction: Motor Programs and Remapping, Locomotion, Social Interaction

Audio: The Physics of Sound, The Physiology of Human Hearing, Auditory Perception, Auditory Rendering

TEXT BOOKS:

1. Augmented Reality: Principles & Practice by Schmalstieg / Hollerer, Pearson Education India;First edition (12 October 2016),ISBN-10: 9332578494
2. Virtual Reality, Steven M. LaValle, Cambridge University Press, 2016

REFERENCE BOOKS:

1. Allan Fowler-AR Game Development‖, 1st Edition, A press Publications, 2018, ISBN 978-1484236178
2. Understanding Virtual Reality: Interface, Application and Design, William R Sherman and Alan B Craig, (The Morgan Kaufmann Series in Computer Graphics)". Morgan Kaufmann Publishers, San Francisco, CA, 2002
3. Developing Virtual Reality Applications: Foundations of Effective Design, Alan B Craig, William R Sherman and Jeffrey D Will, Morgan Kaufmann, 2009
4. Designing for Mixed Reality, Kharis O'Connell Published by O'Reilly Media, Inc., 2016, ISBN: 9781491962381
5. Sanni Siltanen- Theory and applications of marker-based augmented reality. Julkaisija — Utgivare Publisher. 2012. ISBN 978-951-38-7449-0
6. Gerard Jounghyun Kim, "Designing Virtual Systems: The Structured Approach", 2005.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-413-PEC(D): WEB SECURITY
(Professional Elective –V)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

1. Give an Overview of information security
2. Give an overview of Access control of relational databases

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Understand the Web architecture and applications
2. Understand client side and service side programming
3. Understand how common mistakes can be bypassed and exploit the application
4. Identify common application vulnerabilities

Course Contents

UNIT-I	The Web Security	07 Hours

The Web Security Problem, Risk Analysis and Best Practices

Cryptography and the Web: Cryptography and Web Security, Working Cryptographic Systems and Protocols, Legal Restrictions on Cryptography, Digital Identification.

UNIT-II	The Web's War on Your Privacy	08 Hours

Privacy-Protecting Techniques, Backups and Antitheft, Web Server Security, Physical Security for Servers, Host Security for Servers, Securing Web Applications.

UNIT-III	Database Security	08 Hours

Recent Advances in Access Control, Access Control Models for XML, Database Issues in Trust Management and Trust Negotiation, Security in Data Warehouses and OLAP Systems

UNIT-IV	Security Re-engineering for Databases	07 Hours

Concepts and Techniques, Database Watermarking for Copyright Protection, Trustworthy Records Retention, Damage Quarantine and Recovery in Data Processing Systems, Hippocratic Databases: Current Capabilities.

UNIT-V	Future Trends Privacy in Database Publishing	08 Hours

A Bayesian Perspective, Privacy-enhanced Location Based Access Control, Efficiently Enforcing the Security and Privacy Policies in a Mobile Environment.

TEXT BOOKS:

1. Web Security, Privacy and Commerce Simson G Arfinkel, Gene Spafford, O'Reilly.
   Handbook on Database security applications and trends Michael Gertz, Sushil Jajodia.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-413-PEC(e): AD-HOC & SENSOR NETWORKS
(Professional Elective – V)
B.Tech. IV Year II Sem.

Teaching Scheme: TH: - 3 Hours/Week	Credit TH: 03	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Computer Networks, Distributed Systems, Mobile Computing

Course Objective:

- To understand the challenges of routing in ad-hoc and sensor networks
- To understand various broadcast, mutlicast and geocasting protocols in ad hoc and sensor networks
- To understand basics of Wireless sensors, and Lower Layer Issues and Upper Layer Issues of WSN

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the concepts of sensor networks and applications
- Understand and compare the MAC and routing protocols for adhoc networks
- Understand the transport protocols of sensor networks

Course Contents

UNIT-I	Introduction to Ad Hoc Networks	07 Hours

Characteristics of MANETs, Applications of MANETs and Challenges of MANETs.

Routing in MANETs, Criteria for classification, Taxonomy of MANET routing algorithms, Topology-based routing algorithms- Proactive: DSDV, WRP; Reactive: DSR, AODV, TORA; Hybrid: ZRP; Position- based routing algorithms- Location Services-DREAM, Quorum-based, GLS; Forwarding Strategies, Greedy Packet, Restricted Directional Flooding-DREAM, LAR; Other routing algorithms-QoS Routing, CEDAR.

UNIT-II	Data Transmission	08 Hours

Broadcast Storm Problem, Rebroadcasting Schemes-Simple-flooding, Probability-based Methods, Area- based Methods, Neighbour Knowledge-based: SBA, Multipoint Relaying, AHBP. Multicasting: Tree-based: AMRIS, MAODV; Mesh-based: ODMRP, CAMP; Hybrid: AMRoute, MCEDAR.

UNIT-III	Geocasting	08 Hours
Data-transmission Oriented-LBM; Route Creation Oriented-GeoTORA, MGR. TCP over Ad Hoc TCP protocol overview, TCP and MANETs, Solutions for TCP over Ad hoc.		
UNIT-IV	**Basics of Wireless Sensors and Lower Layer Issues**	**07 Hours**
Applications, Classification of sensor networks, Architecture of sensor network, Physical layer, MAC layer, Link layer, Routing Layer.		
UNIT-V	**Upper Layer Issues of WSN**	**08 Hours**
Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs.		

TEXT BOOKS

1. Ad Hoc and Sensor Networks – Theory and Applications, *Carlos Corderio Dharma P.Aggarwal,* World Scientific Publications, March 2006, ISBN — 981-256-681-3
2. Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science, ISBN — 978-1-55860-914-3 (Morgan Kauffman)

REFERENCE BOOKS:

1. C. Siva Ram Murthy, B.S. Manoj Ad Hoc Wireless Networks: Architectures and Protocols.
2. Taieb Znati Kazem Sohraby, Daniel Minoli, Wireless Sensor Networks: Technology, Protocols and Applications, Wiley.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-413-PEC(F): DATA MINING
(Professional Elective – V)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites:

1. Database Management System
2. Probability and Statistics

Course Objective:

- Students will become acquainted with both the strengths and limitations of various data mining techniques like Association, Classification, Cluster and Outlier analysis.

Learning Course Outcome:

After successful completion of the course, students will able to:

- Understand the need of data mining and pre-processing techniques.
- Perform market basket analysis using association rule mining.
- Utilize classification techniques for analysis and interpretation of data.
- Identify appropriate clustering and outlier detection techniques to handle complex data.
- Understand the mining of data from web, text and time series data.

Course Contents

UNIT-I	Introduction to Data Mining	07 Hours

What Data mining? Kinds of Data, Knowledge Discovery process, Data Mining Functionalities, Kinds of Patterns, Major Issues in Data Mining. Data Objects and Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity, Data Pre-processing: Major Tasks in Data Pre-processing, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Data Discretization.

UNIT-II	Association Analysis	08 Hours

Basic Concepts, Market Basket Analysis, Apriori Algorithm, FP-growth, From Association Analysis to Correlation Analysis, Pattern Mining in Multilevel Associations and Multidimensional Associations.

UNIT-III	Classification	08 Hours

Basic Concepts, Decision Tree Induction, Bayes Classification Methods, Rule-Based Classification, Metrics for Evaluating Classifier Performance, Ensemble Methods, Multilayer Feed- Forward Neural Network, Support Vector Machines, k-Nearest-Neighbor Classifiers.

UNIT- IV	Overview of Basic Clustering Methods	07 Hours
Requirements for Cluster Analysis, Overview of Basic Clustering Methods, Partitioning Methods-k-Means, k-Medoids, Hierarchical Methods-AGENES, DIANA, BIRCH, Density-Based Method-DBSCAN, Outlier Analysis: Types of Outliers, Challenges of Outlier Detection, and Overview of Outlier Detection Methods		
UNIT- V	**Advanced Concepts**	**08 Hours**
Web Mining- Web Content Mining, Web Structure Mining, Web Usage Mining, Spatial Mining- Spatial Data Overview, Spatial Data Mining Primitives, Spatial Rules, Spatial Classification Algorithm, Spatial Clustering Algorithms, Temporal Mining- Modeling Temporal Events, Time Series, Pattern Detection, Sequences, Temporal Association Rules.		

TEXT BOOKS:

1. Jiawei Han, Micheline Kamber, Jian Pei., Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann/Elsevier, 2012.
2. Margaret H Dunham, Data Mining Introductory and Advanced Topics, 2nd Edition, Pearson Education, India, 2006.

REFERENCE BOOKS:

1. Data Mining Techniques, Arun K Pujari, 3rd Edition, Universities Press.
2. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne and Vipin Kumar, Introduction to Data Mining, 2nd Edition, Pearson Education India, 2021.
3. Amitesh Sinha, Data Warehousing, Thomson Learning, India, 2007.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-413-L-PEC(F): DATA MINING LAB
(Professional Elective – V)
B.Tech. IV Year II Sem.

Practical Scheme:	Credit	Examination Scheme:
TH: - 4 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: A course on "Database Management System

Course Objective:

- The course is intended to obtain hands-on experience using data mining software.
- Intended to provide practical exposure of the concepts in data mining algorithms

Learning Course Outcome:
After successful completion of the course, students will able to:

1. Apply preprocessing statistical methods for any given raw data.
2. Gain practical experience of constructing a data warehouse.
3. Implement various algorithms for data mining in order to discover interesting patterns from large amounts of data.
4. Apply OLAP operations on data cube construction.

Course Contents

LIST OF EXPERIMENTS

Experiments using Weka/ Pentaho/Python

1. Data Processing Techniques:
   (i) Data cleaning (ii) Data transformation – Normalization (iii) Data integration
2. Partitioning - Horizontal, Vertical, Round Robin, Hash based
3. Data Warehouse schemas – star, snowflake, fact constellation
4. Data cube construction – OLAP operations
5. Data Extraction, Transformations & Loading operations
6. Implementation of Attribute oriented induction algorithm
7. Implementation of apriori algorithm
8. Implementation of FP – Growth algorithm
9. Implementation of Decision Tree Induction
10. Calculating Information gain measures
11. Classification of data using Bayesian approach
12. Classification of data using K – nearest neighbour approach
13. Implementation of K – means algorithm
14. Implementation of BIRCH algorithm

15. Implementation of PAM algorithm
16. Implementation of DBSCAN algorithm

TEXT BOOKS:

1. Data Mining – Concepts and Techniques - JIAWEI HAN &MICHELINE KAMBER, Elsevier.
2. Data Warehousing, Data Mining &OLAP- Alex Berson and Stephen J. Smith- Tata McGraw-Hill Edition, Tenth reprint 2007

REFERENCE BOOK:

1. Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Anuj Karpatne, Introduction to Data Mining, Pearson Education.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-414--PEC(A): ROBOTIC PROCESS AUTOMATION
(Professional Elective – VI)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Introduce robotic process automation, techniques of automation using UIPath RPA tool.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the concepts of Robotic Process Automation.
- Apply the flow chart mechanism in various calculations.
- Applying UIPath tool for debugging process.
- Design system managing techniques.
- Create application for process automation using UIPath tool.

Course Contents

UNIT-I	Robotic Process Automation	07 Hours
Introduction, Scope and techniques of automation, Robotic process automation, Components of RPA, RPA platforms, About UiPath, UIPath Stack Uipath Studio, Uipath Robot, Types of Robots, UiPath Orchestrator, UIPath Studio Projects, User interface The User Interface: Task recorder, Advanced UI interactions: Input methods, Output methods.		
UNIT-II	**Sequence, Flowchart, and Control Flow**	**08 Hours**
Sequencing the workflow, Activities, Control Flow, various types of loops and decision-making. Data Manipulation: Variables and scope, Collections, Arguments – Purpose and use, Data table usage with examples, File operation with systematic example, CSV/Excel to data table and vice versa.		
UNIT-III	**Taking Control of the Controls**	**08 Hours**

Finding and attaching windows, Finding the control, Techniques for waiting for a control, Act on controls – mouse and keyboard activities, Handling events, revisit recorder, When to use OCR, Types of OCR available, How to use OCR Plugins and Extensions: Terminal Plugin, SAP Automation, Citrix automation and Credential management.		
UNIT-IV	**Handling User Events and Assistant Bots**	**07 Hours**
Assistant bots, Monitoring system event triggers, Monitoring image and element triggers, Launching an assistant bot on a keyboard event Exception Handling, Debugging, and Logging: Exception handling, Common exceptions and ways to handle them, Logging and taking screenshots, Debugging techniques, Collecting crash dumps, Error reporting.		
UNIT-V	**Managing and Maintaining the Code**	**08 Hours**
Project organization, nesting workflows, Reusability of workflows, commenting techniques, State Machine, when to use Flowcharts, State Machines, or Sequences, Using config files. Deploying and Maintaining the Bot: Publishing using publish utility, using Orchestration Server to control bots, deploy bots, License Management, Publishing and Managing updates		

TEXT BOOKS:

1. Learning Robotic Process Automation: Create Software robots and automate business processes with the leading RPA tool - UiPath: Create Software robots. with the leading RPA tool — UiPath Kindle Edition

REFERENCE BOOKS:

1. Robotic Process Automation A Complete Guide - 2020 Edition Kindle Edition.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-414-PEC(B): RANDOMIZED ALGORITHMS
(Professional Elective – VI)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- To introduce the power of randomization in the design of algorithms.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Appreciate the fundamentals of randomized algorithm design.
- Understand the fundamentals of Markov chains and the Monte Carlo method.
- Apply high probability analysis to selected randomized algorithms.
- Understand the Fingerprint and Pattern Matching techniques

Course Contents

UNIT-I	Introduction, A Min	07 Hours
Cut algorithm, Las Vegas and Monte Carlo, Binary Planar Partitions, A Probabilistic Recurrence Game-Theoretic Techniques: Game Tree Evaluation, The Minimax Principle		
UNIT-II	**Moments and Deviations**	**08 Hours**
Occupancy Problems, The Markov and Chebyshev Inequalities, Randomized Selection, Two Point sampling, The Coupon Collector's problem. Markov Chains and Random Walks: A 2-SAT example, Markov Chains, Random Walks on Graphs, Graph Connectivity		
UNIT-III	**Algebraic Techniques**	**08 Hours**
Fingerprinting and Freivald's Technique, Verifying Polynomial Identities, Perfect Matching in Graphs, Verifying Equality of Strings, A Comparison of Fingerprinting Techniques, Pattern Matching		
UNIT-IV	**Data Structures**	**07 Hours**

The Fundamental of Data-structures, Random Treaps, Skip Lists, Hash Tables Graph Algorithms: All Pairs Shortest Path, The Min- Cut Problem, Minimum Spanning Trees		
UNIT-V	**Geometric Algorithms**	**08 Hours**
Randomized Incremental Construction, Convex Hulls in the Plane, Duality, Half- Space Intersections, Dalaunay Triangulations, Trapezoidal Decompositions, Parallel and Distributed Algorithms: The PRAM Model, Sorting on a PRAM, Maximal Independent Sets, Perfect Matchings		

TEXT BOOKS:

1. Randomized Algorithms: Rajeev Motwani, Prabhakar Raghavan, cambridge University Press
2. Probability and Computing: Randomization and Probabilistic Techniques in Algorithms and Data Analysis by Eli Upfal and Michael Mitzenmacher.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-414-PEC(C): COGNITIVE COMPUTING
(Professional Elective – VI)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: Probability theory

Course Objective:

- To provide an understanding of the central challenges in realizing aspects of human cognition.
- To provide a basic exposition to the goals and methods of human cognition.
- To develop algorithms that use AI and machine learning along with human interaction and feedback to help humans make choices/decisions.
- To support human reasoning by evaluating data in context and presenting relevant findings along with the evidence that justifies the answers.

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand cognitive computing
- Plan and use the primary tools associated with cognitive computing.
- Plan and execute a project that leverages cognitive computing.
- Understand and develop the business implications of cognitive computing.

Course Contents

UNIT-I	Introduction to Cognitive Science	07 Hours

Understanding Cognition, IBM's Watson, Design for Human Cognition, Augmented Intelligence, Cognition Modeling Paradigms: Declarative/ logic-based computational cognitive modeling, connectionist models of cognition, Bayesian models of cognition, a dynamical systems approach to cognition.

UNIT-II	Cognitive Models	08 Hours

Cognitive Models of memory and language, computational models of episodic and semantic memory, modeling psycholinguistics. Cognitive Modeling: modeling the interaction of language, memory and learning, Modeling select aspects of cognition classical models of rationality, symbolic reasoning and decision-making.

UNIT-III	Formal models	08 Hours

Formal models of inductive generalization, causality, categorization and similarity, the role of analogy in problem solving, Cognitive Development Child concept acquisition. Cognition and Artificial cognitive architectures such as ACT-R, SOAR, OpenCog, CopyCat, Memory Networks.

UNIT-IV	Deep QA Architecture	07 Hours

Unstructured Information Management Architecture (UIMA), Structured Knowledge, Business Implications, Building Cognitive Applications, Application of Cognitive Computing and Systems.

TEXT BOOK:

1. The Cambridge Handbook of Computational Psychology by Ron Sun (ed.), Cambridge University Press.

REFERENCE BOOKS:

1. Judith S. Hurwitz, Marcia Kaufman, Adrian Bowles Cognitive Computing and Big Data Analytics, Wiley
2. Vijay V Raghavan, Venkat N. Gudivada, Venu Govindaraju, Cognitive Computing: Theory and Applications: Volume 35 (Handbook of Statistics), North Hollan.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation

BTDS-414-PEC(D): SEMANTIC WEB
(Professional Elective – VI)
B.Tech. IV Year II Sem.

Teaching Scheme:	Credit	Examination Scheme:
TH: - 3 Hours/Week	TH: 03	In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks

Course Prerequisites: None

Course Objective:

- Introduce Semantic Web Vision and learn Web intelligence
- Understanding about XML, RDF, RDFS, OWL
- Querying Ontology and Ontology Reasoning
- To learn Semantic Web Applications, Services and Technology
- To learn Knowledge Representation for the Semantic Web

Learning Course Outcome:
After successful completion of the course, students will able to:

- Understand the characteristics of the semantic web technology
- Understand the concepts of Web Science, semantics of knowledge resource and ontolog
- Describe logic semantics and inference with OWL.
- Use ontology engineering approaches in semantic applications
- Learn about web graph processing for various applications such as search engine, community detection

Course Contents

UNIT-I	Introduction	07 Hours
Introduction to Semantic Web, the Business Case for the Semantic Web, XML and Its Impact on the Enterprise.		
UNIT-II	**Web Services**	**08 Hours**
Uses, Basics of Web Services, SOAP, UDDI, Orchestrating Web Services, Securing Web Services, Grid Enabled and Semantic Web of Web Services.		
UNIT-III	**Resource Description Framework**	**08 Hours**
Features, Capturing Knowledge with RDF. **XML Technologies:** XPath, The Style Sheet Family: XSL, XSLT, and XSL FO, XQuery, XLink, XPointer, XInclude, XMLBase, XHTML, XForms, SVG.		
UNIT-IV	**Taxonomies and Ontologies**	**07 Hours**

Overview of Taxonomies, Defining the Ontology Spectrum, Topic Maps, Overview of Ontologies, Syntax, Structure, Semantics, and Pragmatics, Expressing Ontologies Logically, Knowledge Representation.

UNIT-V	Semantic Web Application	08 Hours

Semantic Web Services, e-Learning, Semantic Bioinformatics, Enterprise Application Integration, Knowledge Base. **Semantic Search Technology:** Search Engines, Semantic Search, Semantic Search Technology, Web Search Agents, Semantic Methods, Latent Semantic Index Search, TAP, Swoogle.

TEXT BOOKS:

1. The Semantic Web: A Guide to the Future of XML, Web Services, and Knowledge Management by Michael C. Daconta, Leo J. Obrst , Kevin T. Smith, Wiley Publishing, Inc.
2. Peter Mika, Social Networks and the Semantic Web, Springer

REFERENCE BOOKS:

1. Thinking on the Web - Berners Lee, Godel and Turing, Wiley Interscience
2. The Semantic Web: A Guide to the Future of XML, Web Services, and Knowledge Management by Michael C. Daconta, Leo J. Obrst , Kevin T. Smith, Wiley Publishing, Inc.
3. Semantic Web Technologies, Trends and Research in Ontology Based Systems, J. Davies, R. Studer, P. Warren, John Wiley & Sons.
4. Semantic Web and Semantic Web Services - Liyang Lu Chapman and Hall/CRC Publishers, (Taylor & Francis Group)
5. Information Sharing on the semantic Web - Heiner Stuckenschmidt; Frank Van Harmelen, Springer Publications.
6. Programming the Semantic Web, T. Segaran, C. Evans,J. Taylor, O'Reilly, SPD.

CO's-PO's & PSO's MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3
1	3	3	3	2	2	3	3	2
2	-	3	3	1	1	3	3	3
3	-	2	2	2	2	3	2	1
4	-	2	2	2	2	3	2	2
5	-	2	2	3	3	3	3	2
Avg	**3**	**2.4**	**2.4**	**2**	**2**	**3**	**2.6**	**2**

1 - low, 2 - medium, 3 - high, '-' - no correlation